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MATRIX ANALYSIS AND LINEAR ALGEBRA

Selected sections, adapted for

MA1522 Linear Algebra for Computing

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A reading companion to the course, not a replacement for its slides. The sections of Tyrtyshnikov's *Matrix Analysis and Linear Algebra* that bear on the MA1522 syllabus are arranged under the course's chapters and sections and rewritten in the course's notation and terminology. Course topics the book does not treat are supplied in subsections marked ADDED FOR MA1522. The book's material beyond the syllabus is omitted from this printing. Each subsection notes in the source the section of the book it comes from.

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Notation

The notation follows the MA1522 lecture slides.

Notation	Meaning
$\mathbf{u}, \mathbf{v}, \mathbf{w}$; $\mathbf{0}$	vectors in $\mathbb{R}^n$ (columns); the zero vector
$\mathbf{A} = \big(a_{ij}\big)_{m\times n}$	an $m \times n$ matrix with (i,j)-entry a_{ij}
$\mathbf{I}_n$, $\mathbf{0}_{m\times n}$	identity matrix of order n; zero matrix
$\mathbf{A}^T$, $\mathbf{A}^{-1}$	transpose; inverse
$(\mathbf{A} \mid \mathbf{b})$	augmented matrix of the system $\mathbf{Ax} = \mathbf{b}$
$R_i \leftrightarrow R_j$, $R_i + cR_j$, aR_i	elementary row operations ($a \neq 0$)
$\det(\mathbf{A})$, $\lvert\mathbf{A}\rvert$	determinant
$\mathbf{M}_{ij}$, A_{ij}	(i,j) matrix minor (a submatrix); (i,j)-cofactor $A_{ij} = (-1)^{i+j}\det\big(\mathbf{M}_{ij}\big)$
$\operatorname{adj}(\mathbf{A})$	adjoint of $\mathbf{A}$ (transpose of the matrix of cofactors)
$\mathbf{u}\cdot\mathbf{v}$, $\lVert\mathbf{u}\rVert$, $d(\mathbf{u},\mathbf{v})$	dot product, norm, distance $\lVert\mathbf{u}-\mathbf{v}\rVert$ in $\mathbb{R}^n$
$\operatorname{span}\{\mathbf{u}_1, ..., \mathbf{u}_k\}$	linear span
$\dim(V)$	dimension of a subspace V
$[\mathbf{v}]_S$	coordinate vector of $\mathbf{v}$ relative to the basis S
$\mathbf{P}$ with $[\mathbf{w}]_S = \mathbf{P}[\mathbf{w}]_T$	transition matrix from T to S
$\operatorname{Col}(\mathbf{A})$, $\operatorname{Row}(\mathbf{A})$, $\operatorname{Null}(\mathbf{A})$	column space, row space, nullspace
$\operatorname{rank}(\mathbf{A})$, $\operatorname{nullity}(\mathbf{A})$	rank; nullity $= \dim(\operatorname{Null}(\mathbf{A}))$
$V^\perp$	orthogonal complement of a subspace V
$\mathbf{w} = \mathbf{w}_p + \mathbf{w}_n$	orthogonal projection $\mathbf{w}_p$ of $\mathbf{w}$ onto V, and $\mathbf{w}_n \perp V$
$\mathbf{A} = \mathbf{QR}$, $\mathbf{A} = \mathbf{LU}$	QR and LU factorizations
$\operatorname{char}(\mathbf{A}) = \det(x\mathbf{I} - \mathbf{A})$	characteristic polynomial
$E_\lambda = \operatorname{Null}(\lambda\mathbf{I} - \mathbf{A})$	eigenspace of the eigenvalue λ
$\mathbf{A} = \mathbf{PDP}^{-1}$, $\mathbf{A} = \mathbf{PDP}^T$	diagonalization; orthogonal diagonalization
$\mathbf{A} = \mathbf{U\Sigma V}^T$	singular value decomposition
$T : \mathbb{R}^n \to \mathbb{R}^m$	linear transformation, $T(\mathbf{u}) = \mathbf{Au}$ with standard matrix $\mathbf{A}$
$[T]_S$, $\mathrm{R}(T)$, $\ker(T)$	matrix of T relative to a basis S (its columns are $T\big(\mathbf{u}_j\big)$); range; kernel

Some passages work over the complex numbers or in general vector spaces. There $\mathbf{A}^*$ denotes the conjugate transpose ($\mathbf{A}^* = \mathbf{A}^T$ for real matrices), and $\langle \mathbf{u}, \mathbf{v} \rangle$ a general inner product, which for real column vectors is $\mathbf{u} \cdot \mathbf{v} = \mathbf{v}^T \mathbf{u}$.

Chapter 1. Linear Systems

1.1. Introduction to Linear Systems

Linear Systems

A system of equations of the form

$$\begin{cases} a_{11}x_1 + ... + a_{1k}x_k = b_1, \\ ... \\ a_{n1}x_1 + ... + a_{nk}x_k = b_n \end{cases} \tag{1}$$

in the variables (unknowns) $x_1, ..., x_k$ is called a *system of linear equations*, or a *linear system*. Using matrix notation (see Section 2.3), it can be written in the form

$$\mathbf{Ax} = \mathbf{b}, \quad \mathbf{A} = \begin{pmatrix} a_{11} & ... & a_{1k} \\ \vdots & \vdots & \vdots \\ a_{n1} & ... & a_{nk} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

We denote the set of matrices of size $n \times k$ with entries $a_{ij} \in \mathbb{R}$, where $\mathbb{R}$ is the set of real numbers, by $\mathbb{R}^{n\times k}$. In accordance with this notation, $\mathbb{R}^{n\times 1}$ and $\mathbb{R}^{k\times 1}$ are the sets of column matrices having n and k entries, respectively. For brevity we shall write $\mathbb{R}^n = \mathbb{R}^{n\times 1}$ and $\mathbb{R}^k = \mathbb{R}^{k\times 1}$ and call column matrices *vectors*.

The matrix $\mathbf{A} \in \mathbb{R}^{n\times k}$ is called the *coefficient matrix*, the vector $\mathbf{b} \in \mathbb{R}^n$ is called the *constant vector* (the right-hand side), and a vector $\mathbf{x} \in \mathbb{R}^k$ is called a *solution* of the system (1).

Linear Equations, Solutions and Consistency ADDED FOR MA1522

> **Definition.** A *linear equation* in the variables $x_1, ..., x_n$ is an equation that can be written in the *standard form*
>
> $$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$
>
> where the *coefficients* $a_1, ..., a_n$ and the *constant* b are given real numbers. The equation is *homogeneous* if $b = 0$.

Thus each variable enters a linear equation only to the first power and multiplied by a constant. The equation $x_1 = 3x_2 + 1$ is linear (its standard form is $x_1 - 3x_2 = 1$), whereas $x_1x_2 = 1$, $x_1^2 + x_2 = 0$ and $\sin x_1 = x_2$ are not. A linear system is a finite collection of linear equations in the same variables; in standard form, with m equations and n variables, it is the system (1) with m and n in place of n and k. The system is *homogeneous* if all its equations

are homogeneous, that is, $\mathbf{b} = \mathbf{0}$; the solutions of homogeneous systems are studied in Section 4.2.

Definition. Real numbers $c_1, ..., c_n$ give a *solution* $x_1 = c_1, ..., x_n = c_n$ of a linear system in n variables if every equation of the system becomes a true equality when c_j is substituted for x_j, $j = 1, ..., n$; equivalently, $\mathbf{c} = (c_1, ..., c_n)^T$ satisfies $\mathbf{Ac} = \mathbf{b}$. The system is *consistent* if it has at least one solution and *inconsistent* if it has none.

A system with more than one solution is described by a *general solution*, which captures all of its solutions at once: some of the variables are set equal to arbitrary real numbers $s, t, ...$, called *parameters*, and the other variables are expressed in terms of the parameters, so that every choice of the parameters gives a solution and every solution arises from some choice. The parameters are new symbols. An expression of some variables through the others, such as $y = 3x - 2$, only restates an equation of the system and is not a general solution.

Example. Both equations of the system $3x - y = 2$, $6x - 2y = 4$ express the single condition $y = 3x - 2$. Choosing x freely gives the general solution

$$x = s, \quad y = 3s - 2, \quad s \in \mathbb{R};$$

choosing y freely gives another general solution of the same system, $x = (t + 2)/3$, $y = t$, $t \in \mathbb{R}$. In contrast, the system $x + y = 3$, $x - y = 1$, $3x + y = 5$ is inconsistent: its first two equations hold only for $x = 2$, $y = 1$, and then $3x + y = 7 \neq 5$.

Theorem. A linear system has either no solution, or exactly one solution, or infinitely many solutions.

Proof. It suffices to show that a system with two different solutions $\mathbf{u} = (u_1, ..., u_n)^T$ and $\mathbf{v} = (v_1, ..., v_n)^T$ has infinitely many. For $t \in \mathbb{R}$ let $\mathbf{w}_t = \mathbf{u} + t(\mathbf{v} - \mathbf{u})$, whose j-th entry is $(1 - t)u_j + tv_j$. Substituting it into the i-th equation gives

$$\sum_{j=1}^{n} a_{ij}\big((1-t)u_j + tv_j\big) = (1-t)\sum_{j=1}^{n} a_{ij}u_j + t\sum_{j=1}^{n} a_{ij}v_j = (1-t)b_i + tb_i = b_i,$$

so each $\mathbf{w}_t$ is a solution. Since $\mathbf{v} - \mathbf{u} \neq \mathbf{0}$, different values of t give different vectors $\mathbf{w}_t$. □

A more precise description of the set of solutions of a consistent system, as a translate $\mathbf{u} + \mathrm{Null}(\mathbf{A})$ of the nullspace, is given in Section 4.2.

Geometric Picture in Two and Three Variables ADDED FOR MA1522

An equation $ax + by = c$ in two variables with $(a, b) \neq (0, 0)$ describes a straight line in the xy-plane, and the solutions of a system of such equations are the points lying on all of its lines. For a system of two equations

$$a_1x + b_1y = c_1, \quad a_2x + b_2y = c_2$$

there are three possibilities:

1. the two lines are parallel and distinct, and the system has no solution;
2. the lines meet in exactly one point, and the system has a unique solution;
3. the lines coincide, and the system has infinitely many solutions, its general solution having one parameter.

> **Example.** The lines $x + 2y = 4$ and $3x - y = 5$ meet only at the point $x = 2$, $y = 1$. The lines $2x - y = 1$ and $4x - 2y = 3$ are parallel and distinct (the second is the line $2x - y = 3/2$), so this system is inconsistent. The equations $2x - y = 1$ and $6x - 3y = 3$ describe the same line, and the general solution is $x = s$, $y = 2s - 1$, $s \in \mathbb{R}$.

Similarly, an equation $ax + by + cz = d$ with $(a, b, c) \neq (0, 0, 0)$ describes a plane in three-dimensional space, and two planes that are not parallel meet in a straight line. For a system of three such equations:

1. there is no solution if two of the planes are parallel and distinct, or if no two of them are parallel but the three lines in which they meet pairwise are parallel and distinct, so that no point lies on all three planes;
2. there is a unique solution if the three planes have exactly one common point;
3. there are infinitely many solutions, with one parameter in the general solution, if the three planes have a common line but do not all coincide;
4. there are infinitely many solutions, with two parameters in the general solution, if the three planes coincide.

> **Example.** The planes $x + y + z = 6$, $x - y + z = 2$, $x + y - z = 0$ have the single common point $(1, 2, 3)$. In the system $x + y + z = 2$, $x - y = 0$, $2x + z = 2$, the third equation is the sum of the first two, so the third plane contains the line in which the first two meet; the general solution is $x = s$, $y = s$, $z = 2 - 2s$, $s \in \mathbb{R}$. In the system $x + y + z = 1$, $x - y = 0$, $2x + z = 3$ no two planes are parallel, but the sum of the first two equations, $2x + z = 1$, contradicts the third: the system is inconsistent, and the three lines of pairwise intersection are parallel.

1.2. Solving a Linear System and Row-Echelon Form

Augmented Matrix ADDED FOR MA1522

In solving a linear system only its coefficients and constants are manipulated. A system of m equations in n variables, written in standard form, is therefore recorded by its *augmented matrix*

$$(\mathbf{A} \mid \mathbf{b}) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right),$$

whose i-th row lists the coefficients of the i-th equation to the left of the bar and its constant to the right. Once the order of the variables is fixed, a system in standard form and its augmented matrix determine each other. A system must therefore first be brought to standard form, with the variables in the same order in every equation and coefficient 0 for each variable that does not occur in an equation.

Example. Let a be a fixed real number. The system

$$x + 3z = 2y - 1, \quad 4y = z, \quad ax = y + a$$

has the standard form

$$x - 2y + 3z = -1, \quad 4y - z = 0, \quad ax - y = a,$$

and its augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & -2 & 3 & -1 \\ 0 & 4 & -1 & 0 \\ a & -1 & 0 & a \end{array}\right).$$

The number a enters both as a coefficient (of x in the third equation) and as a constant.

Row-Echelon Form

We shall say that a matrix $\mathbf{S} = \left(s_{ij}\right)_{m\times n}$ is in *row-echelon form* (REF) with k nonzero rows if there exist indices $1 \le j_1 < ... < j_k \le n$ for which:

- if $1 \le i \le k$, then $s_{ij} \neq 0$ for $j = j_i$ and $s_{ij} = 0$ for all $1 \le j \le j_i - 1$;
- if $k + 1 \le i \le m$, then $s_{ij} = 0$ for all $1 \le j \le n$.

Thus the first k rows of $\mathbf{S}$ are its nonzero rows and the zero rows are at the bottom; the entry s_{ij_i} is the *leading entry* of the i-th row, and the columns with indices $j_1, ..., j_k$ are the *pivot columns* of $\mathbf{S}$. A matrix $\mathbf{S}$ is said to be in *column-echelon form* with k nonzero columns if $\mathbf{S}^T$ is in row-echelon form with k nonzero rows.

Proposition. If a matrix $\mathbf{S}$ is in row-echelon form with k nonzero rows, or in column-echelon form with k nonzero columns, then $\mathrm{rank}(\mathbf{S}) = k$.

Proof. Recall that $\mathrm{rank}(\mathbf{S}) = \dim(\mathrm{Row}(\mathbf{S})) = \dim(\mathrm{Col}(\mathbf{S}))$ (see Section 4.2). Consider a matrix $\mathbf{S}$ in row-echelon form and let us prove that the dimension of its row space $\mathrm{Row}(\mathbf{S})$ (the span of the rows) is equal to the number k of nonzero rows. The zero rows of $\mathbf{S}$ do not contribute to $\mathrm{Row}(\mathbf{S})$, so we do not lose generality by assuming that $k = m$. Let us prove that the rows of $\mathbf{S}$ are

linearly independent. Set their linear combination with coefficients $\alpha_1, ..., \alpha_k$ equal to zero:

$$(\alpha_1 \quad \alpha_2 \quad \cdots \quad \alpha_k)\mathbf{S} = (0 \quad 0 \quad \cdots \quad 0).$$

From this,

$$\alpha_1 s_{1j_1} = 0 \quad \Rightarrow \quad \alpha_1 = 0.$$

Next,

$$0 \cdot s_{1j_2} + \alpha_2 s_{2j_2} = 0 \quad \Rightarrow \quad \alpha_2 = 0.$$

Continuing in a similar manner, we find $\alpha_1 = ... = \alpha_k = 0$. In the case of a matrix $\mathbf{S}$ in column-echelon form, $\dim(\mathrm{Col}(\mathbf{S}))$ obviously coincides with $\dim(\mathrm{Row}(\mathbf{S}^T))$, where $\mathbf{S}^T$ is in row-echelon form. □

Pivot Columns and Reduced Row-Echelon Form ADDED FOR MA1522

In a matrix (in particular, in an augmented matrix) a *zero row* is a row all of whose entries are 0; every other row is a *nonzero row*, and its first nonzero entry from the left is its *leading entry*. In these terms, the definition of row-echelon form given above says that a matrix is in row-echelon form if and only if

1. all zero rows, if there are any, lie below all nonzero rows, and
2. the leading entry of each nonzero row lies strictly to the right of the leading entry of the row above it.

Indeed, the leading entry of the i-th row is s_{ij_i}, and condition 2 is the inequality $j_1 < j_2 < \cdots < j_k$.

Definition. Let a matrix be in row-echelon form. A column containing a leading entry is a *pivot column*, and a column containing no leading entry is a *non-pivot column*. The matrix is in *reduced row-echelon form* (RREF) if, in addition to conditions 1 and 2,

3. every leading entry is equal to 1, and
4. in each pivot column, all entries other than the leading entry are equal to 0.

A matrix in reduced row-echelon form is, in particular, in row-echelon form. Each nonzero row contains exactly one leading entry, and each pivot column contains exactly one, so a matrix in row-echelon form has as many pivot columns as nonzero rows.

Example. Of the augmented matrices

$$\left(\begin{array}{ccc|c} 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \end{array}\right), \quad \left(\begin{array}{ccc|c} 2 & 1 & -1 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right), \quad \left(\begin{array}{cccc|c} 1 & 3 & 0 & -2 & 5 \\ 0 & 0 & 1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right),$$

the first is not in row-echelon form, since the leading entry of its second row lies to the left of that of the first row. The second is in row-echelon

form, with pivot columns 1 and 3, but not in reduced row-echelon form: its first leading entry is 2, and the pivot column 3 contains the nonzero entry -1 above the leading entry of the second row. The third is in reduced row-echelon form, with pivot columns 1 and 3 and non-pivot columns 2, 4 and 5.

Solutions from REF and RREF ADDED FOR MA1522

When the augmented matrix of a system is in row-echelon form, the system is solved from the bottom up by *back substitution*; when it is in reduced row-echelon form, the solutions are read off directly. In both cases the variables corresponding to the non-pivot columns of the coefficient part are the ones that become parameters.

Theorem. Let the augmented matrix $(\mathbf{A} \mid \mathbf{b})$ of a linear system in n variables be in row-echelon form with k nonzero rows.

1. The system is inconsistent if and only if the last column of $(\mathbf{A} \mid \mathbf{b})$ is a pivot column.
2. If the last column is not a pivot column, then for every choice of values of the variables corresponding to the non-pivot columns of $\mathbf{A}$ the system has exactly one solution in which these variables take the chosen values. Hence the general solution is obtained by making these $n - k$ variables parameters and solving for the remaining k variables; the system has a unique solution if $k = n$ and infinitely many solutions if $k < n$.

Proof. Denote the entries of $(\mathbf{A} \mid \mathbf{b})$ by s_{ij} to the left of the bar and by d_i to the right of it. If the last column is a pivot column, the row containing its leading entry $d_i \neq 0$ is the equation $0x_1 + \cdots + 0x_n = d_i$, which no values of the variables satisfy.

Suppose that the last column is not a pivot column. Then all k pivot columns $j_1 < \cdots < j_k$ are columns of $\mathbf{A}$, and $\mathbf{A}$ has $n - k$ non-pivot columns. The zero rows are the equations $0 = 0$, which impose no condition, and row i $(1 \leq i \leq k)$ is the equation

$$s_{ij_i} x_{j_i} + \sum_{j > j_i} s_{ij} x_j = d_i, \quad s_{ij_i} \neq 0.$$

Each variable x_j with $j > j_i$ is either a non-pivot variable or a pivot variable x_{j_l} with $l > i$. Therefore, once the non-pivot variables are fixed, the equations taken in the order $i = k, k-1, \ldots, 1$ determine $x_{j_k}, x_{j_{k-1}}, \ldots, x_{j_1}$ in turn, uniquely, by

$$x_{j_i} = \frac{1}{s_{ij_i}} \left(d_i - \sum_{j > j_i} s_{ij} x_j \right),$$

and the values so obtained satisfy all the equations. In particular, the system is consistent, which completes the proof of statement 1. If $k < n$, there is at least one parameter, and different values of it give different solutions. $\square$

If $(\mathbf{A} \mid \mathbf{b})$ is in reduced row-echelon form, then $s_{ij_i} = 1$ and $s_{ij_l} = 0$ for $l \neq i$, so the formula becomes $x_{j_i} = d_i - \sum s_{ij} x_j$, the sum being over the non-pivot columns $j > j_i$ of $\mathbf{A}$: each pivot variable is read off from its own row, and no substitution is needed.

Example. (a) The augmented matrix

$$\left(\begin{array}{cccc|c} 1 & 3 & 0 & -2 & 5 \\ 0 & 0 & 1 & 4 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

is in reduced row-echelon form, and the non-pivot columns of its coefficient part are columns 2 and 4. With $x_2 = s$ and $x_4 = t$ the two nonzero rows give $x_1 = 5 - 3s + 2t$ and $x_3 = -1 - 4t$, so the general solution is

$$x_1 = 5 - 3s + 2t, \quad x_2 = s, \quad x_3 = -1 - 4t, \quad x_4 = t, \quad s, t \in \mathbb{R}.$$

(b) The augmented matrix

$$\left(\begin{array}{ccccc|c} 0 & 2 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{array}\right)$$

is in row-echelon form but not in reduced row-echelon form. Its non-pivot columns 1 and 3 give the parameters $x_1 = s$, $x_3 = t$, and back substitution gives $x_5 = -1$ from the third row, $x_4 = 3 - 2x_5 = 5$ from the second, and $2x_2 = 1 - x_3 + x_4 = 6 - t$ from the first. The general solution is

$$x_1 = s, \quad x_2 = 3 - t/2, \quad x_3 = t, \quad x_4 = 5, \quad x_5 = -1, \quad s, t \in \mathbb{R}.$$

(c) In the row-echelon form

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 2 \end{array}\right)$$

the last column is a pivot column: the third row is the equation $0x_1 + 0x_2 + 0x_3 = 2$, and the system is inconsistent.

1.3. Elementary Row Operations

Elementary Row Operations and Row Equivalence ADDED FOR MA1522

Definition. The *elementary row operations* on a matrix are of three types:

1. interchanging two rows, $R_i \leftrightarrow R_j$ $(i \neq j)$;
2. adding a multiple of one row to another row, $R_i + cR_j$ $(i \neq j,\ c \in \mathbb{R})$;
3. multiplying a row by a nonzero constant, aR_i $(a \neq 0)$.

The operation $R_i + cR_j$ replaces the i-th row by the sum of the i-th row and c times the j-th row, and leaves all other rows, the j-th included, unchanged. Thus $R_1 + R_2$ and $R_2 + R_1$ are different operations:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \xrightarrow{R_1+R_2} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \xrightarrow{R_2+R_1} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

An expression such as $2R_2 + R_1$ is not an elementary row operation: it stands for two of them, $2R_2$ followed by $R_2 + R_1$. Operations performed in succession are written from left to right in the order in which they are performed, because in general the result depends on the order:

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \xrightarrow{3R_1} \begin{pmatrix} 3 & 6 \\ 3 & 4 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 3 & 4 \\ 3 & 6 \end{pmatrix},$$

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 3 & 4 \\ 1 & 2 \end{pmatrix} \xrightarrow{3R_1} \begin{pmatrix} 9 & 12 \\ 1 & 2 \end{pmatrix}.$$

Operations that give the same result in either order, such as $R_2 - 2R_1$ and $R_3 + R_1$ (each changes a different row, using a row that neither of them changes), may be written one above the other over a single arrow.

The requirement $a \neq 0$ in the third type is essential. The system $x + y = 3$, $x - y = 1$ has the unique solution $x = 2$, $y = 1$, but "multiplying" its second row by 0,

$$\left(\begin{array}{cc|c} 1 & 1 & 3 \\ 1 & -1 & 1 \end{array}\right) \xrightarrow{0R_2} \left(\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 0 & 0 \end{array}\right),$$

produces a system with the general solution $x = 3 - s$, $y = s$, $s \in \mathbb{R}$.

Proposition. Each elementary row operation is undone by an elementary row operation of the same type, called its *reverse*:

1. the reverse of $R_i \leftrightarrow R_j$ is $R_i \leftrightarrow R_j$;
2. the reverse of $R_i + cR_j$ is $R_i - cR_j$;
3. the reverse of aR_i is $(1/a)R_i$.

Proof. The first and third statements are obvious. Let the rows be $\mathbf{r}_1, ..., \mathbf{r}_m$. The operation $R_i + cR_j$ replaces $\mathbf{r}_i$ by $\mathbf{r}_i + c\mathbf{r}_j$ and does not change $\mathbf{r}_j$, since $i \neq j$; then $R_i - cR_j$ replaces $\mathbf{r}_i + c\mathbf{r}_j$ by $(\mathbf{r}_i + c\mathbf{r}_j) - c\mathbf{r}_j = \mathbf{r}_i$. □

For instance, $R_3 - 2R_1$ is undone by $R_3 + 2R_1$, and $(1/3)R_2$ by $3R_2$.

Definition. Two matrices of the same size are *row equivalent* if one of them can be obtained from the other by a finite sequence of elementary row operations.

If $\mathbf{B}$ is obtained from $\mathbf{A}$ by a sequence of elementary row operations, then $\mathbf{A}$ is obtained from $\mathbf{B}$ by the reverse operations taken in the opposite order. Hence row equivalence is symmetric; it is clearly also reflexive and transitive.

Theorem. If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solutions.

Proof. It suffices to consider a single elementary row operation, transforming a system (S) into a system (S'). Each equation of (S') is an equation of (S), or a nonzero multiple of one, or the sum of an equation of (S) and a multiple of another equation of (S). Hence every solution of (S) satisfies every equation of (S'). Since (S) is obtained from (S') by the reverse operation, which is again an elementary row operation, the same argument shows that every solution of (S') is a solution of (S). □

Since solutions are easily obtained from a row-echelon form (Section 1.2), a linear system is solved by reducing its augmented matrix to row-echelon or reduced row-echelon form by elementary row operations; algorithms for this are given in Section 1.4. In matrix language, each elementary row operation is multiplication on the left by the invertible matrix obtained from $\mathbf{I}$ by the same operation, an elementary matrix (Section 2.5). The subsection below describes the operations $R_i \leftrightarrow R_j$ and $R_i + cR_j$ in this form.

Elementary Row Operations in Matrix Form

Each step of the elimination of unknowns (Gaussian elimination, described in Section 1.4) transforms a linear system $\mathbf{Ax} = \mathbf{b}$ into an equivalent system of the form $(\mathbf{PA})\mathbf{x} = \mathbf{Pb}$, where $\mathbf{P}$ is some invertible matrix. If k steps were required, then as a result a sequence of equivalent systems arises:

$$\mathbf{Ax} = \mathbf{b}, \quad (\mathbf{P}_1\mathbf{A})\mathbf{x} = \mathbf{P}_1\mathbf{b}, \quad (\mathbf{P}_2\mathbf{P}_1\mathbf{A})\mathbf{x} = \mathbf{P}_2\mathbf{P}_1\mathbf{b}, \quad ...,$$
$$..., \quad (\mathbf{P}_k...\mathbf{P}_2\mathbf{P}_1\mathbf{A})\mathbf{x} = \mathbf{P}_k...\mathbf{P}_2\mathbf{P}_1\mathbf{b}.$$

The coefficient matrix of the last system has such a simple form that solving the corresponding system is carried out in an obvious way.

The goal of each step is to obtain additional zeros, or, as is often said, the *elimination of entries* in the matrix of the transformed system. To explain how this is done, consider a matrix $\mathbf{A}$ of size 4×6; entries known to be zero will be denoted by a "zero," arbitrary entries by a "cross," and entries known to be nonzero by a "boxed cross." If $a_{11} \neq 0$, then from each row with index $i > 1$ we can subtract the first row with a coefficient yielding zero in position $(i, 1)$ (the elementary row operations $R_i + c_iR_1$ with $c_i = -a_{i1}/a_{11}$, $i = 2, 3, 4$):

$$\begin{pmatrix} \boxtimes & \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times & \times \end{pmatrix} \mapsto \begin{pmatrix} \boxtimes & \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times & \times \end{pmatrix}.$$

Thereafter the first row and column remain unchanged. If in the second column there is a nonzero entry, we make the corresponding row the second row (a row interchange $R_2 \leftrightarrow R_i$) and use it to eliminate all the remaining entries of the second column (the operations $R_i + c_i R_2$, $i = 3, 4$):

$$\begin{pmatrix} \boxtimes & \times & \times & \times & \times & \times \\ 0 & \boxtimes & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times & \times \\ 0 & \times & \times & \times & \times & \times \end{pmatrix} \mapsto \begin{pmatrix} \boxtimes & \times & \times & \times & \times & \times \\ 0 & \boxtimes & \times & \times & \times & \times \\ 0 & 0 & \times & \times & \times & \times \\ 0 & 0 & \times & \times & \times & \times \end{pmatrix}.$$

The first two columns and the first two rows will no longer change. It may happen that we obtain "unscheduled" zero columns. For instance, if the remaining "crosses" in the third and fourth columns turn out to be zeros, then elimination should be carried out using a nonzero entry in the fifth column (if there is one):

$$\begin{pmatrix} \boxtimes & \times & \times & \times & \times & \times \\ 0 & \boxtimes & \times & \times & \times & \times \\ 0 & 0 & 0 & 0 & \boxtimes & \times \\ 0 & 0 & 0 & 0 & \times & \times \end{pmatrix} \mapsto \begin{pmatrix} \boxtimes & \times & \times & \times & \times & \times \\ 0 & \boxtimes & \times & \times & \times & \times \\ 0 & 0 & 0 & 0 & \boxtimes & \times \\ 0 & 0 & 0 & 0 & 0 & \times \end{pmatrix}.$$

Each step obviously reduces to multiplying the matrix on the left by some matrix $\mathbf{P}_l$. Moreover, each of the matrices $\mathbf{P}_l$, $1 \le l \le k$, can be represented as the product of two matrices:

$$\mathbf{P}_l = \mathbf{Z}_l \mathbf{\Pi}_l,$$

where $\mathbf{Z}_l$ differs from $\mathbf{I}$ (the identity matrix) only in the positions below the main diagonal of some single (say, the j-th) column:

$$\mathbf{Z}_l = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & (\mathbf{Z}_l)_{j+1,j} & \ddots & \\ & & \vdots & & \\ & & (\mathbf{Z}_l)_{nj} & & 1 \end{pmatrix},$$

and $\mathbf{\Pi}_l$ is obtained from $\mathbf{I}$ by a permutation of columns (or rows). The matrix $\mathbf{\Pi}_l$ is called a *permutation matrix*, and $\mathbf{Z}_l$ a *row modification matrix*. Both are products of elementary matrices (matrices obtained from $\mathbf{I}$ by a single elementary row operation): $\mathbf{\Pi}_l$ of elementary matrices corresponding to row interchanges $R_i \leftrightarrow R_j$, and $\mathbf{Z}_l$ of those corresponding to the operations $R_i + (\mathbf{Z}_l)_{ij} R_j$, $i = j+1, ..., n$. Their role in the elimination process is explained by the following facts:

- the matrix $\mathbf{\Pi}_l \mathbf{A}$ differs from $\mathbf{A}$ by a permutation of rows;
- if $\mathbf{Z}_l$ differs from the identity matrix in the j-th column, then the matrix $\mathbf{Z}_l \mathbf{A}$ has the same first j rows as the matrix $\mathbf{A}$, while the i-th row for $i >$

j is the sum of the i-th row and the j-th row of the matrix $\mathbf{A}$ taken with some coefficient (namely $(\mathbf{Z}_l)_{ij}$: the operation $R_i + (\mathbf{Z}_l)_{ij} R_j$).

Proposition 1. Any permutation matrix $\mathbf{\Pi}$ is invertible and, moreover,
$$\mathbf{\Pi}^{-1} = \mathbf{\Pi}^T.$$

Proposition 2. Any row modification matrix $\mathbf{Z} = \mathbf{Z}_l$ is invertible and, moreover, the inverse matrix is obtained from $\mathbf{Z}$ by changing the signs of the subdiagonal entries:
$$\mathbf{Z}^{-1} = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & -(\mathbf{Z}_l)_{j+1,j} & \ddots & \\ & & \vdots & & \\ & & -(\mathbf{Z}_l)_{nj} & & 1 \end{pmatrix}.$$

The proof reduces to a direct verification of the equalities $\mathbf{\Pi}\mathbf{\Pi}^{-1} = \mathbf{I}$, $\mathbf{Z}\mathbf{Z}^{-1} = \mathbf{I}$. (Proposition 2 expresses the fact that the operations $R_i + cR_j$ are reversed by the operations $R_i - cR_j$.)

1.4. Row Reduction, Gaussian and Gauss–Jordan Elimination

Gaussian Elimination

If a linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ is given and one is required to find its general solution or establish its inconsistency, this is most conveniently done by *successive elimination of unknowns* (*Gaussian elimination*): if in some equation the coefficient of x_1 is nonzero, then one can eliminate x_1 from all the other equations by subtracting this equation, previously multiplied by suitably chosen coefficients; if among the equations that no longer contain x_1 there is an equation with a nonzero coefficient of x_2, then x_2 can be eliminated in a similar manner from all the other equations except the given one and the first equation containing x_1, and so on.

At each step of elimination, one obtains a new system that is obviously equivalent to the original one: in terms of the augmented matrix $(\mathbf{A} \mid \mathbf{b})$, each step consists of elementary row operations $R_i + cR_j$ (and row interchanges $R_i \leftrightarrow R_j$ if the equations are reordered), so the new augmented matrix is row equivalent to $(\mathbf{A} \mid \mathbf{b})$. If an equation arises in which all the coefficients of the unknowns are equal to zero while the right-hand side is a nonzero number, then the system has no solutions (in a row-echelon form of $(\mathbf{A} \mid \mathbf{b})$, the last column is then a pivot column). Otherwise, the system is consistent, and the

method described allows one to write out its general solution with ease (by back substitution, with the variables corresponding to the non-pivot columns as parameters).

Reduction to Row-Echelon Form

Theorem 1. For any $m \times n$ matrix $\mathbf{A}$ of rank r, there exists an invertible matrix $\mathbf{P}$ representable as a product of a finite number of permutation and row modification matrices such that the matrix $\mathbf{S} = \mathbf{PA}$ is in row-echelon form with r nonzero rows.

Proof. We follow the steps of Gaussian elimination. Denote by j_1 the index of the first column of the matrix $\mathbf{A}$ in which there is at least one nonzero entry. (If no such column exists, then $\mathbf{A} = \mathbf{0}$ and the theorem is already proved.) By multiplying on the left by some permutation matrix $\mathbf{\Pi}_1$ (a row interchange $R_1 \leftrightarrow R_i$, if necessary), a nonzero entry can be moved to position $(1, j_1)$. Next, by multiplying on the left by some row modification matrix $\mathbf{Z}_1$ (the operations $R_i + c_i R_1$, $i = 2, ..., m$), one can obtain a matrix with zeros in the subdiagonal positions of the j_1-th column while preserving the zeros in the preceding columns. Obviously, the transformed matrix has the block form (by $\mathbf{0}_{p\times q}$ we denote the zero block of size $p \times q$):

$$\mathbf{Z}_1\mathbf{\Pi}_1\mathbf{A} = \begin{pmatrix} \mathbf{0}_{1\times(j_1-1)} & \mathbf{u}^T \\ \mathbf{0}_{(m-1)\times(j_1-1)} & \mathbf{B} \end{pmatrix}, \quad \mathbf{u} \in \mathbb{R}^{(n-j_1+1)\times 1}.$$

Let us make the inductive hypothesis that there exists a matrix $\mathbf{Q}$ which is a product of permutation and row modification matrices of order $m-1$ and is such that $\mathbf{QB}$ is in row-echelon form. Consider the matrix

$$\mathbf{P} = \begin{pmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \mathbf{Q} \end{pmatrix} \mathbf{Z}_1\mathbf{\Pi}_1.$$

It is easy to see that $\mathbf{P}$ is a product of permutation and row modification matrices and, moreover, $\mathbf{S} = \mathbf{PA}$ is in row-echelon form. Let the number of nonzero rows be equal to k. This means that $\dim(\mathrm{Row}(\mathbf{S})) = k \Rightarrow k = \mathrm{rank}(\mathbf{S}) = \mathrm{rank}(\mathbf{A}) = r$. □

Theorem 2. For any $m \times n$ matrix $\mathbf{A}$ of rank r, there exists an invertible matrix $\mathbf{Q}$ representable as a product of a finite number of permutation and row modification matrices such that the matrix $\mathbf{AQ}^T$ is in column-echelon form with r nonzero columns.

Proof. It suffices to apply Theorem 1 to the matrix $\mathbf{A}^T$ and note that if the matrix $\mathbf{QA}^T$ is in row-echelon form, then the matrix $\left(\mathbf{QA}^T\right)^T = \mathbf{AQ}^T$ will be in column-echelon form. □

Gaussian and Gauss–Jordan Elimination as Algorithms ADDED FOR MA1522

The elimination described above is organized as follows; it applies to any matrix, in particular to the augmented matrix of a linear system. A matrix in row-echelon form (reduced row-echelon form) that is row equivalent to $\mathbf{A}$ is called *a row-echelon form* (*a reduced row-echelon form*) *of* $\mathbf{A}$.

Algorithm (Gaussian elimination).

1. Find the leftmost column that does not consist entirely of zeros.
2. If the top entry of this column is 0, interchange the top row with a row that has a nonzero entry in this column.
3. To each row below the top row add a suitable multiple of the top row, so that the entry of that row in this column becomes 0.
4. Cover the top row and apply Steps 1–4 to the submatrix formed by the remaining rows. Stop when no rows remain or the remaining rows are all zero; the matrix is then in row-echelon form.

Algorithm (Gauss–Jordan elimination).

Perform Steps 1–4 of Gaussian elimination, and then:

5. Multiply each nonzero row by the reciprocal of its leading entry, so that all leading entries become 1.
6. Beginning with the lowest nonzero row and working upward, add suitable multiples of each row to the rows above it, so that all entries above its leading entry become 0.

The matrix is then in reduced row-echelon form.

Example. For the system $x + 2y - z = 3$, $2x + 3y + z = 8$, $-x + y + 2z = 1$, Gaussian elimination gives

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 3 & 1 & 8 \\ -1 & 1 & 2 & 1 \end{array}\right) \xrightarrow[R_3+R_1]{R_2-2R_1} \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -1 & 3 & 2 \\ 0 & 3 & 1 & 4 \end{array}\right) \xrightarrow{R_3+3R_2} \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -1 & 3 & 2 \\ 0 & 0 & 10 & 10 \end{array}\right).$$

The last matrix is in row-echelon form, and back substitution gives $z = 1$, then $y = 3z - 2 = 1$ from the second row, and $x = 3 - 2y + z = 2$ from the first. Gauss–Jordan elimination continues with

$$\xrightarrow[-R_2]{(1/10)R_3} \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right) \xrightarrow[R_1+R_3]{R_2+3R_3} \left(\begin{array}{ccc|c} 1 & 2 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right) \xrightarrow{R_1-2R_2} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right),$$

and the unique solution $x = 2$, $y = 1$, $z = 1$ is read off from the reduced row-echelon form.

Theorem 3. Every matrix has a row-echelon form and a reduced row-echelon form: Gaussian elimination produces the former, and Gauss–Jordan elimination the latter.

Proof. We use induction on the number of rows; a matrix with one row, or a zero matrix, is already in row-echelon form. After Steps 1–3 the top row has a nonzero entry in some column j_1, and every other row has zeros in columns $1, \dots, j_1$. By the induction hypothesis, Step 4 brings the submatrix of these rows to row-echelon form; its nonzero rows have their leading entries to the right of column j_1, so the whole matrix is in row-echelon form (cf. Theorem 1).

Step 5 does not move the leading entries, so the matrix stays in row-echelon form and its leading entries become 1. In Step 6, when multiples of the i-th row are added to the rows above it, the i-th row has zeros in all pivot columns other than its own: to the left of its leading entry by the echelon form, and in the pivot columns of the lower rows because these columns have already been cleared. Hence these operations make the entries above the i-th leading entry zero without changing any other entry of a pivot column, and at the end conditions 1–4 of reduced row-echelon form hold. □

A matrix has many row-echelon forms; for example,

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \xrightarrow{R_2 - 3R_1} \begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix} \xrightarrow{2R_1} \begin{pmatrix} 2 & 4 \\ 0 & -2 \end{pmatrix},$$

and both matrices obtained are in row-echelon form. The reduced row-echelon form, however, is unique.

Theorem 4. Every matrix is row equivalent to exactly one matrix in reduced row-echelon form.

Proof. Existence is Theorem 3. Let $\mathbf{R}$ and $\mathbf{R}'$ be reduced row-echelon forms of an $m \times n$ matrix $\mathbf{A}$. Then $\mathbf{R}$ and $\mathbf{R}'$ are row equivalent, and so are $(\mathbf{R} \mid \mathbf{0})$ and $(\mathbf{R}' \mid \mathbf{0})$; by the theorem of Section 1.3 the homogeneous systems $\mathbf{R}\mathbf{x} = \mathbf{0}$ and $\mathbf{R}'\mathbf{x} = \mathbf{0}$ have the same solutions. We prove $\mathbf{R} = \mathbf{R}'$ by induction on n.

If $n = 1$, a matrix in reduced row-echelon form is either the zero column or the first column of $\mathbf{I}_m$. An elementary row operation turns a zero column into a zero column and, being reversible, a nonzero column into a nonzero one; hence $\mathbf{R} = \mathbf{R}'$.

Let $n > 1$, and let $\mathbf{R}_0$, $\mathbf{R}'_0$ and $\mathbf{A}_0$ be obtained from $\mathbf{R}$, $\mathbf{R}'$ and $\mathbf{A}$ by deleting the last column. Deleting the last column preserves conditions 1–4 (a row whose only nonzero entry was in the last column becomes a zero row, but it was the lowest nonzero row), and the operations that reduce $\mathbf{A}$ to $\mathbf{R}$ or $\mathbf{R}'$ reduce $\mathbf{A}_0$ to $\mathbf{R}_0$ or $\mathbf{R}'_0$. By the induction hypothesis, $\mathbf{R}_0 = \mathbf{R}'_0$.

Suppose that $\mathbf{R} \neq \mathbf{R}'$. Then they differ only in the last column, say in the i-th entry: $r_{in} \neq r'_{in}$. For every solution $\mathbf{x}$ of $\mathbf{R}\mathbf{x} = \mathbf{0}$, which is also a solution

of $\mathbf{R}'\mathbf{x} = \mathbf{0}$, the i-th entry of $(\mathbf{R} - \mathbf{R}')\mathbf{x} = \mathbf{0}$ is $(r_{in} - r'_{in})x_n = 0$, so $x_n = 0$. If the last column of $\mathbf{R}$ were a non-pivot column, x_n would be a parameter of the general solution of $\mathbf{R}\mathbf{x} = \mathbf{0}$ (Section 1.2), and there would be a solution with $x_n = 1$. Hence the last column of $\mathbf{R}$ is a pivot column, and likewise the last column of $\mathbf{R}'$. Let p be the number of nonzero rows of $\mathbf{R}_0 = \mathbf{R}'_0$. The leading entry in the last column of $\mathbf{R}$ lies in the first row that is zero in $\mathbf{R}_0$, that is, in row $p + 1$, and by conditions 3 and 4 the last column of $\mathbf{R}$ is the $(p + 1)$-th column of $\mathbf{I}_m$. The same holds for $\mathbf{R}'$, so $\mathbf{R} = \mathbf{R}'$, a contradiction. □

Because of Theorem 4 one speaks of *the* reduced row-echelon form of a matrix.

Corollary. All row-echelon forms of a matrix $\mathbf{A}$ have their leading entries in the same positions, namely those of the reduced row-echelon form of $\mathbf{A}$. In particular, their pivot columns and their number of nonzero rows, equal to $\mathrm{rank}(\mathbf{A})$ by Theorem 1, do not depend on the row-echelon form chosen; they are called the *pivot columns* of $\mathbf{A}$.

Proof. Steps 5 and 6 of Gauss–Jordan elimination, applied to a row-echelon form of $\mathbf{A}$, do not move its leading entries and produce a reduced row-echelon form of $\mathbf{A}$, which by Theorem 4 is the reduced row-echelon form of $\mathbf{A}$. □

1.5. More on Linear Systems

Consistency of a Linear System

Theorem 1. A linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$, $\mathbf{A} = \begin{pmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_k\end{pmatrix}$, is consistent if and only if

$$\mathrm{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} = \mathrm{span}\{\mathbf{a}_1, ..., \mathbf{a}_k, \mathbf{b}\}.$$

Proof. In any case we have

$$\mathrm{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} \subseteq \mathrm{span}\{\mathbf{a}_1, ..., \mathbf{a}_k, \mathbf{b}\}. \qquad (*)$$

If the system is consistent, then $\mathbf{b} \in \mathrm{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$. Consequently,

$$\mathrm{span}\{\mathbf{a}_1, ..., \mathbf{a}_k, \mathbf{b}\} \subseteq \mathrm{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}. \qquad (**)$$

The inclusions $(*)$ and $(**)$ prove the equality of the two spans. If $(**)$ holds, then it is obvious that $\mathbf{b} \in \mathrm{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$, and this is precisely what the consistency of the system $\mathbf{A}\mathbf{x} = \mathbf{b}$ means. □

Theorem 2. If $n = k$, then when the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$ are linearly independent, the linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ is consistent and has a unique solution.

Proof. It is obvious that

$$\mathbf{a}_1, ..., \mathbf{a}_n \in \mathrm{span}\{\mathbf{e}_1, ..., \mathbf{e}_n\},$$

where $\mathbf{e}_1, ..., \mathbf{e}_n$ are the columns of the identity matrix $\mathbf{I}_n$ (in the vector $\mathbf{e}_i$, the i-th entry is 1, and all other entries are equal to 0). By the Basis Extension Lemma (Section 3.7), there exists a basis of $r \geq n$ vectors containing the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$. By the theorem on the dimension of a linear span (Section 3.6), $r \leq n$. For the same reason, the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$ form a basis in $\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}\}$. Therefore $\mathbf{b} \in \text{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}$, which proves the consistency of the system. The uniqueness of the solution follows from Lemma 3 (Section 3.5). □

Problem 1. A linear system of the form

$$\begin{pmatrix} a_0 & a_1 & a_2 \\ a_1 & a_0 & a_1 \\ a_2 & a_1 & a_0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

has a solution with $x_1 \neq 0$. Prove that the columns of the coefficient matrix are linearly independent.

Solving a Linear System: Summary ADDED FOR MA1522

Algorithm (Solving a linear system).

1. Write the system in standard form.
2. Form its augmented matrix $(\mathbf{A} \mid \mathbf{b})$.
3. Reduce $(\mathbf{A} \mid \mathbf{b})$ to a row-echelon form or to its reduced row-echelon form, for instance by Gaussian or Gauss–Jordan elimination (Section 1.4).
4. If the last column is a pivot column, the system is inconsistent. Otherwise it is consistent; make the variables corresponding to the non-pivot columns of $\mathbf{A}$ parameters $s, t, ... \in \mathbb{R}$.
5. From the reduced row-echelon form, read off the remaining variables; from a row-echelon form, find them by back substitution, starting with the lowest nonzero row.
6. Write down the general solution (the unique solution if there are no parameters).

The theorem of Section 1.2, combined with the theorem of Section 1.3 on row-equivalent systems and the corollary of Section 1.4, gives the following criterion, which does not depend on the row-echelon form used.

Theorem 3. Let $\mathbf{Ax} = \mathbf{b}$ be a linear system in n variables.

1. The system is inconsistent if and only if the last column of $(\mathbf{A} \mid \mathbf{b})$ is a pivot column, that is, if and only if $\text{rank}(\mathbf{A}) < \text{rank}((\mathbf{A} \mid \mathbf{b}))$.
2. A consistent system has a unique solution if every column of $\mathbf{A}$ is a pivot column, that is, if $\text{rank}(\mathbf{A}) = n$, and infinitely many solutions otherwise; its general solution has $n - \text{rank}(\mathbf{A})$ parameters, one for each non-pivot column of $\mathbf{A}$.

Proof. Let $\mathbf{R}$ be a row-echelon form of $(\mathbf{A} \mid \mathbf{b})$. The system with augmented matrix $\mathbf{R}$ has the same solutions as $\mathbf{Ax} = \mathbf{b}$ (Section 1.3), and $\mathbf{R}$ without its last column is a row-echelon form of $\mathbf{A}$, with the same pivot columns as $\mathbf{R}$ among the first n columns. By the corollary of Section 1.4, the number of nonzero rows of $\mathbf{R}$ is $\operatorname{rank}((\mathbf{A} \mid \mathbf{b}))$, and that of $\mathbf{R}$ without its last column is $\operatorname{rank}(\mathbf{A})$; these numbers differ (by one) exactly when the last column of $\mathbf{R}$ is a pivot column. It remains to apply the theorem of Section 1.2 to $\mathbf{R}$, whose number of nonzero rows is then $\operatorname{rank}(\mathbf{A})$. □

The rank form of statement 1 is the Kronecker–Capelli theorem of Section 4.2.

Systems with Unknown Coefficients ADDED FOR MA1522

When some coefficients or constants of a system are letters, standing for fixed but unknown real numbers, the same elimination is used, with one precaution: an operation that is legitimate for some values of the letters may be illegitimate for others. The operations $R_i \leftrightarrow R_j$ and $R_i + cR_j$ may be used with any expression c, but multiplying a row by an expression such as a or $1/(a-1)$ is an elementary row operation only for the values of a at which this expression is defined and nonzero. Likewise, an entry such as $a - 1$ can serve as a leading entry only when $a \neq 1$. One therefore reduces as far as possible without such divisions (a row interchange that brings a constant nonzero entry to the top often helps) and then considers separately the values of the letters at which a divisor or a would-be leading entry vanishes. The answer must cover all values of the letters, not one chosen value.

Example. Let $a \in \mathbb{R}$, and consider the system $x + y + az = 1$, $x + ay + z = 1$, $ax + y + z = 1$. Using only operations of the form $R_i + cR_j$, we obtain

$$\left(\begin{array}{ccc|c} 1 & 1 & a & 1 \\ 1 & a & 1 & 1 \\ a & 1 & 1 & 1 \end{array}\right) \xrightarrow[R_3 - aR_1]{R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & a & 1 \\ 0 & a-1 & 1-a & 0 \\ 0 & 1-a & 1-a^2 & 1-a \end{array}\right)$$

$$\xrightarrow{R_3 + R_2} \left(\begin{array}{ccc|c} 1 & 1 & a & 1 \\ 0 & a-1 & 1-a & 0 \\ 0 & 0 & (1-a)(a+2) & 1-a \end{array}\right).$$

If $a = 1$, the last two rows are zero, and the system reduces to the single equation $x + y + z = 1$: it is consistent, with the general solution $x = 1 - s - t$, $y = s$, $z = t$, $s, t \in \mathbb{R}$ (two parameters). If $a = -2$, the last row is $(0, 0, 0 \mid 3)$, so the last column is a pivot column and the system is inconsistent. If $a \neq 1, -2$, the matrix is in row-echelon form with the leading entries 1, $a - 1$, $(1 - a)(a + 2)$ in the three columns of the coefficient part, so the solution is unique. Back substitution gives $z = 1/(a+2)$, then $(a-1)(y-z) = 0$, so $y = z$, and finally $x = 1 - y - az = 1/(a+2)$:

$$x = y = z = \frac{1}{a+2}, \quad a \neq 1, -2.$$

Constructing a System with a Given General Solution ADDED FOR MA1522

Suppose a general solution is given in which every parameter is equal to one of the variables, as is the case for the general solutions obtained in Section 1.2. Replacing each parameter by that variable in the formulas for the other variables, and moving all variables to the left-hand side, gives a linear system. Every vector of the given form satisfies this system; conversely, if $\mathbf{x}$ is a solution of the system, then giving each parameter the value of the corresponding variable in $\mathbf{x}$ shows that $\mathbf{x}$ is of the given form. Hence the system has exactly the given general solution.

Example. For the general solution

$$x_1 = 2 + s - 3t, \quad x_2 = s, \quad x_3 = -1 + 2t, \quad x_4 = t, \quad s, t \in \mathbb{R},$$

substituting $s = x_2$ and $t = x_4$ into the formulas for x_1 and x_3 gives the system

$$x_1 - x_2 + 3x_4 = 2, \quad x_3 - 2x_4 = -1,$$

whose augmented matrix

$$\left(\begin{array}{cccc|c} 1 & -1 & 0 & 3 & 2 \\ 0 & 0 & 1 & -2 & -1 \end{array}\right)$$

is in reduced row-echelon form; its non-pivot columns 2 and 4 give back the general solution above.

A system with more equations and the same solutions is obtained by appending equations that are linear combinations of these, since they impose no new condition. For instance, appending the sum of the two equations, $x_1 - x_2 + x_3 + x_4 = 1$, gives a system of three equations. The operations $R_3 - R_1$ and $R_3 - R_2$ turn its augmented matrix into the matrix above followed by a zero row, so by the theorem of Section 1.3 it has the same general solution.

Chapter 2. Matrix Algebra

2.1. Definition and Special Types of Matrices

Linear Mappings and Matrices

In mathematics and other sciences one constantly studies the dependence of some quantities upon others. Usually such a dependence is described by various types of functions (mappings, operators). The simplest case is that of linear mappings. We shall give rigorous definitions later. For now, let us suppose that the variables $y_1, ..., y_m$ are expressed in terms of $x_1, ..., x_n$ as follows:

$$\begin{cases} y_1 = a_{11}x_1 + ... + a_{1n}x_n, \\ ... \\ y_m = a_{m1}x_1 + ... + a_{mn}x_n, \end{cases} \qquad (*)$$

where the coefficients are assumed to be given constant quantities. Let us collect all the constant coefficients into a rectangular table and denote it by the letter $\mathbf{A}$; let us also form column tables from the quantities $x_1, ..., x_n$ and $y_1, ..., y_m$:

$$\mathbf{A} = \begin{pmatrix} a_{11} & ... & a_{1n} \\ ... & ... & ... \\ a_{m1} & ... & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ ... \\ x_n \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ ... \\ y_m \end{pmatrix}.$$

Such tables are called *matrices*. We have three matrices in all: of sizes $m \times n$, $n \times 1$, and $m \times 1$. The relations $(*)$ describing the dependence of $\mathbf{y}$ on $\mathbf{x}$ may be written symbolically in the following way:

$$\mathbf{y} = \mathbf{A}\mathbf{x}. \qquad (**)$$

One gets the impression that the matrix $\mathbf{A}$ is *multiplied* by the column matrix $\mathbf{x}$, resulting in the column matrix $\mathbf{y}$. And so it will be if we say that the relations $(*)$ constitute the *definition* of the operation $(**)$ of multiplying $\mathbf{A}$ by $\mathbf{x}$.

If $m = n$, the matrix is called *square*. A square matrix of size $n \times n$ is also called a *matrix of order* n.

Notation and Special Types of Matrices ADDED FOR MA1522

A (real) $m \times n$ matrix is a rectangular array of real numbers with m rows and n columns; $m \times n$ (read "m by n") is its *size*. The numbers in the array are the *entries* of the matrix, and the number a_{ij} standing in the i-th row and the j-th column is its (i, j)-*entry*. We write

$$\mathbf{A} = (a_{ij})_{m\times n} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix},$$

and $\mathbf{A} = (a_{ij})_n$ for a square matrix of order n. If the two indices could be confused, they are separated by a comma: $a_{1,23}$ and $a_{12,3}$ are different entries. A matrix may be specified by a rule for its entries; for example, the 2×3 matrix with $a_{ij} = 2i - j$ is

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & -1 \\ 3 & 2 & 1 \end{pmatrix}.$$

Definition.

- An $n \times 1$ matrix is a *column vector*, and a $1 \times n$ matrix is a *row vector*. A vector is understood to be a column vector unless stated otherwise.
- The *zero matrix* $\mathbf{0}_{m\times n}$ is the $m \times n$ matrix all of whose entries are 0; it need not be square.
- In a square matrix $\mathbf{A} = (a_{ij})_n$ the entries $a_{11}, a_{22}, ..., a_{nn}$ are the *diagonal entries*; they form the *main diagonal*.
- A square matrix $\mathbf{D} = (a_{ij})_n$ is *diagonal* if $a_{ij} = 0$ whenever $i \neq j$. With $d_i = a_{ii}$ we write $\mathbf{D} = \operatorname{diag}(d_1, d_2, ..., d_n)$.
- A diagonal matrix with all diagonal entries equal, $\operatorname{diag}(c, c, ..., c)$, is a *scalar matrix*. The scalar matrix with $c = 1$ is the *identity matrix* $\mathbf{I}_n = \operatorname{diag}(1, 1, ..., 1)$, also written $\mathbf{I}$. Using scalar multiplication (Section 2.2), a scalar matrix is $c\mathbf{I}_n$.
- A square matrix $\mathbf{A} = (a_{ij})_n$ is *upper triangular* if $a_{ij} = 0$ for all $i > j$ (all entries below the main diagonal vanish), and *strictly upper triangular* if $a_{ij} = 0$ for all $i \geq j$. It is *lower triangular* if $a_{ij} = 0$ for all $i < j$, and *strictly lower triangular* if $a_{ij} = 0$ for all $i \leq j$. Upper and lower triangular matrices are called *triangular*.
- A square matrix $\mathbf{A} = (a_{ij})_n$ is *symmetric* if $a_{ij} = a_{ji}$ for all i and j, that is, if its entries are symmetric about the main diagonal.

Example. The matrices

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 2 & 5 & -1 \\ 0 & 0 & 4 \\ 0 & 0 & 7 \end{pmatrix}, \quad \begin{pmatrix} 1 & 4 & -2 \\ 4 & 0 & 5 \\ -2 & 5 & 3 \end{pmatrix}$$

are diagonal (namely $\operatorname{diag}(3, -1, 0)$), upper triangular, and symmetric, respectively. The zero matrix $\mathbf{0}_{n\times n}$ is diagonal, and scalar with $c = 0$, but it is not an identity matrix. A matrix that is both symmetric and upper triangular is diagonal.

2.2. Matrix Algebra

Matrix Multiplication

Suppose that $y_1, ..., y_m$ are expressed in terms of $x_1, ..., x_n$, and at the same time $x_1, ..., x_n$ are expressed in terms of $z_1, ..., z_k$ as follows:

$$\begin{cases} y_1 = a_{11}x_1 + ... + a_{1n}x_n, \\ ... \\ y_m = a_{m1}x_1 + ... + a_{mn}x_n, \end{cases} \qquad \begin{cases} x_1 = b_{11}z_1 + ... + b_{1k}z_k, \\ ... \\ x_n = b_{n1}z_1 + ... + b_{nk}z_k. \end{cases}$$

Clearly, $y_1, ..., y_m$ are expressed in terms of $z_1, ..., z_k$ in a similar manner. Let us denote the matrix of constant coefficients of this dependence by $\mathbf{C}$. Then

$$\mathbf{y} = \mathbf{Ax}, \quad \mathbf{x} = \mathbf{Bz} \quad \text{and} \quad \mathbf{y} = \mathbf{Cz}.$$

To obtain the entries of the matrix $\mathbf{C}$, one must substitute the expressions for $x_1, ..., x_n$ in terms of $z_1, ..., z_k$ into the formulas expressing $y_1, ..., y_m$ in terms of $x_1, ..., x_n$, and collect the coefficients of the quantities $z_1, ..., z_k$. The result is:

$$\mathbf{C} = (c_{ij}), \quad \text{where} \quad c_{ij} = \sum_{l=1}^{n} a_{il}b_{lj}. \qquad (*)$$

Definition. The matrix $\mathbf{C}$ of the form $(*)$ is called the *product* of the matrices $\mathbf{A}$ and $\mathbf{B}$ and is denoted by $\mathbf{C} = \mathbf{AB}$.

Corollary. $\mathbf{y} = \mathbf{A}(\mathbf{Bz}) = (\mathbf{AB})\mathbf{z}$.

It is often said that matrices are multiplied according to the "row-by-column" rule. The number of columns in the first factor must, of course, coincide with the number of rows in the second. When we write $\mathbf{C} = \mathbf{AB}$, we automatically mean that the matrices $\mathbf{A}$ and $\mathbf{B}$ are not altogether arbitrary.

Associativity of Matrix Multiplication

Theorem. $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$.

Proof. Let $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ have sizes $m \times n$, $n \times k$, $k \times l$. Then

$$\{(\mathbf{AB})\mathbf{C}\}_{ij} = \sum_{p=1}^{k} \{\mathbf{AB}\}_{ip}c_{pj} = \sum_{p=1}^{k} \left(\sum_{q=1}^{n} a_{iq}b_{qp} \right) c_{pj} =$$

$$= \sum_{q=1}^{n} a_{iq} \left(\sum_{p=1}^{k} b_{qp}c_{pj} \right) = \{\mathbf{A}(\mathbf{BC})\}_{ij}. \qquad \square$$

Noncommutativity of Matrix Multiplication

In the general case $\mathbf{AB} \neq \mathbf{BA}$ even for square matrices. For example,

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$
$$\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Matrix Addition and Multiplication by a Scalar

A matrix $\mathbf{C} = (c_{ij})$ is called the *sum* of the matrices $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$ if

$$c_{ij} = a_{ij} + b_{ij} \quad \text{for all} \quad i, j.$$

The matrices $\mathbf{A}, \mathbf{B}$ and $\mathbf{C} = \mathbf{A} + \mathbf{B}$ have the same dimensions. The operation of matrix addition possesses two pleasant properties at once:

$$\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C} \quad \text{(associativity)},$$
$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A} \quad \text{(commutativity)}.$$

It is also useful to introduce the operation of multiplying a matrix by a scalar (number). If α is a scalar, then the matrix $\mathbf{C} = \alpha\mathbf{A}$ is defined as the matrix of the same dimensions with entries $c_{ij} = \alpha a_{ij}$.

Equality and the Laws of Matrix Arithmetic ADDED FOR MA1522

Matrices $\mathbf{A} = (a_{ij})_{m\times n}$ and $\mathbf{B} = (b_{ij})_{k\times l}$ are *equal*, $\mathbf{A} = \mathbf{B}$, if they have the same size ($m = k$, $n = l$) and $a_{ij} = b_{ij}$ for all i, j. Matrices of different sizes are never equal, and matrix addition is defined only for matrices of the same size. The *negative* of $\mathbf{A}$ is $-\mathbf{A} = (-1)\mathbf{A}$, and the *difference* of matrices of the same size is $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-1)\mathbf{B}$.

Theorem. Let $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ be $m \times n$ matrices and $\alpha, \beta \in \mathbb{R}$. Then:

(i) $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$ (commutativity);
(ii) $\mathbf{A} + (\mathbf{B} + \mathbf{C}) = (\mathbf{A} + \mathbf{B}) + \mathbf{C}$ (associativity);
(iii) $\mathbf{0}_{m\times n} + \mathbf{A} = \mathbf{A}$ (additive identity);
(iv) $\mathbf{A} + (-\mathbf{A}) = \mathbf{0}_{m\times n}$ (additive inverse);
(v) $\alpha(\mathbf{A} + \mathbf{B}) = \alpha\mathbf{A} + \alpha\mathbf{B}$ (distributive law);
(vi) $(\alpha + \beta)\mathbf{A} = \alpha\mathbf{A} + \beta\mathbf{A}$;
(vii) $(\alpha\beta)\mathbf{A} = \alpha(\beta\mathbf{A})$;
(viii) if $\alpha\mathbf{A} = \mathbf{0}_{m\times n}$, then $\alpha = 0$ or $\mathbf{A} = \mathbf{0}_{m\times n}$.

Proof. Both operations act entrywise, so each identity reduces to the corresponding law for real numbers applied to the (i, j)-entries. For instance, the (i, j)-entry of $\alpha(\mathbf{A} + \mathbf{B})$ is $\alpha(a_{ij} + b_{ij}) = \alpha a_{ij} + \alpha b_{ij}$, which is the (i, j)-entry of $\alpha\mathbf{A} + \alpha\mathbf{B}$; and the (i, j)-entry of $(\alpha\beta)\mathbf{A}$ is $(\alpha\beta)a_{ij} = \alpha(\beta a_{ij})$. For (viii): if $\alpha a_{ij} = 0$ for all i, j and $\alpha \neq 0$, then all $a_{ij} = 0$. □

By associativity, a sum $\mathbf{A}_1 + \mathbf{A}_2 + \cdots + \mathbf{A}_k$ of matrices of the same size needs no brackets. In a product $\mathbf{AB}$ we say that $\mathbf{A}$ *pre-multiplies* $\mathbf{B}$ (is multiplied on the left) and $\mathbf{B}$ *post-multiplies* $\mathbf{A}$; since $\mathbf{AB} \neq \mathbf{BA}$ in general, the side matters.

Theorem. Let $\alpha \in \mathbb{R}$, and let the matrices have sizes for which the operations below are defined. Then:

(i) $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$ (associativity);
(ii) $\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB} + \mathbf{AC}$ (left distributive law);
(iii) $(\mathbf{A}+\mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$ (right distributive law);
(iv) $\alpha(\mathbf{AB}) = (\alpha\mathbf{A})\mathbf{B} = \mathbf{A}(\alpha\mathbf{B})$;
(v) $\mathbf{I}_m\mathbf{A} = \mathbf{A} = \mathbf{A}\mathbf{I}_n$ for every $m \times n$ matrix $\mathbf{A}$ (multiplicative identity);
(vi) there exist matrices $\mathbf{A} \neq \mathbf{0}_{m\times p}$ and $\mathbf{B} \neq \mathbf{0}_{p\times n}$ with $\mathbf{AB} = \mathbf{0}_{m\times n}$ (zero divisors);
(vii) $\mathbf{A}\mathbf{0}_{n\times p} = \mathbf{0}_{m\times p}$ and $\mathbf{0}_{p\times m}\mathbf{A} = \mathbf{0}_{p\times n}$ for every $m \times n$ matrix $\mathbf{A}$.

Proof. Property (i) is the theorem on associativity proved above. For (ii), the (i,j)-entry of $\mathbf{A}(\mathbf{B}+\mathbf{C})$ is

$$\sum_k a_{ik}\big(b_{kj} + c_{kj}\big) = \sum_k a_{ik}b_{kj} + \sum_k a_{ik}c_{kj},$$

the (i,j)-entry of $\mathbf{AB} + \mathbf{AC}$; (iii) is proved in the same way, and (iv) follows from $\alpha\sum_k a_{ik}b_{kj} = \sum_k (\alpha a_{ik})b_{kj} = \sum_k a_{ik}\big(\alpha b_{kj}\big)$. For (v), the (i,k)-entry of $\mathbf{I}_m$ is 1 for $k = i$ and 0 otherwise, so the (i,j)-entry of $\mathbf{I}_m\mathbf{A}$ is a_{ij}; similarly $\mathbf{A}\mathbf{I}_n = \mathbf{A}$. In (vii) every term of every entry contains a zero factor. For (vi), take

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 2 & -4 \\ -1 & 2 \end{pmatrix} : \quad \mathbf{AB} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \qquad \square$$

So $\mathbf{AB} = \mathbf{0}$ does not imply that $\mathbf{A} = \mathbf{0}$ or $\mathbf{B} = \mathbf{0}$; nor does it imply $\mathbf{BA} = \mathbf{0}$: for the matrices just used, $\mathbf{BA} = \begin{pmatrix} -6 & -12 \\ 3 & 6 \end{pmatrix}$. By associativity, a product $\mathbf{A}_1\mathbf{A}_2\cdots\mathbf{A}_k$ of matrices of compatible sizes is also written without brackets.

Powers and the Transpose of a Matrix ADDED FOR MA1522

For a square matrix $\mathbf{A}$ of order n, the *powers* of $\mathbf{A}$ are defined inductively:

$$\mathbf{A}^0 = \mathbf{I}_n, \quad \mathbf{A}^k = \mathbf{A}\mathbf{A}^{k-1}, \quad k \geq 1.$$

Thus $\mathbf{A}^k$ is the product of k factors equal to $\mathbf{A}$, and by associativity $\mathbf{A}^k\mathbf{A}^l = \mathbf{A}^{k+l}$ for all nonnegative integers k, l. Powers are defined only for square matrices, since $\mathbf{AA}$ requires $\mathbf{A}$ to have as many columns as rows. For example, induction on k gives

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}, \quad \text{because} \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & k-1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}.$$

Since multiplication is not commutative, the identities of school algebra need care: $(\mathbf{A}+\mathbf{B})^2 = \mathbf{A}^2 + \mathbf{AB} + \mathbf{BA} + \mathbf{B}^2$, and this equals $\mathbf{A}^2 + 2\mathbf{AB} + \mathbf{B}^2$ if and only if $\mathbf{AB} = \mathbf{BA}$ (as happens, for example, for diagonal matrices).

The *transpose* of an $m \times n$ matrix $\mathbf{A} = \big(a_{ij}\big)_{m\times n}$ is the $n \times m$ matrix $\mathbf{A}^T = \big(b_{ij}\big)_{n\times m}$ with $b_{ij} = a_{ji}$: the i-th row of $\mathbf{A}$ becomes the i-th column of $\mathbf{A}^T$. (The definition is repeated in Section 2.10.) For example,

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

Theorem. For matrices of sizes for which the operations are defined and every $\alpha \in \mathbb{R}$:
(i) $(\mathbf{A}^T)^T = \mathbf{A}$;
(ii) $(\alpha\mathbf{A})^T = \alpha\mathbf{A}^T$;
(iii) $(\mathbf{A}+\mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$;
(iv) $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$.

Proof. The (i,j)-entries of the two sides of (i), (ii) and (iii) are a_{ij}, αa_{ji} and $a_{ji} + b_{ji}$, respectively. Property (iv) is proved in Section 2.4 (subsection "Inversion and Transposition"). □

Note the reversed order in (iv): if $\mathbf{A}$ is $m \times p$ and $\mathbf{B}$ is $p \times n$, then $\mathbf{A}^T\mathbf{B}^T$ is not even defined unless $m = n$. A square matrix $\mathbf{A}$ is symmetric if and only if $\mathbf{A}^T = \mathbf{A}$. By (ii) and (iii), scalar multiples and sums of symmetric matrices are symmetric. A product of symmetric matrices need not be: by (iv), $(\mathbf{AB})^T = \mathbf{BA}$ for symmetric $\mathbf{A}$ and $\mathbf{B}$, so $\mathbf{AB}$ is symmetric exactly when $\mathbf{AB} = \mathbf{BA}$; for instance, $\operatorname{diag}(1,2)\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$.

2.3. Linear System and Matrix Equation

Multiplication of Block Matrices

Suppose that the matrices $\mathbf{A}$ and $\mathbf{B}$ are partitioned into blocks $\mathbf{A}_{ij}$ and $\mathbf{B}_{ij}$:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \dots & \mathbf{A}_{1q} \\ \dots & \dots & \dots \\ \mathbf{A}_{p1} & \dots & \mathbf{A}_{pq} \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} \mathbf{B}_{11} & \dots & \mathbf{B}_{1r} \\ \dots & \dots & \dots \\ \mathbf{B}_{q1} & \dots & \mathbf{B}_{qr} \end{pmatrix},$$

where $\mathbf{A}_{ij}$ is $m_i \times n_j$ and $\mathbf{B}_{ij}$ is $n_i \times k_j$. Then the product $\mathbf{C} = \mathbf{AB}$ exists and can be computed using the operations of multiplication and addition of block submatrices:

$$\mathbf{C} = \begin{pmatrix} \mathbf{C}_{11} & \dots & \mathbf{C}_{1r} \\ \dots & \dots & \dots \\ \mathbf{C}_{p1} & \dots & \mathbf{C}_{pr} \end{pmatrix}, \quad \text{where} \quad \mathbf{C}_{ij} = \sum_{l=1}^{q} \mathbf{A}_{il}\mathbf{B}_{lj} \quad \text{is} \quad m_i \times k_j.$$

Prove this!

One may say that block matrices are multiplied by the "block-row by block-column" rule. We shall see very soon how useful block multiplication can be.

Linear Combinations

For understanding the essence of the matter, the following interpretation of the linear system (1) of Section 1.1, $\mathbf{Ax} = \mathbf{b}$, is exceptionally useful. By the definition of multiplying a matrix by a scalar, if $\alpha \in \mathbb{R}$, then

$$\alpha \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \equiv \begin{pmatrix} \alpha b_1 \\ \vdots \\ \alpha b_n \end{pmatrix}.$$

Let $\mathbf{a}_1, ..., \mathbf{a}_k$ be the columns of the matrix $\mathbf{A}$:

$$\mathbf{A} = (\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \cdots \ \ \mathbf{a}_k), \quad \mathbf{a}_1, ..., \mathbf{a}_k \in \mathbb{R}^n.$$

Then the relations (1) are equivalent to the vector equality

$$x_1\mathbf{a}_1 + ... + x_k\mathbf{a}_k = \mathbf{b}. \tag{2}$$

The expression $x_1\mathbf{a}_1 + ... + x_k\mathbf{a}_k$ is called a *linear combination* of the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$, and the numbers $x_1, ..., x_k$ are called the *coefficients of the linear combination*. The set of all possible linear combinations of the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$,

$$\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} = \{\alpha_1\mathbf{a}_1 + ... + \alpha_k\mathbf{a}_k : \alpha_1, ..., \alpha_k \in \mathbb{R}\},$$

is called the *linear span* of the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$.

Thus, the equality (2) means that

$$\mathbf{b} \in \text{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}. \tag{3}$$

In other words, the system (1) has a solution (is consistent) if and only if the right-hand side $\mathbf{b}$ belongs to the linear span (is a linear combination) of the columns of the coefficient matrix.

Matrix Equations and Homogeneous Systems ADDED FOR MA1522

A linear system of m equations in n variables, $a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = b_i$, $i = 1, ..., m$, is written as the *matrix equation* $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A} = \big(a_{ij}\big)_{m\times n}$ is the coefficient matrix, $\mathbf{x} = \big(x_j\big)_{n\times 1}$ is the *variable vector*, and $\mathbf{b} = (b_i)_{m\times 1}$ is the constant vector; or, as in (2), as the *vector equation*

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b},$$

in which the j-th column $\mathbf{a}_j$ of $\mathbf{A}$ is called the *coefficient vector* of the variable x_j. A solution of the system is the same as a vector $\mathbf{u} \in \mathbb{R}^n$ with $\mathbf{Au} = \mathbf{b}$.

The system is homogeneous when $\mathbf{b} = \mathbf{0}$. A homogeneous system $\mathbf{Ax} = \mathbf{0}$ is always consistent, since $\mathbf{A0} = \mathbf{0}$: the zero vector is a solution, called the *trivial solution*, and a solution $\mathbf{u} \neq \mathbf{0}$ is called a *nontrivial solution*. Conversely, if $\mathbf{x} = \mathbf{0}$ is a solution of $\mathbf{Ax} = \mathbf{b}$, then $\mathbf{b} = \mathbf{A0} = \mathbf{0}$, so the system is homogeneous.

Theorem. A homogeneous linear system has infinitely many solutions if and only if it has a nontrivial solution.

Proof. If there are infinitely many solutions, at least one of them is nonzero. Conversely, let $\mathbf{Au} = \mathbf{0}$ with $\mathbf{u} \neq \mathbf{0}$. For every $s \in \mathbb{R}$ we have $\mathbf{A}(s\mathbf{u}) = s(\mathbf{Au}) = \mathbf{0}$ (Section 2.2), and the vectors $s\mathbf{u}$, $s \in \mathbb{R}$, are pairwise distinct. □

Hence a homogeneous system has either only the trivial solution or infinitely many solutions; in particular, if its solution is unique, it is the trivial one. For example, the system

$$x_1 + x_2 - 2x_3 = 0, \quad 2x_1 - x_2 - x_3 = 0$$

has the nontrivial solution $(1, 1, 1)^T$, and therefore all the solutions $s(1, 1, 1)^T$, $s \in \mathbb{R}$. There are no others: adding the two equations gives $x_1 = x_3$, and then the first equation gives $x_2 = x_3$.

The relation between the solutions of $\mathbf{Ax} = \mathbf{b}$ *and those of* $\mathbf{Ax} = \mathbf{0}$ *(if* $\mathbf{v}$ *is a particular solution of* $\mathbf{Ax} = \mathbf{b}$*, its solutions are the vectors* $\mathbf{v} + \mathbf{u}$ *with* $\mathbf{Au} = \mathbf{0}$*; the difference of two solutions solves* $\mathbf{Ax} = \mathbf{0}$*) is treated in Section 4.2.*

Submatrices of a Product ADDED FOR MA1522

A *submatrix* of a matrix $\mathbf{A}$ is formed by the entries at the intersections of some chosen rows and columns of $\mathbf{A}$ (see also Section 2.8); in particular, each row and each column of $\mathbf{A}$ is a submatrix. The following special case of the rule for multiplying block matrices is used constantly.

Theorem (Block Multiplication)**.** Let $\mathbf{A}$ be an $m \times p$ matrix and $\mathbf{B}$ a $p \times n$ matrix. Let $\mathbf{A}_1$ be the submatrix of $\mathbf{A}$ formed by its rows $m_1, m_1 + 1, ..., m_2$, and $\mathbf{B}_1$ the submatrix of $\mathbf{B}$ formed by its columns $n_1, n_1 + 1, ..., n_2$. Then $\mathbf{A}_1\mathbf{B}_1$ is the $(m_2 - m_1 + 1) \times (n_2 - n_1 + 1)$ submatrix of $\mathbf{AB}$ formed by its rows $m_1, ..., m_2$ and columns $n_1, ..., n_2$.

Proof. The two matrices have the same size. The entry of $\mathbf{A}_1\mathbf{B}_1$ that stands in the place of the (i, j)-entry of $\mathbf{AB}$ is the product of the row of $\mathbf{A}_1$ coming from the i-th row of $\mathbf{A}$ and the column of $\mathbf{B}_1$ coming from the j-th column of $\mathbf{B}$, that is, $\sum_{k=1}^{p} a_{ik}b_{kj}$, which is the (i, j)-entry of $\mathbf{AB}$. □

In particular, write $\mathbf{B} = (\mathbf{b}_1 \;\; \mathbf{b}_2 \;\; \cdots \;\; \mathbf{b}_n)$ by its columns and let $\mathbf{r}_1, ..., \mathbf{r}_m$ be the rows of $\mathbf{A}$. Taking for $\mathbf{B}_1$ a single column, or for $\mathbf{A}_1$ a single row, we get

$$\mathbf{AB} = (\mathbf{Ab}_1 \;\; \mathbf{Ab}_2 \;\; \cdots \;\; \mathbf{Ab}_n), \qquad \mathbf{AB} = \begin{pmatrix} \mathbf{r}_1\mathbf{B} \\ \mathbf{r}_2\mathbf{B} \\ \vdots \\ \mathbf{r}_m\mathbf{B} \end{pmatrix}:$$

the j-th column of $\mathbf{AB}$ is $\mathbf{A}$ times the j-th column of $\mathbf{B}$ (the *column rule*), and the i-th row of $\mathbf{AB}$ is the i-th row of $\mathbf{A}$ times $\mathbf{B}$ (the *row rule*). For example, for

$$\mathbf{A}=\begin{pmatrix}1&0&2\\-1&3&1\\2&1&0\end{pmatrix},\quad \mathbf{B}=\begin{pmatrix}1&2&0&1\\0&1&-1&2\\3&0&1&1\end{pmatrix},\quad \mathbf{AB}=\begin{pmatrix}7&2&2&3\\2&1&-2&6\\2&5&-1&4\end{pmatrix},$$

rows 2 and 3 of $\mathbf{A}$ times columns 2 and 3 of $\mathbf{B}$ give

$$\begin{pmatrix}-1&3&1\\2&1&0\end{pmatrix}\begin{pmatrix}2&0\\1&-1\\0&1\end{pmatrix}=\begin{pmatrix}1&-2\\5&-1\end{pmatrix},$$

which is indeed the submatrix of $\mathbf{AB}$ in rows 2, 3 and columns 2, 3.

Solving Matrix Equations ADDED FOR MA1522

Let $\mathbf{A}$ be an $m\times n$ matrix and $\mathbf{B}=\begin{pmatrix}\mathbf{b}_1&\mathbf{b}_2&\cdots&\mathbf{b}_p\end{pmatrix}$ an $m\times p$ matrix. By the column rule, an $n\times p$ matrix $\mathbf{X}=\begin{pmatrix}\mathbf{x}_1&\mathbf{x}_2&\cdots&\mathbf{x}_p\end{pmatrix}$ satisfies the matrix equation $\mathbf{AX}=\mathbf{B}$ if and only if its columns solve the p linear systems

$$\mathbf{Ax}_1=\mathbf{b}_1,\quad \mathbf{Ax}_2=\mathbf{b}_2,\quad ...,\quad \mathbf{Ax}_p=\mathbf{b}_p.$$

All these systems have the same coefficient matrix, so the same row operations reduce each of them. It therefore suffices to reduce once the *combined augmented matrix*

$$(\mathbf{A}\mid\mathbf{b}_1\mid\mathbf{b}_2\mid\cdots\mid\mathbf{b}_p),$$

also written $(\mathbf{A}\mid\mathbf{B})$, and to read the solutions of all p systems off its (reduced) row-echelon form.

Example. Let us solve $\mathbf{AX}=\mathbf{B}$ for

$$\mathbf{A}=\begin{pmatrix}1&1&1\\0&1&2\\1&2&4\end{pmatrix},\quad \mathbf{B}=\begin{pmatrix}2&2\\2&1\\5&4\end{pmatrix}.$$

The operations R_3-R_1 and R_3-R_2 produce a row-echelon form, and R_2-2R_3, R_1-R_3, R_1-R_2 then produce the reduced row-echelon form:

$$\left(\begin{array}{ccc|c|c}1&1&1&2&2\\0&1&2&2&1\\1&2&4&5&4\end{array}\right)\longrightarrow\left(\begin{array}{ccc|c|c}1&1&1&2&2\\0&1&2&2&1\\0&0&1&1&1\end{array}\right)\longrightarrow\left(\begin{array}{ccc|c|c}1&0&0&1&2\\0&1&0&0&-1\\0&0&1&1&1\end{array}\right).$$

Both systems have unique solutions, $\mathbf{x}_1=(1,0,1)^T$ and $\mathbf{x}_2=(2,-1,1)^T$, so $\mathbf{X}=\begin{pmatrix}1&2\\0&-1\\1&1\end{pmatrix}$ is the unique solution of $\mathbf{AX}=\mathbf{B}$.

2.4. Inverse of Matrices

Inverse Matrix

A matrix $\mathbf{A}$ of order n is called *invertible* if there exists a matrix $\mathbf{X}$ of order n such that

$$\mathbf{AX} = \mathbf{XA} = \mathbf{I},$$

where $\mathbf{I}$ is the identity matrix of order n; $\mathbf{X}$ is called the *inverse matrix* for $\mathbf{A}$.

There can exist only one inverse matrix: if $\mathbf{AX} = \mathbf{XA} = \mathbf{I}$ and $\mathbf{AY} = \mathbf{YA} = \mathbf{I}$, then $\mathbf{X} = \mathbf{X}(\mathbf{AY}) = (\mathbf{XA})\mathbf{Y} = \mathbf{Y}$.

Notation for the inverse matrix: $\mathbf{X} = \mathbf{A}^{-1}$.

Problem 1. Let $\mathbf{A}, \mathbf{B}$ be arbitrary matrices of order n; $\mathbf{I}$ and $\mathbf{0}$ be the identity and zero matrices of order n. Prove that

$$\begin{pmatrix} \mathbf{I} & \mathbf{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} & \mathbf{B} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{pmatrix}^{-1} = \begin{pmatrix} \mathbf{I} & -\mathbf{A} & \mathbf{AB} \\ \mathbf{0} & \mathbf{I} & -\mathbf{B} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{pmatrix}.$$

(It follows from this that any algorithm for computing the inverse of a matrix of order n with operation count $s(n)$ generates an algorithm for multiplying matrices of order n with operation count $s(3n)$.)

Problem 2. Given a square matrix $\mathbf{A}$ such that $\mathbf{A}^3 = \mathbf{0}$. Prove that the matrix $\mathbf{I} - \mathbf{A}$ is invertible.

Problem 3. Find all invertible matrices $\mathbf{A}$ of order n for which all entries of $\mathbf{A}$ and $\mathbf{A}^{-1}$ are nonnegative. Prove that the set of all such matrices forms a group with respect to the operation of matrix multiplication.

Problem 4. Given matrices $\mathbf{P}_1, ..., \mathbf{P}_n$ of order n, each differing from the identity matrix by a permutation of columns (such matrices are called *permutation matrices*). Let $\mathbf{P}_1 + ... + \mathbf{P}_n = \mathbf{E}$, where $\mathbf{E}$ is the matrix all of whose entries are equal to 1. Furthermore, let $\mathbf{P}_i\mathbf{P}_j = \mathbf{P}_j\mathbf{P}_i$ for all i, j. Prove that the set of matrices $\mathbf{P}_1, ..., \mathbf{P}_n$ forms a group with respect to the operation of matrix multiplication.

Inversion and Transposition

Proposition. $(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$.

Proof. $(\mathbf{AB})_{ij} = \sum_k (\mathbf{A})_{ik}(\mathbf{B})_{kj} = \sum_k (\mathbf{B})_{kj}(\mathbf{A})_{ik} = \sum_k \left(\mathbf{B}^T\right)_{jk}\left(\mathbf{A}^T\right)_{ki} = \left(\mathbf{B}^T\mathbf{A}^T\right)_{ji}$. □

From the equality $\mathbf{AX} = \mathbf{XA} = \mathbf{I}$ we obtain: $\mathbf{X}^T\mathbf{A}^T = \mathbf{A}^T\mathbf{X}^T = \mathbf{I}$. Thus, *a matrix is invertible if and only if its transpose is invertible.* Moreover,

$$\mathbf{X}^T = \left(\mathbf{A}^T\right)^{-1} = \left(\mathbf{A}^{-1}\right)^T.$$

Notation: $\mathbf{A}^{-T} \equiv \left(\mathbf{A}^{-1}\right)^T$.

As a corollary, we obtain the "row" analogue of the matrix invertibility criterion (Section 2.6): *the invertibility of a square matrix is equivalent to the linear independence of its rows.*

Group of Invertible Matrices

The set of invertible $(n \times n)$-matrices forms a group with respect to the operation of multiplication. For the proof we have everything except the fact

that the product of invertible matrices is invertible. But this is verified directly: if $\mathbf{A}$ and $\mathbf{B}$ are invertible, then

$$(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) = \mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B} = \mathbf{B}^{-1}\mathbf{IB} = \mathbf{I},$$
$$(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{A}(\mathbf{BB}^{-1})\mathbf{A}^{-1} = \mathbf{AIA}^{-1} = \mathbf{I}.$$

Hence

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}.$$

Problem 5. Let H be the group of invertible upper triangular matrices of order n, and G the group of all invertible matrices of order n. Is it true that $\mathbf{ABA}^{-1} \in H$ for any matrices $\mathbf{B} \in H$ and $\mathbf{A} \in G$ (in which case the subgroup H is called a *normal subgroup* of the group G)?

Singular Matrices and the Cancellation Law ADDED FOR MA1522

Only square matrices can be invertible in the sense of the definition above. A square matrix that is not invertible is called *singular*. (Some books call every non-invertible matrix, square or not, singular.) Non-square matrices may have one-sided inverses; see Section 2.6.

For real numbers, $ab = ac$ with $a \neq 0$ implies $b = c$. For matrices this *cancellation* fails even when $\mathbf{A} \neq \mathbf{0}$:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{B} = \mathbf{I}_2, \quad \mathbf{C} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}:$$
$$\mathbf{AB} = \mathbf{AC} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \text{but} \quad \mathbf{B} \neq \mathbf{C}.$$

Invertibility restores it.

Theorem (Cancellation Law)**.** Let $\mathbf{A}$ be an invertible matrix of order n.

(i) (Left cancellation) If $\mathbf{B}$ and $\mathbf{C}$ are $n \times p$ matrices with $\mathbf{AB} = \mathbf{AC}$, then $\mathbf{B} = \mathbf{C}$.

(ii) (Right cancellation) If $\mathbf{B}$ and $\mathbf{C}$ are $p \times n$ matrices with $\mathbf{BA} = \mathbf{CA}$, then $\mathbf{B} = \mathbf{C}$.

Proof. $\mathbf{B} = \mathbf{A}^{-1}\mathbf{AB} = \mathbf{A}^{-1}\mathbf{AC} = \mathbf{C}$, and $\mathbf{B} = \mathbf{BAA}^{-1} = \mathbf{CAA}^{-1} = \mathbf{C}$. □

The matrix must be cancelled on the same side: $\mathbf{AB} = \mathbf{CA}$ only gives $\mathbf{B} = \mathbf{A}^{-1}\mathbf{CA}$. For instance,

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{B} = \operatorname{diag}(1, 2), \quad \mathbf{C} = \operatorname{diag}(2, 1):$$
$$\mathbf{AB} = \mathbf{CA} = \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix}, \quad \text{but} \quad \mathbf{B} \neq \mathbf{C}.$$

Theorem. A 2×2 matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is invertible if and only if $ad - bc \neq 0$, and in this case

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Proof. Let $\mathbf{B} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. Direct multiplication gives $\mathbf{AB} = \mathbf{BA} = (ad - bc)\mathbf{I}_2$. Hence, if $ad - bc \neq 0$, the matrix $(ad - bc)^{-1}\mathbf{B}$ is the inverse of $\mathbf{A}$. If $ad - bc = 0$, then $\mathbf{AB} = \mathbf{0}$; were $\mathbf{A}$ invertible, we would get $\mathbf{B} = \mathbf{A}^{-1}\mathbf{AB} = \mathbf{0}$, so $a = b = c = d = 0$ and $\mathbf{A} = \mathbf{0}$, which is not invertible—a contradiction. □

For instance,

$$\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}^{-1} = \frac{1}{5} \begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix}.$$

Here $ad - bc = \det(\mathbf{A})$, and the matrix $\mathbf{B}$ of the proof is the adjoint $\operatorname{adj}(\mathbf{A})$; so the formula is the case $n = 2$ of the adjoint formula for the inverse (Section 2.10).

Theorem. Let $\mathbf{A}$ and $\mathbf{B}$ be invertible matrices of order n, and let $\alpha \neq 0$ be a real number. Then:

(i) $\mathbf{A}^{-1}$ is invertible and $\left(\mathbf{A}^{-1}\right)^{-1} = \mathbf{A}$;
(ii) $\alpha\mathbf{A}$ is invertible and $(\alpha\mathbf{A})^{-1} = \alpha^{-1}\mathbf{A}^{-1}$;
(iii) $\mathbf{A}^T$ is invertible and $\left(\mathbf{A}^T\right)^{-1} = \left(\mathbf{A}^{-1}\right)^T$;
(iv) $\mathbf{AB}$ is invertible and $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.

Proof. (i) The equalities $\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ say that $\mathbf{A}$ is an inverse of $\mathbf{A}^{-1}$. (ii) $(\alpha^{-1}\mathbf{A}^{-1})(\alpha\mathbf{A}) = (\alpha^{-1}\alpha)\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$, and likewise in the other order. Statements (iii) and (iv) are proved in the subsections "Inversion and Transposition" (where it is also shown that $\mathbf{A}$ is invertible if and only if $\mathbf{A}^T$ is) and "Group of Invertible Matrices". □

By (iv) and induction, a product of invertible matrices $\mathbf{A}_1, \mathbf{A}_2, ..., \mathbf{A}_k$ of order n is invertible and

$$(\mathbf{A}_1\mathbf{A}_2 \cdots \mathbf{A}_k)^{-1} = \mathbf{A}_k^{-1} \cdots \mathbf{A}_2^{-1}\mathbf{A}_1^{-1};$$

note the reversed order: in general $(\mathbf{AB})^{-1} \neq \mathbf{A}^{-1}\mathbf{B}^{-1}$. *Negative powers* of an invertible matrix are defined by $\mathbf{A}^{-k} = \left(\mathbf{A}^{-1}\right)^k$ for $k > 0$. By (iii), the inverse of an invertible symmetric matrix is symmetric: $\left(\mathbf{A}^{-1}\right)^T = \left(\mathbf{A}^T\right)^{-1} = \mathbf{A}^{-1}$.

Invertible Matrices and Linear Systems ADDED FOR MA1522

Theorem. If $\mathbf{A}$ is an invertible matrix of order n, then for every $\mathbf{b} \in \mathbb{R}^n$ the system $\mathbf{Ax} = \mathbf{b}$ has a unique solution, namely $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$.

Proof. The vector $\mathbf{u} = \mathbf{A}^{-1}\mathbf{b}$ is a solution: $\mathbf{Au} = \mathbf{AA}^{-1}\mathbf{b} = \mathbf{b}$. If $\mathbf{v}$ is also a solution, then $\mathbf{Av} = \mathbf{b} = \mathbf{Au}$, and left cancellation gives $\mathbf{v} = \mathbf{u}$. □

Corollary. If $\mathbf{A}$ is invertible, then the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution.

Section 2.6 shows that both properties characterize invertible matrices. It also shows that the reduced row-echelon form of an invertible matrix $\mathbf{A}$ is $\mathbf{I}$; then the reduced row-echelon form of $(\mathbf{A} \mid \mathbf{b})$ is $(\mathbf{I} \mid \mathbf{c})$, where $\mathbf{c}$ is the unique solution, so that

$$(\mathbf{A} \mid \mathbf{b}) \longrightarrow \cdots \longrightarrow (\mathbf{I} \mid \mathbf{A}^{-1}\mathbf{b}).$$

Similarly, the inverse $\mathbf{A}^{-1}$ is the unique solution $\mathbf{X}$ of the matrix equation $\mathbf{AX} = \mathbf{I}$. By Section 2.3, it is found by reducing the combined augmented matrix $(\mathbf{A} \mid \mathbf{e}_1 \mid \cdots \mid \mathbf{e}_n) = (\mathbf{A} \mid \mathbf{I})$; for an invertible $\mathbf{A}$, Gauss–Jordan elimination gives

$$(\mathbf{A} \mid \mathbf{I}) \longrightarrow \cdots \longrightarrow (\mathbf{I} \mid \mathbf{A}^{-1}).$$

Strictly speaking, solving $\mathbf{AX} = \mathbf{I}$ produces a right inverse of $\mathbf{A}$; Section 2.6 shows that for a square matrix a right inverse is automatically the inverse, and turns this computation into an algorithm that also detects singular matrices.

2.5. Elementary Matrices

The material on the three types of elementary row operations and their reverses, row equivalence, the fact that row-equivalent augmented matrices give linear systems with the same solutions, and elementary row operations in matrix form (permutation and row modification matrices) is treated in Section 1.3.

Elementary Matrices and Row Operations ADDED FOR MA1522

Definition. A square matrix $\mathbf{E}$ of order n is an *elementary matrix* if it can be obtained from the identity matrix $\mathbf{I}_n$ by a single elementary row operation r, $\mathbf{I}_n \xrightarrow{r} \mathbf{E}$. The operation r is called the row operation *corresponding* to $\mathbf{E}$.

For example, for $n = 3$ the operations $R_2 - 4R_1$, $R_1 \leftrightarrow R_3$ and $5R_2$ correspond to the elementary matrices

$$\mathbf{E}_1 = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{E}_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{E}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

In the language of Section 1.3, the elementary matrices corresponding to $R_i + cR_j$ with $i > j$ are row modification matrices, and those corresponding to $R_i \leftrightarrow R_j$ are permutation matrices.

Theorem. Let $\mathbf{A}$ be an $n \times p$ matrix and $\mathbf{E}$ the elementary matrix of order n corresponding to an elementary row operation r. Then $\mathbf{EA}$ is the matrix obtained from $\mathbf{A}$ by performing r: $\mathbf{A} \xrightarrow{r} \mathbf{EA}$.

Proof. By the row rule (Section 2.3), the i-th row of $\mathbf{EA}$ is the i-th row of $\mathbf{E}$ times $\mathbf{A}$, and $\mathbf{e}_k^T\mathbf{A}$ is the k-th row of $\mathbf{A}$. If r does not change the i-th row of $\mathbf{I}_n$, then the i-th row of $\mathbf{E}$ is $\mathbf{e}_i^T$, and the i-th row of $\mathbf{EA}$ is the i-th row of $\mathbf{A}$. The rows that r changes are: for $R_i \leftrightarrow R_j$, the i-th and j-th rows, which become $\mathbf{e}_j^T$ and $\mathbf{e}_i^T$; for cR_i, the i-th row, which becomes $c\mathbf{e}_i^T$; for $R_i + cR_j$, the i-th row, which becomes $\mathbf{e}_i^T + c\mathbf{e}_j^T$. Multiplied by $\mathbf{A}$, these rows give, respectively, the j-th and i-th rows of $\mathbf{A}$; c times its i-th row; and its i-th row plus c times its j-th row. □

For instance, pre-multiplication by $\mathbf{E}_1$ performs $R_2 - 4R_1$:

$$\mathbf{E}_1\begin{pmatrix}1&0&2\\4&1&3\\0&2&1\end{pmatrix} = \begin{pmatrix}1&0&2\\0&1&-5\\0&2&1\end{pmatrix}.$$

Thus performing an elementary row operation is the same as pre-multiplying by the corresponding elementary matrix. If $\mathbf{B}$ is row equivalent to $\mathbf{A}$, obtained by the operations $r_1, r_2, ..., r_k$ performed in this order, and $\mathbf{E}_i$ is the elementary matrix corresponding to r_i, then applying the theorem k times gives

$$\mathbf{A} \xrightarrow{r_1} \mathbf{E}_1\mathbf{A} \xrightarrow{r_2} \mathbf{E}_2\mathbf{E}_1\mathbf{A} \xrightarrow{r_3} \cdots \xrightarrow{r_k} \mathbf{E}_k\cdots\mathbf{E}_2\mathbf{E}_1\mathbf{A} = \mathbf{B}.$$

Theorem. Every elementary matrix $\mathbf{E}$ is invertible, and $\mathbf{E}^{-1}$ is the elementary matrix corresponding to the reverse of the row operation corresponding to $\mathbf{E}$:

(i) if $\mathbf{E}$ corresponds to $R_i + cR_j$, then $\mathbf{E}^{-1}$ corresponds to $R_i - cR_j$;
(ii) if $\mathbf{E}$ corresponds to $R_i \leftrightarrow R_j$, then $\mathbf{E}^{-1}$ corresponds to $R_i \leftrightarrow R_j$, that is, $\mathbf{E}^{-1} = \mathbf{E}$;
(iii) if $\mathbf{E}$ corresponds to cR_i $(c \neq 0)$, then $\mathbf{E}^{-1}$ corresponds to $(1/c)R_i$.

Proof. Let r be the operation corresponding to $\mathbf{E}$, let r' be its reverse (Section 1.3), and let $\mathbf{E}'$ be the elementary matrix corresponding to r'. Performing r on $\mathbf{I}$ and then r' returns $\mathbf{I}$: $\mathbf{I} \xrightarrow{r} \mathbf{E} \xrightarrow{r'} \mathbf{I}$. By the preceding theorem the second step is pre-multiplication by $\mathbf{E}'$, so $\mathbf{E}'\mathbf{E} = \mathbf{I}$. Since r is the reverse of r', the same argument gives $\mathbf{E}\mathbf{E}' = \mathbf{I}$. □

For the matrices above,

$$\mathbf{E}_1^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{E}_2^{-1} = \mathbf{E}_2, \quad \mathbf{E}_3^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/5 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

corresponding to $R_2 + 4R_1$, $R_1 \leftrightarrow R_3$ and $(1/5)R_2$. If $\mathbf{B} = \mathbf{E}_k \cdots \mathbf{E}_1 \mathbf{A}$, then $\mathbf{A} = \mathbf{E}_1^{-1} \cdots \mathbf{E}_k^{-1} \mathbf{B}$, in accordance with the symmetry of row equivalence. In this matrix form, the theorem of Section 1.3 on row-equivalent augmented matrices reads: if $(\mathbf{B} \mid \mathbf{c}) = \mathbf{P}(\mathbf{A} \mid \mathbf{b}) = (\mathbf{PA} \mid \mathbf{Pb})$ with $\mathbf{P} = \mathbf{E}_k \cdots \mathbf{E}_1$, then $\mathbf{Au} = \mathbf{b}$ implies $\mathbf{Bu} = \mathbf{PAu} = \mathbf{Pb} = \mathbf{c}$, and $\mathbf{Bu} = \mathbf{c}$ implies $\mathbf{Au} = \mathbf{P}^{-1}\mathbf{Bu} = \mathbf{P}^{-1}\mathbf{c} = \mathbf{b}$.

2.6. Equivalent Statements for Invertibility

Invertibility Criterion for a Matrix

Theorem. A square matrix is invertible if and only if its columns form a linearly independent system.

Proof. Let a matrix $\mathbf{A}$ of order n have linearly independent columns. According to the results of Section 1.5 on the consistency of linear systems, each of the systems

$$\mathbf{Ax}_1 = \mathbf{e}_1, \quad \mathbf{Ax}_2 = \mathbf{e}_2, \quad ..., \quad \mathbf{Ax}_n = \mathbf{e}_n,$$

where $\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_n$ are the columns of the identity matrix, has a unique solution. Let $\mathbf{X} = (\mathbf{x}_1 \ \ \mathbf{x}_2 \ \ \cdots \ \ \mathbf{x}_n)$. Then $\mathbf{AX} = \mathbf{I}$.

The columns of the matrix $\mathbf{X}$ are linearly independent. Indeed, suppose some linear combination of these columns is equal to zero:

$$\alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + ... + \alpha_n \mathbf{x}_n = \mathbf{0}.$$

This means that

$$\mathbf{X} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \mathbf{0}.$$

Consequently,

$$\mathbf{AX} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix} = \mathbf{0}.$$

Hence $\alpha_1 = \alpha_2 = ... = \alpha_n = 0$. Thus, for $\mathbf{X}$ there exists a matrix $\mathbf{Y}$ such that $\mathbf{XY} = \mathbf{I}$. Let us prove that $\mathbf{Y} = \mathbf{A}$. Indeed, $\mathbf{A} = \mathbf{A}(\mathbf{XY}) = (\mathbf{AX})\mathbf{Y} = \mathbf{Y}$.[1]

Now suppose that $\mathbf{A} = (\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \cdots \ \ \mathbf{a}_n)$ is an invertible matrix, and consider a linear combination of its columns equal to zero:

$$\alpha_1\mathbf{a}_1 + \alpha_2\mathbf{a}_2 + ... + \alpha_n\mathbf{a}_n = \mathbf{0}.$$

Let us write this equality in the following form:

$$\mathbf{A}\begin{pmatrix}\alpha_1\\ \alpha_2\\ \vdots\\ \alpha_n\end{pmatrix} = \mathbf{0}.$$

Multiplying both sides on the left by $\mathbf{A}^{-1}$, we find: $\alpha_1 = ... = \alpha_n = 0$. Consequently, the columns $\mathbf{a}_1, ..., \mathbf{a}_n$ are linearly independent. □

Invertibility and the Determinant

Theorem. A square matrix $\mathbf{A}$ is invertible if and only if $\det(\mathbf{A}) \neq 0$.

Proof. Suppose that for the matrix $\mathbf{A}$ there exists an inverse matrix $\mathbf{A}^{-1}$. Then $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ and, by virtue of the theorem on the determinant of a product of matrices (Section 2.10),

$$\det(\mathbf{A}) \cdot \det(\mathbf{A}^{-1}) = \det(\mathbf{I}) = 1 \quad \Rightarrow \quad \det(\mathbf{A}) \neq 0.$$

If $\det(\mathbf{A}) \neq 0$, then $\mathbf{A}$ is invertible by the theorem on the inversion of a matrix with nonzero determinant (the adjoint formula, Section 2.10). □

Corollary. The columns of a matrix $\mathbf{A}$ are linearly independent if and only if $\det(\mathbf{A}) \neq 0$.

Problem 1. Let $\mathbf{I}_n$ and $\mathbf{I}_m$ be the identity matrices of orders n and m, respectively. Prove that for any matrices $\mathbf{A}$ of size $m \times n$ and $\mathbf{B}$ of size $n \times m$, the invertibility of $\mathbf{I}_m - \mathbf{A}\mathbf{B}$ implies the invertibility of $\mathbf{I}_n - \mathbf{B}\mathbf{A}$. Prove also that the invertibility of each of these matrices is equivalent to the invertibility of the matrix of order $m + n$ with the block partitioning of the form

$$\begin{pmatrix}\mathbf{I}_m & \mathbf{A}\\ \mathbf{B} & \mathbf{I}_n\end{pmatrix}.$$

Problem 2. Given numbers a and b such that $1 - ab \neq 0$. Prove that the matrix

$$\mathbf{A} = \begin{pmatrix}1 & a & a^2 & ... & a^n\\ b & 1 & a & ... & a^{n-1}\\ b^2 & b & 1 & ... & a^{n-2}\\ \vdots & \vdots & \vdots & \ddots & \vdots\\ b^n & b^{n-1} & b^{n-2} & ... & 1\end{pmatrix}$$

is invertible and its inverse matrix is tridiagonal.

[1] In essence, this reproduces part of the proof of a more general proposition connected with the redundancy of the definition of a group (see Tyrtyshnikov, Supplement to Lecture 2, §$\mathcal{D}$2.5).

Problem 3. Suppose that in the statement of the preceding problem a and b are square blocks $\mathbf{a}$ and $\mathbf{b}$ of order n, and 1 is replaced by the identity matrix $\mathbf{I}$ of the same order. Prove that if the block $\mathbf{I}-\mathbf{ab}$ is an invertible matrix, then the block matrix $\mathbf{A}$ with blocks of order n is invertible and, moreover, its inverse matrix is block-tridiagonal.

Equivalent Statements for Invertibility ADDED FOR MA1522

Definition. Let $\mathbf{A}$ be an $n\times m$ matrix. An $m\times n$ matrix $\mathbf{B}$ is a *left inverse* of $\mathbf{A}$ if $\mathbf{BA}=\mathbf{I}_m$, and a *right inverse* of $\mathbf{A}$ if $\mathbf{AB}=\mathbf{I}_n$.

Thus $\mathbf{B}$ is a left inverse of $\mathbf{A}$ if and only if $\mathbf{A}$ is a right inverse of $\mathbf{B}$, and the inverse of an invertible matrix is both a left and a right inverse. A non-square matrix may have a one-sided inverse; for example,

$$\mathbf{A}=\begin{pmatrix}1&0&2\\0&1&-1\end{pmatrix},\quad \mathbf{B}=\begin{pmatrix}1&0\\0&1\\0&0\end{pmatrix}:\quad \mathbf{AB}=\mathbf{I}_2,\quad \mathbf{BA}=\begin{pmatrix}1&0&2\\0&1&-1\\0&0&0\end{pmatrix}\neq\mathbf{I}_3,$$

so $\mathbf{B}$ is a right inverse of $\mathbf{A}$ but not a left inverse. A non-square matrix cannot have both. Indeed, let $\mathbf{A}$ be $n\times m$, $\mathbf{BA}=\mathbf{I}_m$ and $\mathbf{AC}=\mathbf{I}_n$. Then $\mathbf{B}=\mathbf{B}(\mathbf{AC})=(\mathbf{BA})\mathbf{C}=\mathbf{C}$. Moreover, the *trace* $\operatorname{tr}(\mathbf{X})$ of a square matrix, the sum of its diagonal entries, satisfies $\operatorname{tr}(\mathbf{AB})=\sum_{i,k}a_{ik}b_{ki}=\operatorname{tr}(\mathbf{BA})$; hence $n=\operatorname{tr}(\mathbf{I}_n)=\operatorname{tr}(\mathbf{AB})=\operatorname{tr}(\mathbf{BA})=\operatorname{tr}(\mathbf{I}_m)=m$.

Theorem (Equivalent Statements for Invertibility)**.** Let $\mathbf{A}$ be a square matrix of order n. The following statements are equivalent.

(i) $\mathbf{A}$ is invertible.
(ii) $\mathbf{A}^T$ is invertible.
(iii) (Left inverse) There is a matrix $\mathbf{B}$ such that $\mathbf{BA}=\mathbf{I}$.
(iv) (Right inverse) There is a matrix $\mathbf{B}$ such that $\mathbf{AB}=\mathbf{I}$.
(v) The reduced row-echelon form of $\mathbf{A}$ is the identity matrix.
(vi) $\mathbf{A}$ can be expressed as a product of elementary matrices.
(vii) The homogeneous system $\mathbf{Ax}=\mathbf{0}$ has only the trivial solution.
(viii) For every $\mathbf{b}\in\mathbb{R}^n$, the system $\mathbf{Ax}=\mathbf{b}$ has a unique solution.
(ix) The determinant of $\mathbf{A}$ is nonzero, $\det(\mathbf{A})\neq 0$.

Proof. The equivalence (i) ⇔ (ii) is shown in the subsection "Inversion and Transposition" of Section 2.4, and (i) ⇔ (ix) is the theorem of the subsection "Invertibility and the Determinant" above. We prove the cycle (i) ⇒ (vii) ⇒ (v) ⇒ (vi) ⇒ (i), and then (i) ⇒ (iii) ⇒ (vii), (i) ⇒ (iv) ⇒ (i), and (i) ⇒ (viii) ⇒ (iv).

(i) ⇒ (vii) is the corollary in the subsection "Invertible Matrices and Linear Systems" of Section 2.4.

(vii) ⇒ (v). Let $\mathbf{R}$ be the reduced row-echelon form of $\mathbf{A}$. If all n columns of $\mathbf{R}$ are pivot columns, its leading entries lie on the main diagonal and $\mathbf{R}=\mathbf{I}_n$. Otherwise $\mathbf{R}$ has a non-pivot column; the corresponding variable is

a parameter of the general solution of $\mathbf{Rx} = \mathbf{0}$ (Section 1.2), so this system has a nontrivial solution, and so does $\mathbf{Ax} = \mathbf{0}$, which has the same solutions (Section 1.3). This contradicts (vii).

(v) $\Rightarrow$ (vi). If $\mathbf{E}_k \cdots \mathbf{E}_2 \mathbf{E}_1 \mathbf{A} = \mathbf{I}$ with elementary matrices $\mathbf{E}_i$ (Section 2.5), then multiplying on the left successively by $\mathbf{E}_k^{-1}, ..., \mathbf{E}_1^{-1}$ gives $\mathbf{A} = \mathbf{E}_1^{-1}\mathbf{E}_2^{-1} \cdots \mathbf{E}_k^{-1}$, and the inverse of an elementary matrix is elementary.

(vi) $\Rightarrow$ (i). Elementary matrices are invertible, and a product of invertible matrices is invertible (Section 2.4).

(i) $\Rightarrow$ (iii) and (i) $\Rightarrow$ (iv): take $\mathbf{B} = \mathbf{A}^{-1}$.

(iii) $\Rightarrow$ (vii). If $\mathbf{BA} = \mathbf{I}$ and $\mathbf{Au} = \mathbf{0}$, then $\mathbf{u} = \mathbf{BAu} = \mathbf{B0} = \mathbf{0}$.

(iv) $\Rightarrow$ (i). Transposing $\mathbf{AB} = \mathbf{I}$ gives $\mathbf{B}^T\mathbf{A}^T = \mathbf{I}^T = \mathbf{I}$. Thus $\mathbf{A}^T$ has a left inverse; by the implications (iii) $\Rightarrow$ (vii) $\Rightarrow$ (i) already proved, applied to $\mathbf{A}^T$, the matrix $\mathbf{A}^T$ is invertible, and therefore so is $\mathbf{A}$.

(i) $\Rightarrow$ (viii) is the theorem in the subsection "Invertible Matrices and Linear Systems" of Section 2.4.

(viii) $\Rightarrow$ (iv). Let $\mathbf{b}_j$ be the solution of $\mathbf{Ax} = \mathbf{e}_j$, $j = 1, ..., n$, and $\mathbf{B} = (\mathbf{b}_1 \;\; \mathbf{b}_2 \;\; \cdots \;\; \mathbf{b}_n)$. By the column rule (Section 2.3),

$$\mathbf{AB} = (\mathbf{Ab}_1 \;\; \mathbf{Ab}_2 \;\; \cdots \;\; \mathbf{Ab}_n) = (\mathbf{e}_1 \;\; \mathbf{e}_2 \;\; \cdots \;\; \mathbf{e}_n) = \mathbf{I}. \qquad \square$$

The textbook's criteria—the columns of $\mathbf{A}$ are linearly independent (subsection "Invertibility Criterion for a Matrix"), the rows of $\mathbf{A}$ are linearly independent (Section 2.4)—can be added to this list; Sections 3.6 and 4.2 extend it further.

Corollary. Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices of order n.
(i) If $\mathbf{BA} = \mathbf{I}$ or $\mathbf{AB} = \mathbf{I}$, then $\mathbf{A}$ is invertible and $\mathbf{B} = \mathbf{A}^{-1}$.
(ii) If $\mathbf{AB}$ is invertible, then both $\mathbf{A}$ and $\mathbf{B}$ are invertible.

Proof. (i) By the theorem, $\mathbf{A}$ is invertible, and then $\mathbf{B} = \mathbf{B}(\mathbf{AA}^{-1}) = (\mathbf{BA})\mathbf{A}^{-1} = \mathbf{A}^{-1}$, or $\mathbf{B} = (\mathbf{A}^{-1}\mathbf{A})\mathbf{B} = \mathbf{A}^{-1}(\mathbf{AB}) = \mathbf{A}^{-1}$. (ii) Let $\mathbf{C} = (\mathbf{AB})^{-1}$. Then $\mathbf{A}(\mathbf{BC}) = \mathbf{I}$ and $(\mathbf{CA})\mathbf{B} = \mathbf{I}$, so $\mathbf{A}$ has a right inverse and $\mathbf{B}$ has a left inverse. $\square$

In particular, a matrix $\mathbf{X}$ found by solving $\mathbf{AX} = \mathbf{I}$ for a square matrix $\mathbf{A}$ is the inverse of $\mathbf{A}$, not merely a right inverse.

Finding the Inverse by Row Reduction ADDED FOR MA1522

Algorithm (Inverse by row reduction).

Let $\mathbf{A}$ be a square matrix of order n.

1. Form the $n \times 2n$ matrix $(\mathbf{A} \mid \mathbf{I}_n)$.
2. Reduce it by elementary row operations to a row-echelon form, or to the reduced row-echelon form $(\mathbf{R} \mid \mathbf{B})$.

3. If the left half of the row-echelon form has a zero row (equivalently, $\mathbf{R} \neq \mathbf{I}_n$), then $\mathbf{A}$ is singular. Otherwise $\mathbf{A}$ is invertible; the reduced row-echelon form is then $(\mathbf{I}_n \mid \mathbf{B})$, and $\mathbf{A}^{-1} = \mathbf{B}$.

Row operations act on the two halves of $(\mathbf{A} \mid \mathbf{I})$ independently, so the left half of every matrix obtained is row equivalent to $\mathbf{A}$. A square matrix in row-echelon form has a zero row if and only if it has fewer than n pivot columns; then $\mathbf{Ax} = \mathbf{0}$ has a nontrivial solution, as in the proof of (vii) $\Rightarrow$ (v), and $\mathbf{A}$ is singular. Otherwise the reduced row-echelon form of $\mathbf{A}$ is $\mathbf{I}$; the columns of $\mathbf{B}$ then solve the systems $\mathbf{Ax} = \mathbf{e}_j$, so $\mathbf{AB} = \mathbf{I}$ by the column rule, and $\mathbf{B} = \mathbf{A}^{-1}$ by the corollary above.

Example. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

The operations $R_2 - R_1$ and $R_3 - R_2$ give a row-echelon form

$$(\mathbf{A} \mid \mathbf{I}) = \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -1 & 1 \end{array}\right)$$

whose left half has no zero row, so $\mathbf{A}$ is invertible. The operations $(1/2)R_3$, $R_2 + R_3$, $R_1 - R_3$ then give the reduced row-echelon form

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 1/2 & -1/2 \\ 0 & 1 & 0 & -1/2 & 1/2 & 1/2 \\ 0 & 0 & 1 & 1/2 & -1/2 & 1/2 \end{array}\right), \quad \text{so} \quad \mathbf{A}^{-1} = \frac{1}{2}\begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}.$$

On the other hand, for

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 1 & 3 & 4 \end{pmatrix}$$

the operations $R_3 - R_1$ and $R_3 - R_2$ give

$$(\mathbf{A} \mid \mathbf{I}) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right),$$

whose left half is a row-echelon form with a zero row; this matrix $\mathbf{A}$ is singular (its third row is the sum of the first two).

Invertible Matrices and Row Equivalence ADDED FOR MA1522

Theorem. Let $\mathbf{A}$ and $\mathbf{B}$ be $n \times m$ matrices. Then $\mathbf{B}$ is row equivalent to $\mathbf{A}$ if and only if $\mathbf{B} = \mathbf{PA}$ for some invertible matrix $\mathbf{P}$ of order n.

Proof. If $\mathbf{B}$ is obtained from $\mathbf{A}$ by row operations with elementary matrices $\mathbf{E}_1, ..., \mathbf{E}_k$, then $\mathbf{B} = \mathbf{PA}$ with $\mathbf{P} = \mathbf{E}_k \cdots \mathbf{E}_1$, which is invertible. Conversely, an invertible $\mathbf{P}$ is a product of elementary matrices, $\mathbf{P} = \mathbf{E}_k \cdots \mathbf{E}_1$ (statement (vi)), and then $\mathbf{PA} = \mathbf{E}_k(\cdots(\mathbf{E}_1\mathbf{A}))$ is obtained from $\mathbf{A}$ by the corresponding row operations (Section 2.5). □

This makes precise the remark that elementary row operations are the building blocks of all reversible row operations: every operation on n-row matrices of the form $\mathbf{A} \mapsto \mathbf{PA}$ with an invertible $\mathbf{P}$—each new row is a fixed linear combination of the old rows, and the operation can be undone—is a composition of elementary row operations.

The uniqueness of the reduced row-echelon form of a matrix is treated in Section 1.4.

Corollary. Two $m \times n$ matrices are row equivalent if and only if they have the same reduced row-echelon form.

Proof. If $\mathbf{A}$ and $\mathbf{B}$ have the same reduced row-echelon form $\mathbf{R}$, then $\mathbf{A}$ is row equivalent to $\mathbf{R}$, and $\mathbf{R}$ to $\mathbf{B}$. Conversely, if $\mathbf{A}$ and $\mathbf{B}$ are row equivalent and $\mathbf{R}$ is the reduced row-echelon form of $\mathbf{A}$, then $\mathbf{B}$ is row equivalent to $\mathbf{R}$ as well, since row equivalence is symmetric and transitive; by uniqueness, $\mathbf{R}$ is the reduced row-echelon form of $\mathbf{B}$. □

2.7. LU Factorization

Gaussian Elimination and LU Factorization

The method of successive elimination of unknowns described in Section 1.4 is usually called *Gaussian elimination* (or *Gauss's method*). Suppose it is applied to a linear system $\mathbf{Ax} = \mathbf{b}$ with an invertible matrix $\mathbf{A}$. In this case, the row-echelon form to which the matrix $\mathbf{A}$ is reduced turns out to be an upper triangular matrix.

Gaussian elimination can be interpreted as a method of eliminating entries of a matrix with the aim of reducing it to a simpler form. If one can do without permuting equations (row interchanges $R_i \leftrightarrow R_j$), then Gaussian elimination for a matrix of order n consists in successively eliminating the entries in columns from the 1-st to the $(n-1)$-st and leads to an equivalent system

$$(\mathbf{Z}_{n-1} \ldots \mathbf{Z}_2 \mathbf{Z}_1 \mathbf{A})\mathbf{x} = \mathbf{Z}_{n-1} \ldots \mathbf{Z}_2 \mathbf{Z}_1 \mathbf{b}, \qquad (*)$$

where $\mathbf{Z}_1, ..., \mathbf{Z}_{n-1}$ are row modification matrices (Section 1.3), with $\mathbf{Z}_i$ differing from $\mathbf{I}$ precisely in the i-th column. Each of the matrices $\mathbf{Z}_1, ..., \mathbf{Z}_{n-1}$ is lower triangular, and therefore their product

$$\hat{\mathbf{L}} = \mathbf{Z}_{n-1}...\mathbf{Z}_1$$

is also a lower triangular matrix. The coefficient matrix of the system $(*)$

$$\mathbf{U} = \mathbf{Z}_{n-1}...\mathbf{Z}_1\mathbf{A} = \hat{\mathbf{L}}\mathbf{A}$$

is upper triangular. The matrix $\mathbf{L} = \hat{\mathbf{L}}^{-1}$ is lower triangular. Consequently, Gaussian elimination generates a decomposition of the matrix $\mathbf{A}$ into a product of a lower and an upper triangular matrix:

$$\mathbf{A} = \mathbf{LU}.$$

Moreover, $\mathbf{L}$ has ones on the main diagonal (it is a *unit lower triangular* matrix), and $\mathbf{U}$ is an invertible matrix (by virtue of the invertibility of $\mathbf{A}$). Such a decomposition, with $\mathbf{L}$ unit lower triangular and $\mathbf{U}$ a row-echelon form of $\mathbf{A}$, is called an *LU factorization* of $\mathbf{A}$.

Let us count the number of arithmetic operations in reducing $\mathbf{A}$ to the upper triangular matrix $\mathbf{U}$. At the i-th step, it is required to obtain $n-i$ zeros below the diagonal in the i-th column. When obtaining a zero at the intersection of the i-th column and the l-th row for $l > i$, from the l-th row one subtracts the i-th row, previously multiplied by a coefficient whose choice ensures obtaining this zero. Since in the rows under consideration there can be only $n-i$ nonzero entries (beyond the i-th position), the number of multiplications (and subtractions) in obtaining zeros in the i-th column is equal to $(n-i)^2$. In total, this requires

$$(n-1)^2 + (n-2)^2 + ... + 1^2 = \frac{1}{3}n^3 + \mathcal{O}(n^2)$$

multiplications and the same number of subtractions; by $\mathcal{O}(n^2)$ we denote a polynomial in n of degree 2.

To find the solution of the system $\mathbf{Ax} = \mathbf{b}$, one must perform two more steps:

- compute the vector $\mathbf{Z}_{n-1}...\mathbf{Z}_1\mathbf{b}$;
- find the solution of the system with the upper triangular matrix $\mathbf{U}$.

Each of these steps requires only $\mathcal{O}(n^2)$ arithmetic operations—an order of magnitude less than the reduction to upper triangular form.

Problem 1. An invertible matrix and its inverse are partitioned into blocks of identical dimensions:

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}, \quad \mathbf{A}^{-1} = \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix}.$$

Prove that the block $\mathbf{A}_{11}$ is invertible if and only if the block $\mathbf{B}_{22}$ is invertible.

LU Factorization of an $m \times n$ Matrix ADDED FOR MA1522

The construction above extends to rectangular matrices, without assuming invertibility.

Definition. A square matrix $\mathbf{L}$ is *unit lower triangular* if it is lower triangular and all its diagonal entries are equal to 1. An *LU factorization* of an $m \times n$ matrix $\mathbf{A}$ is a factorization

$$\mathbf{A} = \mathbf{LU},$$

where $\mathbf{L}$ is an $m \times m$ unit lower triangular matrix and $\mathbf{U}$ is a row-echelon form of $\mathbf{A}$. If $\mathbf{A}$ has an LU factorization, it is said to be *LU factorizable*.

Lemma. The product of two unit lower triangular matrices of the same order is unit lower triangular.

Proof. Let $\mathbf{A} = \left(a_{ij}\right)_n$ and $\mathbf{B} = \left(b_{ij}\right)_n$ be unit lower triangular. A term $a_{ik}b_{kj}$ of $(\mathbf{AB})_{ij}$ can be nonzero only if $k \leq i$ and $k \geq j$. Hence $(\mathbf{AB})_{ij} = 0$ for $i < j$, and $(\mathbf{AB})_{ii} = a_{ii}b_{ii} = 1$. □

The elementary matrix corresponding to $R_i + cR_j$ with $i > j$ differs from $\mathbf{I}$ only in its (i, j)-entry, which is c and lies below the diagonal; so it is unit lower triangular, and so is its inverse, which corresponds to $R_i - cR_j$ (Section 2.5).

Algorithm (LU factorization).

Let $\mathbf{A}$ be an $m \times n$ matrix.

1. Reduce $\mathbf{A}$ to a row-echelon form $\mathbf{U}$ using only operations $R_i + cR_j$ with $i > j$ (adding a multiple of a row to a row below it), in the order of Gaussian elimination: first the operations that use R_1, then those that use R_2, and so on, each entry being eliminated once. If a row interchange is needed, the algorithm fails.
2. If $r_1, ..., r_k$ are the operations used and $\mathbf{E}_1, ..., \mathbf{E}_k$ the corresponding elementary matrices, then $\mathbf{E}_k \cdots \mathbf{E}_2\mathbf{E}_1\mathbf{A} = \mathbf{U}$, and therefore
$$\mathbf{A} = \mathbf{LU}, \quad \mathbf{L} = \mathbf{E}_1^{-1}\mathbf{E}_2^{-1}\cdots\mathbf{E}_k^{-1},$$
where $\mathbf{L}$ is unit lower triangular by the lemma.
3. In practice $\mathbf{L}$ is written down at once: if r_l is the operation $R_i + c_lR_j$, then the (i, j)-entry of $\mathbf{L}$ is $-c_l$; all other entries below the diagonal are 0.

Step 3 holds because $\mathbf{E}_l^{-1} = \mathbf{I} - c_l\mathbf{e}_{i_l}\mathbf{e}_{j_l}^T$ when r_l is $R_{i_l} + c_lR_{j_l}$. In the order of step 1, $j_l \leq j_s < i_s$ for $l < s$, so $\mathbf{e}_{j_l}^T\mathbf{e}_{i_s} = 0$, and when the product $\mathbf{E}_1^{-1}\cdots\mathbf{E}_k^{-1}$ is multiplied out, every product of two or more of the terms $c_l\mathbf{e}_{i_l}\mathbf{e}_{j_l}^T$ vanishes. Hence $\mathbf{L} = \mathbf{I} - \sum_l c_l\mathbf{e}_{i_l}\mathbf{e}_{j_l}^T$.

Example. For

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & -1 & 3 \\ 4 & 5 & 1 & 4 \\ -2 & 2 & 4 & -1 \end{pmatrix}$$

the operations $R_2 - 2R_1$, $R_3 + R_1$ and then $R_3 - R_2$ give

$$\mathbf{A} \longrightarrow \begin{pmatrix} 2 & 1 & -1 & 3 \\ 0 & 3 & 3 & -2 \\ 0 & 3 & 3 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 1 & -1 & 3 \\ 0 & 3 & 3 & -2 \\ 0 & 0 & 0 & 4 \end{pmatrix} = \mathbf{U}.$$

The multipliers c_l are -2, 1, -1, so $\mathbf{L}$ has the entries 2, -1, 1 in positions $(2,1)$, $(3,1)$, $(3,2)$:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & -1 & 3 \\ 0 & 3 & 3 & -2 \\ 0 & 0 & 0 & 4 \end{pmatrix}.$$

An LU factorization replaces the system $\mathbf{Ax} = \mathbf{b}$ by two easy ones. Put $\mathbf{y} = \mathbf{Ux}$. First solve $\mathbf{Ly} = \mathbf{b}$ by forward substitution; since $\mathbf{L}$ is unit lower triangular, this system has a unique solution, found from the first equation down. Then solve $\mathbf{Ux} = \mathbf{y}$ by back substitution, since $\mathbf{U}$ is in row-echelon form. For $\mathbf{A}$ as in the example and $\mathbf{b} = (2, 11, 9)^T$, the system $\mathbf{Ly} = \mathbf{b}$ gives

$$y_1 = 2, \quad y_2 = 11 - 2y_1 = 7, \quad y_3 = 9 + y_1 - y_2 = 4,$$

and $\mathbf{Ux} = \mathbf{y}$ gives $x_4 = 1$, $x_2 = 3 - x_3$, $x_1 = x_3 - 2$; with the parameter $x_3 = t$ the general solution is

$$\mathbf{x} = (t - 2, 3 - t, t, 1)^T, \quad t \in \mathbb{R}.$$

Not every matrix is LU factorizable. Every 2×2 matrix in row-echelon form is upper triangular, so an equality

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ l & 1 \end{pmatrix} \begin{pmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{pmatrix}$$

would give, on comparing the first columns, $u_{11} = 0$ and $lu_{11} = 1$, which is impossible. (A characterization of the invertible LU factorizable matrices, beyond the course, is given in Tyrtyshnikov, §8.8.) Every matrix, however, has an *LU factorization with pivoting* (an *LUP factorization*)

$$\mathbf{A} = \mathbf{PLU},$$

where $\mathbf{L}$ is unit lower triangular, $\mathbf{U}$ is a row-echelon form of $\mathbf{A}$, and $\mathbf{P}$ is a *permutation matrix*: a square matrix with exactly one entry 1 in each row and each column and zeros elsewhere, or, equivalently, a product of elementary matrices corresponding to row interchanges (Section 1.3). Indeed, Gaussian elimination as in the proof of Theorem 1 of Section 1.4 gives $\mathbf{Z}_k\mathbf{\Pi}_k \cdots \mathbf{Z}_1\mathbf{\Pi}_1\mathbf{A} = \mathbf{U}$, where $\mathbf{\Pi}_l$ interchanges the l-th row with a lower one and $\mathbf{Z}_l = \mathbf{I} + \mathbf{z}_l\mathbf{e}_l^T$, with $\mathbf{z}_l$ having zeros in its first l positions, uses the l-th row to eliminate below it. For $s > l$ we have $\mathbf{\Pi}_s\mathbf{e}_l = \mathbf{e}_l$, hence $\mathbf{\Pi}_s\mathbf{Z}_l = \big(\mathbf{I} + (\mathbf{\Pi}_s\mathbf{z}_l)\mathbf{e}_l^T\big)\mathbf{\Pi}_s$, a matrix of the same kind as $\mathbf{Z}_l$ times $\mathbf{\Pi}_s$. Moving all the permutations to the right in this way gives $\mathbf{L}'\mathbf{\Pi}\mathbf{A} = \mathbf{U}$ with $\mathbf{\Pi} = \mathbf{\Pi}_k \cdots \mathbf{\Pi}_1$, where $\mathbf{L}'$ is a product of matrices $\mathbf{I} + \mathbf{w}\mathbf{e}_l^T$ with the first l entries of $\mathbf{w}$ equal to zero. Each such factor has the unit lower triangular inverse $\mathbf{I} - \mathbf{w}\mathbf{e}_l^T$, so $\mathbf{L}'$ is invertible and $\mathbf{L}'^{-1}$ is unit lower

triangular by the lemma; thus $\mathbf{A} = \mathbf{\Pi}^T\mathbf{L}\mathbf{U}$ with $\mathbf{L} = \mathbf{L}'^{-1}$, and $\mathbf{P} = \mathbf{\Pi}^T$ is a permutation matrix.

2.8. Determinant by Cofactor Expansion

Linear Dependence Indicator

Consider a system of vectors $\mathbf{a}_1, ..., \mathbf{a}_n \in \mathbb{R}^n$ and let us try to construct a linear dependence indicator—a function $f(\mathbf{a}_1, ..., \mathbf{a}_n)$ that equals zero when the given system is linearly dependent. At the same time, the function f should have as simple a form as possible: let f be *linear* in each argument when the values of the remaining arguments are fixed.

Let us give a precise formulation of the requirements on the function f:

- A) for every $1 \le i \le n$, the function is linear in the i-th argument (the function must have a "simple form"):
$$f(\mathbf{a}_1, ..., \mathbf{a}_{i-1}, \alpha\mathbf{a} + \beta\mathbf{b}, \mathbf{a}_{i+1}, ..., \mathbf{a}_n) =$$
$$= \alpha f(\mathbf{a}_1, ..., \mathbf{a}_{i-1}, \mathbf{a}, \mathbf{a}_{i+1}, ..., \mathbf{a}_n) +$$
$$+\beta f(\mathbf{a}_1, ..., \mathbf{a}_{i-1}, \mathbf{b}, \mathbf{a}_{i+1}, ..., \mathbf{a}_n)$$
for any vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ and scalars $\alpha, \beta \in \mathbb{R}$;
- B) if the system of vectors $\mathbf{a}_1, ..., \mathbf{a}_n$ is linearly dependent, then $f(\mathbf{a}_1, ..., \mathbf{a}_n) = 0$;
- C) the function takes a prescribed nonzero value on a given linearly independent system (normalization condition):
$$f(\mathbf{e}_1, ..., \mathbf{e}_n) = 1,$$
where $\mathbf{e}_1, ..., \mathbf{e}_n$ are the columns of the identity matrix of order n.

A function f with the stated properties will be called a linear dependence indicator. To construct it, we shall need the concept of a permutation.

Permutations

An invertible mapping $\sigma : \boldsymbol{N} \to \boldsymbol{N}$, where $\boldsymbol{N} = \{1, 2, ..., n\}$, is called a *permutation* (also a *substitution*) of degree n. To denote a permutation σ, one often uses the table

$$\sigma = \begin{pmatrix} 1 & 2 & ... & n \\ \sigma(1) & \sigma(2) & ... & \sigma(n) \end{pmatrix},$$

in which the numbers $\sigma(1), \sigma(2), ..., \sigma(n)$ are the numbers $1, 2, ..., n$ written in some order (this is equivalent to the invertibility of the mapping σ).

Let us define the product of permutations a and b as the mapping obtained by the sequential execution (composition) of the mappings b and a:

$$(ab)(i) = a(b(i)), \quad i \in \boldsymbol{N}.$$

This is an algebraic operation on the set of all permutations of degree n, with respect to which it is a group. Indeed, associativity is obvious (the composition

of mappings always possesses this property). The role of the identity element is played by the identity mapping

$$e = \begin{pmatrix} 1 & 2 & \dots & n \\ 1 & 2 & \dots & n \end{pmatrix},$$

and the inverse element for σ is the inverse mapping σ^{-1}.

The group of permutations of degree n is called the *symmetric group* of degree n and is denoted by S_n. This is one of the most important examples of finite groups (groups with a finite number of elements; the number of elements is called the *order* of the group). It is not hard to verify that the order of the group S_n is equal to $n! = 1 \cdot 2 \cdot \dots \cdot n$.

The name of the symmetric group is inspired by the definition of *symmetric functions*: a function $F(x_1, \dots, x_n)$ is so called if it is invariant under all permutations of its arguments:

$$F(x_1, \dots, x_n) = F\big(x_{\sigma(1)}, \dots, x_{\sigma(n)}\big) \quad \forall \sigma \in S_n.$$

An example of a symmetric function (determined by a numerical parameter k):

$$F_k(x_1, \dots, x_n) = \sum_{i=1}^{n} x_i^k.$$

Problem 1. Given two systems of numbers $x_1 \le x_2 \le \dots \le x_n$ and $y_1 \le y_2 \le \dots \le y_n$, prove that for any permutation $\sigma \in S_n$ the inequality

$$\sum_{i=1}^{n} |x_i - y_i| \le \sum_{i=1}^{n} \big|x_i - y_{\sigma(i)}\big|$$

holds.

Transpositions ADDED FOR MA1522

The group S_n is also called the *n-permutation group*; its elements are the permutations of the n objects $1, 2, \dots, n$. MA1522 writes a permutation in *one-line notation*, listing only the bottom row of its table: $\sigma = (\sigma(1), \sigma(2), \dots, \sigma(n))$. For example, S_2 consists of $(1, 2)$, the identity, and $(2, 1)$; and $(2, 4, 1, 3) \in S_4$ is the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 1 & 3 \end{pmatrix}.$$

A *transposition* is a permutation that exchanges two numbers $i \ne j$ and leaves all the others fixed. In this book the transposition exchanging i and j is denoted by (i, j):

$$(i, j)(i) = j, \quad (i, j)(j) = i, \quad (i, j)(l) = l \quad \text{for } l \ne i, j.$$

This two-entry symbol is not one-line notation: the transposition $(1, 2) \in S_3$ is $(2, 1, 3)$ in one-line notation. To avoid ambiguity, one-line notation is not used below; permutations are given by their tables. Clearly $\tau\tau = e$, that is, $\tau^{-1} = \tau$, for every transposition τ.

Theorem. Every permutation is a composition (product) of transpositions.

Proof. We use induction on the number m of indices moved by $\sigma \in S_n$, that is, of indices i with $\sigma(i) \neq i$. If $m = 0$, then $\sigma = e = \tau\tau$ for any transposition τ (for $n = 1$, e is regarded as the empty product). Let $m > 0$, choose i with $j = \sigma(i) \neq i$, and let $\tau = (i, j)$. Then $(\tau\sigma)(i) = \tau(j) = i$. If $\sigma(l) = l$, then $l \neq i$, and also $l \neq j$, since $\sigma(j) = j$ would contradict $\sigma(i) = j$; hence $(\tau\sigma)(l) = \tau(l) = l$. Thus $\tau\sigma$ fixes i and every index fixed by σ, so it moves fewer than m indices and is a product of transpositions by the induction hypothesis. Hence so is $\sigma = \tau(\tau\sigma)$. □

For the permutation σ above, $\sigma = (1, 2)(2, 4)(3, 4)$ (recall that the right-hand factor acts first). A permutation has many such decompositions, but the number of factors always has the same parity (subsection "Parity of a Permutation" below). This is what makes the *sign* $\operatorname{sgn}(\sigma) = (-1)^k$, where k is the number of transpositions in a decomposition of σ, well defined; it is introduced formally in the subsection "Uniqueness of the Linear Dependence Indicator". For the σ above, $\operatorname{sgn}(\sigma) = (-1)^3 = -1$.

Parity of a Permutation

A permutation can be decomposed into a product of transpositions in many ways. For example,

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 5 & 3 & 1 & 2 & 4 & 6 \end{pmatrix} =$$

$$= (1, 7)(7, 6)(6, 4)(2, 5) = (1, 7)(7, 6)(6, 4)(7, 2)(7, 5)(7, 2).$$

However, the number of transpositions in any decomposition of one and the same permutation possesses the following important property.

Lemma (Number of Transpositions)**.** The parity of the number of transpositions does not depend on the way a permutation is represented as a product of transpositions.

Proof. For a given permutation $\sigma \in S_n$,

$$\sigma = \begin{pmatrix} 1 & 2 & \dots & n \\ \sigma(1) & \sigma(2) & \dots & \sigma(n) \end{pmatrix},$$

let us call a pair (i, j) an *inversion* if $i < j$ but $\sigma(i) > \sigma(j)$. Let $\delta(\sigma)$ be the total number of inversions for σ. Let us prove that for any transposition τ, the difference $\delta(\sigma\tau) - \delta(\sigma)$ is an odd number.

Let $\tau = (i, j)$, $i < j$. Then

$$\sigma\tau = \begin{pmatrix} 1 & \dots & i-1 & i & i+1 & \dots & j-1 & j & j+1 & \dots & n \\ \sigma(1) & \dots & \sigma(i-1) & \sigma(j) & \sigma(i+1) & \dots & \sigma(j-1) & \sigma(i) & \sigma(j+1) & \dots & \sigma(n) \end{pmatrix}.$$

Suppose that the permutation σ has k inversions among pairs of the form

$$(i,l), \quad \text{where} \quad l \in \{i+1, i+2, \dots, j-1\}, \tag{†}$$

m inversions among pairs of the form

$$(l,j), \quad \text{where} \quad l \in \{i+1, i+2, \dots, j-1\}, \tag{‡}$$

and s more inversions among all other pairs. Then $\sigma\tau$ will have $j-i-1-m$ inversions among pairs of the form (†) and $j-i-1-k$ inversions among pairs of the form (‡). Furthermore, among all other pairs the permutation $\sigma\tau$ will have $s+1$ inversions if the pair (i,j) was not an inversion, and $s-1$ inversions otherwise. Thus,

$$\delta(\sigma) = k + m + s,$$
$$\delta(\sigma\tau) = (j-i-1-k) + (j-i-1-m) + s \pm 1.$$

Hence

$$\delta(\sigma\tau) - \delta(\sigma) = 2(j-i-1-k-m) \pm 1. \qquad \square$$

Corollary. The parity of the number of transpositions in a decomposition of a permutation coincides with the parity of its number of inversions.

Definition. A permutation is called *even* if it is a product of an even number of transpositions, and *odd* otherwise.

Remark. Consider the function

$$\Delta(x_1, \dots, x_n) = \prod_{i<j} (x_j - x_i).$$

Then for any permutation $\sigma \in S_n$ one of the two holds: either

$$\Delta(x_1, \dots, x_n) = \Delta\big(x_{\sigma(1)}, \dots, x_{\sigma(n)}\big)$$

or

$$\Delta(x_1, \dots, x_n) = -\Delta\big(x_{\sigma(1)}, \dots, x_{\sigma(n)}\big).$$

Even permutations, and only they, preserve the sign (the first case); odd permutations, and only they, reverse the sign (the second case).

The set of all even permutations in S_n forms a subgroup (prove this!), which is called the *alternating group* of degree n and is denoted by A_n.

Problem 2. Prove that every even permutation of degree $n \geq 3$ can be represented as a product of cycles of length 3 (permutations that map $i_1 \mapsto i_2 \mapsto i_3 \mapsto i_1$ for some pairwise distinct indices i_1, i_2, i_3 and fix all other indices).

Uniqueness of the Linear Dependence Indicator

Let us return to the construction of the linear dependence indicator—a function $f(\mathbf{a}_1, ..., \mathbf{a}_n)$ of the vectors

$$\mathbf{a}_1 = \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{n2} \end{pmatrix}, \quad ..., \quad \mathbf{a}_n = \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{nn} \end{pmatrix},$$

satisfying requirements (A), (B), (C). It is easy to see that

$$\mathbf{a}_1 = \sum_{i_1=1}^{n} a_{i_1 1}\mathbf{e}_{i_1}, \quad \mathbf{a}_2 = \sum_{i_2=1}^{n} a_{i_2 2}\mathbf{e}_{i_2}, \quad ..., \quad \mathbf{a}_n = \sum_{i_n=1}^{n} a_{i_n n}\mathbf{e}_{i_n},$$

where $\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_n$ are the columns of the identity matrix of size $n \times n$.

If the sought function f exists, then property (A) of linearity in each argument leads to the expression

$$f(\mathbf{a}_1, ..., \mathbf{a}_n) = \sum_{i_1=1}^{n} ... \sum_{i_n=1}^{n} a_{i_1 1} a_{i_2 2} ... a_{i_n n} f\big(\mathbf{e}_{i_1}, \mathbf{e}_{i_2}, ..., \mathbf{e}_{i_n}\big).$$

According to requirement (B), $f = 0$ on any linearly dependent system of vectors. Obviously, the system of vectors $\mathbf{e}_{i_1}, \mathbf{e}_{i_2}, ..., \mathbf{e}_{i_n}$ is linearly dependent if and only if there are equal vectors among them (if all these vectors are pairwise distinct, then they form a permutation of the columns of the identity matrix). Consequently, excluding the terms that are certainly zero from the summation, we find

$$f(\mathbf{a}_1, ..., \mathbf{a}_n) = \sum_{\sigma \in S_n} a_{\sigma(1)1} a_{\sigma(2)2} ... a_{\sigma(n)n} \times$$
$$\times f\big(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, ..., \mathbf{e}_{\sigma(n)}\big).$$

Furthermore, from requirements (A) and (B) it follows that f must change sign when any two arguments are interchanged. Let us prove this, for example, for the first and second arguments. Taking into account that $f = 0$ in the case of equal arguments, and using linearity in each argument, we have

$$\begin{aligned} 0 &= f(\mathbf{a}_1 + \mathbf{a}_2, \mathbf{a}_1 + \mathbf{a}_2, \mathbf{a}_3, ..., \mathbf{a}_n) = \\ &= f(\mathbf{a}_1, \mathbf{a}_1, \mathbf{a}_3, ..., \mathbf{a}_n) + f(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, ..., \mathbf{a}_n) + \\ &+ f(\mathbf{a}_2, \mathbf{a}_1, \mathbf{a}_3, ..., \mathbf{a}_n) + f(\mathbf{a}_2, \mathbf{a}_2, \mathbf{a}_3, ..., \mathbf{a}_n). \end{aligned}$$

The first and fourth terms have coinciding vectors and are therefore equal to zero. Hence

$$f(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, ..., \mathbf{a}_n) = -f(\mathbf{a}_2, \mathbf{a}_1, \mathbf{a}_3, ..., \mathbf{a}_n).$$

Consequently, if the permutation σ is a transposition, then

$$f\big(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, ..., \mathbf{e}_{\sigma(n)}\big) = -f(\mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_n) = -1.$$

In the general case, a permutation σ can be decomposed into a product of transpositions. Let $\delta(\sigma)$ be the number of transpositions in any one of the decompositions. Then

$$f\big(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, ..., \mathbf{e}_{\sigma(n)}\big) = (-1)^{\delta(\sigma)}.$$

By the lemma on the number of transpositions, the parity of the number $\delta(\sigma)$ does not depend on the specific decomposition into a product of transpositions; therefore the quantity $(-1)^{\delta(\sigma)}$ depends only on σ. We call it the *sign of the permutation* and denote it by $\operatorname{sgn}(\sigma)$. Finally:

$$f(\mathbf{a}_1, ..., \mathbf{a}_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} ... a_{\sigma(n)n}. \qquad (*)$$

We have proved an important result:

Proposition. If a linear dependence indicator function exists, then it is given by formula $(*)$.

The Determinant

Definition. A function of the form $(*)$ is called the *determinant* of the matrix $\mathbf{A}$ with columns $\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_n$ and is denoted by $\det(\mathbf{A})$ or $|\mathbf{A}|$.

Thus, if $\mathbf{A} = \big(a_{ij}\big)$ is a matrix of size $n \times n$, then

$$\det(\mathbf{A}) = |\mathbf{A}| = \begin{vmatrix} a_{11} & ... & a_{1n} \\ \vdots & \vdots & \vdots \\ a_{n1} & ... & a_{nn} \end{vmatrix} = \\ = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} ... a_{\sigma(n)n}. \qquad (\triangle)$$

Special cases:

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12},$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - \\ - a_{13}a_{22}a_{31} - a_{12}a_{21}a_{33} - a_{11}a_{23}a_{32}.$$

In the general case, the sum $(\triangle)$ (the *Leibniz formula* for the determinant) contains $n!$ terms; in each of them n entries of the matrix are multiplied together, and no two entries in a single product belong to the same row or the same column.

Despite the fact that the determinant is introduced as a function of a matrix, historically the concept of a determinant took shape in the 18th century (first in the works of Leibniz and Cramer, and then the theory of

determinants was developed in the works of Vandermonde, Laplace, Cauchy, and C. Jacobi)—much earlier than the concept of a matrix, which was introduced into algebra by Hamilton and Cayley in the middle of the 19th century.

Of course, from the very beginning the determinant was associated with a square $n \times n$ table of numbers (which is why one spoke of a determinant of *order* n). These were, in particular, the coefficient tables of a "square" linear system. Yet such tables came to be called matrices only later—when the operation of multiplication was introduced for them.

Submatrices and $k \times k$ Minors

For a given matrix $\mathbf{A} = \left(a_{ij}\right)$, one can select some of its rows and columns and form the table of entries located at the intersections of the selected rows and columns. Such a table is called a *submatrix* of the matrix $\mathbf{A}$.

Let $\mathbf{A}$ be a square matrix of order n. To specify a square submatrix of order k, one must indicate the indices of the rows $1 \le i_1 < ... < i_k \le n$ and columns $1 \le j_1 < ... < j_k \le n$ containing it. Let us denote by $\boldsymbol{N}_k$ the set of all systems of indices $(i_1, ..., i_k)$ ordered in ascending order: $1 \le i_1 < ... < i_k \le n$. Then specifying a submatrix is equivalent to choosing two specific systems of indices

$$I = (i_1, ..., i_k) \in \boldsymbol{N}_k, \quad J = (j_1, ..., j_k) \in \boldsymbol{N}_k.$$

The submatrix on the rows with indices from I and columns with indices from J is denoted by

$$\mathbf{A}(I, J) = \left(a_{i_p j_q}\right), \quad 1 \le p \le k, \quad 1 \le q \le k.$$

Let $I' = (i'_1, ..., i'_m)$ be another system of indices ordered in ascending order: $1 \le i'_1 < ... < i'_m \le n$. We call the system I' *complementary* to $I = (i_1, ..., i_k)$ if

$$\{i_1, ..., i_k\} \cap \{i'_1, ..., i'_m\} = \varnothing,$$

$$\{i_1, ..., i_k\} \cup \{i'_1, ..., i'_m\} = \{1, ..., n\}.$$

Obviously, in this case $k + m = n$.

Suppose we are given systems of row and column indices $I, J \in \boldsymbol{N}_k$, and let I' and J' be the complementary systems for I and J, respectively. The submatrix $\mathbf{A}(I', J')$ of order $m = n - k$ is called the *complementary* submatrix with respect to the submatrix $\mathbf{A}(I, J)$ of order k.

The determinant $\det(\mathbf{A}(I, J))$ of a submatrix of order k is called a $k \times k$ *minor* (or a *minor of order* k), and the determinant of the corresponding complementary submatrix is called the *complementary minor*. A $k \times k$ minor is a number; it must not be confused with the course's (i, j) *matrix minor* $\mathbf{M}_{ij}$, which is the submatrix obtained from $\mathbf{A}$ by deleting the i-th row and the j-th column. In the notation above, $\mathbf{M}_{ij} = \mathbf{A}(I', J')$ is the complementary submatrix of the 1×1 submatrix $\left(a_{ij}\right) = \mathbf{A}(I, J)$, $I = (i)$, $J = (j)$.

Laplace's Theorem

Theorem (Laplace's Theorem). Let $\mathbf{A}$ be a square matrix of order n. Fix any system of k columns by choosing $J \in \boldsymbol{N}_k$. Then the computation of the determinant of the matrix $\mathbf{A}$ reduces to computing the $k \times k$ minors on the fixed k columns and their complementary minors:

$$\det(\mathbf{A}) = \sum_{I \in \boldsymbol{N}_k} \det(\mathbf{A}(I, J)) \det(\mathbf{A}(I', J'))(-1)^{\nu(I)+\nu(J)},$$

where $\nu(I) = i_1 + ... + i_k$ for $I = (i_1, ..., i_k)$, and likewise $\nu(J) = j_1 + ... + j_k$.

Proof. Relying on the lemma on partitioning the set of permutations of degree n into the subsets of permutations $\sigma(\pi, \tau)$, $\pi \in S_k$, $\tau \in S_m$ (see Tyrtyshnikov, §5.6), we find

$$\begin{aligned}
\det(\mathbf{A}) = \sum_{I \in \boldsymbol{N}_k} \Bigg(& \sum_{\pi \in S_k} \sum_{\tau \in S_m} \left(a_{i_{\pi(1)} j_1} ... a_{i_{\pi(k)} j_k} \right) \times \\
& \times \left(a_{i'_{\tau(1)} j'_1} ... a_{i'_{\tau(m)} j'_m} \right) \operatorname{sgn}(\sigma(\pi, \tau)) \Bigg) = \\
= \sum_{I \in \boldsymbol{N}_k} \Bigg(& \sum_{\pi \in S_k} \sum_{\tau \in S_m} \left(a_{i_{\pi(1)} j_1} ... a_{i_{\pi(k)} j_k} \right) \times \\
& \times \left(a_{i'_{\tau(1)} j'_1} ... a_{i'_{\tau(m)} j'_m} \right) \operatorname{sgn}(\pi) \operatorname{sgn}(\tau) \Bigg) (-1)^{\nu(I)+\nu(J)} = \\
= \sum_{I \in \boldsymbol{N}_k} & \left(\sum_{\pi \in S_k} \left(a_{i_{\pi(1)} j_1} ... a_{i_{\pi(k)} j_k} \right) \operatorname{sgn}(\pi) \right) \times \\
& \times \left(\sum_{\tau \in S_m} \left(a_{i'_{\tau(1)} j'_1} ... a_{i'_{\tau(m)} j'_m} \right) \operatorname{sgn}(\tau) \right) (-1)^{\nu(I)+\nu(J)}.
\end{aligned}$$

It remains to note that the first and second parenthesized sums yield $\det(\mathbf{A}(I, J))$ and $\det(\mathbf{A}(I', J'))$, respectively. □

The quantity $\det(\mathbf{A}(I', J'))(-1)^{\nu(I)+\nu(J)}$ is called the *cofactor* (or *algebraic complement*) of the $k \times k$ minor $\det(\mathbf{A}(I, J))$. Thus, Laplace's theorem asserts that for any choice of a system of columns, *the determinant of a matrix is equal to the sum of all possible $k \times k$ minors located in the given columns, multiplied by their cofactors.*

Since $\det(\mathbf{A}) = \det(\mathbf{A}^T)$ (Section 2.10), the following version of Laplace's theorem also holds: *for any choice of a system of rows, the determinant of a matrix is equal to the sum of all possible $k \times k$ minors located in the given rows, multiplied by their cofactors.*

For $k = 1$, Laplace's theorem is the *cofactor expansion*. Take $J = (j)$ and $I = (i)$: the 1×1 minor is the entry a_{ij}, its complementary submatrix is the (i, j) matrix minor $\mathbf{M}_{ij}$, and its cofactor is the (i, j)-*cofactor* $A_{ij} = (-1)^{i+j} \det(\mathbf{M}_{ij})$. Hence

$$\det(\mathbf{A}) = a_{1j}A_{1j} + a_{2j}A_{2j} + ... + a_{nj}A_{nj}$$

(cofactor expansion along the j-th column) and, by the row version,

$$\det(\mathbf{A}) = a_{i1}A_{i1} + a_{i2}A_{i2} + ... + a_{in}A_{in}$$

(cofactor expansion along the i-th row). MA1522 takes the cofactor expansion as the definition of the determinant; Laplace's theorem shows that it agrees with formula ($\triangle$).

Problem 3. A matrix $\mathbf{B}$ with determinant $b = \det(\mathbf{B})$ is obtained from a matrix $\mathbf{A}$ with determinant $a = \det(\mathbf{A})$ by adding a number $c \neq 0$ to each entry. Find the sums of the cofactors of all entries (submatrices of order 1) for $\mathbf{A}$ and for $\mathbf{B}$.

Problem 4. In an $(n \times n)$-matrix there is a unique $r \times r$ minor, $r < n$, that is nonzero. Prove that all $k \times k$ minors with $k \geq r + 1$ are equal to zero.

Problem 5. Let $\mathbf{A}$ be a matrix of order n with entries $a_{ij} = \pm 1$. Prove that if $n = 4$, then $|\det(\mathbf{A})| \leq 16$, and construct a matrix for which $\det(\mathbf{A}) = 16$.[2]

The Determinant by Cofactor Expansion ADDED FOR MA1522

MA1522 defines the determinant $\det(\mathbf{A}) = |\mathbf{A}|$ of a square matrix $\mathbf{A} = \left(a_{ij}\right)_n$ by induction on the order n:

- for $n = 1$, $\mathbf{A} = (a)$ and $\det(\mathbf{A}) = a$;
- for $n = 2$, $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$;
- for $n \geq 2$, once determinants of order $n - 1$ are defined, let $\mathbf{M}_{ij}$ be the (i, j) matrix minor (the submatrix of order $n - 1$ obtained by deleting the i-th row and the j-th column) and $A_{ij} = (-1)^{i+j} \det(\mathbf{M}_{ij})$ the (i, j)-cofactor; then

$$\det(\mathbf{A}) = a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in} = \sum_{k=1}^{n} a_{ik}A_{ik}$$

$$= a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj} = \sum_{k=1}^{n} a_{kj}A_{kj},$$

the *cofactor expansion* along the i-th row, or along the j-th column.

This is at once a theorem and a definition: the theorem is that all $2n$ expansions give the same number. It is the case $k = 1$ of Laplace's theorem

[2] In the general case one can prove that if $|a_{ij}| \leq 1$, then $|\det(\mathbf{A})| \leq n^{\frac{n}{2}}$ (see Section 5.4). Matrices with entries ± 1 for which $|\det(\mathbf{A})| = n^{\frac{n}{2}}$ are called *Hadamard matrices*. It is not hard to establish that Hadamard matrices do not exist for all n. There is a conjecture that Hadamard matrices exist for all n divisible by 4.

(end of the preceding subsection): every expansion equals the Leibniz sum ($\triangle$). Hence the course's determinant is the determinant studied in this chapter, and the case $n = 2$ agrees with the general rule. The sign $(-1)^{i+j}$ of the (i, j) -cofactor follows a checkerboard pattern beginning with $+$ in position $(1, 1)$:

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

For $n = 3$, expansion along the first row gives

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a\begin{vmatrix} e & f \\ h & i \end{vmatrix} - b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + c\begin{vmatrix} d & e \\ g & h \end{vmatrix} = aei - afh - bdi + bfg + cdh - ceg,$$

which is the formula for a determinant of order 3 in the subsection "The Determinant", written in other letters.

Example. Let

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \\ 1 & 2 & -2 \end{pmatrix}.$$

Expanding along the second row, which contains a zero, we need

$$A_{22} = (+1)\begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -7, \quad A_{23} = (-1)\begin{vmatrix} 2 & -1 \\ 1 & 2 \end{vmatrix} = -5,$$

and obtain $\det(\mathbf{A}) = 0 \cdot A_{21} + 4 \cdot (-7) + 1 \cdot (-5) = -33$. Along the first column, $A_{11} = \begin{vmatrix} 4 & 1 \\ 2 & -2 \end{vmatrix} = -10$ and $A_{31} = \begin{vmatrix} -1 & 3 \\ 4 & 1 \end{vmatrix} = -13$ give the same value, $\det(\mathbf{A}) = 2 \cdot (-10) + 0 \cdot A_{21} + 1 \cdot (-13) = -33$.

In the course's Appendix the determinant is also defined by the *Leibniz formula*, written with the permutations acting on the column indices:

$$\det(\mathbf{A}) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}.$$

The right-hand side is formula ($\triangle$) applied to $\mathbf{A}^T$, so it equals $\det(\mathbf{A}^T) = \det(\mathbf{A})$ (Section 2.10; the first line of the proof there is exactly this sum). For $n = 2$ the identity permutation and the transposition $(1, 2)$ give $a_{11}a_{22} - a_{12}a_{21}$. Together with Laplace's theorem this shows that the Leibniz formula and the cofactor expansion along any row or column define the same function.

Theorem. The determinant of a triangular matrix $\mathbf{A} = \left(a_{ij}\right)_n$ is the product of its diagonal entries:

$$\det(\mathbf{A}) = a_{11}a_{22} \cdots a_{nn}.$$

Proof. Induction on n; for $n = 1$ there is nothing to prove. If $\mathbf{A}$ is upper triangular, the only entry of its first column that can be nonzero is a_{11}, so the expansion along the first column gives $\det(\mathbf{A}) = a_{11}A_{11} = a_{11}\det(\mathbf{M}_{11})$. The matrix minor $\mathbf{M}_{11}$ is upper triangular with diagonal entries $a_{22}, ..., a_{nn}$, and the induction hypothesis gives the result. For a lower triangular matrix, expand along the first row in the same way. □

For example,

$$\begin{vmatrix} 2 & 7 & -1 \\ 0 & 3 & 5 \\ 0 & 0 & -4 \end{vmatrix} = 2 \cdot 3 \cdot (-4) = -24,$$

and in particular $\det(\mathbf{I}_n) = 1$. Determinants of triangular matrices are therefore easy to compute, and Section 2.9 reduces the general case to this one.

2.9. Determinant by Reduction

The Determinant as a Function of the Columns (Rows) of a Matrix

Property 1. The determinant as a function of the columns of a matrix is a linear function with respect to each column: if $\mathbf{A} = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_n)$ and $\mathbf{a}_i = \alpha\mathbf{p} + \beta\mathbf{q}$ is a linear combination of the columns $\mathbf{p}$ and $\mathbf{q}$, then

$$\det(\mathbf{A}) = \alpha\det\big(\mathbf{A}_p\big) + \beta\det\big(\mathbf{A}_q\big),$$

where the matrices $\mathbf{A}_p$ and $\mathbf{A}_q$ are obtained from $\mathbf{A}$ by replacing the column $\mathbf{a}_i$ with $\mathbf{p}$ and $\mathbf{q}$, respectively.

Proof. In accordance with the definition,

$$\begin{aligned}
\det(\mathbf{A}) &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1}a_{\sigma(2)2}...a_{\sigma(i)i}...a_{\sigma(n)n} = \\
&= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1}a_{\sigma(2)2}...\big(\alpha p_{\sigma(i)i} + \beta q_{\sigma(i)i}\big)...a_{\sigma(n)n} = \\
&= \alpha \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1}a_{\sigma(2)2}...p_{\sigma(i)i}...a_{\sigma(n)n} + \\
&+\beta \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1}a_{\sigma(2)2}...q_{\sigma(i)i}...a_{\sigma(n)n} = \\
&= \alpha\det\big(\mathbf{A}_p\big) + \beta\det\big(\mathbf{A}_q\big).
\end{aligned}$$

□

Property 2. The determinant changes sign when two columns are interchanged.

Proof. Suppose the matrix $\mathbf{B} = \big(b_{ij}\big)$ differs from $\mathbf{A}$ by the interchange of the columns $\mathbf{a}_k$ and $\mathbf{a}_l$. Then for any permutation $\sigma \in S_n$,

$$a_{\sigma(1)1}...a_{\sigma(n)n} = b_{(\sigma\tau)(1)1}...b_{(\sigma\tau)(n)n},$$

where $\tau = (k, l)$, and since a transposition changes the sign of a permutation,

$$\operatorname{sgn}(\sigma\tau) = -\operatorname{sgn}(\sigma).$$

It is easy to see that the mapping $\sigma \to \sigma\tau$ defines a one-to-one correspondence between permutations. Each term of the sum of the form ($\triangle$) of Section 2.8 is determined by one and only one permutation. The permutations σ and $\sigma\tau$ in the expansions of $\det(\mathbf{A})$ and $\det(\mathbf{B})$ determine terms with the product of the very same entries (in a different order), but with opposite signs. Hence $\det(\mathbf{A}) = -\det(\mathbf{B})$. □

Property 3. If the columns of a matrix are linearly dependent, then its determinant is equal to zero.

Proof. First of all, let us note that a determinant with two equal columns is equal to zero, since by statement (2) it is equal to itself with the opposite sign.

If the columns $\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_n$ are linearly dependent, then at least one of them is linearly expressible in terms of the others. Suppose

$$\mathbf{a}_i = \sum_{k \neq i} \alpha_k \mathbf{a}_k.$$

Denote by $\mathbf{B}$ the matrix obtained from $\mathbf{A}$ by replacing the column $\mathbf{a}_i$ with

$$\mathbf{a}_i - \sum_{k \neq i} \alpha_k \mathbf{a}_k = \mathbf{0}.$$

Relying on property (1) already established, we find

$$0 = \det(\mathbf{B}) = \det(\mathbf{A}) - \sum_{k \neq i} \alpha_k \det(\mathbf{A}_k),$$

where the matrix $\mathbf{A}_k$ is obtained from $\mathbf{A}$ by replacing the i-th column with $\mathbf{a}_k$. Clearly, in $\mathbf{A}_k$ the i-th and k-th columns are equal, so $\det(\mathbf{A}_k) = 0$. Thus, $\det(\mathbf{A}) = \det(\mathbf{B}) = 0$. □

Property 4. The determinant as a function of the rows of a matrix possesses properties analogous to (1), (2), (3).

Proof. It suffices to take into account that $\det(\mathbf{A}) = \det(\mathbf{A}^T)$ (Section 2.10) and consider $\det(\mathbf{A})$ as a function of the columns of the matrix $\mathbf{A}^T$. □

Problem 1. Given column matrices $\mathbf{u}_1, ..., \mathbf{u}_k, \mathbf{v}_1, ..., \mathbf{v}_k \in \mathbb{R}^n$ and $\mathbf{A} = \mathbf{u}_1\mathbf{v}_1^T + ... + \mathbf{u}_k\mathbf{v}_k^T$, prove that $\det(\mathbf{A}) = 0$ if $k < n$.

Problem 2. Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ and let $\mathbf{I}$ be the identity matrix. Prove that $\det(\mathbf{I} + \mathbf{u}\mathbf{v}^T) = 1 + \mathbf{v}^T\mathbf{u}$.

Existence of the Linear Dependence Indicator

Theorem. The linear dependence indicator (a function endowed with properties (A), (B), (C) from Section 2.8) exists, is unique, and is the determinant.

Properties (A) and (B) of the linear dependence indicator coincide with properties (1) and (3) of the determinant established above. Property (C) means that the determinant of the identity matrix is equal to 1, and is a consequence of the following more general statement.

Proposition. The determinant of a diagonal matrix is equal to the product of its diagonal entries:

$$\det \begin{pmatrix} a_{11} & & 0 \\ & \ddots & \\ 0 & & a_{nn} \end{pmatrix} = a_{11} \cdot \ldots \cdot a_{nn}.$$

Proof. For a diagonal matrix, in the sum ($\triangle$) of Section 2.8 for its determinant there is only one nonzero term, equal to the product of the entries on the main diagonal. □

Determinant and Elementary Row Operations ADDED FOR MA1522

Theorem. Let $\mathbf{A}$ be a square matrix, and let $\mathbf{B}$ be obtained from $\mathbf{A}$ by a single elementary row operation. Then

(i) $\det(\mathbf{B}) = \det(\mathbf{A})$ if the operation is $R_i + aR_j$ $(i \neq j)$;
(ii) $\det(\mathbf{B}) = c\det(\mathbf{A})$ if the operation is cR_i;
(iii) $\det(\mathbf{B}) = -\det(\mathbf{A})$ if the operation is $R_i \leftrightarrow R_j$.

Proof. By Property 4, the determinant is a linear function of each row, changes sign when two rows are interchanged, and vanishes when the rows are linearly dependent, in particular when two rows are equal. Statement (iii) is the second of these facts, and (ii) is linearity in the i-th row. For (i), let $\mathbf{r}_1, \ldots, \mathbf{r}_n$ be the rows of $\mathbf{A}$; the i-th row of $\mathbf{B}$ is $\mathbf{r}_i + a\mathbf{r}_j$, and its other rows are those of $\mathbf{A}$. By linearity in the i-th row, $\det(\mathbf{B}) = \det(\mathbf{A}) + a\det(\mathbf{C})$, where $\mathbf{C}$ is obtained from $\mathbf{A}$ by replacing its i-th row with $\mathbf{r}_j$. The matrix $\mathbf{C}$ has two equal rows, the i-th and the j-th, so $\det(\mathbf{C}) = 0$. □

Corollary. Let $\mathbf{E}$ be an elementary matrix. If $\mathbf{E}$ corresponds to $R_i + aR_j$, then $\det(\mathbf{E}) = 1$; if it corresponds to cR_i, then $\det(\mathbf{E}) = c$; if it corresponds to $R_i \leftrightarrow R_j$, then $\det(\mathbf{E}) = -1$. In every case $\det(\mathbf{E}) \neq 0$, and

$$\det(\mathbf{EA}) = \det(\mathbf{E})\det(\mathbf{A})$$

for every square matrix $\mathbf{A}$ of the same order.

Proof. Apply the theorem to $\mathbf{I}$, whose determinant is 1, to obtain $\det(\mathbf{E})$. Since $\mathbf{EA}$ is obtained from $\mathbf{A}$ by the row operation corresponding to $\mathbf{E}$ (Section 2.5), the theorem gives $\det(\mathbf{EA}) = \det(\mathbf{E})\det(\mathbf{A})$ in each of the three cases. □

Theorem. If $\mathbf{R} = \mathbf{E}_k \cdots \mathbf{E}_2 \mathbf{E}_1 \mathbf{A}$ for a square matrix $\mathbf{A}$ and elementary matrices $\mathbf{E}_1, ..., \mathbf{E}_k$, then

$$\det(\mathbf{R}) = \det(\mathbf{E}_k) \cdots \det(\mathbf{E}_2)\det(\mathbf{E}_1)\det(\mathbf{A}).$$

Proof. Apply the corollary k times. □

Since the determinants of elementary matrices are nonzero, $\det(\mathbf{A})$ and $\det(\mathbf{R})$ determine each other for row-equivalent matrices $\mathbf{A}$ and $\mathbf{R}$. Let

$$\mathbf{A} \xrightarrow{r_1} \cdots \xrightarrow{r_k} \mathbf{R},$$

where $\mathbf{R}$ is a row-echelon form of $\mathbf{A}$, and let $\mathbf{E}_i$ be the elementary matrix corresponding to r_i. A square matrix in row-echelon form is upper triangular; if $d_1, ..., d_n$ are its diagonal entries, the theorem on triangular matrices (Section 2.8) gives $\det(\mathbf{R}) = d_1 d_2 \cdots d_n$, and hence

$$\det(\mathbf{A}) = \frac{d_1 d_2 \cdots d_n}{\det(\mathbf{E}_k) \cdots \det(\mathbf{E}_2)\det(\mathbf{E}_1)}.$$

This is the computation of a determinant *by reduction*; for large n it is far cheaper than cofactor expansion.

Example. Let us reduce

$$\mathbf{A} = \begin{pmatrix} 0 & 2 & 4 \\ 1 & 1 & 1 \\ 2 & 3 & 5 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 2 & 3 & 5 \end{pmatrix} \xrightarrow{R_3 - 2R_1} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 1 & 3 \end{pmatrix}$$

and further

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 1 & 3 \end{pmatrix} \xrightarrow{(1/2)R_2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 1 & 3 \end{pmatrix} \xrightarrow{R_3 - R_2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

The elementary matrices used have determinants -1, 1, $1/2$, 1, and the final matrix has determinant 1. Hence

$$\det(\mathbf{A}) = \frac{1}{(1)(1/2)(1)(-1)} = -2,$$

in agreement with the cofactor expansion along the first column: $0 - 1 \cdot (2 \cdot 5 - 4 \cdot 3) + 2 \cdot (2 \cdot 1 - 4 \cdot 1) = -2$.

2.10. Properties of Determinant

Determinant of the Transpose Matrix

Suppose we have a rectangular matrix of size $m \times n$:

$$\mathbf{A} = \big(a_{ij}\big), \quad 1 \le i \le m, \quad 1 \le j \le n.$$

If we interchange its rows and columns, we obtain a new matrix—of size $n \times m$. It is called the *transpose* with respect to $\mathbf{A}$ and is denoted by $\mathbf{A}^T$:

$$\mathbf{A}^T = \big(a_{ji}\big), \quad 1 \le j \le n, \quad 1 \le i \le m.$$

Proposition. For any square matrix, $\det(\mathbf{A}^T) = \det(\mathbf{A})$.

Proof. By the definition of the transpose matrix and formula ($\triangle$) of Section 2.8 for the determinant of a matrix of order n,

$$\begin{aligned}\det(\mathbf{A}^T) &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)}\ldots a_{n\sigma(n)} = \\ &= \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{\sigma^{-1}(1)1}\ldots a_{\sigma^{-1}(n)n} = \\ &= \sum_{\sigma^{-1}\in S_n} \operatorname{sgn}(\sigma^{-1}) a_{\sigma^{-1}(1)1}\ldots a_{\sigma^{-1}(n)n} = \det(\mathbf{A}).\end{aligned}$$

In the last equality we took into account that $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma)$. □

Problem 1. Prove that the columns of a real rectangular matrix $\mathbf{A}$ are linearly independent if and only if the columns of the matrix $\mathbf{A}^T\mathbf{A}$ are linearly independent.

Determinant of a Product of Matrices

Theorem. The determinant of a product of square matrices is equal to the product of their determinants.

Proof. Let $\mathbf{A} = \begin{pmatrix}\mathbf{a}_1 & \cdots & \mathbf{a}_n\end{pmatrix}$ be a matrix of order n with columns $\mathbf{a}_1, \ldots, \mathbf{a}_n$ and $\mathbf{B} = \big(b_{ij}\big)$. Then any column of the matrix $\mathbf{AB}$ is a linear combination of the columns of the matrix $\mathbf{A}$ with coefficients from the corresponding column of the matrix $\mathbf{B}$:

$$\mathbf{AB} = \Big(\textstyle\sum_{i_1=1}^n b_{i_1 1}\mathbf{a}_{i_1} \quad \cdots \quad \sum_{i_n=1}^n b_{i_n n}\mathbf{a}_{i_n}\Big).$$

Using the linearity of the determinant in each column, we obtain

$$\begin{aligned}\det(\mathbf{AB}) &= \sum_{i_1=1}^n \ldots \sum_{i_n=1}^n b_{i_1 1}\ldots b_{i_n n} \det\begin{pmatrix}\mathbf{a}_{i_1} & \cdots & \mathbf{a}_{i_n}\end{pmatrix} = \\ &= \sum_{\sigma\in S_n} b_{\sigma(1)1}\ldots b_{\sigma(n)n} \det\begin{pmatrix}\mathbf{a}_{\sigma(1)} & \cdots & \mathbf{a}_{\sigma(n)}\end{pmatrix} = \\ &= \left(\sum_{\sigma\in S_n} b_{\sigma(1)1}\ldots b_{\sigma(n)n} \operatorname{sgn}(\sigma)\right) \det(\mathbf{A}) = \\ &= \det(\mathbf{B}) \cdot \det(\mathbf{A}).\end{aligned}$$

□

Problem 2. Let $\mathbf{P}$ be an invertible matrix of order n. Prove that for any columns $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{n\times 1}$ the equality $\det(\mathbf{I}+\mathbf{u}\mathbf{v}^T) = \det\Big(\mathbf{I}+(\mathbf{P}\mathbf{u})\big(\mathbf{P}^{-T}\mathbf{v}\big)^T\Big)$ holds.

Problem 3. Prove that the determinant of a tridiagonal matrix does not change if every superdiagonal entry is multiplied and every subdiagonal entry is divided by one and the same nonzero number.

The Adjoint Formula for the Inverse

In this subsection we consider square matrices with a nonzero determinant.

Let $\tilde{\mathbf{A}} = \big(A_{ij}\big)$ be the matrix of order n in which the entry A_{ij} is the cofactor of the entry a_{ij} in the matrix $\mathbf{A}$, that is, the (i,j)-cofactor $A_{ij} = (-1)^{i+j}\det\big(\mathbf{M}_{ij}\big)$ (Section 2.8). The matrix $\operatorname{adj}(\mathbf{A}) = \tilde{\mathbf{A}}^T$, whose (i,j)-entry is the (j,i)-cofactor A_{ji}, is called the *adjoint* (also the *adjugate* or *classical adjoint*) of the matrix $\mathbf{A}$.

Theorem. If a matrix $\mathbf{A}$ of order n has $\det(\mathbf{A}) \neq 0$, then it is invertible and, moreover,

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})}\operatorname{adj}(\mathbf{A}),$$

where $\operatorname{adj}(\mathbf{A})$ is the adjoint of $\mathbf{A}$.

Proof. Note that

$$\sum_{j=1}^{n} a_{ij}A_{kj} = \begin{cases} \det(\mathbf{A}), & k = i, \\ 0, & k \neq i. \end{cases} \qquad (*)$$

For $k = i$, equality $(*)$ is obtained by applying Laplace's theorem, that is, by the cofactor expansion of the determinant of the matrix $\mathbf{A}$ along the k-th row. For $k \neq i$, the left-hand side of equality $(*)$ represents the cofactor expansion along the k-th row of the determinant of the matrix obtained from $\mathbf{A}$ by replacing the k-th row with the i-th row. Such a determinant is equal to zero, as the determinant of a matrix with two identical rows. For example, if $n = 3$, $i = 1$, $k = 2$, then

$$a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23} =$$

$$= a_{11}\left(-\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix}\right) + a_{12}\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\left(-\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix}\right) =$$

$$= \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 0.$$

Furthermore,

$$\sum_{j=1}^{n} a_{ij}A_{kj} = \sum_{j=1}^{n} a_{ij}(\operatorname{adj}(\mathbf{A}))_{jk}, \quad 1 \le i, k \le n.$$

Therefore, by virtue of the relations $(*)$,

$$\mathbf{A}\operatorname{adj}(\mathbf{A}) = \begin{pmatrix} \det(\mathbf{A}) & & \\ & \ddots & \\ & & \det(\mathbf{A}) \end{pmatrix} = \det(\mathbf{A})\mathbf{I}.$$

Using Laplace's theorem for expanding the determinant along the k-th column (cofactor expansion along a column), we find

$$\sum_{i=1}^{n} a_{ij}A_{ik} = \begin{cases} \det(\mathbf{A}), & k = j, \\ 0, & k \ne j \end{cases} \quad \Rightarrow$$

$$\Rightarrow \quad \operatorname{adj}(\mathbf{A})\mathbf{A} = \begin{pmatrix} \det(\mathbf{A}) & & 0 \\ & \ddots & \\ 0 & & \det(\mathbf{A}) \end{pmatrix} = \det(\mathbf{A})\mathbf{I}. \qquad \square$$

Problem 4. Prove that any invertible matrix can be made singular by changing just one of its entries.

Cramer's Rule

Theorem. Let $\mathbf{A}$ be a matrix of order n with $\det(\mathbf{A}) \ne 0$. Then the linear system $\mathbf{Ax} = \mathbf{b}$ has a unique solution $\mathbf{x}$ with components

$$x_i = \frac{\det(\mathbf{A}_i(\mathbf{b}))}{\det(\mathbf{A})}, \quad 1 \le i \le n,$$

where $\mathbf{A}_i(\mathbf{b})$ is the matrix obtained from $\mathbf{A}$ by replacing the i-th column with $\mathbf{b}$.

Proof. According to the theorem on the inversion of a matrix with nonzero determinant (the adjoint formula),

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \frac{1}{\det(\mathbf{A})}\operatorname{adj}(\mathbf{A})\mathbf{b} \quad \Rightarrow \quad x_i = \frac{1}{\det(\mathbf{A})}\sum_{j=1}^{n} A_{ji}b_j, \quad 1 \le i \le n.$$

It remains to note that the sum $\sum_{j=1}^{n} A_{ji}b_j$ is the cofactor expansion along the i-th column of the determinant of the matrix $\mathbf{A}_i(\mathbf{b})$. $\square$

Further Properties of the Determinant ADDED FOR MA1522

By induction on k, the theorem on the determinant of a product extends to any number of factors:

$$\det(\mathbf{A}_1\mathbf{A}_2\cdots\mathbf{A}_k) = \det(\mathbf{A}_1)\det(\mathbf{A}_2)\cdots\det(\mathbf{A}_k).$$

For instance, if $\mathbf{A} = \mathbf{LU}$ is an LU factorization of a square matrix (Section 2.7), then $\det(\mathbf{L}) = 1$ and $\det(\mathbf{A}) = \det(\mathbf{U})$ is the product of the diagonal entries of $\mathbf{U}$.

Theorem. Let $\mathbf{A}$ be a square matrix of order n.
(i) If $\mathbf{A}$ is invertible, then $\det(\mathbf{A}^{-1}) = \det(\mathbf{A})^{-1}$.
(ii) For every $c \in \mathbb{R}$, $\det(c\mathbf{A}) = c^n \det(\mathbf{A})$.

Proof. (i) $1 = \det(\mathbf{I}) = \det(\mathbf{A}^{-1}\mathbf{A}) = \det(\mathbf{A}^{-1})\det(\mathbf{A})$, and $\det(\mathbf{A}) \neq 0$ since $\mathbf{A}$ is invertible. (ii) $c\mathbf{A} = (c\mathbf{I}_n)\mathbf{A}$, and the scalar matrix $c\mathbf{I}_n$ is diagonal with determinant c^n; hence $\det(c\mathbf{A}) = \det(c\mathbf{I}_n)\det(\mathbf{A}) = c^n \det(\mathbf{A})$. □

For example, if $\mathbf{A}$ and $\mathbf{B}$ are of order 3, $\det(\mathbf{A}) = 2$ and $\det(\mathbf{B}) = -1$, then

$$\det(2\mathbf{A}^T\mathbf{B}^{-1}) = 2^3 \det(\mathbf{A}^T)\det(\mathbf{B}^{-1}) = 8 \cdot 2 \cdot (-1)^{-1} = -16.$$

The proof of the adjoint formula above never uses the assumption $\det(\mathbf{A}) \neq 0$: the relations $(*)$ and the two matrix identities derived from them hold for every square matrix. Thus, for every square matrix $\mathbf{A}$,

$$\mathbf{A}\,\mathrm{adj}(\mathbf{A}) = \mathrm{adj}(\mathbf{A})\mathbf{A} = \det(\mathbf{A})\mathbf{I},$$

and the *adjoint formula for the inverse*, $\mathbf{A}^{-1} = \det(\mathbf{A})^{-1}\,\mathrm{adj}(\mathbf{A})$ for an invertible $\mathbf{A}$, is the special case $\det(\mathbf{A}) \neq 0$; for $n = 2$ it is the formula for the inverse of a 2×2 matrix in Section 2.4. For a singular $\mathbf{A}$ the identity gives $\mathbf{A}\,\mathrm{adj}(\mathbf{A}) = \mathbf{0}$; for example,

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \quad \mathrm{adj}(\mathbf{A}) = \begin{pmatrix} 4 & -2 \\ -2 & 1 \end{pmatrix}, \quad \mathbf{A}\,\mathrm{adj}(\mathbf{A}) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Finally, the condition $\det(\mathbf{A}) \neq 0$ is statement (ix) in the list of equivalent statements for invertibility (Section 2.6).

Chapter 3. Euclidean Vector Spaces

3.1. Euclidean Vector Spaces

Vectors in $\mathbb{R}^n$ ADDED FOR MA1522

An *n-vector* (or simply a *vector*) is an ordered collection of n real numbers written as a column,

$$\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}, \quad v_1, ..., v_n \in \mathbb{R},$$

that is, a matrix of size $n \times 1$ (Section 1.1); the number v_i is the *i-th coordinate* (or entry) of $\mathbf{v}$. In running text we write $\mathbf{v} = (v_1, ..., v_n)^T$.

Definition. The *Euclidean n-space* $\mathbb{R}^n$ is the set of all n-vectors:

$$\mathbb{R}^n = \left\{ (v_1, v_2, ..., v_n)^T \mid v_1, v_2, ..., v_n \in \mathbb{R} \right\}.$$

Vectors are added and multiplied by real numbers entrywise, as $n \times 1$ matrices (Section 2.2):

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, ..., u_n + v_n)^T, \quad c\mathbf{u} = (cu_1, ..., cu_n)^T.$$

The *zero vector* is $\mathbf{0} = (0, ..., 0)^T$, the *negative* of $\mathbf{v}$ is $-\mathbf{v} = (-1)\mathbf{v}$, and $\mathbf{u} - \mathbf{v} = \mathbf{u} + (-\mathbf{v})$.

For $n = 2$ and $n = 3$ a vector $\mathbf{v}$ has a double geometric meaning: it is the point with coordinates $v_1, ..., v_n$ in a Cartesian coordinate system, and it is the arrow (directed segment) from the origin O to that point. Thus $(1, 1, 1)^T \in \mathbb{R}^3$ is both the point $(1, 1, 1)$ and the arrow from O to it. The sum $\mathbf{u} + \mathbf{v}$ is then the arrow obtained by placing the tail of $\mathbf{v}$ at the head of $\mathbf{u}$, and $c\mathbf{u}$ is the arrow $\mathbf{u}$ stretched by the factor $|c|$ and reversed when $c < 0$. These are the geometric operations on free vectors described in the next subsection; in Cartesian coordinates they act entrywise, as above.

Theorem (Vector algebra in $\mathbb{R}^n$). Let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$. Then:

(i) $\mathbf{u} + \mathbf{v} \in \mathbb{R}^n$;
(ii) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (commutativity);
(iii) $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ (associativity);
(iv) $\mathbf{0} + \mathbf{v} = \mathbf{v}$ (zero vector);
(v) $-\mathbf{v} \in \mathbb{R}^n$ and $\mathbf{v} - \mathbf{v} = \mathbf{0}$ (negative);
(vi) $a\mathbf{v} \in \mathbb{R}^n$;

(vii) $a(\mathbf{u}+\mathbf{v}) = a\mathbf{u} + a\mathbf{v}$ (distributivity);
(viii) $(a+b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$ (distributivity);
(ix) $(ab)\mathbf{u} = a(b\mathbf{u})$ (associativity of scalar multiplication);
(x) if $a\mathbf{u} = \mathbf{0}$, then $a = 0$ or $\mathbf{u} = \mathbf{0}$.

Proof. Since the operations act entrywise, (i)–(ix) reduce to the corresponding laws for real numbers in each coordinate; for instance, the i-th entry of $a(\mathbf{u}+\mathbf{v})$ is $a(u_i + v_i) = au_i + av_i$, the i-th entry of $a\mathbf{u} + a\mathbf{v}$. For (x), if $a \neq 0$, then $au_i = 0$ gives $u_i = 0$ for every i. □

Together with the obvious rule $1\mathbf{u} = \mathbf{u}$, properties (i)–(ix) are exactly the axioms of a vector space listed in the subsection "Abstract Vector Spaces" below. Hence $\mathbb{R}^n$ is a vector space, and property (x) is also a consequence of the axioms (Proposition 4 there).

Linear Operations on Vectors

Sum of vectors: let $\overrightarrow{AB} \in \mathbf{a}$ and $\overrightarrow{BC} \in \mathbf{b}$; then $\mathbf{c} = \mathbf{a} + \mathbf{b}$ is defined as the vector generated by the directed segment $\overrightarrow{AC}$.

It is important that the vector $\mathbf{c}$ obtained in this way does not depend on the choice of the point A. Indeed, let $\overrightarrow{PQ} \in \mathbf{a}$ and $\overrightarrow{QR} \in \mathbf{b}$. Then from the congruence of the triangles $\triangle ABC$ and $\triangle PQR$ there follows the equality of lengths and codirectionality of the directed segments $\overrightarrow{AC}$ and $\overrightarrow{PR}$, and hence their equality.

The set of free vectors forms an abelian group with respect to the operation of vector addition. The role of the identity element is played by the zero vector $\mathbf{0} = \overrightarrow{AA}$. For $\mathbf{a} = \overrightarrow{AB}$, the inverse element is $\mathbf{b} = \overrightarrow{BA}$. In this context, the vector $\mathbf{b}$ is called the *opposite vector* for $\mathbf{a}$ and is denoted by $\mathbf{b} = -\mathbf{a}$.

Multiplication of a vector by a scalar: let $\overrightarrow{AB} \in \mathbf{a}$; then $\alpha\mathbf{a}$ is defined as the vector generated by the directed segment $\overrightarrow{AC}$ that has length $|AC| = |\alpha|\ |AB|$ and, if $\alpha \neq 0$, is codirectional with $\overrightarrow{AB}$ for $\alpha > 0$ and oppositely directed for $\alpha < 0$. It is not difficult to verify that the vector $\alpha\mathbf{a}$ does not depend on the choice of the point A.

One can check that $\alpha(\beta\mathbf{a}) = (\alpha\beta)\mathbf{a}$ for any real numbers α, β. It is also useful to note that $1\mathbf{a} = \mathbf{a}$, and the vector $(-1)\mathbf{a}$ is the opposite of $\mathbf{a}$.

Furthermore, the operations of vector addition and multiplication of a vector by a scalar are linked by the following distributive laws:

$$(\alpha+\beta)\mathbf{a} = (\alpha\mathbf{a}) + (\beta\mathbf{a}), \quad \alpha(\mathbf{a}+\mathbf{b}) = (\alpha\mathbf{a}) + (\alpha\mathbf{b}).$$

Abstract Vector Spaces

When studying the linear dependence of vectors, linear spans, bases, and dimensions (Sections 2.3 and 3.5–3.7), we assume that vectors are column vectors with real entries. However, when studying the rank of a matrix (Section 4.2), we shall also speak of the linear dependence and independence of the rows of a matrix. Of course, from a formal standpoint, rows can be transposed so

that we once again deal with column vectors. Yet all the concepts listed above and many of the facts obtained can be applied without any changes to the case where vectors are understood to be matrices of any fixed size. This fact alone prompts us to consider introducing a more general (and more abstract) concept of a vector.

Moreover, when studying bases and dimensions, we deal exclusively with linear spans of vectors, and this is not always convenient: for example, the set of all solutions of a homogeneous linear system $\mathbf{Ax} = \mathbf{0}$ is, of course, the linear span of the vectors of a basis for the solution space, but it would be useful to have the right to study the properties of this set without mentioning the system of vectors that generates it.

Let us say that vectors are elements of some nonempty set V on which two operations are defined: addition of vectors (if $\mathbf{a}, \mathbf{b} \in V$, then $\mathbf{a} + \mathbf{b} \in V$) and multiplication of vectors by real numbers (if $\mathbf{a} \in V$ and $\alpha \in \mathbb{R}$, then $\alpha\mathbf{a} \in V$). Let us require that these operations possess the following properties:

- $(\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c}) \quad \forall \mathbf{a}, \mathbf{b}, \mathbf{c} \in V$ (associativity of vector addition);
- there exists a special vector $\mathbf{0}$, called the *zero vector*, such that
$$\mathbf{a} + \mathbf{0} = \mathbf{0} + \mathbf{a} = \mathbf{a} \quad \forall \mathbf{a} \in V;$$
- for any vector $\mathbf{a} \in V$ there exists a vector $\mathbf{b} \in V$ such that
$$\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a} = \mathbf{0};$$
- $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a} \quad \forall \mathbf{a}, \mathbf{b} \in V$ (commutativity of vector addition);
- $\alpha(\beta\mathbf{a}) = (\alpha\beta)\mathbf{a} \quad \forall \alpha, \beta \in \mathbb{R}, \forall \mathbf{a} \in V$;
- $(\alpha + \beta)\mathbf{a} = \alpha\mathbf{a} + \beta\mathbf{a} \quad \forall \alpha, \beta \in \mathbb{R}, \forall \mathbf{a} \in V$ (distributivity);
- $\alpha(\mathbf{a} + \mathbf{b}) = \alpha\mathbf{a} + \alpha\mathbf{b} \quad \forall \alpha \in \mathbb{R}, \forall \mathbf{a}, \mathbf{b} \in V$ (distributivity);
- $1\mathbf{a} = \mathbf{a} \quad \forall \mathbf{a} \in V$.[3]

A set V possessing all these properties is called a (real) *vector space*, or a vector space over $\mathbb{R}$. A frequently encountered synonymous term is *linear space*.

Note that the set V with respect to the operation of vector addition is an abelian group. The role of the identity element is played by the zero vector. A vector $\mathbf{b}$ such that $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a} = \mathbf{0}$ is called the *opposite* (or *negative*) of the vector $\mathbf{a}$ and is denoted by $\mathbf{b} = -\mathbf{a}$.

Certain familiar properties of the operations we have introduced, which were previously applied freely to column vectors, *require proofs* in the more abstract setting considered here.

Proposition 1. $0\mathbf{a} = \mathbf{0} \quad \forall \mathbf{a} \in V$.

Proof. By distributivity,
$$0\mathbf{a} = (0 + 0)\mathbf{a} = (0\mathbf{a}) + (0\mathbf{a}).$$

[3]This property is equivalent to saying that every vector $\mathbf{a}$ can be represented in the form $\mathbf{a} = \alpha\mathbf{b}$ for some vector $\mathbf{b}$ and some number α. Indeed, if this property holds, one can take $\mathbf{b} = \mathbf{a}$ and $\alpha = 1$. On the other hand, suppose this property is not assumed to hold, but it is known that $\mathbf{a} = \alpha\mathbf{b}$. Then, using the axiom $\alpha(\beta\mathbf{a}) = (\alpha\beta)\mathbf{a}$, we obtain $1(\alpha\mathbf{b}) = (1 \cdot \alpha)\mathbf{b} = \alpha\mathbf{b} \Rightarrow 1\mathbf{a} = \mathbf{a}$.

Further, let $\mathbf{b} = -(0\mathbf{a})$ (the vector opposite to the vector $0\mathbf{a}$). Then $\mathbf{0} = \mathbf{b} + (0\mathbf{a}) = (\mathbf{b} + (0\mathbf{a})) + (0\mathbf{a}) \Rightarrow \mathbf{0} = 0\mathbf{a}$. □

Proposition 2. $\alpha\mathbf{0} = \mathbf{0} \quad \forall \alpha \in \mathbb{R}$.

Proof. $\alpha\mathbf{0} = \alpha(\mathbf{0} + \mathbf{0}) = \alpha\mathbf{0} + \alpha\mathbf{0} \Rightarrow \alpha\mathbf{0} = \mathbf{0}$. □

Proposition 3. $(-1)\mathbf{a} = -\mathbf{a} \quad \forall \mathbf{a} \in V$.

Proof. By Proposition 1 and distributivity,

$$\mathbf{0} = 0\mathbf{a} = (1 + (-1))\mathbf{a} = (1\mathbf{a}) + ((-1)\mathbf{a}) = \mathbf{a} + ((-1)\mathbf{a}).$$ □

Proposition 4. If $\alpha\mathbf{a} = \mathbf{0}$, then either $\alpha = 0$ or $\mathbf{a} = \mathbf{0}$.

Proof. Suppose[4] $\alpha \neq 0$. Then

$$\mathbf{a} = 1\mathbf{a} = \left(\frac{1}{\alpha}\alpha\right)\mathbf{a} = \frac{1}{\alpha}(\alpha\mathbf{a}) = \frac{1}{\alpha}\mathbf{0} = \mathbf{0}.$$ □

As before, for any numbers $\alpha_1, ..., \alpha_n$, a vector $\mathbf{w}$ of the form

$$\mathbf{w} = \alpha_1\mathbf{a}_1 + ... + \alpha_n\mathbf{a}_n$$

is called a linear combination of the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$, and the set of all linear combinations with all possible values of the coefficients $\alpha_1, ..., \alpha_n$ is called the linear span of the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$ and is denoted by $\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}$.

Note that the zero vector is denoted by $\mathbf{0}$, to distinguish it from the number 0.

3.2. Dot Product, Norm, Distance

Dot Product of Geometric Vectors

The length of a vector $\mathbf{a}$ is defined as the length of a directed segment generating it (directed segments generating one and the same vector are equal and therefore have the same length). Notation for the length: $\|\mathbf{a}\|$. The angle θ between nonzero vectors $\mathbf{a} = \overrightarrow{OA}$, $\mathbf{b} = \overrightarrow{OB}$ is defined as the angle between the sides OA and OB in the triangle OAB.

The *dot product* (or *scalar product*) of vectors $\mathbf{a}$ and $\mathbf{b}$ is the quantity

$$\mathbf{a} \cdot \mathbf{b} = \begin{cases} \|\mathbf{a}\|\|\mathbf{b}\| \cos\theta, & \mathbf{a} \neq \mathbf{0} \text{ and } \mathbf{b} \neq \mathbf{0}, \\ 0, & \mathbf{a} = \mathbf{0} \text{ or } \mathbf{b} = \mathbf{0}. \end{cases}$$

By virtue of the definition, it is obvious that

$$\mathbf{a} \cdot \mathbf{a} > 0 \quad \text{for} \quad \mathbf{a} \neq \mathbf{0}; \quad \mathbf{a} \cdot \mathbf{a} = 0 \Leftrightarrow \mathbf{a} = \mathbf{0}. \tag{1}$$

[4] This proposition cannot be obtained without the axiom $1\mathbf{a} = \mathbf{a}$. Indeed, take any abelian group V with zero element $\mathbf{0}$ and define multiplication by a number by the rule $\alpha\mathbf{a} = \mathbf{0}$ for all numbers α and vectors $\mathbf{a} \in V$. Then all the axioms of a vector space will be satisfied except this one.

It is also obvious that

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a} \quad \forall \mathbf{a}, \mathbf{b}. \tag{2}$$

If the vectors $\mathbf{a} = \overrightarrow{OA}$, $\mathbf{b} = \overrightarrow{OB}$ are noncollinear, then in the plane passing through the points O, A, B one can introduce a Cartesian coordinate system with origin at the point O and first axis coinciding with the line OB and giving the point B a positive coordinate. Then the quantity $\|\mathbf{a}\| \cos\theta$ will be precisely the coordinate of the point A on this axis. From this there immediately follow the important properties of linearity of the dot product in the first argument:

$$(\mathbf{a} + \mathbf{b}) \cdot \mathbf{c} = \mathbf{a} \cdot \mathbf{c} + \mathbf{b} \cdot \mathbf{c} \quad \forall \mathbf{a}, \mathbf{b}, \mathbf{c}, \tag{3}$$

$$(\alpha \mathbf{a}) \cdot \mathbf{b} = \alpha(\mathbf{a} \cdot \mathbf{b}) \quad \forall \mathbf{a}, \mathbf{b}, \quad \forall \alpha \in \mathbb{R}. \tag{4}$$

Property (2) immediately yields the analogous properties of linearity of the dot product in the second argument as well.

Vectors are called *orthogonal* if their dot product is equal to zero.

Dot Product and Coordinates

Suppose a Cartesian coordinate system with basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ is given. Then

$$\mathbf{e}_i \cdot \mathbf{e}_j = \begin{cases} 0, & i \neq j, \\ 1, & i = j. \end{cases} \tag{5}$$

Theorem. Suppose that in a given Cartesian coordinate system the vector $\mathbf{a}$ has coordinates a_1, a_2, a_3, and the vector $\mathbf{b}$ has coordinates b_1, b_2, b_3. Then the formula

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 \tag{6}$$

holds.

Proof. Let us write

$$\mathbf{a} = a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_3, \quad \mathbf{b} = b_1 \mathbf{e}_1 + b_2 \mathbf{e}_2 + b_3 \mathbf{e}_3.$$

Relying on properties (2)–(4) of the dot product and relations (5), we find

$$\mathbf{a} \cdot \mathbf{b} = (a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_3) \cdot (b_1 \mathbf{e}_1 + b_2 \mathbf{e}_2 + b_3 \mathbf{e}_3) =$$

$$= \sum_{i=1}^{3} \sum_{j=1}^{3} a_i b_j (\mathbf{e}_i \cdot \mathbf{e}_j) = a_1 b_1 + a_2 b_2 + a_3 b_3. \qquad \square$$

Remark 1. If in some coordinate system the dot product of any vectors $\mathbf{a}$ and $\mathbf{b}$ is computed by formula (6), then that coordinate system is Cartesian.

Remark 2. In the case of a Cartesian coordinate system for vectors in a plane, formula (6) takes the form

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2.$$

On Generalizations

Formula (6) and properties (1)–(4) provide a basis for introducing an inner product in more general cases—for objects that are no longer vectors in geometric space. For example, if $\mathbf{a} = (a_1, ..., a_n)^T$, $\mathbf{b} = (b_1, ..., b_n)^T$ are column vectors in $\mathbb{R}^n$, one can define their dot product by analogy with formula (6):

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + ... + a_n b_n. \tag{7}$$

There is also another idea of a more general character—to take properties (1)–(4) as the foundation and call any function of column vectors $\mathbf{a}, \mathbf{b}$ satisfying axioms (1)–(4) an inner product.

For geometric vectors, the dot product was defined on the basis of concepts such as the length of a vector and the angle between vectors. In more general cases, it is simpler to introduce an inner product in some way and then use it to introduce the concepts of length (norm) and angle:

$$\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}}, \quad \cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|}.$$

For example, relying on (7), one can introduce length and angle in this way for vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$. Here it is important that

$$|\mathbf{a} \cdot \mathbf{b}| \le \sqrt{\mathbf{a} \cdot \mathbf{a}}\sqrt{\mathbf{b} \cdot \mathbf{b}}.$$

This inequality (known as the Cauchy–Bunyakovsky–Schwarz inequality) is easily deduced from (7), but in fact it holds for all conceivable ways of specifying an inner product. A short proof for the dot product on $\mathbb{R}^n$ is given in the subsection "Norm, Distance and Angle in $\mathbb{R}^n$" below; for a general inner product, see Tyrtyshnikov, §24.4.

Euclidean Space

Let V be a real vector space on which to each ordered pair of vectors $\mathbf{x}, \mathbf{y} \in V$ there is assigned a real number $\langle \mathbf{x}, \mathbf{y} \rangle$ in such a way that:

(i) $\langle \mathbf{x}, \mathbf{x} \rangle \ge 0 \quad \forall \mathbf{x} \in V; \quad \langle \mathbf{x}, \mathbf{x} \rangle = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}$;
(ii) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle \quad \forall \mathbf{x}, \mathbf{y} \in V$;
(iii) $\langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle \quad \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in V$;
(iv) $\langle \alpha \mathbf{x}, \mathbf{y} \rangle = \alpha \langle \mathbf{x}, \mathbf{y} \rangle \quad \forall \alpha \in \mathbb{R}, \quad \forall \mathbf{x} \in V$.

The number $\langle \mathbf{x}, \mathbf{y} \rangle$ is called the *inner product* (or *scalar product*) of the vectors $\mathbf{x}$ and $\mathbf{y}$. A real vector space with an inner product is called *Euclidean*.

In $\mathbb{R}^n$, the inner product of vectors $\mathbf{x} = (x_1, ..., x_n)^T$ and $\mathbf{y} = (y_1, ..., y_n)^T$ is often introduced as the sum of pairwise products of their coordinates:

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^{n} x_i y_i = \mathbf{y}^T \mathbf{x}. \tag{8}$$

It is called the *natural* inner product on $\mathbb{R}^n$; this is the dot product.

However, on $\mathbb{R}^n$ an inner product can also be introduced in many other ways: for example, if we fix numbers $\lambda_1, ..., \lambda_n > 0$, then the expression

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{n} \lambda_i x_i y_i = \mathbf{y}^T \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \mathbf{x}$$

possesses properties (i)–(iv) and, consequently, defines an inner product.

Norm, Distance and Angle in $\mathbb{R}^n$ ADDED FOR MA1522

In $\mathbb{R}^n$ we always use the dot product (7): $\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + ... + u_n v_n = \mathbf{u}^T \mathbf{v}$.

By Pythagoras' theorem, the point $\mathbf{u} = (x, y)^T$ of the plane lies at distance $\sqrt{x^2 + y^2}$ from the origin. In space, the point $\mathbf{v} = (x, y, z)^T$ lies above (or below) the point $(x, y, 0)$, whose distance from the origin is $\sqrt{x^2 + y^2}$; applying Pythagoras' theorem once more, now in a vertical right triangle, we find the distance $\sqrt{\left(\sqrt{x^2 + y^2}\right)^2 + z^2} = \sqrt{x^2 + y^2 + z^2}$. In both cases the distance from the origin equals $\sqrt{\mathbf{u} \cdot \mathbf{u}}$, which suggests the general definition.

Definition. The *norm* (also *length* or *magnitude*) of a vector $\mathbf{u} = (u_1, ..., u_n)^T \in \mathbb{R}^n$ is

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}} = \sqrt{u_1^2 + u_2^2 + ... + u_n^2}.$$

Theorem. For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and $a, b, c \in \mathbb{R}$:

(i) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ (symmetry);
(ii) $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$;
(iii) $\mathbf{u} \cdot (a\mathbf{v} + b\mathbf{w}) = a(\mathbf{u} \cdot \mathbf{v}) + b(\mathbf{u} \cdot \mathbf{w})$ (distributivity);
(iv) $\mathbf{u} \cdot \mathbf{u} \geq 0$, and $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$ (positive definiteness);
(v) $\|c\mathbf{u}\| = |c| \|\mathbf{u}\|$.

Proof. Properties (i)–(iii) follow at once from $\mathbf{u} \cdot \mathbf{v} = \sum_i u_i v_i$; for example, $\mathbf{u} \cdot (a\mathbf{v} + b\mathbf{w}) = \sum_i u_i (a v_i + b w_i) = a \sum_i u_i v_i + b \sum_i u_i w_i$. (iv) The number $\mathbf{u} \cdot \mathbf{u} = u_1^2 + ... + u_n^2$ is a sum of nonnegative terms; it vanishes if and only if every $u_i^2 = 0$, that is, $\mathbf{u} = \mathbf{0}$. (v) By (ii), $\|c\mathbf{u}\|^2 = (c\mathbf{u}) \cdot (c\mathbf{u}) = c^2 (\mathbf{u} \cdot \mathbf{u})$, and $\sqrt{c^2} = |c|$. □

Thus the dot product satisfies the axioms (i)–(iv) of an inner product given in the preceding subsection, and $\mathbb{R}^n$ with the dot product is a Euclidean space.

Definition. A vector $\mathbf{u} \in \mathbb{R}^n$ is a *unit vector* if $\|\mathbf{u}\| = 1$.

Every column $\mathbf{e}_i$ of the identity matrix $\mathbf{I}_n$ is a unit vector, and so is $\frac{1}{2}(1, 1, 1, 1)^T \in \mathbb{R}^4$. If $\mathbf{u} \neq \mathbf{0}$, then $\mathbf{u}/\|\mathbf{u}\|$ (that is, $(1/\|\mathbf{u}\|)\mathbf{u}$) is a unit vector pointing in the same direction as $\mathbf{u}$, since by property (v) its norm is $\|\mathbf{u}\|/\|\mathbf{u}\| = 1$. Passing from $\mathbf{u}$ to $\mathbf{u}/\|\mathbf{u}\|$ is called *normalizing* $\mathbf{u}$. For example, $\mathbf{u} = (2, -1, 2)^T$ has $\|\mathbf{u}\| = \sqrt{4 + 1 + 4} = 3$, and normalizing it gives the unit vector $\left(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\right)^T$.

By Pythagoras' theorem, the distance between the points (x_1, y_1) and (x_2, y_2) of the plane is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$; in space a third term

$(z_1 - z_2)^2$ appears under the root. In both cases the distance is the norm of the difference of the two vectors.

Definition. The *distance* between vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.

For example, if $\mathbf{u} = (1, 0, 2, -1)^T$ and $\mathbf{v} = (0, 2, 2, 1)^T$, then $\mathbf{u} - \mathbf{v} = (1, -2, 0, -2)^T$ and $d(\mathbf{u}, \mathbf{v}) = \sqrt{1 + 4 + 0 + 4} = 3$.

To define angles we need the Cauchy–Bunyakovsky–Schwarz inequality, which in $\mathbb{R}^n$ has a two-line proof. Let $\mathbf{v} \neq \mathbf{0}$. For every real t, by the theorem above,

$$0 \leq \|\mathbf{u} + t\mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2t(\mathbf{u} \cdot \mathbf{v}) + t^2\|\mathbf{v}\|^2,$$

so the discriminant of this quadratic polynomial in t is nonpositive: $(\mathbf{u} \cdot \mathbf{v})^2 \leq \|\mathbf{u}\|^2\|\mathbf{v}\|^2$. Hence

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\| \quad \text{for all } \mathbf{u}, \mathbf{v} \in \mathbb{R}^n$$

(for $\mathbf{v} = \mathbf{0}$ both sides vanish).

Definition. The *angle* between nonzero vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is the number θ, $0 \leq \theta \leq \pi$, such that

$$\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}.$$

By the inequality just proved, the right-hand side lies in $[-1, 1]$, so θ exists and is unique. For $n = 2, 3$ it is the usual angle between the arrows $\mathbf{u}$ and $\mathbf{v}$: by formula (6) (and Remark 2 for the plane), the geometric dot product $\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$ of the first subsection equals $\mathbf{u} \cdot \mathbf{v}$ in Cartesian coordinates. For example, for $(1, 1, 0)^T$ and $(1, 0, 1)^T$ we get $\cos\theta = 1/\left(\sqrt{2}\sqrt{2}\right) = \frac{1}{2}$, so $\theta = \pi/3$; and since $(1, 2, -1)^T \cdot (3, -1, 1)^T = 0$, the angle between these two vectors is $\pi/2$: they are orthogonal (see Section 5.1).

3.3. Linear Combinations and Linear Spans

The definitions of a linear combination and of the linear span $\operatorname{span}\{\mathbf{u}_1, ..., \mathbf{u}_k\}$*, and the fact that* $\mathbf{Ax} = \mathbf{b}$ *is consistent if and only if* $\mathbf{b}$ *lies in the span of the columns of* $\mathbf{A}$*, is treated in Section 2.3.*

Checking Linear Combinations and Spanning Sets ADDED FOR MA1522

Throughout, $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ is a set of vectors in $\mathbb{R}^n$, $\operatorname{span}(S) = \operatorname{span}\{\mathbf{u}_1, ..., \mathbf{u}_k\}$, and $\mathbf{A} = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \end{pmatrix}$ is the $n \times k$ matrix with these columns. Since $\mathbf{Ac} = c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k$ for $\mathbf{c} = (c_1, ..., c_k)^T$ (Section 2.3), the vector equation $c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k = \mathbf{v}$ is the same as the linear system $\mathbf{Ac} = \mathbf{v}$.

Algorithm (membership in $\operatorname{span}(S)$**).**

1. Form the augmented matrix $(\mathbf{A} \mid \mathbf{v}) = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_k \mid \mathbf{v})$ and reduce it to row-echelon (or reduced row-echelon) form.
2. If the last column is a pivot column, the system is inconsistent and $\mathbf{v} \notin \operatorname{span}(S)$.
3. Otherwise $\mathbf{v} \in \operatorname{span}(S)$, and the solutions $(c_1, \ldots, c_k)^T$ of $\mathbf{A}\mathbf{c} = \mathbf{v}$ are exactly the coefficients with $\mathbf{v} = c_1\mathbf{u}_1 + \ldots + c_k\mathbf{u}_k$; they are unique if and only if the system has a unique solution.

The algorithm combines the fact just recalled with the consistency criterion of Section 1.4.

Example. Let $\mathbf{u}_1 = (1, 2, 0)^T$, $\mathbf{u}_2 = (0, 1, 1)^T$, $\mathbf{u}_3 = (1, 3, 1)^T$. For $\mathbf{v} = (2, 3, -1)^T$,

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 2 & 1 & 3 & 3 \\ 0 & 1 & 1 & -1 \end{array}\right) \xrightarrow{\text{RREF}} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right),$$

so $\mathbf{v} \in \operatorname{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$. The general solution $c_1 = 2 - s$, $c_2 = -1 - s$, $c_3 = s$ $(s \in \mathbb{R})$ gives $\mathbf{v} = (2 - s)\mathbf{u}_1 - (1 + s)\mathbf{u}_2 + s\mathbf{u}_3$ for every s (for instance, $\mathbf{v} = 2\mathbf{u}_1 - \mathbf{u}_2$): here the coefficients are not unique. For an arbitrary vector $(x, y, z)^T$ the operations $R_2 - 2R_1$ and $R_3 - R_2$ give

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & x \\ 2 & 1 & 3 & y \\ 0 & 1 & 1 & z \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 0 & 1 & x \\ 0 & 1 & 1 & y - 2x \\ 0 & 0 & 0 & 2x - y + z \end{array}\right),$$

so $(x, y, z)^T \in \operatorname{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ if and only if $2x - y + z = 0$. The span is a plane through the origin, and, for example, $(1, 0, 0)^T$ does not lie in it.

Instead of a single vector $\mathbf{v}$ one may ask whether *every* vector of $\mathbb{R}^n$ lies in $\operatorname{span}(S)$, that is, whether $\operatorname{span}(S) = \mathbb{R}^n$.

Theorem. $\operatorname{span}(S) = \mathbb{R}^n$ if and only if the reduced row-echelon form of $\mathbf{A} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_k)$ has no zero rows.

Proof. We have $\operatorname{span}(S) = \mathbb{R}^n$ if and only if the system $\mathbf{A}\mathbf{x} = \mathbf{v}$ is consistent for every $\mathbf{v} \in \mathbb{R}^n$. Let $\mathbf{R}$ be the reduced row-echelon form of $\mathbf{A}$. Then $\mathbf{R} = \mathbf{E}\mathbf{A}$, where $\mathbf{E}$ is an invertible matrix, a product of elementary matrices (Section 1.3), and the same row operations take $(\mathbf{A} \mid \mathbf{v})$ to $(\mathbf{R} \mid \mathbf{E}\mathbf{v})$. If $\mathbf{R}$ has no zero rows, every row of $(\mathbf{R} \mid \mathbf{E}\mathbf{v})$ has its leading entry to the left of the bar, so the last column is not a pivot column and the system is consistent, whatever $\mathbf{v}$ is. If $\mathbf{R}$ has a zero row, then its last row is zero; for $\mathbf{v} = \mathbf{E}^{-1}\mathbf{e}_n$ the matrix $(\mathbf{A} \mid \mathbf{v})$ reduces to $(\mathbf{R} \mid \mathbf{e}_n)$, whose last row is $(0, \ldots, 0 \mid 1)$. This system is inconsistent, and $\mathbf{v} \notin \operatorname{span}(S)$. $\square$

Since every row-echelon form of $\mathbf{A}$ has exactly $\operatorname{rank}(\mathbf{A})$ nonzero rows (by the proposition of Section 1.2, as row operations do not change the rank,

Section 4.2), the condition can also be checked on any row-echelon form, and it means that $\text{rank}(\mathbf{A}) = n$.

Corollary. If $k < n$, then $\text{span}\{\mathbf{u}_1, ..., \mathbf{u}_k\} \neq \mathbb{R}^n$: fewer than n vectors never span $\mathbb{R}^n$.

Proof. Each nonzero row of the reduced row-echelon form of the $n \times k$ matrix $\mathbf{A}$ has its leading entry in a different column, so there are at most $k < n$ nonzero rows, and there is a zero row. □

Example (the standard basis)**.** The identity matrix $\mathbf{I}_n = (\mathbf{e}_1 \ \cdots \ \mathbf{e}_n)$ is in reduced row-echelon form and has no zero rows, so its columns span $\mathbb{R}^n$; explicitly, every $\mathbf{w} = (w_1, ..., w_n)^T$ equals $w_1\mathbf{e}_1 + w_2\mathbf{e}_2 + ... + w_n\mathbf{e}_n$. The set $E = \{\mathbf{e}_1, ..., \mathbf{e}_n\}$ is called the *standard basis* of $\mathbb{R}^n$; it is linearly independent (Section 3.5) and hence a basis in the sense of Section 3.6.

Example. For $S = \{(1,1,0)^T, (0,1,1)^T, (1,0,1)^T, (1,1,1)^T\}$,

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 0 & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} \end{pmatrix}$$

has no zero rows, so $\text{span}(S) = \mathbb{R}^3$. In contrast, the reduced row-echelon form of $(\mathbf{u}_1 \ \ \mathbf{u}_2 \ \ \mathbf{u}_3)$ in the previous example has a zero row, in accordance with the fact that these three vectors span only a plane.

Properties of Linear Spans ADDED FOR MA1522

Theorem. Let $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\} \subseteq \mathbb{R}^n$. Then:
(i) $\mathbf{0} \in \text{span}(S)$;
(ii) $\text{span}(S)$ is closed under scalar multiplication: if $\mathbf{v} \in \text{span}(S)$ and $\alpha \in \mathbb{R}$, then $\alpha\mathbf{v} \in \text{span}(S)$;
(iii) $\text{span}(S)$ is closed under addition: if $\mathbf{u}, \mathbf{v} \in \text{span}(S)$, then $\mathbf{u} + \mathbf{v} \in \text{span}(S)$.

Proof. $\mathbf{0} = 0\mathbf{u}_1 + ... + 0\mathbf{u}_k$. If $\mathbf{u} = \sum_i c_i\mathbf{u}_i$ and $\mathbf{v} = \sum_i d_i\mathbf{u}_i$, then $\alpha\mathbf{v} = \sum_i (\alpha d_i)\mathbf{u}_i$ and $\mathbf{u} + \mathbf{v} = \sum_i (c_i + d_i)\mathbf{u}_i$. □

Properties (ii) and (iii) together are equivalent to a single property: (ii′) *if* $\mathbf{u}, \mathbf{v} \in \text{span}(S)$ *and* $\alpha, \beta \in \mathbb{R}$*, then* $\alpha\mathbf{u} + \beta\mathbf{v} \in \text{span}(S)$, that is, $\text{span}(S)$ is closed under linear combinations. Applying (ii′) repeatedly (induction on m), we see that $c_1\mathbf{v}_1 + ... + c_m\mathbf{v}_m \in \text{span}(S)$ whenever $\mathbf{v}_1, ..., \mathbf{v}_m \in \text{span}(S)$. Since the coefficients c_i are arbitrary, we obtain

Corollary. If $\mathbf{v}_1, ..., \mathbf{v}_m \in \text{span}(S)$, then $\text{span}\{\mathbf{v}_1, ..., \mathbf{v}_m\} \subseteq \text{span}(S)$.

(This is the transitivity of linear dependence stated in Section 3.5.) Now let $T = \{\mathbf{v}_1, ..., \mathbf{v}_m\}$ be a second set of vectors in $\mathbb{R}^n$. By the corollary, $\text{span}(T) \subseteq$

$\operatorname{span}(S)$ as soon as each $\mathbf{v}_j \in \operatorname{span}(S)$, that is, as soon as each of the m systems $\left(\mathbf{u}_1 \ \cdots \ \mathbf{u}_k \,\middle|\, \mathbf{v}_j\right)$ is consistent. These systems have the same coefficient matrix, so they can be solved simultaneously.

Theorem. Let $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ and $T = \{\mathbf{v}_1, ..., \mathbf{v}_m\}$ be sets of vectors in $\mathbb{R}^n$. Then $\operatorname{span}(T) \subseteq \operatorname{span}(S)$ if and only if the augmented matrix

$$(S \mid T) = \left(\mathbf{u}_1 \ \cdots \ \mathbf{u}_k \,\middle|\, \mathbf{v}_1 \ \cdots \ \mathbf{v}_m\right)$$

is consistent, that is, each of the systems $\left(\mathbf{u}_1 \ \cdots \ \mathbf{u}_k\right)\mathbf{x} = \mathbf{v}_j$, $j = 1, ..., m$, is consistent. Equivalently, no row of the reduced row-echelon form of $(S \mid T)$ is zero to the left of the bar and nonzero to the right of it.

Proof. If $\operatorname{span}(T) \subseteq \operatorname{span}(S)$, then every $\mathbf{v}_j \in \operatorname{span}(S)$, so every system is consistent; the converse is the corollary above. The same row operations reduce $(S \mid T)$ and each $\left(\mathbf{u}_1 \ \cdots \ \mathbf{u}_k \,\middle|\, \mathbf{v}_j\right)$; the j-th system is inconsistent exactly when some row of the reduced matrix has zeros to the left of the bar and a nonzero entry in the column of $\mathbf{v}_j$. □

Consequently, $\operatorname{span}(S) = \operatorname{span}(T)$ if and only if both $(S \mid T)$ and $(T \mid S) = \left(\mathbf{v}_1 \ \cdots \ \mathbf{v}_m \,\middle|\, \mathbf{u}_1 \ \cdots \ \mathbf{u}_k\right)$ are consistent.

Example. Let $S = \{(1,2,0)^T, (0,1,1)^T\}$ and $T = \{(1,3,1)^T, (2,3,-1)^T\}$. Then

$$(S \mid T) = \left(\begin{array}{cc|cc} 1 & 0 & 1 & 2 \\ 2 & 1 & 3 & 3 \\ 0 & 1 & 1 & -1 \end{array}\right) \xrightarrow{\text{RREF}} \left(\begin{array}{cc|cc} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right),$$

$$(T \mid S) = \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 3 & 2 & 1 \\ 1 & -1 & 0 & 1 \end{array}\right) \xrightarrow{\text{RREF}} \left(\begin{array}{cc|cc} 1 & 0 & \frac{1}{3} & \frac{2}{3} \\ 0 & 1 & \frac{1}{3} & -\frac{1}{3} \\ 0 & 0 & 0 & 0 \end{array}\right).$$

Both are consistent, so $\operatorname{span}(S) = \operatorname{span}(T)$: it is the plane $2x - y + z = 0$ of the preceding subsection. The first reduction shows, moreover, that $(1,3,1)^T$ and $(2,3,-1)^T$ are, respectively, the sum and the combination $2\mathbf{u}_1 - \mathbf{u}_2$ of the vectors $\mathbf{u}_1 = (1,2,0)^T$ and $\mathbf{u}_2 = (0,1,1)^T$ of S.

3.4. Subspaces

Subspaces of a Vector Space

A nonempty set $W \subseteq V$ is called a *subspace* of the vector space V if it is itself a vector space with respect to the operations acting in V. Clearly, in order for W to be a subspace, it is necessary and sufficient that for any vectors $\mathbf{a}, \mathbf{b} \in W$ and any number α the inclusions $\mathbf{a} + \mathbf{b} \in W$ and $\alpha\mathbf{a} \in W$ hold.

If the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$ belong to a subspace W, then

$$\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\} \subseteq W.$$

Example. Consider the set V of all free vectors in the plane with a coordinate system centered at the point O. Since each free vector is generated by one and only one position vector, any subset of free vectors can be identified with a subset of position vectors $\overrightarrow{OA}$ or their endpoints—the points A.

The set V is obviously a vector space. Any line passing through the origin is a subspace in V. At the same time, if l is a line not passing through the origin, then it is not a subspace: let $A, B \in l$ and $\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{OB}$; clearly, $C \notin l$.

Problem 1. Prove that the vector space $\mathbb{R}^n$ cannot be represented as the union of a finite number of sets, each of which does not coincide with $\mathbb{R}^n$ and is a subspace of it.

Subspaces of $\mathbb{R}^n$ ADDED FOR MA1522

Definition. A subset $V \subseteq \mathbb{R}^n$ is a *subspace* of $\mathbb{R}^n$ if
(i) V contains the zero vector: $\mathbf{0} \in V$;
(ii) V is closed under scalar multiplication: if $\mathbf{v} \in V$ and $\alpha \in \mathbb{R}$, then $\alpha\mathbf{v} \in V$;
(iii) V is closed under addition: if $\mathbf{u}, \mathbf{v} \in V$, then $\mathbf{u} + \mathbf{v} \in V$.

This is the definition of the preceding subsection for $\mathbb{R}^n$: a set satisfying (i)–(iii) is nonempty, and a nonempty set closed under the two operations contains $\mathbf{0} = 0\mathbf{v}$ for any of its vectors $\mathbf{v}$. Accordingly, condition (i) may be replaced by (i′) *V is nonempty*, and conditions (ii) and (iii) together may be replaced by (ii′) *V is closed under linear combinations: $\alpha\mathbf{u} + \beta\mathbf{v} \in V$ for all $\mathbf{u}, \mathbf{v} \in V$ and $\alpha, \beta \in \mathbb{R}$*. A subset satisfying (i)–(iii) satisfies all the axioms of a vector space, because the operations do not lead out of it and the axioms hold in $\mathbb{R}^n$ (Section 3.1). The *zero space* $\{\mathbf{0}\}$ and $\mathbb{R}^n$ itself are subspaces of $\mathbb{R}^n$.

Theorem. A subset $V \subseteq \mathbb{R}^n$ is a subspace if and only if $V = \operatorname{span}(S)$ for some finite set $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ of vectors in $\mathbb{R}^n$.

Proof. If $V = \operatorname{span}(S)$, conditions (i)–(iii) are the properties of linear spans (Section 3.3). Conversely, let V be a subspace. If $V = \{\mathbf{0}\}$, take $S = \{\mathbf{0}\}$. Otherwise choose $\mathbf{u}_1 \in V$, $\mathbf{u}_1 \neq \mathbf{0}$, and, as long as $\operatorname{span}\{\mathbf{u}_1, ..., \mathbf{u}_i\} \neq V$, choose $\mathbf{u}_{i+1} \in V$ outside this span. All these spans lie in V, since V is closed under linear combinations. No $\mathbf{u}_j$ is a linear combination of the preceding vectors, so by Lemma 1 of Section 3.5 the vectors $\mathbf{u}_1, ..., \mathbf{u}_i$ are linearly independent; as they lie in $\mathbb{R}^n = \operatorname{span}\{\mathbf{e}_1, ..., \mathbf{e}_n\}$, Lemma 4 of Section 3.5 gives $i \leq n$. Hence the process stops after at most n steps, and then $\operatorname{span}\{\mathbf{u}_1, ..., \mathbf{u}_i\} = V$. □

Remark. To show that $V \subseteq \mathbb{R}^n$ is a subspace, either find a finite set S with $V = \operatorname{span}(S)$ or verify conditions (i)–(iii). To show that V is *not* a subspace, it suffices to exhibit one of the following: $\mathbf{0} \notin V$; a vector $\mathbf{v} \in V$ and a scalar α with $\alpha\mathbf{v} \notin V$; vectors $\mathbf{u}, \mathbf{v} \in V$ with $\mathbf{u} + \mathbf{v} \notin V$.

Example. (a) $V = \{(x+2y, y, x-y)^T \mid x, y \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^3$, since $(x+2y, y, x-y)^T = x(1,0,1)^T + y(2,1,-1)^T$, so that $V = \text{span}\{(1,0,1)^T, (2,1,-1)^T\}$.

(b) The plane $x + 2y - z = 3$ in $\mathbb{R}^3$ is not a subspace: it does not contain $\mathbf{0}$.

(c) $V = \{(t, |t|)^T \mid t \in \mathbb{R}\} \subset \mathbb{R}^2$ is not a subspace: $(1,1)^T \in V$, but $(-1)(1,1)^T = (-1,-1)^T \notin V$.

(d) $V = \{(x,y)^T \mid xy = 0\}$, the union of the two coordinate axes of $\mathbb{R}^2$, contains $\mathbf{0}$ and is closed under scalar multiplication, but it is not closed under addition: $(1,0)^T + (0,1)^T = (1,1)^T \notin V$. So conditions (i) and (ii) do not imply (iii).

Proposition. The subspaces of $\mathbb{R}^2$ are: the zero space $\{\mathbf{0}\}$ (a point); the lines through the origin $\text{span}\{\mathbf{u}\}$, $\mathbf{u} \neq \mathbf{0}$; and $\mathbb{R}^2$ itself. The subspaces of $\mathbb{R}^3$ are: $\{\mathbf{0}\}$; the lines through the origin $\text{span}\{\mathbf{u}\}$, $\mathbf{u} \neq \mathbf{0}$; the planes through the origin $\text{span}\{\mathbf{u}, \mathbf{v}\}$, where neither of $\mathbf{u}, \mathbf{v}$ is a scalar multiple of the other; and $\mathbb{R}^3$ itself.

Proof. Each listed set is a span and hence a subspace. Conversely, let V be a subspace of $\mathbb{R}^3$ and $d = \dim(V)$; by Section 3.7, $d \leq \dim(\mathbb{R}^3) = 3$, and $d = 3$ only for $V = \mathbb{R}^3$. If $d = 0$, then $V = \{\mathbf{0}\}$. If $d = 1$ or $d = 2$, then V is the span of a basis consisting of one nonzero vector, or of two linearly independent vectors, that is, two vectors neither of which is a multiple of the other (Section 3.5). The case of $\mathbb{R}^2$ is the same. □

Lines and planes that do not pass through the origin are not subspaces (they do not contain $\mathbf{0}$). Note also that, strictly speaking, $\mathbb{R}^2$ is not a subset of $\mathbb{R}^3$, since its elements have two entries, not three; but the plane $z = 0$ in $\mathbb{R}^3$ is a subspace that is identified with $\mathbb{R}^2$ by taking coordinates relative to its basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ (Section 3.6).

Solution Sets of Linear Systems ADDED FOR MA1522

The solution set of a linear system $\mathbf{Ax} = \mathbf{b}$ in n variables is a subset of $\mathbb{R}^n$ (empty if the system is inconsistent). It can be described *implicitly*, by the equations,

$$\{\mathbf{u} \in \mathbb{R}^n \mid \mathbf{Au} = \mathbf{b}\},$$

or *explicitly*, by the general solution (Section 4.2),

$$\{\mathbf{u} + s_1\mathbf{v}_1 + \ldots + s_k\mathbf{v}_k \mid s_1, \ldots, s_k \in \mathbb{R}\}.$$

For example, solving $x_1 - x_2 + 2x_3 = 1$, $x_2 - x_3 = 2$ with the parameter $x_3 = t$ gives $x_2 = 2 + t$ and $x_1 = 3 - t$, so the solution set is the line $\{(3,2,0)^T + t(-1,1,1)^T \mid t \in \mathbb{R}\}$. In the opposite direction, the set $W = \{(1,0,2)^T + s(1,1,0)^T + t(0,1,1)^T \mid s, t \in \mathbb{R}\}$ consists of the vectors with $x = 1 + s$, $y = s + t$, $z = 2 + t$; eliminating $s = x - 1$ and $t = z - 2$, we obtain the implicit description $W = \{(x,y,z)^T \mid x - y + z = 3\}$.

Theorem. The solution set $V = \{\mathbf{u} \in \mathbb{R}^n \mid \mathbf{Au} = \mathbf{b}\}$ of a linear system $\mathbf{Ax} = \mathbf{b}$ is a subspace of $\mathbb{R}^n$ if and only if $\mathbf{b} = \mathbf{0}$, that is, the system is homogeneous.

Proof. If V is a subspace, then $\mathbf{0} \in V$, so $\mathbf{b} = \mathbf{A0} = \mathbf{0}$. Conversely, let $\mathbf{b} = \mathbf{0}$. Then $\mathbf{0} \in V$; and if $\mathbf{Au} = \mathbf{Av} = \mathbf{0}$ and $\alpha \in \mathbb{R}$, then $\mathbf{A}(\alpha\mathbf{v}) = \alpha\mathbf{Av} = \mathbf{0}$ and $\mathbf{A}(\mathbf{u} + \mathbf{v}) = \mathbf{Au} + \mathbf{Av} = \mathbf{0}$. □

Definition. The solution set of a homogeneous linear system $\mathbf{Ax} = \mathbf{0}$ is called its *solution space*. It is the nullspace $\text{Null}(\mathbf{A})$ of the matrix $\mathbf{A}$ (Section 4.1).

If $s_1\mathbf{v}_1 + ... + s_k\mathbf{v}_k$ $(s_1, ..., s_k \in \mathbb{R})$ is the general solution of $\mathbf{Ax} = \mathbf{0}$, the solution space is $\text{span}\{\mathbf{v}_1, ..., \mathbf{v}_k\}$, which gives a second proof that it is a subspace.

The solution space of a homogeneous system as the span of the (linearly independent) vectors of its general solution is treated in Section 4.2.

For $\mathbf{b} \neq \mathbf{0}$ the solution set W of $\mathbf{Ax} = \mathbf{b}$ is never a subspace, since $\mathbf{0} \notin W$. If the system is consistent and $\mathbf{u}$ is a particular solution, then

$$W = \mathbf{u} + V = \{\mathbf{u} + \mathbf{v} \mid \mathbf{v} \in V\},$$

where $V = \text{Null}(\mathbf{A})$ is the solution space of the associated homogeneous system (the theorem on the general solution in Section 4.2). Thus W is the translate of the subspace V by the vector $\mathbf{u}$ (an affine set in the sense of the subsection "Affine Sets" below), "parallel" to V and passing through $\mathbf{u}$ instead of $\mathbf{0}$. In the example above, the solution set of $x_1 - x_2 + 2x_3 = 1$, $x_2 - x_3 = 2$ is the line through $(3, 2, 0)^T$ parallel to the line $V = \text{span}\{(-1, 1, 1)^T\}$, the solution space of $x_1 - x_2 + 2x_3 = 0$, $x_2 - x_3 = 0$.

Affine Sets (Linear Manifolds)

Let W be a subspace in a vector space V and $\mathbf{x}$ be some vector from V. A set of vectors of the form

$$M = \{\mathbf{v} = \mathbf{x} + \mathbf{w} \mid \mathbf{w} \in W\},$$

that is, the translate of the subspace W by the vector $\mathbf{x}$, is called an *affine set* (or *linear manifold*) in V. Notation: $M = \mathbf{x} + W$.

The subspace W is called the *direction subspace* for M and is uniquely determined by the set M.

Indeed, let

$$M = \mathbf{x}_1 + W_1 = \mathbf{x}_2 + W_2.$$

From this, $\mathbf{x}_1 - \mathbf{x}_2 \in W_1 \cap W_2$. Let $\mathbf{y} \in W_1$. Then $\mathbf{y} + (\mathbf{x}_1 - \mathbf{x}_2) \in W_2 \Rightarrow \mathbf{y} \in W_2$. Similarly, if $\mathbf{y} \in W_2$, then $\mathbf{y} + (\mathbf{x}_2 - \mathbf{x}_1) \in W_1 \Rightarrow \mathbf{y} \in W_1$. □

For $M = \mathbf{x} + W$, the vector $\mathbf{x}$ is called a *shift vector*. As the shift vector one can take any vector from M:

$$M = \mathbf{x} + W = \mathbf{y} + W \quad \forall \mathbf{y} \in M.$$

Indeed, let $\mathbf{y} = \mathbf{x} + \mathbf{w}_0$ for some $\mathbf{w}_0 \in W$. Then, if $\mathbf{z} = \mathbf{x} + \mathbf{w}$ for some $\mathbf{w} \in W$, we have $\mathbf{z} = \mathbf{y} + (\mathbf{x} - \mathbf{y}) + \mathbf{w} = \mathbf{y} + (-\mathbf{w}_0 + \mathbf{w})$. Hence, $\mathbf{x} + W \subseteq \mathbf{y} + W$. The reverse inclusion is proved similarly. □

If W is a finite-dimensional space, then its dimension is also called the *dimension of the affine set* $M = \mathbf{x} + W$.

Example 1. The set of solutions of a linear system $\mathbf{Ax} = \mathbf{b}$ with an $m \times n$ matrix $\mathbf{A}$ of rank r is an affine set $\mathbf{v} + W$, where $\mathbf{v}$ is a particular solution of this system and $W = \mathrm{Null}(\mathbf{A})$ is the solution space of the associated homogeneous system $\mathbf{Ax} = \mathbf{0}$.

Example 2. A line in the plane or in three-dimensional space is an affine set of dimension 1. A plane in three-dimensional space is an affine set of dimension 2.

When studying affine sets, the elements of a vector space are usually called *points*. By analogy with geometric space, one can think of vectors as position vectors laid off from a common initial point identified with the zero vector.

Problem 2. Prove that two affine sets $\mathbf{a}_1 + L_1$ and $\mathbf{a}_2 + L_2$ with direction subspaces L_1 and L_2 intersect if and only if $\mathbf{a}_1 - \mathbf{a}_2 \in L_1 + L_2$, where $L_1 + L_2 = \{\mathbf{l}_1 + \mathbf{l}_2 \mid \mathbf{l}_1 \in L_1, \mathbf{l}_2 \in L_2\}$ is the sum of the subspaces.

3.5. Linear Independence

Linear Dependence

Vectors all of whose entries are equal to zero are called *zero vectors*, and sometimes simply *zeros*. We shall denote any zero vector by the symbol $\mathbf{0}$.

A linear combination of vectors is called *nontrivial* if at least one of its coefficients is nonzero. A system (in other words, a nonempty ordered collection of a finite number) of vectors is called *linearly dependent* if there exists for them a nontrivial linear combination equal to the zero vector. (MA1522 speaks of a *set* $\{\mathbf{a}_1, ..., \mathbf{a}_k\}$; a system lists its vectors in a fixed order and may contain repeated vectors.)

Lemma 1. If $\mathbf{a}_1, ..., \mathbf{a}_k$ is a linearly dependent system of $k > 1$ nonzero vectors, then there exists in it a vector $\mathbf{a}_m$, $m > 1$, that is a linear combination of the preceding vectors:

$$\mathbf{a}_m \in \mathrm{span}\{\mathbf{a}_1, ..., \mathbf{a}_{m-1}\}.$$

Proof. Consider a nontrivial linear combination equal to zero:

$$\alpha_1 \mathbf{a}_1 + ... + \alpha_k \mathbf{a}_k = \mathbf{0},$$

and let m be the largest index such that $\alpha_m \neq 0$. If $m = 1$, then $\alpha_1 \mathbf{a}_1 = \mathbf{0}$ and, since $\alpha_1 \neq 0$, we obtain $\mathbf{a}_1 = \mathbf{0}$, which contradicts the hypothesis of the lemma. Consequently, $m > 1$. Then

$$\alpha_1 \mathbf{a}_1 + ... + \alpha_m \mathbf{a}_m = \mathbf{0} \quad \Rightarrow \quad \mathbf{a}_m = \left(-\frac{\alpha_1}{\alpha_m}\right)\mathbf{a}_1 + ... + \left(-\frac{\alpha_{m-1}}{\alpha_m}\right)\mathbf{a}_{m-1}. \ \square$$

Linear Independence

A system of vectors is called *linearly independent* if it is not linearly dependent. Thus, if the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$ are linearly independent, then

$$\alpha_1 \mathbf{a}_1 + ... + \alpha_k \mathbf{a}_k = \mathbf{0} \quad \Rightarrow \quad \alpha_1 = ... = \alpha_k = 0.$$

Lemma 2. Every subsystem of a linearly independent system is linearly independent.

Proof. Suppose that a subsystem is linearly dependent. This means that there exists a nontrivial linear combination of the vectors of this subsystem equal to zero. Then the linear combination of the vectors of the original system with the same coefficients for the vectors from the subsystem and zero coefficients for the other vectors is a nontrivial linear combination equal to zero. We obtain a contradiction with the linear independence of the original system. $\square$

Lemma 3. If a vector is a linear combination of linearly independent vectors, then the coefficients of this linear combination are uniquely determined.

Proof. Suppose the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$ are linearly independent and

$$\mathbf{b} = \alpha_1 \mathbf{a}_1 + ... + \alpha_k \mathbf{a}_k = \beta_1 \mathbf{a}_1 + ... + \beta_k \mathbf{a}_k.$$

Hence

$$(\alpha_1 - \beta_1)\mathbf{a}_1 + ... + (\alpha_k - \beta_k)\mathbf{a}_k = \mathbf{0} \quad \Rightarrow \quad \alpha_1 - \beta_1 = ... = \alpha_k - \beta_k = 0. \ \square$$

Problem 1. For each n, find all values of the parameter a for which the columns of the tridiagonal matrix

$$\mathbf{A} = \begin{pmatrix} a & 1 & & & \\ -1 & a & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & a & 1 \\ & & & -1 & a \end{pmatrix}$$

of order n are linearly independent.

Problem 2. An $(n+1) \times n$ matrix has entries $a_{ij} > 0$ for $i = j$ and $a_{ij} < 0$ for $i \neq j$. Prove that for $n = 3$ its columns are linearly independent. Is this true for $n = 4$?

Transitivity of Linear Dependence

An important (though obvious) property: if

$$\operatorname{span}\{\mathbf{c}_1, ..., \mathbf{c}_r\} \subseteq \operatorname{span}\{\mathbf{b}_1, ..., \mathbf{b}_m\}$$
$$\text{and} \quad \operatorname{span}\{\mathbf{b}_1, ..., \mathbf{b}_m\} \subseteq \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\},$$

then

$$\operatorname{span}\{\mathbf{c}_1, ..., \mathbf{c}_r\} \subseteq \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}.$$

Monotonicity of the Number of Linearly Independent Vectors

Lemma 4. Suppose that each of the systems of vectors $\mathbf{b}_1, ..., \mathbf{b}_m$ and $\mathbf{a}_1, ..., \mathbf{a}_k$ is linearly independent, and suppose that

$$\operatorname{span}\{\mathbf{b}_1, ..., \mathbf{b}_m\} \subseteq \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}. \qquad (*)$$

Then $m \leq k$.

Proof. According to $(*)$, the system

$$\mathbf{b}_1, \mathbf{a}_1, ..., \mathbf{a}_k$$

is linearly dependent. By Lemma 1, there exists a vector that is a linear combination of the preceding vectors; let this be the vector

$$\mathbf{a}_k \in \operatorname{span}\{\mathbf{b}_1, \mathbf{a}_1, ..., \mathbf{a}_{k-1}\}.$$

From this it follows that

$$\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} \subseteq \operatorname{span}\{\mathbf{b}_1, \mathbf{a}_1, ..., \mathbf{a}_{k-1}\}.$$

By the transitivity of linear dependence,

$$\operatorname{span}\{\mathbf{b}_1, ..., \mathbf{b}_m\} \subseteq \operatorname{span}\{\mathbf{b}_1, \mathbf{a}_1, ..., \mathbf{a}_{k-1}\},$$

and therefore the system

$$\mathbf{b}_2, \mathbf{b}_1, \mathbf{a}_1, ..., \mathbf{a}_{k-1}$$

is linearly dependent. By Lemma 1, in this system too there exists a vector that is linearly expressible in terms of the preceding ones, and this vector cannot be $\mathbf{b}_1$ (the vectors $\mathbf{b}_1, \mathbf{b}_2$ are linearly independent as a subsystem of a linearly independent system (Lemma 2)). Without loss of generality, we may assume that

$$\mathbf{a}_{k-1} \in \operatorname{span}\{\mathbf{b}_2, \mathbf{b}_1, \mathbf{a}_1, ..., \mathbf{a}_{k-2}\}.$$

Suppose that $m > k$. Then, continuing the preceding constructions, at the k-th step we obtain

$$\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} \subseteq \operatorname{span}\{\mathbf{b}_k, \mathbf{b}_{k-1}, ..., \mathbf{b}_1\}.$$

Consequently, $\mathbf{b}_{k+1} \in \operatorname{span}\{\mathbf{b}_k, \mathbf{b}_{k-1}, ..., \mathbf{b}_1\}$, which contradicts the assumption that the vectors $\mathbf{b}_1, ..., \mathbf{b}_m$ are linearly independent. The resulting contradiction proves that $m \leq k$. □

Note that the lemma remains valid even without the assumption that the system $\mathbf{a}_1, ..., \mathbf{a}_k$ is linearly independent—prove this!

Checking Linear Independence ADDED FOR MA1522

In MA1522 linear independence is a property of a set: $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ is *linearly independent* if the only coefficients with $c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + ... + c_k\mathbf{u}_k = \mathbf{0}$ are $c_1 = c_2 = ... = c_k = 0$, and *linearly dependent* otherwise. The idea behind the definition is redundancy: a vector of S that is a linear combination of the others can be dropped without changing the span (Section 3.3). Testing each vector separately would require k linear systems; a single homogeneous system tests all of them at once.

Proposition. A set $\{\mathbf{u}_1, ..., \mathbf{u}_k\}$ with $k \geq 2$ is linearly dependent if and only if one of its vectors is a linear combination of the others.

Proof. If $\mathbf{u}_i = \sum_{j \neq i} c_j\mathbf{u}_j$, then $\sum_{j \neq i} c_j\mathbf{u}_j - \mathbf{u}_i = \mathbf{0}$ is a linear combination equal to zero in which the coefficient of $\mathbf{u}_i$ is $-1 \neq 0$. Conversely, if $c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k = \mathbf{0}$ with, say, $c_i \neq 0$, then $\mathbf{u}_i = \sum_{j \neq i}(-c_j/c_i)\mathbf{u}_j$. □

Theorem. Let $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\} \subseteq \mathbb{R}^n$ and $\mathbf{A} = \begin{pmatrix}\mathbf{u}_1 & \cdots & \mathbf{u}_k\end{pmatrix}$. The following statements are equivalent:

(i) S is linearly independent;
(ii) the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution;
(iii) the reduced row-echelon form of $\mathbf{A}$ has no non-pivot columns.

Proof. Since $\mathbf{Ac} = c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k$, statements (i) and (ii) say the same thing. Row operations do not change the solution set, so $\mathbf{Ax} = \mathbf{0}$ is equivalent to $\mathbf{Rx} = \mathbf{0}$, where $\mathbf{R}$ is the reduced row-echelon form of $\mathbf{A}$. If $\mathbf{R}$ has a non-pivot column j, the variable x_j is free: setting $x_j = 1$ and the other free variables equal to 0, and solving for the variables of the pivot columns, we obtain a nontrivial solution. If all k columns of $\mathbf{R}$ are pivot columns, then $\mathbf{R}$ consists of the identity matrix $\mathbf{I}_k$ followed by $n - k$ zero rows, and $\mathbf{Rx} = \mathbf{0}$ forces $\mathbf{x} = \mathbf{0}$. □

Algorithm (linear independence).

1. Form the matrix $\mathbf{A} = \begin{pmatrix}\mathbf{u}_1 & \cdots & \mathbf{u}_k\end{pmatrix}$ and reduce it to row-echelon (or reduced row-echelon) form; both have the same pivot columns.
2. If every column is a pivot column, S is linearly independent. If some column is a non-pivot column, S is linearly dependent, and any nontrivial solution of $\mathbf{Ax} = \mathbf{0}$ gives a linear relation among the vectors of S.

Example. For $S = \{(1,2,1)^T, (2,1,0)^T, (0,3,2)^T\}$,

$$\begin{pmatrix}1 & 2 & 0\\ 2 & 1 & 3\\ 1 & 0 & 2\end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix}1 & 0 & 2\\ 0 & 1 & -1\\ 0 & 0 & 0\end{pmatrix}.$$

The third column is a non-pivot column, so S is linearly dependent; taking $x_3 = 1$ we get $x_1 = -2$, $x_2 = 1$, that is, $(0,3,2)^T = 2(1,2,1)^T - (2,1,0)^T$. On the other hand, the vectors $(1,0,1,1)^T$, $(0,1,1,0)^T$, $(1,1,0,1)^T$ are linearly independent: the reduced row-echelon form of the 4×3 matrix with these columns consists of $\mathbf{I}_3$ followed by a zero row. Likewise, $\mathbf{I}_n$ has only pivot columns, so the standard basis $\{\mathbf{e}_1, ..., \mathbf{e}_n\}$ is linearly independent; directly, $c_1\mathbf{e}_1 + ... + c_n\mathbf{e}_n = (c_1, ..., c_n)^T$ is zero only for $c_1 = ... = c_n = 0$.

Proposition (special cases).

(i) The empty set $\varnothing$ is linearly independent (vacuously: there is no linear combination to check).
(ii) The set $\{\mathbf{0}\}$ is linearly dependent, since $1 \cdot \mathbf{0} = \mathbf{0}$.
(iii) If $\mathbf{v} \neq \mathbf{0}$, then $\{\mathbf{v}\}$ is linearly independent.
(iv) A set $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly dependent if and only if one of the two vectors is a scalar multiple of the other.
(v) Every set of $k > n$ vectors in $\mathbb{R}^n$ is linearly dependent.

Proof. (iii) If $c\mathbf{v} = \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$, then $c = 0$ (property (x) of Section 3.1). (iv) is the case $k = 2$ of the proposition above. (v) The reduced row-echelon form of the $n \times k$ matrix $\mathbf{A}$ has at most n nonzero rows and hence at most $n < k$ pivot columns, so it has a non-pivot column. (Alternatively, apply Lemma 4 with $\mathbb{R}^n = \operatorname{span}\{\mathbf{e}_1, ..., \mathbf{e}_n\}$.) □

Adding and Removing Vectors ADDED FOR MA1522

Theorem. If $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ is linearly dependent, then $S \cup \{\mathbf{u}\}$ is linearly dependent for every vector $\mathbf{u}$. In particular, every set containing the zero vector is linearly dependent.

Proof. A nontrivial linear combination $c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k = \mathbf{0}$ remains nontrivial and equal to $\mathbf{0}$ when the term $0\mathbf{u}$ is added. The second assertion follows because $\{\mathbf{0}\}$ is linearly dependent. □

Theorem. If $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ is linearly independent and $\mathbf{u} \notin \operatorname{span}(S)$, then $S \cup \{\mathbf{u}\} = \{\mathbf{u}_1, ..., \mathbf{u}_k, \mathbf{u}\}$ is linearly independent.

Proof. Let $c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k + c\mathbf{u} = \mathbf{0}$. If $c \neq 0$, then $\mathbf{u} = -(c_1/c)\mathbf{u}_1 - ... - (c_k/c)\mathbf{u}_k \in \operatorname{span}(S)$, a contradiction. Hence $c = 0$, and then $c_1 = ... = c_k = 0$ because S is linearly independent. □

In terms of the algorithm: all columns of the reduced row-echelon form of $(\mathbf{u}_1 \ \cdots \ \mathbf{u}_k)$ are pivot columns, and $\mathbf{u} \notin \operatorname{span}(S)$ means that the last column of the reduced row-echelon form of $(\mathbf{u}_1 \ \cdots \ \mathbf{u}_k \,|\, \mathbf{u})$ is a pivot column too. In the other direction, removing vectors preserves linear independence: every subset of a linearly independent set is linearly independent (Lemma 2).

3.6. Basis and Coordinates

Basis and Dimension

A linearly independent set of vectors $\{\mathbf{b}_1, ..., \mathbf{b}_m\} \subseteq V = \text{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$ is called a *basis* for the linear span V if $\text{span}\{\mathbf{b}_1, ..., \mathbf{b}_m\} = V$.

Theorem (Basis Theorem)**.** Any bases for a linear span V contain one and the same number of vectors.

Proof. Let $\{\mathbf{b}_1, ..., \mathbf{b}_m\}$ and $\{\mathbf{c}_1, ..., \mathbf{c}_r\}$ be two bases for the given linear span. Clearly,

$$\text{span}\{\mathbf{b}_1, ..., \mathbf{b}_m\} = \text{span}\{\mathbf{c}_1, ..., \mathbf{c}_r\}.$$

Applying Lemma 4 (Section 3.5) twice, we obtain two inequalities: $m \leq r$ and $r \leq m$. Hence $m = r$. □

Definition. The number of vectors in the bases for a linear span V is called its *dimension* and is denoted by $\dim(V)$.

Theorem (Dimension of a Linear Span)**.**

$$\dim(\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}) \leq k.$$

Proof. It suffices to note that as a basis for the linear span of a given system of vectors one can choose a maximal linearly independent subsystem of it. □

As a basis for the linear span $\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}$, one can always choose some subsystem of the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$. A maximal linearly independent subsystem is called a *base* of the given system.

Proposition. For a subsystem of the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$ to be a basis for $V = \text{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}$, it is necessary and sufficient that it be a base.

Proof. Without loss of generality, we may assume that the subsystem is formed by the first k vectors $\mathbf{a}_1, ..., \mathbf{a}_k$. If this is a base, then each of the vectors $\mathbf{a}_{k+1}, ..., \mathbf{a}_n$ is linearly expressible in terms of $\mathbf{a}_1, ..., \mathbf{a}_k \Rightarrow V \subseteq \text{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\} \subseteq V \Rightarrow V = \text{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$. Thus, $\{\mathbf{a}_1, ..., \mathbf{a}_k\}$ is a basis for V.

If the chosen subsystem is a basis for V, then by the preceding argument and the Basis Theorem, no base can have a larger number of vectors. □

Problem 1. The vectors $\mathbf{a}_1, ..., \mathbf{a}_{k+1}$ are linearly independent. Prove that in the linear span $\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_{k+1}\}$ there exists a basis containing no vector from the linear span $\text{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$.

Bases of Subspaces of $\mathbb{R}^n$ ADDED FOR MA1522

Definition. Let V be a subspace of $\mathbb{R}^n$. A set $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\} \subseteq V$ is a *basis* for V if
(i) S spans V: $\text{span}(S) = V$; and
(ii) S is linearly independent.

Every subspace of $\mathbb{R}^n$ is a linear span (Section 3.4), so this is the definition given above for linear spans. In particular, all bases for V contain the same number of vectors (Basis Theorem), and every vector $\mathbf{v} \in V$ can be written as a linear combination of the vectors of a basis S in exactly one way: such a combination exists because S spans V, and its coefficients are unique by Lemma 3 of Section 3.5. Conversely, if every vector of V is a linear combination of the vectors of S in exactly one way, then S spans V and, by the uniqueness of the representation of $\mathbf{0}$, is linearly independent.

Example. The standard basis $E = \{\mathbf{e}_1, ..., \mathbf{e}_n\}$ is a basis for $\mathbb{R}^n$ (Sections 3.3 and 3.5). The plane $V = \{(x, y, z)^T \mid x - y + 2z = 0\}$ in $\mathbb{R}^3$ is the solution space of a single equation; with the parameters $y = s$, $z = t$ its general solution is $s(1, 1, 0)^T + t(-2, 0, 1)^T$. Hence $S = \{(1, 1, 0)^T, (-2, 0, 1)^T\}$ spans V, and it is linearly independent because neither vector is a multiple of the other: S is a basis for V. The set $T = \{(-1, 1, 1)^T, (3, 1, -1)^T\}$ is another basis for V: both vectors satisfy $x - y + 2z = 0$ and are not multiples of each other, so T is a basis by (B1) of Section 3.7. Thus a subspace $V \neq \{\mathbf{0}\}$ has many bases.

The fact that the vectors $\mathbf{v}_1, ..., \mathbf{v}_k$ of the general solution $s_1\mathbf{v}_1 + ... + s_k\mathbf{v}_k$ of a homogeneous system form a basis for its solution space is treated in Section 4.2.

Remark (basis for the zero space)**.** The zero space $\{\mathbf{0}\}$ has no basis made of vectors: the set $\{\mathbf{0}\}$ spans it but is linearly dependent, and a set containing a nonzero vector spans more than $\{\mathbf{0}\}$. MA1522 takes the empty set $\varnothing$ as the basis for $\{\mathbf{0}\}$. To make sense of $\text{span}(\varnothing)$, one regards the span of a set S as the *smallest subspace of $\mathbb{R}^n$ containing S*; for a nonempty finite set S this is the set of all linear combinations (it is a subspace containing S, and every subspace containing S contains all linear combinations of vectors of S). The smallest subspace containing $\varnothing$ is $\{\mathbf{0}\}$, so $\text{span}(\varnothing) = \{\mathbf{0}\}$; and $\varnothing$ is linearly independent vacuously. Accordingly, $\dim(\{\mathbf{0}\}) = 0$ (Section 3.7).

Bases of $\mathbb{R}^n$ and Invertibility ADDED FOR MA1522

In general, linear independence and spanning are unrelated properties. For n vectors in $\mathbb{R}^n$, however, they coincide.

Theorem. For a square matrix $\mathbf{A}$ of order n the following are equivalent: (a) $\mathbf{A}$ is invertible; (b) the columns of $\mathbf{A}$ are linearly independent; (c) the columns of $\mathbf{A}$ span $\mathbb{R}^n$.

Proof. (a) $\Leftrightarrow$ (b) is the invertibility criterion of Section 2.6. (a) $\Rightarrow$ (c): for every $\mathbf{b} \in \mathbb{R}^n$ the system $\mathbf{Ax} = \mathbf{b}$ has the solution $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$, so $\mathbf{b}$ is a linear combination of the columns. (c) $\Rightarrow$ (b): suppose the columns span $\mathbb{R}^n$ but are linearly dependent. Then one of them is a linear combination of the others (Section 3.5) and can be removed without changing the span (Section 3.3), leaving $n-1$ vectors that span $\mathbb{R}^n$; this is impossible, since fewer than n vectors never span $\mathbb{R}^n$ (Section 3.3). (For $n = 1$, a linearly dependent column is $\mathbf{0}$, which does not span $\mathbb{R}^1$.) □

In terms of the reduced row-echelon form $\mathbf{R}$ of $\mathbf{A}$: (b) says that $\mathbf{R}$ has no non-pivot columns and (c) that it has no zero rows (Sections 3.5 and 3.3); for a square matrix each of these means $\mathbf{R} = \mathbf{I}_n$.

Corollary 1. A set of exactly n vectors in $\mathbb{R}^n$ is linearly independent if and only if it spans $\mathbb{R}^n$.

Corollary 2. Let $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\} \subseteq \mathbb{R}^n$ and $\mathbf{A} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_k)$. Then S is a basis for $\mathbb{R}^n$ if and only if $k = n$ and $\mathbf{A}$ is invertible.

Proof. If $k < n$, then S does not span $\mathbb{R}^n$ (Section 3.3); if $k > n$, then S is linearly dependent (Section 3.5). Hence a basis for $\mathbb{R}^n$ has exactly n vectors, and for $k = n$ the theorem shows that S is a basis if and only if $\mathbf{A}$ is invertible. □

Since $\mathbf{A}$ is invertible if and only if $\mathbf{A}^T$ is (Section 2.4), and the rows of $\mathbf{A}$ are the columns of $\mathbf{A}^T$, the theorem holds for rows as well: *a square matrix $\mathbf{A}$ of order n is invertible if and only if its rows are linearly independent, if and only if its rows span $\mathbb{R}^n$, if and only if its rows form a basis for $\mathbb{R}^n$.* For example, $S = \{(1,1,0)^T, (0,1,1)^T, (1,0,1)^T\}$ is a basis for $\mathbb{R}^3$, since the matrix with these columns has determinant $2 \neq 0$.

Theorem (Equivalent statements for invertibility)**.** For a square matrix $\mathbf{A}$ of order n the following statements are equivalent:

(i) $\mathbf{A}$ is invertible;
(ii) $\mathbf{A}^T$ is invertible;
(iii) there is a matrix $\mathbf{B}$ with $\mathbf{BA} = \mathbf{I}$ (left inverse);
(iv) there is a matrix $\mathbf{B}$ with $\mathbf{AB} = \mathbf{I}$ (right inverse);
(v) the reduced row-echelon form of $\mathbf{A}$ is $\mathbf{I}_n$;
(vi) $\mathbf{A}$ is a product of elementary matrices;
(vii) the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution;
(viii) for every $\mathbf{b} \in \mathbb{R}^n$ the system $\mathbf{Ax} = \mathbf{b}$ has a unique solution;
(ix) $\det(\mathbf{A}) \neq 0$;
(x) the columns of $\mathbf{A}$ (equivalently, its rows) are linearly independent;
(xi) the columns of $\mathbf{A}$ (equivalently, its rows) span $\mathbb{R}^n$.

The equivalence of (i)–(ix) is proved in Chapter 2 (the equivalent statements for invertibility; see Sections 2.4–2.6 and 2.10); statements (x) and (xi) are added by the theorem above and the remark on rows following Corollary 2.

Basis and Dimension of an Abstract Vector Space

Let V be a finite-dimensional vector space. By definition, it is the linear span of a finite number of its vectors:

$$V = \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}.$$

The concepts of linearly dependent and linearly independent systems of vectors in the abstract case do not differ in any way from the same concepts in the case of column vectors. The same holds with regard to basis and dimension:

- V can be represented as the linear span of some linearly independent subsystem of the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$;
- a basis for the space V is defined as any linearly independent set of vectors for which V is the linear span; any two bases for V contain the same number of vectors; the number of vectors in a basis is called the dimension of the space V and is denoted by $\dim(V)$;
- any linearly independent set of vectors from V can be extended to a basis for V; moreover, this can be done using some of the vectors $\mathbf{a}_1, ..., \mathbf{a}_n$.

The proofs of these propositions repeat the proofs given in Sections 3.5–3.7 for the special case where vectors are understood to be column vectors.

Expansion in a Basis

Let V be a real vector space of dimension n and $T = \{\mathbf{f}_1, ..., \mathbf{f}_n\}$ be some basis for it. Then any vector $\mathbf{v} \in V$ has a unique expansion in this basis:

$$\mathbf{v} = x_1\mathbf{f}_1 + ... + x_n\mathbf{f}_n.$$

The coefficients $x_1, ..., x_n$ are called the coordinates of the vector $\mathbf{v}$ relative to the basis T. Clearly, between the elements of the vector space V and the set of columns $\mathbb{R}^n$ there is a one-to-one correspondence:

$$\mathbf{v} \leftrightarrow [\mathbf{v}]_T = \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Upon choosing another basis $S = \{\mathbf{g}_1, ..., \mathbf{g}_n\}$, there arises one more one-to-one correspondence between the same sets:

$$\mathbf{v} = y_1\mathbf{g}_1 + ... + y_n\mathbf{g}_n \leftrightarrow [\mathbf{v}]_S = \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}.$$

Consider the expansions

$$\begin{aligned} \mathbf{f}_1 &= p_{11}\mathbf{g}_1 + ... + p_{n1}\mathbf{g}_n, \\ &... \\ \mathbf{f}_n &= p_{1n}\mathbf{g}_1 + ... + p_{nn}\mathbf{g}_n \end{aligned} \qquad (*)$$

and introduce the $n \times n$ matrix $\mathbf{P} = (p_{ij})$; its jth column is the coordinate vector $[\mathbf{f}_j]_S$, that is,

$$\mathbf{P} = \begin{pmatrix} [\mathbf{f}_1]_S & \cdots & [\mathbf{f}_n]_S \end{pmatrix}.$$

Substituting $(*)$ into the expansion of the vector $\mathbf{v}$ in the basis T, we find

$$[\mathbf{v}]_S = \mathbf{P}[\mathbf{v}]_T, \quad \text{i.e.} \quad \mathbf{y} = \mathbf{P}\mathbf{x}. \qquad (**)$$

This relation makes it possible to pass from the coordinates of a vector relative to the basis T to the coordinates of the same vector relative to the basis S.

By virtue of $(**)$, the matrix $\mathbf{P}$ is called the *transition matrix from T to S*. If $\mathbf{f}_i$ and $\mathbf{g}_i$ are the columns of coordinates of the corresponding vectors in some third basis, then according to $(*)$ we have

$$(\mathbf{f}_1 \ \cdots \ \mathbf{f}_n) = (\mathbf{g}_1 \ \cdots \ \mathbf{g}_n)\mathbf{P},$$

whence follows the invertibility of the matrix $\mathbf{P}$ and the fact that $\mathbf{P}^{-1}$ is the transition matrix from S to T. Since this equality expresses the vectors $\mathbf{f}_j$ through the vectors $\mathbf{g}_i$, some books (the full textbook among them) use the opposite name and call $\mathbf{P}$ the transition matrix from $\{\mathbf{g}_i\}$ to $\{\mathbf{f}_i\}$, that is, from S to T. However, the name is not what matters—what is important is that the matrix $\mathbf{P}$ be used properly when recalculating coordinates!

Computing Coordinates Relative to a Basis ADDED FOR MA1522

Let V be a subspace of $\mathbb{R}^n$ and $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ a basis for V. As in the preceding subsection, every $\mathbf{v} \in V$ is written uniquely as $\mathbf{v} = c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k$, and the column

$$[\mathbf{v}]_S = \begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix} \in \mathbb{R}^k$$

is called the *coordinate vector* of $\mathbf{v}$ (the coordinates of $\mathbf{v}$) relative to S. Thus a basis identifies V with $\mathbb{R}^k$ through the one-to-one correspondence $\mathbf{v} \leftrightarrow [\mathbf{v}]_S$; note that $[\mathbf{v}]_S$ has k entries even though $\mathbf{v}$ itself has n. Three remarks:

- for the standard basis E of $\mathbb{R}^n$ we have $\mathbf{w} = w_1\mathbf{e}_1 + ... + w_n\mathbf{e}_n$, so $[\mathbf{w}]_E = \mathbf{w}$;
- the correspondence is one-to-one only because S is a basis: for a linearly dependent spanning set, different coefficient columns give the same vector (see the first example in Section 3.3);
- $[\mathbf{v}]_S$ depends on the order in which the vectors of S are listed. For $S = \{\mathbf{e}_1, \mathbf{e}_2\}$ and $T = \{\mathbf{e}_2, \mathbf{e}_1\}$ in $\mathbb{R}^2$ and $\mathbf{v} = (1, 2)^T$, we get $[\mathbf{v}]_S = (1, 2)^T$ but $[\mathbf{v}]_T = (2, 1)^T$. So a basis is always understood as an *ordered* set.

Algorithm (coordinates relative to a basis).

1. To find $[\mathbf{v}]_S$ for $\mathbf{v} \in V$, solve $c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k = \mathbf{v}$: reduce the augmented matrix $(\mathbf{u}_1 \ \cdots \ \mathbf{u}_k \mid \mathbf{v})$ to reduced row-echelon form.

2. Since the columns $\mathbf{u}_1, ..., \mathbf{u}_k$ are linearly independent and $\mathbf{v} \in \operatorname{span}(S)$, the result has the form

$$\left(\begin{array}{c|c} \mathbf{I}_k & \mathbf{c} \\ \mathbf{0} & \mathbf{0} \end{array}\right)$$

(Sections 3.3 and 3.5), and $[\mathbf{v}]_S = \mathbf{c}$. (If the system is inconsistent, then $\mathbf{v} \notin V$.)

Example. Take the basis $S = \{(1,1,0)^T, (-2,0,1)^T\}$ of the plane $V = \{(x,y,z)^T \mid x - y + 2z = 0\}$ (see the example in the subsection on bases of subspaces above) and $\mathbf{v} = (1,3,1)^T \in V$. Then

$$\left(\begin{array}{cc|c} 1 & -2 & 1 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{array}\right) \xrightarrow{\text{RREF}} \left(\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{array}\right), \quad \text{so} \quad [\mathbf{v}]_S = \begin{pmatrix} 3 \\ 1 \end{pmatrix};$$

indeed, $3(1,1,0)^T + (-2,0,1)^T = (1,3,1)^T$.

Theorem. Let B be a basis for a subspace V of $\mathbb{R}^n$. Then:
(i) for $\mathbf{u}, \mathbf{v} \in V$, $\mathbf{u} = \mathbf{v}$ if and only if $[\mathbf{u}]_B = [\mathbf{v}]_B$;
(ii) for $\mathbf{v}_1, ..., \mathbf{v}_m \in V$ and $c_1, ..., c_m \in \mathbb{R}$,

$$[c_1\mathbf{v}_1 + ... + c_m\mathbf{v}_m]_B = c_1[\mathbf{v}_1]_B + ... + c_m[\mathbf{v}_m]_B.$$

Proof. Let $B = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ and $\mathbf{U} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_k)$; by definition, $\mathbf{v} = \mathbf{U}[\mathbf{v}]_B$ for every $\mathbf{v} \in V$. (i) If $[\mathbf{u}]_B = [\mathbf{v}]_B$, then $\mathbf{u} = \mathbf{U}[\mathbf{u}]_B = \mathbf{U}[\mathbf{v}]_B = \mathbf{v}$; the converse holds by the uniqueness of coordinates. (ii) We have $\mathbf{U}\big(c_1[\mathbf{v}_1]_B + ... + c_m[\mathbf{v}_m]_B\big) = c_1\mathbf{v}_1 + ... + c_m\mathbf{v}_m$, so the column $c_1[\mathbf{v}_1]_B + ... + c_m[\mathbf{v}_m]_B$ consists of coefficients expressing $c_1\mathbf{v}_1 + ... + c_m\mathbf{v}_m$ through B; by uniqueness, it is the coordinate vector of this linear combination. □

3.7. Dimensions

Completion to a Basis

Lemma (Basis Extension Lemma)**.** Every linearly independent set of vectors $\{\mathbf{b}_1, ..., \mathbf{b}_m\} \subseteq \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$ is a subset of some basis for the given linear span.

Proof. It suffices to consider the case where the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$ are linearly independent. The system of vectors $\mathbf{b}_1, ..., \mathbf{b}_m, \mathbf{a}_1, ..., \mathbf{a}_k$ is linearly dependent. By Lemma 1 (Section 3.5), there exists in it a vector that is linearly expressible in terms of the preceding ones. Let us remove this vector and consider the remaining subsystem. If it is linearly independent, then it is a basis for the linear span $\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_k\}$. If not, there is a vector in it that is linearly

expressible in terms of the preceding ones. We remove it from the system as well, consider the remaining subsystem, and so on. Ultimately, the system of vectors $\mathbf{b}_1, ..., \mathbf{b}_m$ will be completed to a basis by some of the vectors $\mathbf{a}_1, ..., \mathbf{a}_k$. □

Existence of a Basis

Does a basis exist for every linear span? By the Basis Extension Lemma, a basis exists if the linear span contains a linearly independent subsystem of vectors. This will be the case if there exists at least one nonzero vector.

Thus, there is no basis only in the case of the zero space $\{\mathbf{0}\}$, the linear span which contains a single vector—the zero vector. By definition, the dimension of the zero space is equal to zero. (MA1522 regards the empty set $\varnothing$ as the basis for the zero space; this agrees with $\dim(\{\mathbf{0}\}) = 0$.)

Dimension of Subspaces of $\mathbb{R}^n$ ADDED FOR MA1522

A basis B of k vectors identifies a subspace V of $\mathbb{R}^n$ with $\mathbb{R}^k$ through coordinates (Section 3.6). This identification respects linear independence and spanning.

> **Theorem.** Let B be a basis for a subspace V of $\mathbb{R}^n$ consisting of k vectors, and let $\mathbf{v}_1, ..., \mathbf{v}_m \in V$. Then:
> (i) $\mathbf{v}_1, ..., \mathbf{v}_m$ are linearly independent (dependent) if and only if $[\mathbf{v}_1]_B, ..., [\mathbf{v}_m]_B$ are linearly independent (dependent) in $\mathbb{R}^k$;
> (ii) $\{\mathbf{v}_1, ..., \mathbf{v}_m\}$ spans V if and only if $\left\{[\mathbf{v}_1]_B, ..., [\mathbf{v}_m]_B\right\}$ spans $\mathbb{R}^k$.

Proof. (i) By the properties of coordinates (Section 3.6), $c_1\mathbf{v}_1 + ... + c_m\mathbf{v}_m = \mathbf{0}$ if and only if $c_1[\mathbf{v}_1]_B + ... + c_m[\mathbf{v}_m]_B = [\mathbf{0}]_B = \mathbf{0}$: the two collections satisfy the same linear relations.

(ii) Let $B = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$. Suppose the coordinate vectors span $\mathbb{R}^k$, and let $\mathbf{v} \in V$. Then $[\mathbf{v}]_B = c_1[\mathbf{v}_1]_B + ... + c_m[\mathbf{v}_m]_B = [c_1\mathbf{v}_1 + ... + c_m\mathbf{v}_m]_B$ for some $c_1, ..., c_m$, whence $\mathbf{v} = c_1\mathbf{v}_1 + ... + c_m\mathbf{v}_m$. Conversely, suppose $\mathbf{v}_1, ..., \mathbf{v}_m$ span V, and let $\mathbf{w} = (w_1, ..., w_k)^T \in \mathbb{R}^k$. The vector $\mathbf{v} = w_1\mathbf{u}_1 + ... + w_k\mathbf{u}_k \in V$ has $[\mathbf{v}]_B = \mathbf{w}$; writing $\mathbf{v} = c_1\mathbf{v}_1 + ... + c_m\mathbf{v}_m$, we get $\mathbf{w} = c_1[\mathbf{v}_1]_B + ... + c_m[\mathbf{v}_m]_B$. □

> **Corollary.** Let V be a subspace of $\mathbb{R}^n$ with a basis of k vectors, and let $S = \{\mathbf{v}_1, ..., \mathbf{v}_m\} \subseteq V$. If $m > k$, then S is linearly dependent; if $m < k$, then S does not span V.

Proof. More than k vectors in $\mathbb{R}^k$ are linearly dependent (Section 3.5), and fewer than k vectors do not span $\mathbb{R}^k$ (Section 3.3); now apply the theorem to the coordinate vectors $[\mathbf{v}_1]_B, ..., [\mathbf{v}_m]_B$. □

In particular, any two bases for V have the same number of vectors (this is the Basis Theorem of Section 3.6), and the *dimension* $\dim(V)$ is the number of vectors in any basis for V: V is k-dimensional exactly when coordinates

relative to a basis identify V with $\mathbb{R}^k$. For example, $\dim(\mathbb{R}^n) = n$, since the standard basis has n vectors; $\dim(\{\mathbf{0}\}) = 0$, its basis being $\varnothing$; and the plane $x - y + 2z = 0$ of Section 3.6 has dimension 2.

Theorem (dimension of the solution space)**.** Let $\mathbf{A}$ be an $m \times n$ matrix. The dimension of the solution space $V = \{\mathbf{u} \in \mathbb{R}^n \mid \mathbf{Au} = \mathbf{0}\}$ equals the number of non-pivot columns of the reduced row-echelon form of $\mathbf{A}$, that is, the number of parameters in the general solution of $\mathbf{Ax} = \mathbf{0}$.

Proof. Let $\mathbf{R}$ be the reduced row-echelon form of $\mathbf{A}$. Since $\mathbf{R} = \mathbf{EA}$ with $\mathbf{E}$ invertible, $\operatorname{rank}(\mathbf{R}) = \operatorname{rank}(\mathbf{A})$ (Section 4.2), and $\operatorname{rank}(\mathbf{R})$ is the number of nonzero rows of $\mathbf{R}$ (Section 1.2), which is the number of its pivot columns. So the number of non-pivot columns is $n - \operatorname{rank}(\mathbf{A}) = \operatorname{nullity}(\mathbf{A}) = \dim(V)$ by the Rank–Nullity Theorem (Section 4.2). In the general solution one parameter is assigned to the variable of each non-pivot column, and the vectors multiplying the parameters form a basis for V (Section 4.2). □

For instance, a *hyperplane* in $\mathbb{R}^n$, the solution space $V = \{\mathbf{x} \in \mathbb{R}^n \mid a_1x_1 + a_2x_2 + \ldots + a_nx_n = 0\}$ of one equation with not all $a_i = 0$, has dimension $n - 1$: the reduced row-echelon form of the nonzero $1 \times n$ coefficient matrix has one pivot column and $n - 1$ non-pivot columns. Hyperplanes in $\mathbb{R}^3$ are the planes through the origin.

Example. For

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & -1 & 3 \\ 2 & 4 & 1 & 0 & 1 \\ 1 & 2 & 1 & 1 & -2 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

the pivot columns are the first and the third, and there are three non-pivot columns; hence $\dim(\operatorname{Null}(\mathbf{A})) = 3$. With the parameters $x_2 = s$, $x_4 = t$, $x_5 = r$, the general solution of $\mathbf{Ax} = \mathbf{0}$ is

$$\mathbf{x} = s(-2, 1, 0, 0, 0)^T + t(1, 0, -2, 1, 0)^T + r(-3, 0, 5, 0, 1)^T, \quad s, t, r \in \mathbb{R},$$

and these three vectors form a basis for the solution space.

The following two theorems restate results proved above for linear spans in the form used in MA1522.

Theorem (spanning set theorem)**.** Let S be a finite set of vectors in $\mathbb{R}^n$ such that $V = \operatorname{span}(S) \neq \{\mathbf{0}\}$. Then some subset of S is a basis for V.

Proof. If S is linearly dependent, one of its vectors is a linear combination of the others (Section 3.5); removing it does not change the span (Section 3.3). Repeating this, we arrive at a linearly independent subset of S with the same span V, that is, at a basis for V; it is not empty, since $V \neq \{\mathbf{0}\}$. (Compare the proposition on bases and maximal linearly independent subsystems in Section 3.6.) □

Theorem (linear independence theorem)**.** Let V be a subspace of $\mathbb{R}^n$ and $S \subseteq V$ a linearly independent set. Then there is a basis T for V such that $S \subseteq T$.

Proof. As long as $\text{span}(S) \neq V$, add to S a vector of V that is not in $\text{span}(S)$; the enlarged set is still linearly independent (Section 3.5, "Adding and Removing Vectors") and lies in V. The subspace V has a basis (it is a span by Section 3.4; use the spanning set theorem, or $\varnothing$ if $V = \{\mathbf{0}\}$), so by the corollary above a linearly independent subset of V has at most $\dim(V)$ vectors. Hence the process stops, and the final set T spans V. (This is also the Basis Extension Lemma of the first subsection, applied to the linear span V.) □

Theorem (dimension and subspaces)**.** Let U and V be subspaces of $\mathbb{R}^n$ with $U \subseteq V$. Then $\dim(U) \leq \dim(V)$, and $\dim(U) = \dim(V)$ if and only if $U = V$. In particular, $\dim(U) < \dim(V)$ if U is a proper subset of V.

Proof. A basis S for U is a linearly independent subset of V, so by the linear independence theorem $S \subseteq T$ for some basis T for V. Hence $\dim(U) = |S| \leq |T| = \dim(V)$. If $\dim(U) = \dim(V)$, then $T = S$ and $V = \text{span}(T) = \text{span}(S) = U$. □

To check that a set S is a basis for a subspace V one must, by definition, verify that $\text{span}(S) = V$ and that S is linearly independent. When $\dim(V)$ is known and S has exactly $\dim(V)$ vectors, one of the two conditions suffices.

Theorem (B1)**.** Let V be a subspace of $\mathbb{R}^n$ with $\dim(V) = k$, and let $S \subseteq V$ be a linearly independent set of k vectors. Then S is a basis for V.

Proof. The subspace $U = \text{span}(S)$ lies in V, and S is a basis for U, so $\dim(U) = k = \dim(V)$. By the dimension theorem above, $U = V$. □

Theorem (B2)**.** Let V be a subspace of $\mathbb{R}^n$ with $\dim(V) = k$, and let S be a set of k vectors in $\mathbb{R}^n$ such that $V \subseteq \text{span}(S)$. Then S is a basis for V.

Proof. Let $U = \text{span}(S)$. By the theorem on the dimension of a linear span (Section 3.6), $\dim(U) \leq k$, and $V \subseteq U$ gives $k = \dim(V) \leq \dim(U)$. Hence $\dim(U) = \dim(V)$, and $V = U = \text{span}(S)$ by the dimension theorem above. If S were linearly dependent, one of its vectors could be removed without changing the span, and then $\dim(U) \leq k - 1$, a contradiction. So S is linearly independent. □

In summary, each of the following three sets of conditions means that S is a basis for V:

Definition	**(B1)**	**(B2)**
$\operatorname{span}(S) = V$; S is linearly independent	$S \subseteq V$; $\lvert S\rvert = \dim(V)$; S is linearly independent	$\lvert S\rvert = \dim(V)$; $V \subseteq \operatorname{span}(S)$

With (B1) one need not check that S spans V; with (B2) one need not check that S is linearly independent.

Example. (a) Let $V = \{\mathbf{x} \in \mathbb{R}^4 \mid x_1 + x_2 + x_3 = 0, x_2 - x_4 = 0\}$. The coefficient matrix reduces as

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & -1 \end{pmatrix},$$

with two non-pivot columns, so $\dim(V) = 2$. The vectors $(1, 0, -1, 0)^T$ and $(1, 1, -2, 1)^T$ satisfy both equations, and neither is a multiple of the other; by (B1) they form a basis for V, and there is no need to check that they span V.

(b) Let $V = \operatorname{span}(T)$, where $T = \{(1, 0, 0, 1)^T, (0, 1, 0, 1)^T, (0, 0, 1, 1)^T\}$; looking at the first three entries, we see that T is linearly independent, so $\dim(V) = 3$. For $S = \{(1, 1, 0, 2)^T, (0, 1, 1, 2)^T, (1, 0, 1, 2)^T\}$,

$$(S \mid T) = \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 2 & 2 & 2 & 1 & 1 & 1 \end{array}\right) \xrightarrow{\text{RREF}} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

is consistent, so $V = \operatorname{span}(T) \subseteq \operatorname{span}(S)$ (Section 3.3), and S is a basis for V by (B2); its linear independence need not be checked separately.

3.8. Transition Matrices

The definition of the transition matrix from T to S and the formula $[\mathbf{w}]_S = \mathbf{P}[\mathbf{w}]_T$ (subsection "Expansion in a Basis") is treated in Section 3.6.

Computing Transition Matrices ADDED FOR MA1522

Let $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ and $T = \{\mathbf{v}_1, ..., \mathbf{v}_k\}$ be two bases for a subspace V of $\mathbb{R}^n$. By Section 3.6, the *transition matrix from T to S* (sometimes written $\mathbf{P}_{T \to S}$) is the $k \times k$ matrix

$$\mathbf{P} = \begin{pmatrix} [\mathbf{v}_1]_S & [\mathbf{v}_2]_S & \cdots & [\mathbf{v}_k]_S \end{pmatrix}$$

whose columns are the coordinate vectors of the vectors of T relative to S, and

$$[\mathbf{w}]_S = \mathbf{P}[\mathbf{w}]_T \quad \text{for every } \mathbf{w} \in V.$$

(The derivation in Section 3.6 is written for an abstract vector space and applies verbatim to a subspace of $\mathbb{R}^n$.) An easy way to remember which matrix

is which: $\left[\mathbf{v}_j\right]_T = \mathbf{e}_j$, so $\mathbf{P}\left[\mathbf{v}_j\right]_T = \mathbf{P}\mathbf{e}_j$ is the j-th column of $\mathbf{P}$, and it must equal $\left[\mathbf{v}_j\right]_S$.

Algorithm (transition matrix from T to S).

1. Form the $n \times 2k$ matrix $(S \mid T) = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_k \mid \mathbf{v}_1 \ \cdots \ \mathbf{v}_k)$.
2. Reduce it to reduced row-echelon form. The result has the form

$$\left(\begin{array}{c|c} \mathbf{I}_k & \mathbf{P} \\ \mathbf{0}_{(n-k)\times k} & \mathbf{0}_{(n-k)\times k} \end{array}\right),$$

and the right-hand block $\mathbf{P}$ is the transition matrix from T to S.

Indeed, the j-th column of $\mathbf{P}$, $\left[\mathbf{v}_j\right]_S$, is found by solving $(\mathbf{u}_1 \ \cdots \ \mathbf{u}_k \mid \mathbf{v}_j)$ (Section 3.6), and the k systems, having the same coefficient matrix, are solved simultaneously. Since S is linearly independent, the left block reduces to $\mathbf{I}_k$ followed by $n-k$ zero rows (Section 3.5); since every $\mathbf{v}_j \in V = \operatorname{span}(S)$, all systems are consistent, so the last $n-k$ rows are zero.

Example. In the plane $V = \{(x,y,z)^T \mid x - y + 2z = 0\}$ take the bases $S = \{(1,1,0)^T, (-2,0,1)^T\}$ and $T = \{(-1,1,1)^T, (3,1,-1)^T\}$ of Section 3.6. Then

$$(S \mid T) = \left(\begin{array}{cc|cc} 1 & -2 & -1 & 3 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{array}\right) \xrightarrow{\text{RREF}} \left(\begin{array}{cc|cc} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right), \quad \text{so} \quad \mathbf{P} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

For $\mathbf{v} = (1,3,1)^T$ we have $\mathbf{v} = 2(-1,1,1)^T + (3,1,-1)^T$, so $[\mathbf{v}]_T = (2,1)^T$, and $\mathbf{P}[\mathbf{v}]_T = (3,1)^T$, which is $[\mathbf{v}]_S$ as found in Section 3.6.

Theorem. If $\mathbf{P}$ is the transition matrix from T to S, then $\mathbf{P}$ is invertible, and $\mathbf{P}^{-1}$ is the transition matrix from S to T.

Proof. Let $\mathbf{Q}$ be the transition matrix from S to T. For every $\mathbf{w} \in V$ we have $[\mathbf{w}]_S = \mathbf{P}[\mathbf{w}]_T = \mathbf{P}\mathbf{Q}[\mathbf{w}]_S$ and, similarly, $[\mathbf{w}]_T = \mathbf{Q}\mathbf{P}[\mathbf{w}]_T$. Every $\mathbf{c} \in \mathbb{R}^k$ is a coordinate vector relative to S, namely $\mathbf{c} = [c_1\mathbf{u}_1 + \ldots + c_k\mathbf{u}_k]_S$, and likewise relative to T. Hence $\mathbf{P}\mathbf{Q}\mathbf{c} = \mathbf{c}$ and $\mathbf{Q}\mathbf{P}\mathbf{c} = \mathbf{c}$ for all $\mathbf{c} \in \mathbb{R}^k$; taking $\mathbf{c} = \mathbf{e}_1, \ldots, \mathbf{e}_k$, we obtain $\mathbf{P}\mathbf{Q} = \mathbf{Q}\mathbf{P} = \mathbf{I}_k$. □

In the example,

$$\mathbf{P}^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix};$$

this is also the right-hand block of the reduced row-echelon form of $(T \mid S)$.

Remark (geometric meaning)**.** Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be a basis for V. If $T = \{k_1\mathbf{u}_1, k_2\mathbf{u}_2, k_3\mathbf{u}_3\}$ with nonzero k_1, k_2, k_3, then the transition matrix from T to S is $\operatorname{diag}(k_1, k_2, k_3)$: the "axes" $\mathbf{u}_i$ of S are rescaled to give those

of T. If $T = \{a_1\mathbf{u}_1 + a_2\mathbf{u}_2, b_1\mathbf{u}_1 + b_2\mathbf{u}_2, k_3\mathbf{u}_3\}$, the transition matrix from T to S is

$$\begin{pmatrix} a_1 & b_1 & 0 \\ a_2 & b_2 & 0 \\ 0 & 0 & k_3 \end{pmatrix};$$

its upper left block describes how the first two axes are turned (and stretched) within the plane $\operatorname{span}\{\mathbf{u}_1, \mathbf{u}_2\}$, while the third axis is only rescaled.

Transformation of Coordinates

Suppose we have two affine coordinate systems: the first centered at the point O with basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and the second centered at the point O' with basis vectors $\mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3$. Let us write

$$\begin{aligned} \mathbf{e}'_1 &= p_{11}\mathbf{e}_1 + p_{21}\mathbf{e}_2 + p_{31}\mathbf{e}_3, \\ \mathbf{e}'_2 &= p_{12}\mathbf{e}_1 + p_{22}\mathbf{e}_2 + p_{32}\mathbf{e}_3, \\ \mathbf{e}'_3 &= p_{13}\mathbf{e}_1 + p_{23}\mathbf{e}_2 + p_{33}\mathbf{e}_3 \end{aligned}$$

and form the *transition matrix* from the basis $\{\mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3\}$ to the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ (named as in Section 3.6)

$$\mathbf{P} = \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{pmatrix}.$$

If there is some third system of basis vectors and the vectors $\mathbf{e}_i, \mathbf{e}'_i$ are regarded as columns of the coordinates of their expansions in this third system, then the equality

$$(\mathbf{e}'_1 \quad \mathbf{e}'_2 \quad \mathbf{e}'_3) = (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3)\mathbf{P}$$

holds. From this follows the invertibility of the matrix $\mathbf{P}$.

Suppose a point M has coordinates (x, y, z) in the first system and (x', y', z') in the second system. Obviously, $\overrightarrow{OM} = \overrightarrow{OO'} + \overrightarrow{O'M}$. Let the coordinates of the point O' in the first system be (x_0, y_0, z_0). Then

$$\begin{aligned} x\mathbf{e}_1 + y\mathbf{e}_2 + z\mathbf{e}_3 = {} & (x_0\mathbf{e}_1 + y_0\mathbf{e}_2 + z_0\mathbf{e}_3) + x'(p_{11}\mathbf{e}_1 + p_{21}\mathbf{e}_2 + p_{31}\mathbf{e}_3) + \\ & + y'(p_{12}\mathbf{e}_1 + p_{22}\mathbf{e}_2 + p_{32}\mathbf{e}_3) + z'(p_{13}\mathbf{e}_1 + p_{23}\mathbf{e}_2 + p_{33}\mathbf{e}_3) = \\ & = (x_0 + p_{11}x' + p_{12}y' + p_{13}z')\mathbf{e}_1 + \\ & + (y_0 + p_{21}x' + p_{22}y' + p_{23}z')\mathbf{e}_2 + \\ & + (z_0 + p_{31}x' + p_{32}y' + p_{33}z')\mathbf{e}_3. \end{aligned}$$

Consequently, the coordinates of one and the same point in the first and second coordinate systems are related by the following formula:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} + \mathbf{P} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}.$$

From this it is easy to understand, for example, how the general equations of one and the same plane in different affine coordinate systems are related. If in the first coordinate system we have the equation $Ax + By + Cz + D = 0$, then, writing it in matrix form, we find

$$(A \quad B \quad C) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = -D \quad \Leftrightarrow \quad (A \quad B \quad C) \left(\begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} + \mathbf{P} \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} \right) = -D.$$

Thus, the equation of the same plane in the second system takes the form

$$(A' \quad B' \quad C') \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = -D',$$

$$(A' \quad B' \quad C') = (A \quad B \quad C)\mathbf{P}, \quad D' = D + (Ax_0 + By_0 + Cz_0).$$

Chapter 4. Subspaces Associated to a Matrix

4.1. Column Space, Row Space, and Nullspace

Examples of Finite-Dimensional Vector Spaces

Vector spaces that are the span of some finite number of their vectors are called *finite-dimensional*.

1. *The set of polynomials of order n.*

   A polynomial in x of order n is an expression of the form

   $$f(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + ... + a_1x + a_0.$$

   If $a_k \neq 0$ and $a_{k+1} = ... = a_{n-1} = 0$, then k is called the *degree* of the polynomial $f(x)$. Expressions of the form ax^i are called *monomials*.

   Let us regard $f(x)$ as a function of x. Then the sum of polynomials and the multiplication of a polynomial by a number are defined in the same way as in the case of general functions. In doing so, it is clear that the results of these operations remain polynomials. Obviously, the vector space of all polynomials of order n is the span of the monomials of the form

   $$x^{n-1}, x^{n-2}, ..., x^1, x^0 \equiv 1.$$

2. *The set of $m \times n$ matrices with fixed dimensions m and n.*

   In this case, vector addition is defined as matrix addition, and multiplication of a vector by a number as multiplication of a matrix by a number.

   Denote by $\mathbf{E}^{kl} = \left(\left(\mathbf{E}^{kl}\right)_{ij}\right)$ the $m \times n$ matrix with entries of the form

   $$\left(\mathbf{E}^{kl}\right)_{ij} = \begin{cases} 1, & i = k, j = l, \\ 0 & \text{otherwise.} \end{cases}$$

   There are exactly mn such matrices, and obviously the entire space of $m \times n$ matrices is their span.

3. *The set of all solutions of a homogeneous linear system $\mathbf{Ax} = \mathbf{0}$.*

   If the rank of the $m \times n$ matrix $\mathbf{A}$ equals r, then a basis for the solution space of this homogeneous system contains $n - r$ vectors, and the entire set of solutions coincides with their span. This vector space is called the *nullspace* of the matrix $\mathbf{A}$. Notation: Null($\mathbf{A}$) (in some books $\ker \mathbf{A}$ or null $\mathbf{A}$). Its dimension is called the *nullity* of $\mathbf{A}$: nullity($\mathbf{A}$) = dim(Null($\mathbf{A}$)).

4. *The set of all vectors of the form $\mathbf{y} = \mathbf{Ax}$ (for a given matrix $\mathbf{A}$).*

This is a set well familiar to us, coinciding with the span of the columns of the matrix $\mathbf{A}$. It is called the *column space* of the matrix $\mathbf{A}$ (in some books, the *image* of $\mathbf{A}$, im $\mathbf{A}$). Notation: $\mathrm{Col}(\mathbf{A})$.

Row Space and Column Space ADDED FOR MA1522

Let $\mathbf{A} = (a_{ij})_{m\times n}$. Each of its m rows $(a_{i1} \;\; a_{i2} \;\; \cdots \;\; a_{in})$ has n entries, and each of its n columns is a vector of $\mathbb{R}^m$.

Definition. The *row space* of an $m \times n$ matrix $\mathbf{A}$ is the subspace of $\mathbb{R}^n$ spanned by the rows of $\mathbf{A}$:

$$\mathrm{Row}(\mathbf{A}) = \mathrm{span}\{(a_{11} \;\; \cdots \;\; a_{1n}), (a_{21} \;\; \cdots \;\; a_{2n}), ..., (a_{m1} \;\; \cdots \;\; a_{mn})\}.$$

The *column space* of $\mathbf{A}$ is the subspace of $\mathbb{R}^m$ spanned by the columns of $\mathbf{A}$:

$$\mathrm{Col}(\mathbf{A}) = \mathrm{span}\left\{\begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix}, \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix}, ..., \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix}\right\}.$$

Thus $\mathrm{Row}(\mathbf{A}) \subseteq \mathbb{R}^n$, since a row has as many entries as $\mathbf{A}$ has columns, while $\mathrm{Col}(\mathbf{A}) \subseteq \mathbb{R}^m$. The vectors of the row space are written here as rows, but it is often convenient to write them as columns, that is, to work with the transposed rows; in this form the row space of $\mathbf{A}$ is the column space of $\mathbf{A}^T$. Together with the nullspace $\mathrm{Null}(\mathbf{A}) = \{\mathbf{v} \in \mathbb{R}^n \mid \mathbf{Av} = \mathbf{0}\} \subseteq \mathbb{R}^n$ (item 3 above), these are the *subspaces associated to the matrix* $\mathbf{A}$. That $\mathrm{Row}(\mathbf{A})$ and $\mathrm{Col}(\mathbf{A})$ always have the same dimension is proved in Section 4.2.

In the language of column spaces, the consistency criterion of Section 2.3 reads as follows.

Proposition (Column space and consistency)**.** Let $\mathbf{A} = (\mathbf{a}_1 \;\; \mathbf{a}_2 \;\; \cdots \;\; \mathbf{a}_n)$ be an $m \times n$ matrix and $\mathbf{v} \in \mathbb{R}^m$. Then $\mathbf{v} \in \mathrm{Col}(\mathbf{A})$ if and only if the linear system $\mathbf{Ax} = \mathbf{v}$ is consistent. Consequently,

$$\mathrm{Col}(\mathbf{A}) = \{\mathbf{Au} \mid \mathbf{u} \in \mathbb{R}^n\} = \{\mathbf{v} \in \mathbb{R}^m \mid \mathbf{Ax} = \mathbf{v} \text{ is consistent}\}.$$

Proof. For $\mathbf{u} = (c_1, ..., c_n)^T$ we have $\mathbf{Au} = c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + ... + c_n\mathbf{a}_n$. Hence $\mathbf{v}$ is a linear combination of the columns of $\mathbf{A}$ exactly when $\mathbf{v} = \mathbf{Au}$ for some $\mathbf{u} \in \mathbb{R}^n$, that is, exactly when some $\mathbf{u}$ is a solution of $\mathbf{Ax} = \mathbf{v}$. □

Bases for the Row Space, Column Space and Nullspace ADDED FOR MA1522

Recall that $\mathbf{A}$ and $\mathbf{B}$ are *row equivalent* if $\mathbf{B}$ can be obtained from $\mathbf{A}$ by a finite sequence of elementary row operations. Each elementary row operation is multiplication on the left by an elementary matrix, which is invertible (Sections 1.3 and 2.5). Hence, if the $m \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$ are row equivalent, then

$$\mathbf{B} = \mathbf{PA}, \quad \mathbf{A} = \mathbf{P}^{-1}\mathbf{B}$$

for some invertible matrix $\mathbf{P}$ of order m (a product of elementary matrices).

Theorem (Row operations preserve the row space)**.** If $\mathbf{A}$ and $\mathbf{B}$ are row equivalent matrices, then $\mathrm{Row}(\mathbf{A}) = \mathrm{Row}(\mathbf{B})$.

Proof. Let $\mathbf{B} = \mathbf{PA}$ with $\mathbf{P} = (p_{ij})$ invertible. Row i of $\mathbf{PA}$ is row i of $\mathbf{P}$ times $\mathbf{A}$, that is, the linear combination of the rows of $\mathbf{A}$ with the coefficients $p_{i1}, ..., p_{im}$. So every row of $\mathbf{B}$ lies in the subspace $\mathrm{Row}(\mathbf{A})$, and therefore $\mathrm{Row}(\mathbf{B}) \subseteq \mathrm{Row}(\mathbf{A})$. The same argument applied to $\mathbf{A} = \mathbf{P}^{-1}\mathbf{B}$ gives $\mathrm{Row}(\mathbf{A}) \subseteq \mathrm{Row}(\mathbf{B})$. □

Theorem (Basis for the row space)**.** Let $\mathbf{R}$ be the reduced row-echelon form of a matrix $\mathbf{A}$. Then the nonzero rows of $\mathbf{R}$ form a basis for $\mathrm{Row}(\mathbf{R}) = \mathrm{Row}(\mathbf{A})$.

Proof. By the preceding theorem $\mathrm{Row}(\mathbf{A}) = \mathrm{Row}(\mathbf{R})$, and the zero rows of $\mathbf{R}$ do not contribute to the span, so the nonzero rows $\mathbf{r}_1, ..., \mathbf{r}_k$ of $\mathbf{R}$ span $\mathrm{Row}(\mathbf{A})$. Let the leading entry of $\mathbf{r}_i$ lie in column j_i. In the reduced row-echelon form, this leading entry is 1 and all the other entries of column j_i are 0. Hence, if $\alpha_1\mathbf{r}_1 + ... + \alpha_k\mathbf{r}_k = \mathbf{0}$, then the entry of the left-hand side in position j_i equals α_i, so $\alpha_i = 0$ for $i = 1, ..., k$. Thus $\mathbf{r}_1, ..., \mathbf{r}_k$ are linearly independent. □

The nonzero rows of *any* row-echelon form of $\mathbf{A}$ also form a basis for $\mathrm{Row}(\mathbf{A})$: they span $\mathrm{Row}(\mathbf{A})$ by the same argument, and they are linearly independent, as shown in the proof of the proposition in Section 1.2.

Theorem (Row operations preserve linear relations between columns)**.** Let $\mathbf{A} = (\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \cdots \ \ \mathbf{a}_n)$ and $\mathbf{B} = (\mathbf{b}_1 \ \ \mathbf{b}_2 \ \ \cdots \ \ \mathbf{b}_n)$ be row equivalent $m \times n$ matrices. Then for any scalars $c_1, c_2, ..., c_n$,

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + ... + c_n\mathbf{a}_n = \mathbf{0} \quad \Leftrightarrow \quad c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + ... + c_n\mathbf{b}_n = \mathbf{0}.$$

Proof. Put $\mathbf{c} = (c_1, ..., c_n)^T$ and let $\mathbf{B} = \mathbf{PA}$ with $\mathbf{P}$ invertible. The two equalities read $\mathbf{Ac} = \mathbf{0}$ and $\mathbf{Bc} = \mathbf{P}(\mathbf{Ac}) = \mathbf{0}$. The first implies the second, and multiplying the second on the left by $\mathbf{P}^{-1}$ gives the first. □

In other words, row equivalent matrices have the same nullspace, $\mathrm{Null}(\mathbf{A}) = \mathrm{Null}(\mathbf{B})$. In particular, some columns of $\mathbf{A}$ are linearly independent if and only if the corresponding columns of $\mathbf{B}$ are, and a column of $\mathbf{A}$ is a given linear combination of other columns of $\mathbf{A}$ if and only if the corresponding column of $\mathbf{B}$ is the same linear combination of the corresponding columns of $\mathbf{B}$.

Theorem (Basis for the column space)**.** Let $\mathbf{R} = (r_{ij})$ be the reduced row-echelon form of $\mathbf{A} = (\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \cdots \ \ \mathbf{a}_n)$, and let $j_1 < j_2 < ... < j_k$ be

the indices of the pivot columns of $\mathbf{R}$. Then the columns $\mathbf{a}_{j_1}, ..., \mathbf{a}_{j_k}$ of $\mathbf{A}$ form a basis for $\mathrm{Col}(\mathbf{A})$. Moreover, every column of $\mathbf{A}$ is expressed through them by the entries of the corresponding column of $\mathbf{R}$:

$$\mathbf{a}_j = r_{1j}\mathbf{a}_{j_1} + r_{2j}\mathbf{a}_{j_2} + ... + r_{kj}\mathbf{a}_{j_k}, \quad j = 1, ..., n.$$

Proof. Denote the columns of $\mathbf{R}$ by $\mathbf{r}_1, ..., \mathbf{r}_n$. The pivot column $\mathbf{r}_{j_i}$ is the column $\mathbf{e}_i \in \mathbb{R}^m$ of the identity matrix, so $\mathbf{r}_{j_1}, ..., \mathbf{r}_{j_k}$ are linearly independent. Every column $\mathbf{r}_j$ has zero entries below row k, so

$$\mathbf{r}_j = r_{1j}\mathbf{e}_1 + ... + r_{kj}\mathbf{e}_k = r_{1j}\mathbf{r}_{j_1} + ... + r_{kj}\mathbf{r}_{j_k}.$$

Both facts are linear relations between columns of $\mathbf{R}$, and by the preceding theorem they hold for the corresponding columns of $\mathbf{A}$: the columns $\mathbf{a}_{j_1}, ..., \mathbf{a}_{j_k}$ are linearly independent, and each $\mathbf{a}_j$ is the stated combination of them. The latter shows that $\mathrm{Col}(\mathbf{A}) = \mathrm{span}\{\mathbf{a}_{j_1}, ..., \mathbf{a}_{j_k}\}$. □

Remark (What row operations do not preserve)**.** Row operations change the column space in general, and they do not preserve linear relations between the rows. For example,

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \mathbf{B} = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix},$$

but $\mathrm{Col}(\mathbf{A}) = \mathrm{span}\{(1,2)^T\} \neq \mathrm{span}\{(1,0)^T\} = \mathrm{Col}(\mathbf{B})$; the second row of $\mathbf{A}$ is twice the first, while the second row of $\mathbf{B}$ is not. This is why the basis for $\mathrm{Col}(\mathbf{A})$ must be taken from the columns of $\mathbf{A}$ itself, not from those of $\mathbf{R}$. Likewise, the rows of $\mathbf{A}$ standing in the positions of the nonzero rows of $\mathbf{R}$ need not form a basis for $\mathrm{Row}(\mathbf{A})$, because the reduction may interchange rows:

$$\begin{pmatrix} 1 & 1 \\ 2 & 2 \\ 0 & 1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix},$$

yet the first two rows of the matrix on the left are linearly dependent.

Algorithm (Bases for the subspaces associated to a matrix).

1. Reduce $\mathbf{A}$ to its reduced row-echelon form $\mathbf{R}$ by Gauss–Jordan elimination (Section 1.4).
2. A basis for $\mathrm{Row}(\mathbf{A})$: the nonzero rows of $\mathbf{R}$.
3. A basis for $\mathrm{Col}(\mathbf{A})$: the columns of $\mathbf{A}$ in the positions of the pivot columns of $\mathbf{R}$.
4. A basis for $\mathrm{Null}(\mathbf{A})$: solve $\mathbf{Rx} = \mathbf{0}$ (which has the same solutions as $\mathbf{Ax} = \mathbf{0}$), assigning a parameter s_i to each variable that corresponds to a non-pivot column of $\mathbf{R}$, and write the general solution as $s_1\mathbf{v}_1 + ... + s_p\mathbf{v}_p$. The vectors $\mathbf{v}_1, ..., \mathbf{v}_p$ form a basis for $\mathrm{Null}(\mathbf{A})$: they span

it, and they are linearly independent because $\mathbf{v}_i$ has the entry 1 in the position of the i-th parameter and 0 in the positions of the other parameters (see Homogeneous Linear Systems in Section 4.2).

Example (Bases for the three subspaces)**.** Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & -1 & 0 \\ 2 & 4 & 1 & 0 & 0 \\ -1 & -2 & 1 & 3 & 1 \\ 1 & 2 & 1 & 1 & 2 \end{pmatrix} \xrightarrow{\text{RREF}} \mathbf{R} = \begin{pmatrix} 1 & 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivot columns of $\mathbf{R}$ are its columns 1, 3 and 5. Hence

$$\text{Row}(\mathbf{A}) = \text{span}\{(1 \;\; 2 \;\; 0 \;\; -1 \;\; 0), (0 \;\; 0 \;\; 1 \;\; 2 \;\; 0), (0 \;\; 0 \;\; 0 \;\; 0 \;\; 1)\},$$

$$\text{Col}(\mathbf{A}) = \text{span}\{\mathbf{a}_1, \mathbf{a}_3, \mathbf{a}_5\} = \text{span}\left\{ \begin{pmatrix} 1 \\ 2 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\},$$

and the vectors displayed are bases. The non-pivot columns of $\mathbf{R}$ give the relations $\mathbf{a}_2 = 2\mathbf{a}_1$ and $\mathbf{a}_4 = -\mathbf{a}_1 + 2\mathbf{a}_3$ between the columns of $\mathbf{A}$. The columns $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ of $\mathbf{R}$ span a different subspace of $\mathbb{R}^4$: all its vectors have last entry 0, whereas $\mathbf{a}_5 = (0, 0, 1, 2)^T$ does not. Finally, with the parameters $x_2 = s$ and $x_4 = t$, the system $\mathbf{Rx} = \mathbf{0}$ gives $x_1 = -2s + t$, $x_3 = -2t$, $x_5 = 0$, so

$$\mathbf{x} = s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix},$$

and these two vectors form a basis for $\text{Null}(\mathbf{A})$; thus $\text{nullity}(\mathbf{A}) = 2$.

4.2. Rank

Dimensions of the Column and Row Spaces ADDED FOR MA1522

The bases found in Section 4.1 show that the column space and the row space of every matrix have the same dimension, and they show how to compute it.

Theorem. Let $\mathbf{R}$ be the reduced row-echelon form of a matrix $\mathbf{A}$. Then

$$\begin{aligned} \dim(\text{Col}(\mathbf{A})) &= \text{number of pivot columns of } \mathbf{R} \\ &= \text{number of nonzero rows of } \mathbf{R} = \dim(\text{Row}(\mathbf{A})). \end{aligned}$$

Proof. By the theorems of Section 4.1, the columns of $\mathbf{A}$ in the positions of the pivot columns of $\mathbf{R}$ form a basis for $\mathrm{Col}(\mathbf{A})$, and the nonzero rows of $\mathbf{R}$ form a basis for $\mathrm{Row}(\mathbf{A})$. Each nonzero row of $\mathbf{R}$ contains exactly one leading entry, and each pivot column contains exactly one leading entry; so both numbers equal the number of leading entries of $\mathbf{R}$. □

This common dimension is the *rank* of $\mathbf{A}$, $\mathrm{rank}(\mathbf{A}) = \dim(\mathrm{Col}(\mathbf{A})) = \dim(\mathrm{Row}(\mathbf{A}))$ (see Matrix Rank below). Thus $\mathrm{rank}(\mathbf{A})$ is the number of pivot columns, or of nonzero rows, of the reduced row-echelon form of $\mathbf{A}$; by the observation following the theorem on a basis for the row space (Section 4.1), it is also the number of nonzero rows of *any* row-echelon form of $\mathbf{A}$.

Corollary.
1. If $\mathbf{A}$ and $\mathbf{B}$ are row equivalent, then $\mathrm{rank}(\mathbf{A}) = \mathrm{rank}(\mathbf{B})$.
2. $\mathrm{rank}(\mathbf{A}) = 0$ if and only if $\mathbf{A} = \mathbf{0}$.

Proof.
1. Row equivalent matrices have the same row space (Section 4.1), hence the same rank. (This is also a special case of Proposition 3 in Properties of Rank below.)
2. $\mathrm{rank}(\mathbf{A}) = 0$ means $\mathrm{Col}(\mathbf{A}) = \{\mathbf{0}\}$, that is, every column of $\mathbf{A}$ is zero.

□

Example. For the 4×5 matrix $\mathbf{A}$ of the example in Section 4.1, the reduced row-echelon form has three pivot columns and three nonzero rows, so $\mathrm{rank}(\mathbf{A}) = 3$. Its column space is a 3-dimensional subspace of $\mathbb{R}^4$, and its row space a 3-dimensional subspace of $\mathbb{R}^5$.

Matrix Rank

The dimension of the column space $\mathrm{Col}(\mathbf{A})$ (row space $\mathrm{Row}(\mathbf{A})$) of a matrix $\mathbf{A}$ is sometimes called its column (row) rank. Since the column and row ranks coincide, it is natural to call their common value simply the *rank* of the matrix; this is the definition used in MA1522:

$$\mathrm{rank}(\mathbf{A}) = \dim(\mathrm{Col}(\mathbf{A})) = \dim(\mathrm{Row}(\mathbf{A})).$$

This value is also equal to the smallest number of matrices of the "column-by-row" form $\mathbf{u}\mathbf{v}^T$ whose sum yields the given matrix (this follows from the skeleton decomposition; see Tyrtyshnikov, §§7.1–7.3).

Nevertheless, another definition is often given: the *rank* of a matrix is the largest k for which the matrix has a nonzero $k \times k$ minor. Here and throughout this section, a $k \times k$ *minor* (or *minor of order* k) means the determinant of a $k \times k$ submatrix; it is a number, not to be confused with the (i, j) matrix minor $\mathbf{M}_{ij}$ of Section 2.8, which is a submatrix. A nonzero minor of this largest order and the corresponding submatrix are called a *basis minor* and a *basis submatrix*. Since a square matrix is invertible if and only if its determinant is

nonzero (Section 2.6), the rank in this sense is equal to the largest order of an invertible square submatrix of the given matrix. Notation: $\operatorname{rank}(\mathbf{A})$.

Two obvious properties of the rank of an $m \times n$ matrix $\mathbf{A}$:

$$\operatorname{rank}(\mathbf{A}) \le \min\{m, n\}, \quad \operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{A}^T).$$

It is less obvious that: *the largest order of a nonzero minor of a matrix coincides with its column and row rank*. Let us prove this.

Basis Minor Theorem

Theorem. The columns (rows) containing a basis minor are linearly independent, and any column (any row) of the given matrix is a linear combination of them.

Proof. Without loss of generality, suppose that the basis submatrix $\mathbf{P}$ of order k is located in the upper left corner of the $m \times n$ matrix $\mathbf{A}$. Thus,

$$\mathbf{A} = \begin{pmatrix} \mathbf{P} & \mathbf{v}_{k+1} & \dots & \mathbf{v}_n \\ \mathbf{u}_{k+1}^T & a_{k+1,k+1} & \dots & a_{k+1,n} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{u}_m^T & a_{m,k+1} & \dots & a_{mn} \end{pmatrix},$$

$$\mathbf{u}_{k+1}, \dots, \mathbf{u}_m, \mathbf{v}_{k+1}, \dots, \mathbf{v}_n \in \mathbb{R}^{k \times 1}.$$

By the hypothesis of the theorem, any submatrix of order $k+1$ of the form

$$\mathbf{M} = \begin{pmatrix} \mathbf{P} & \mathbf{v}_j \\ \mathbf{u}_i^T & a_{ij} \end{pmatrix}, \quad i, j > k,$$

is noninvertible. Such a matrix $\mathbf{M}$, obtained from $\mathbf{P}$ by adding one row and one column, is called a *bordering* of $\mathbf{P}$. By the lemma on noninvertible bordering (proved in Tyrtyshnikov, §7.4: if the submatrix $\mathbf{P}$ of such a bordering is invertible while the bordering itself is not, then its last column is the linear combination of its first k columns with the coefficients $\mathbf{P}^{-1}\mathbf{v}_j$, where $\mathbf{v}_j$ consists of the first k entries of that column), the last column in $\mathbf{M}$ is a linear combination of the first k columns. Moreover, the coefficients of this linear combination do not depend on i (since they are determined by the vector $\mathbf{P}^{-1}\mathbf{v}_j$). Hence, the j-th column of the matrix $\mathbf{A}$ for $j > k$ is a linear combination of the first k columns. The linear independence of the first k columns is proved as follows: suppose their linear combination with coefficients $\alpha_1, \dots, \alpha_k$ is equal to $\mathbf{0}$; then

$$\mathbf{P}\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_k \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_k \end{pmatrix} = \mathbf{0}.$$

The assertion of the theorem concerning the rows is proved by passing to the transposed matrix. □

Corollary 1. A matrix has rank r if and only if some $r \times r$ minor is nonzero, while all $(r+1) \times (r+1)$ minors bordering it are equal to zero.

Corollary 2. The largest order of a nonzero minor of a matrix $\mathbf{A}$ coincides with its column and row rank, that is, with $\operatorname{rank}(\mathbf{A}) = \dim(\operatorname{Col}(\mathbf{A})) = \dim(\operatorname{Row}(\mathbf{A}))$.

Remark. The basis minor theorem does not use the previously proved theorem on the equality of the column and row ranks. In essence, it provides yet another proof of this fact.

Thus the columns containing a basis minor form a basis for $\operatorname{Col}(\mathbf{A})$, just as the columns of $\mathbf{A}$ corresponding to the pivot columns of its reduced row-echelon form do; likewise, the rows containing a basis minor form a basis for $\operatorname{Row}(\mathbf{A})$.

Problem 1. Let $\mathbf{A}$ be an $n \times n$ matrix of rank k, and $\mathbf{B}$ any invertible $k \times k$ submatrix. Denote by $\mathbf{R}$ the $k \times n$ submatrix consisting of the rows of the matrix $\mathbf{A}$ containing the submatrix $\mathbf{B}$, and by $\mathbf{C}$ the $n \times k$ submatrix consisting of the columns containing $\mathbf{B}$. Prove that

$$\mathbf{A} = \mathbf{C}\mathbf{B}^{-1}\mathbf{R}.$$

Problem 2. Prove that the submatrix located at the intersection of r linearly independent rows and r linearly independent columns of a matrix of rank r is invertible.

Problem 3. It is known that $\mathbf{A}^T = -\mathbf{A}$. Prove that the rank of the matrix $\mathbf{A}$ is an even number.

Properties of Rank

Proposition 1. The rank of a sum of matrices does not exceed the sum of their ranks:

$$\operatorname{rank}(\mathbf{A} + \mathbf{B}) \leq \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(\mathbf{B}).$$

Proof. Obviously, $\mathbf{A}$ and $\mathbf{B}$ must have the same number of columns:

$$\mathbf{A} = (\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n), \quad \mathbf{B} = (\mathbf{b}_1 \quad \mathbf{b}_2 \quad \cdots \quad \mathbf{b}_n).$$

Clearly, $\operatorname{span}\{\mathbf{a}_1 + \mathbf{b}_1, ..., \mathbf{a}_n + \mathbf{b}_n\} \subseteq \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}_1, ..., \mathbf{b}_n\}$. In the smaller span, let us choose some set of vectors forming a basis. According to the Basis Extension Lemma (Section 3.7), a basis for the larger span can be obtained by extending this set by some vectors from the larger span. Therefore,

$$\begin{aligned}\operatorname{rank}(\mathbf{A} + \mathbf{B}) &= \dim(\operatorname{span}\{\mathbf{a}_1 + \mathbf{b}_1, ..., \mathbf{a}_n + \mathbf{b}_n\}) \leq \\ &\leq \dim(\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}_1, ..., \mathbf{b}_n\}).\end{aligned}$$

Let $p = \operatorname{rank}(\mathbf{A})$, $q = \operatorname{rank}(\mathbf{B})$, and suppose without loss of generality that a basis for $\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\}$ is formed by the first p vectors, and a basis for $\operatorname{span}\{\mathbf{b}_1, ..., \mathbf{b}_n\}$ by the first q vectors. Then $\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}_1, ..., \mathbf{b}_n\} = \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_p, \mathbf{b}_1, ..., \mathbf{b}_q\} \Rightarrow$

$$\dim(\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}_1, ..., \mathbf{b}_n\}) \leq p + q. \qquad \square$$

Proposition 2. The rank of a product of matrices does not exceed the rank of each of the factors:

$$\operatorname{rank}(\mathbf{AB}) \leq \min\{\operatorname{rank}(\mathbf{A}), \operatorname{rank}(\mathbf{B})\}.$$

Proof. It suffices to note that each of the columns of the matrix $\mathbf{AB}$ is a linear combination of the columns of the matrix $\mathbf{A}$. Therefore, the column space of the matrix $\mathbf{AB}$ is contained in the column space of the matrix $\mathbf{A}$: $\operatorname{Col}(\mathbf{AB}) \subseteq \operatorname{Col}(\mathbf{A})$. Consequently, $\operatorname{rank}(\mathbf{AB}) \leq \operatorname{rank}(\mathbf{A})$. Furthermore,

$$\operatorname{rank}(\mathbf{AB}) = \operatorname{rank}((\mathbf{AB})^T) = \operatorname{rank}(\mathbf{B}^T\mathbf{A}^T) \leq \operatorname{rank}(\mathbf{B}^T) = \operatorname{rank}(\mathbf{B}). \qquad \square$$

Proposition 3. The rank of a matrix does not change when it is multiplied on the left or on the right by an invertible matrix.

Proof. Let $\mathbf{B} = \mathbf{PAQ}$, where $\mathbf{P}$ and $\mathbf{Q}$ are invertible matrices.[5] By virtue of the preceding proposition, $\operatorname{rank}(\mathbf{B}) \leq \operatorname{rank}(\mathbf{A})$. At the same time, $\mathbf{A} = \mathbf{P}^{-1}\mathbf{BQ}^{-1} \Rightarrow \operatorname{rank}(\mathbf{A}) \leq \operatorname{rank}(\mathbf{B})$. $\square$

This proposition is useful when computing the rank. Usually this is done with the aid of *elementary row and column operations*, simplifying the form of the matrix by eliminating its entries; these operations reduce to multiplying the matrix on the left and on the right by certain invertible matrices (elementary matrices) and therefore preserve the rank.

Proposition 4. When k rows (columns) of a matrix are changed, its rank cannot change by more than k.

Proof. It suffices to consider the case $k = 1$. Let the rank be equal to r, and let the columns $\mathbf{a}_{j_1}, ..., \mathbf{a}_{j_r}$ be basis columns. If the column $\mathbf{a}_j$ coincides with one of these columns, then under any change to it, this set of columns will have a linearly independent subset of $r - 1$ columns. Therefore, the rank of the new matrix is no less than $r - 1$ and, of course, no greater than $r + 1$. If the column $\mathbf{a}_j$ does not coincide with any of the chosen basis columns, then every column of the new matrix is a linear combination of the columns $\mathbf{a}_{j_1}, ..., \mathbf{a}_{j_r}, \mathbf{a}_j$, so its rank is no greater than $r + 1$ and no less than r, since there is a set of r linearly independent columns $\mathbf{a}_{j_1}, ..., \mathbf{a}_{j_r}$. $\square$

[5] Of course, the order of $\mathbf{P}$ is equal to the number of rows, and the order of $\mathbf{Q}$ to the number of columns of the matrix $\mathbf{A}$.

Corollary. When k columns are added (or removed), the rank of a matrix cannot change by more than k.

Problem 4. Let $\mathbf{A}$ and $\mathbf{B}$ be matrices of rank 1. Prove that if $\mathbf{AB} \neq \mathbf{0}$, then the rank of the matrix $\mathbf{A} + \mathbf{B}$ is at most 1.

Problem 5. Given vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, with $\mathbf{x} \neq \mathbf{0}$. Prove that there exists a symmetric matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ such that $\operatorname{rank}(\mathbf{A}) \leq 2$ and $\mathbf{Ax} = \mathbf{y}$.

Problem 6. A matrix $\mathbf{A}$ has r columns, and a matrix $\mathbf{B}$ has r rows. Prove that $r \geq \operatorname{rank}(\mathbf{A}) + \operatorname{rank}(\mathbf{B}) - \operatorname{rank}(\mathbf{AB})$.

Problem 7. Prove that any matrix of rank r with nonnegative entries can be represented as the difference of matrices with nonnegative entries and rank not exceeding 2^r.

Homogeneous Linear Systems

A linear system with a zero right-hand side

$$\mathbf{Ax} = \mathbf{0} \qquad (*)$$

is called *homogeneous*. Suppose that the given system has m equations and n variables. Then the coefficient matrix $\mathbf{A}$ has size $m \times n$. Let us write $\mathbf{A}$ by its columns, $\mathbf{A} = (\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \cdots \ \ \mathbf{a}_n)$, and suppose that its rank is equal to r. Without loss of generality, we shall assume that a basis submatrix in $\mathbf{A}$ is located in the first r columns—we shall call them *basis* columns. The components of the solution $x_1, ..., x_r$ corresponding to the basis columns will also be called *basis* (or *basic*) variables, and the remaining components $x_{r+1}, ..., x_n$ will be called *free* variables. (Under this assumption the basis columns are exactly the columns of $\mathbf{A}$ corresponding to the pivot columns of its reduced row-echelon form, and the free variables are those to which parameters are assigned in the general solution.) Thus, the solution vector has the form

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_r \\ x_{r+1} \\ \vdots \\ x_n \end{pmatrix}.$$

The system $(*)$ is equivalent to the equality

$$x_1\mathbf{a}_1 + ... + x_r\mathbf{a}_r = -x_{r+1}\mathbf{a}_{r+1} - ... - x_n\mathbf{a}_n. \qquad (**)$$

By the basis minor theorem, the columns $\mathbf{a}_{r+1}, ..., \mathbf{a}_n$ belong to $\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_r\}$. Therefore, for any choice of values of the free variables, the values of the basis variables satisfying equality $(**)$ exist and are uniquely determined. Thus, there exist $n - r$ vectors of the form

$$\mathbf{v}_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{r1} \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} x_{12} \\ \vdots \\ x_{r2} \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad ..., \quad \mathbf{v}_{n-r} = \begin{pmatrix} x_{1,n-r} \\ \vdots \\ x_{r,n-r} \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix},$$

each of which is a solution of the system $(*)$:

$$\mathbf{A}\mathbf{v}_1 = \mathbf{0}, \quad ..., \quad \mathbf{A}\mathbf{v}_{n-r} = \mathbf{0}.$$

The vectors $\mathbf{v}_1, ..., \mathbf{v}_{n-r}$ are linearly independent:

$$\alpha_1\mathbf{v}_1 + ... + \alpha_{n-r}\mathbf{v}_{n-r} = \begin{pmatrix} * \\ \vdots \\ * \\ \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_{n-r} \end{pmatrix} = \mathbf{0} \quad \Rightarrow \quad \alpha_1 = ... = \alpha_{n-r} = 0.$$

Moreover, any linear combination of these vectors is a solution of the system $(*)$ and, furthermore, if $\mathbf{x}$ is an arbitrary solution of the system $(*)$, then

$$\mathbf{x} = x_{r+1}\mathbf{v}_1 + ... + x_n\mathbf{v}_{n-r}.$$

Thus, we have proved the following important statement.

Theorem. The set of solutions of the homogeneous linear system $(*)$ coincides with $\text{span}\{\mathbf{v}_1, ..., \mathbf{v}_{n-r}\}$, the span of the linearly independent vectors $\mathbf{v}_1, ..., \mathbf{v}_{n-r}$; that is, $\{\mathbf{v}_1, ..., \mathbf{v}_{n-r}\}$ is a basis for the nullspace $\text{Null}(\mathbf{A})$.

Corollary (Rank–Nullity Theorem)**.** For the $m \times n$ matrix $\mathbf{A}$ of rank r, $\text{nullity}(\mathbf{A}) = \dim(\text{span}\{\mathbf{v}_1, ..., \mathbf{v}_{n-r}\}) = n - r$; in other words,

$$\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n.$$

A linearly independent set of solutions $\{\mathbf{w}_1, ..., \mathbf{w}_k\}$ of the system $\mathbf{A}\mathbf{x} = \mathbf{0}$ is called a *basis for the solution space* (classically, a *fundamental system of solutions*) if $\text{span}\{\mathbf{w}_1, ..., \mathbf{w}_k\}$ coincides with the set of all solutions of the homogeneous system $\mathbf{A}\mathbf{x} = \mathbf{0}$.

Corollary. The number of vectors in any basis for the solution space of $\mathbf{A}\mathbf{x} = \mathbf{0}$ is equal to $n - r$, where $r = \text{rank}(\mathbf{A})$.

For the proof, it suffices to note that the span of the vectors of such a basis also has a basis consisting of the vectors $\mathbf{v}_1, ..., \mathbf{v}_{n-r}$ constructed above.

Problem 8. Given square matrices $\mathbf{A}$ and $\mathbf{B}$ of order n such that $\mathbf{AB} = \mathbf{0}$ and the matrix $\mathbf{A} + \mathbf{B}$ is invertible. Prove that $\operatorname{rank}(\mathbf{A}) + \operatorname{rank}(\mathbf{B}) = n$.

Rank and Consistency (Kronecker–Capelli Theorem)
Consider a linear system $\mathbf{Ax} = \mathbf{b}$ with an $m \times n$ matrix $\mathbf{A} = (\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \cdots \ \ \mathbf{a}_n)$. The matrix $(\mathbf{A} \mid \mathbf{b}) = (\mathbf{a}_1 \ \ \mathbf{a}_2 \ \ \cdots \ \ \mathbf{a}_n \mid \mathbf{b})$ is called the *augmented matrix* of the given system.

Theorem. The linear system $\mathbf{Ax} = \mathbf{b}$ is consistent if and only if the rank of the coefficient matrix coincides with the rank of the augmented matrix:

$$\operatorname{rank}(\mathbf{A}) = \operatorname{rank}((\mathbf{A} \mid \mathbf{b})).$$

Proof. We already know (see Section 1.5) that the consistency of the system $\mathbf{Ax} = \mathbf{b}$ is equivalent to the equality of the spans $\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\} = \operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}\}$. It remains to note that $\operatorname{rank}(\mathbf{A}) = \dim(\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n\})$ and $\operatorname{rank}((\mathbf{A} \mid \mathbf{b})) = \dim(\operatorname{span}\{\mathbf{a}_1, ..., \mathbf{a}_n, \mathbf{b}\})$. □

General Solution of a Linear System
If U and V are two sets of vectors in $\mathbb{R}^n$, then the sum $U + V$ is defined as the set consisting of all possible sums of vectors of the form $\mathbf{u} + \mathbf{v}$, where $\mathbf{u} \in U$, $\mathbf{v} \in V$.

Theorem. Suppose that the system $\mathbf{Ax} = \mathbf{b}$ is consistent, and fix an arbitrary particular solution $\mathbf{u}$ ($\mathbf{Au} = \mathbf{b}$). Then the set of all solutions of the system $\mathbf{Ax} = \mathbf{b}$ has the form $\mathbf{u} + V$, where $V = \operatorname{Null}(\mathbf{A})$ is the set of all solutions of the corresponding homogeneous system $\mathbf{Ax} = \mathbf{0}$.

Proof. Let $\mathbf{x}$ be an arbitrary solution of the system $\mathbf{Ax} = \mathbf{b}$. Then, obviously, $\mathbf{A}(\mathbf{x} - \mathbf{u}) = \mathbf{0} \Rightarrow \mathbf{x} - \mathbf{u} \in V \Rightarrow \mathbf{x} \in \mathbf{u} + V$. Next, take an arbitrary vector $\mathbf{x} \in \mathbf{u} + V \Rightarrow \mathbf{x} = \mathbf{u} + \mathbf{v}$, $\mathbf{v} \in V \Rightarrow \mathbf{A}(\mathbf{u} + \mathbf{v}) = \mathbf{Au} + \mathbf{Av} = \mathbf{b} + \mathbf{0} = \mathbf{b}$. □

Corollary. The general solution of a consistent system $\mathbf{Ax} = \mathbf{b}$ has the form

$$\mathbf{x} = \mathbf{u} + c_1\mathbf{v}_1 + ... + c_{n-r}\mathbf{v}_{n-r},$$

where $\mathbf{u}$ is an arbitrary particular solution of the given system, $\mathbf{v}_1, ..., \mathbf{v}_{n-r}$ are linearly independent solutions of the corresponding homogeneous system (a basis for $\operatorname{Null}(\mathbf{A})$), $r = \operatorname{rank}(\mathbf{A})$, and the coefficients $c_1, ..., c_{n-r}$ are arbitrary real numbers (the parameters of the general solution).

Problem 9. What can be said about the matrix and right-hand side of a system $\mathbf{Ax} = \mathbf{b}$ with respect to a vector $\mathbf{x} \in \mathbb{R}^n$ if every vector in $\mathbb{R}^n$ is a solution of it?

Summary of the Subspaces Associated to a Matrix ADDED FOR MA1522

Let $\mathbf{A}$ be an $m \times n$ matrix with reduced row-echelon form $\mathbf{R}$. The basis for $\text{Null}(\mathbf{A})$ constructed in Section 4.1 contains one vector for each variable to which a parameter is assigned, that is, for each non-pivot column of $\mathbf{R}$. Hence

$$\text{nullity}(\mathbf{A}) = \text{number of non-pivot columns of } \mathbf{R},$$

and the rank–nullity theorem merely counts the columns of $\mathbf{R}$: the $\text{rank}(\mathbf{A})$ pivot columns and the $\text{nullity}(\mathbf{A})$ non-pivot columns together make up all n columns. The following table collects the results on the three subspaces associated to $\mathbf{A}$.

Subspace	Subspace of	Basis	Dimension
$\text{Col}(\mathbf{A})$	$\mathbb{R}^m$	the columns of $\mathbf{A}$ in the positions of the pivot columns of $\mathbf{R}$	$\text{rank}(\mathbf{A})$ = number of pivot columns of $\mathbf{R}$
$\text{Row}(\mathbf{A})$	$\mathbb{R}^n$	the nonzero rows of $\mathbf{R}$	$\text{rank}(\mathbf{A})$ = number of nonzero rows of $\mathbf{R}$
$\text{Null}(\mathbf{A})$	$\mathbb{R}^n$	the vectors multiplying the parameters in the general solution of $\mathbf{Ax} = \mathbf{0}$	$\text{nullity}(\mathbf{A})$ = number of non-pivot columns of $\mathbf{R}$

Example. For the 4×5 matrix $\mathbf{A}$ of the example in Section 4.1, $\text{Col}(\mathbf{A})$ is a subspace of $\mathbb{R}^4$, while $\text{Row}(\mathbf{A})$ and $\text{Null}(\mathbf{A})$ are subspaces of $\mathbb{R}^5$. We found $\text{rank}(\mathbf{A}) = 3$ (pivot columns 1, 3, 5) and $\text{nullity}(\mathbf{A}) = 2$ (non-pivot columns 2, 4), and indeed $3 + 2 = 5$ is the number of columns of $\mathbf{A}$.

Full Rank ADDED FOR MA1522

Let $\mathbf{A}$ be an $m \times n$ matrix with reduced row-echelon form $\mathbf{R}$. Since $\text{rank}(\mathbf{A})$ is the number of pivot columns of $\mathbf{R}$, it is at most n; since it is also the number of nonzero rows of $\mathbf{R}$, it is at most m. Thus $\text{rank}(\mathbf{A}) \leq \min\{m, n\}$: the largest rank an $m \times n$ matrix can have is the smaller of its two dimensions.

Definition. An $m \times n$ matrix $\mathbf{A}$ is of *full rank* if $\text{rank}(\mathbf{A}) = \min\{m, n\}$, that is, if its rank equals the number of its rows or the number of its columns, whichever is smaller.

Example. The matrices

$$\mathbf{A}_1 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{A}_2 = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad \mathbf{A}_3 = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

have the reduced row-echelon forms

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}.$$

Thus $\text{rank}(\mathbf{A}_1) = 3$ is the number of columns and $\text{rank}(\mathbf{A}_2) = 2$ is the number of rows: both matrices are of full rank. The square matrix $\mathbf{A}_3$ has rank $1 < 2$ and is not of full rank; neither is the 4×5 matrix of the example in Section 4.1, whose rank is $3 < 4$.

For a square matrix of order n, full rank means $\text{rank}(\mathbf{A}) = n$. This is one more condition equivalent to invertibility, and we add it, together with a condition on the nullity, to the equivalent statements for invertibility collected in Sections 2.6 and 3.6.

Theorem (Equivalent statements for invertibility)**.** Let $\mathbf{A}$ be a square matrix of order n. The following statements are equivalent.

(i) $\mathbf{A}$ is invertible.
(ii) $\mathbf{A}^T$ is invertible.
(iii) (left inverse) There is a matrix $\mathbf{B}$ such that $\mathbf{BA} = \mathbf{I}$.
(iv) (right inverse) There is a matrix $\mathbf{B}$ such that $\mathbf{AB} = \mathbf{I}$.
(v) The reduced row-echelon form of $\mathbf{A}$ is the identity matrix.
(vi) $\mathbf{A}$ can be expressed as a product of elementary matrices.
(vii) The homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution.
(viii) For every $\mathbf{b} \in \mathbb{R}^n$, the system $\mathbf{Ax} = \mathbf{b}$ has a unique solution.
(ix) The determinant of $\mathbf{A}$ is nonzero, $\det(\mathbf{A}) \neq 0$.
(x) The columns of $\mathbf{A}$ are linearly independent; the rows of $\mathbf{A}$ are linearly independent.
(xi) The columns of $\mathbf{A}$ span $\mathbb{R}^n$; the rows of $\mathbf{A}$ span $\mathbb{R}^n$.
(xii) $\text{rank}(\mathbf{A}) = n$, that is, $\mathbf{A}$ is of full rank.
(xiii) $\text{nullity}(\mathbf{A}) = 0$.

In (x) and (xi), the statement about the columns and the statement about the rows are each equivalent to all the other statements.

Proof. The equivalence of (i)–(ix) is established in Section 2.6. We show that each of the remaining statements is equivalent to (xii), and that (xii) is equivalent to (vii). Recall that $\text{rank}(\mathbf{A}) = \dim(\text{Col}(\mathbf{A})) = \dim(\text{Row}(\mathbf{A}))$, where $\text{Col}(\mathbf{A})$ is spanned by the n columns and $\text{Row}(\mathbf{A})$ by the n rows of $\mathbf{A}$, both subspaces of $\mathbb{R}^n$. For any n vectors $\mathbf{u}_1, ..., \mathbf{u}_n \in \mathbb{R}^n$ with span V, we have:

- $\mathbf{u}_1, ..., \mathbf{u}_n$ are linearly independent if and only if $\dim(V) = n$, since a maximal linearly independent subsystem of them is a basis for V (Section 3.6);
- $V = \mathbb{R}^n$ if and only if $\dim(V) = n$. Indeed, $\dim(\mathbb{R}^n) = n$; conversely, if $\dim(V) = n$, a basis for V is a linearly independent set in $\mathbb{R}^n$, so by the Basis Extension Lemma (Section 3.7) it is part of a basis for $\mathbb{R}^n$; as every basis for $\mathbb{R}^n$ has n vectors (Section 3.6), it is itself a basis for $\mathbb{R}^n$, and $V = \mathbb{R}^n$.

Applied to the columns and to the rows of $\mathbf{A}$, these two facts give (x) $\Leftrightarrow$ (xii) and (xi) $\Leftrightarrow$ (xii). Next, (vii) says that $\text{Null}(\mathbf{A}) = \{\mathbf{0}\}$, which holds if and only if $\text{nullity}(\mathbf{A}) = 0$; thus (vii) $\Leftrightarrow$ (xiii). Finally, (xii) $\Leftrightarrow$ (xiii) by the rank–nullity theorem $\text{rank}(\mathbf{A}) + \text{nullity}(\mathbf{A}) = n$. □

When $\mathbf{A}$ is not square, a matrix of full rank satisfies either $\text{rank}(\mathbf{A}) = n < m$ or $\text{rank}(\mathbf{A}) = m < n$, and in each case part of the list above survives. A *left inverse* of an $m \times n$ matrix $\mathbf{A}$ is an $n \times m$ matrix $\mathbf{B}$ with $\mathbf{BA} = \mathbf{I}_n$; a *right inverse* is an $n \times m$ matrix $\mathbf{C}$ with $\mathbf{AC} = \mathbf{I}_m$.

Lemma. For every $m \times n$ matrix $\mathbf{A}$, $\text{Null}(\mathbf{A}) = \text{Null}(\mathbf{A}^T\mathbf{A})$.

Proof. If $\mathbf{Au} = \mathbf{0}$, then $\mathbf{A}^T\mathbf{Au} = \mathbf{A}^T\mathbf{0} = \mathbf{0}$; hence $\text{Null}(\mathbf{A}) \subseteq \text{Null}(\mathbf{A}^T\mathbf{A})$. Conversely, let $\mathbf{A}^T\mathbf{Au} = \mathbf{0}$. Then

$$\|\mathbf{Au}\|^2 = (\mathbf{Au}) \cdot (\mathbf{Au}) = (\mathbf{Au})^T(\mathbf{Au}) = \mathbf{u}^T(\mathbf{A}^T\mathbf{Au}) = \mathbf{u}^T\mathbf{0} = 0,$$

so $\mathbf{Au} = \mathbf{0}$. Hence $\text{Null}(\mathbf{A}^T\mathbf{A}) \subseteq \text{Null}(\mathbf{A})$. □

Theorem (Full rank equal to the number of columns)**.** Let $\mathbf{A}$ be an $m \times n$ matrix. The following statements are equivalent.

(i) $\text{rank}(\mathbf{A}) = n$, that is, $\mathbf{A}$ is of full rank and its rank equals the number of its columns.
(ii) The rows of $\mathbf{A}$ span $\mathbb{R}^n$, that is, $\text{Row}(\mathbf{A}) = \mathbb{R}^n$.
(iii) The columns of $\mathbf{A}$ are linearly independent.
(iv) The homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution, that is, $\text{Null}(\mathbf{A}) = \{\mathbf{0}\}$.
(v) $\mathbf{A}^T\mathbf{A}$ is an invertible matrix of order n.
(vi) $\mathbf{A}$ has a left inverse.

Proof. (i) $\Leftrightarrow$ (ii): $\text{Row}(\mathbf{A})$ is a subspace of $\mathbb{R}^n$ of dimension $\text{rank}(\mathbf{A})$, and a subspace of $\mathbb{R}^n$ coincides with $\mathbb{R}^n$ if and only if its dimension is n (see the proof of the preceding theorem).

(i) $\Leftrightarrow$ (iii): $\text{rank}(\mathbf{A})$ is the dimension of the span of the n columns of $\mathbf{A}$, and it equals n if and only if these columns are linearly independent (ibid.).

(iii) $\Leftrightarrow$ (iv): if $\mathbf{a}_1, ..., \mathbf{a}_n$ are the columns of $\mathbf{A}$ and $\mathbf{c} = (c_1, ..., c_n)^T$, then $c_1\mathbf{a}_1 + ... + c_n\mathbf{a}_n = \mathbf{Ac}$.

(iv) ⇒ (v): by the lemma, $\text{Null}(\mathbf{A}^T\mathbf{A}) = \text{Null}(\mathbf{A}) = \{\mathbf{0}\}$, so the square matrix $\mathbf{A}^T\mathbf{A}$ of order n is invertible by the equivalent statements for invertibility.

(v) ⇒ (vi): put $\mathbf{B} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$. Then $\mathbf{B}$ is a left inverse of $\mathbf{A}$, since $\mathbf{BA} = (\mathbf{A}^T\mathbf{A})^{-1}(\mathbf{A}^T\mathbf{A}) = \mathbf{I}_n$.

(vi) ⇒ (i): if $\mathbf{BA} = \mathbf{I}_n$, then by Proposition 2 of Properties of Rank, $n = \text{rank}(\mathbf{I}_n) = \text{rank}(\mathbf{BA}) \leq \text{rank}(\mathbf{A}) \leq n$. □

Example (A left inverse)**.** The matrix $\mathbf{A}_1$ of the example above has rank 3, the number of its columns. Accordingly,

$$\mathbf{A}_1^T\mathbf{A}_1 = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \quad (\mathbf{A}_1^T\mathbf{A}_1)^{-1} = \frac{1}{4}\begin{pmatrix} 3 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 3 \end{pmatrix},$$

and the matrix

$$\mathbf{L} = (\mathbf{A}_1^T\mathbf{A}_1)^{-1}\mathbf{A}_1^T = \frac{1}{4}\begin{pmatrix} 3 & 1 & -1 & 1 \\ -2 & 2 & 2 & -2 \\ 1 & -1 & 1 & 3 \end{pmatrix}$$

satisfies $\mathbf{LA}_1 = \mathbf{I}_3$, as is checked directly.

Theorem (Full rank equal to the number of rows)**.** Let $\mathbf{A}$ be an $m \times n$ matrix. The following statements are equivalent.

(i) $\text{rank}(\mathbf{A}) = m$, that is, $\mathbf{A}$ is of full rank and its rank equals the number of its rows.
(ii) The columns of $\mathbf{A}$ span $\mathbb{R}^m$, that is, $\text{Col}(\mathbf{A}) = \mathbb{R}^m$.
(iii) The rows of $\mathbf{A}$ are linearly independent.
(iv) The linear system $\mathbf{Ax} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^m$.
(v) $\mathbf{AA}^T$ is an invertible matrix of order m.
(vi) $\mathbf{A}$ has a right inverse.

Proof. Apply the preceding theorem to the $n \times m$ matrix $\mathbf{A}^T$, which has m columns. Here $\text{rank}(\mathbf{A}^T) = \text{rank}(\mathbf{A})$; the rows of $\mathbf{A}^T$ are the transposed columns of $\mathbf{A}$, so $\text{Row}(\mathbf{A}^T) = \mathbb{R}^m$ means $\text{Col}(\mathbf{A}) = \mathbb{R}^m$; the columns of $\mathbf{A}^T$ are the transposed rows of $\mathbf{A}$; $(\mathbf{A}^T)^T\mathbf{A}^T = \mathbf{AA}^T$; and $\mathbf{CA}^T = \mathbf{I}_m$ if and only if $\mathbf{AC}^T = \mathbf{I}_m$, so $\mathbf{A}^T$ has a left inverse if and only if $\mathbf{A}$ has a right inverse. This proves that (i), (ii), (iii), (v) and (vi) are equivalent. Finally, (ii) ⇔ (iv), because the system $\mathbf{Ax} = \mathbf{b}$ is consistent exactly when $\mathbf{b} \in \text{Col}(\mathbf{A})$ (Section 4.1). □

Transposing the left inverse $(\mathbf{AA}^T)^{-1}\mathbf{A}$ of $\mathbf{A}^T$ given by the proof, we obtain an explicit right inverse of $\mathbf{A}$: $\mathbf{A}^T(\mathbf{AA}^T)^{-1}$. Indeed, $\mathbf{AA}^T(\mathbf{AA}^T)^{-1} = \mathbf{I}_m$.

Example (A right inverse)**.** The matrix $\mathbf{A}_2$ of the example above has rank 2, the number of its rows. Here

$$\mathbf{A}_2\mathbf{A}_2^T = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}, \quad (\mathbf{A}_2\mathbf{A}_2^T)^{-1} = \frac{1}{6}\begin{pmatrix} 2 & -2 \\ -2 & 5 \end{pmatrix},$$

and

$$\mathbf{A}_2^T(\mathbf{A}_2\mathbf{A}_2^T)^{-1} = \frac{1}{6}\begin{pmatrix} 2 & -2 \\ 2 & 1 \\ -2 & 5 \end{pmatrix}$$

is a right inverse of $\mathbf{A}_2$: multiplying $\mathbf{A}_2$ by it gives $\mathbf{I}_2$.

Remark (A left inverse does not produce solutions)**.** Let $\operatorname{rank}(\mathbf{A}) = n < m$, and let $\mathbf{L}$ be a left inverse of $\mathbf{A}$. If the system $\mathbf{Ax} = \mathbf{b}$ is consistent, then its solution is unique (as $\operatorname{Null}(\mathbf{A}) = \{\mathbf{0}\}$; see General Solution of a Linear System above), and multiplying $\mathbf{Ax} = \mathbf{b}$ on the left by $\mathbf{L}$ shows that this solution is $\mathbf{x} = \mathbf{Lb}$. The argument presupposes that a solution exists; it does not make $\mathbf{Lb}$ a solution for every $\mathbf{b}$, and it cannot, since by the last theorem ($\operatorname{rank}(\mathbf{A}) = n < m$) the system is inconsistent for some $\mathbf{b} \in \mathbb{R}^m$. For the matrix $\mathbf{A}_1$ and its left inverse $\mathbf{L}$ above, $\mathbf{b} = (1, 3, 1, -1)^T$ gives $\mathbf{Lb} = (1, 2, -1)^T$, and indeed $\mathbf{A}_1(1, 2, -1)^T = \mathbf{b}$; but $\mathbf{Le}_1 = (3/4, -1/2, 1/4)^T$, while $\mathbf{A}_1\mathbf{Le}_1 = (3/4, 1/4, -1/4, 1/4)^T \neq \mathbf{e}_1$, so the system $\mathbf{A}_1\mathbf{x} = \mathbf{e}_1$ is inconsistent. Note also that a matrix with both a left and a right inverse has $\operatorname{rank}(\mathbf{A}) = n$ and $\operatorname{rank}(\mathbf{A}) = m$, so it must be square.

Chapter 5. Orthogonality, Projection, and Least Squares

5.1. Orthogonality

Orthogonality of Vectors

An inner product makes it possible to introduce the general notion of orthogonality of vectors: $\mathbf{x}$ and $\mathbf{y}$ are called *orthogonal* if $\mathbf{x} \cdot \mathbf{y} = 0$. Notation: $\mathbf{x} \perp \mathbf{y}$.

In Sections 5.1–5.3 the inner product is written as the dot product $\mathbf{x} \cdot \mathbf{y}$ of vectors in $\mathbb{R}^n$. Unless the text says otherwise, the definitions and results hold verbatim in any space with an inner product $\langle \mathbf{x}, \mathbf{y} \rangle$, real (a *Euclidean* space, see Section 3.2) or complex (a *unitary* space), once $\mathbf{x} \cdot \mathbf{y}$ is replaced by $\langle \mathbf{x}, \mathbf{y} \rangle$ and $\mathbb{R}^n$ by that space.

Note that in one and the same space an inner product can be introduced in many different ways (for $\mathbb{R}^n$, see Section 3.2), and vectors that are orthogonal with respect to one inner product may fail to be orthogonal with respect to another inner product.

In a Euclidean space, one can also introduce the general notion of the *angle* $\theta = \theta(\mathbf{x}, \mathbf{y})$ between vectors $\mathbf{x}$ and $\mathbf{y}$. By definition,

$$\cos\theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}.$$

It should be noted that the right-hand side does not exceed 1 in modulus (by virtue of the Cauchy–Bunyakovsky–Schwarz inequality). For orthogonal vectors, $\theta = \frac{\pi}{2}$. For an obvious reason, this definition of angle *does not carry over* to the case of unitary spaces. However, the notion of orthogonality works, of course, there as well.

> **Theorem** (Pythagorean Theorem)**.** If $\mathbf{x} \perp \mathbf{y}$, then $\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$.

Proof. Let $\mathbf{x} \cdot \mathbf{y} = 0$. Then $(\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \mathbf{x} \cdot \mathbf{x} + \mathbf{x} \cdot \mathbf{y} + \mathbf{y} \cdot \mathbf{x} + \mathbf{y} \cdot \mathbf{y} = \mathbf{x} \cdot \mathbf{x} + \mathbf{y} \cdot \mathbf{y}$. □

> **Remark.** In a Euclidean (but not in a unitary!) space, the Pythagorean theorem can be inverted: if $\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$, then $\mathbf{x} \cdot \mathbf{y} = 0$—this is obvious since $\mathbf{x} \cdot \mathbf{y} + \mathbf{y} \cdot \mathbf{x} = 2\mathbf{x} \cdot \mathbf{y}$. However, in a unitary space the equality $\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{x} \rangle = 2\langle \mathbf{x}, \mathbf{y} \rangle$ is not true for arbitrary vectors.

Orthogonal Vectors and Orthogonal Sets ADDED FOR MA1522

Since $\mathbf{0} \cdot \mathbf{v} = 0$ for every $\mathbf{v} \in \mathbb{R}^n$, the zero vector is orthogonal to every vector, itself included. If $\mathbf{u} \neq \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$, then $\mathbf{u} \cdot \mathbf{v} = 0$ means $\cos\theta = 0$, that is, $\theta = \pi/2$. Thus two vectors of $\mathbb{R}^n$ are orthogonal if and only if one of them is the zero vector or they are perpendicular. Clearly, if $\mathbf{u} \perp \mathbf{v}$, then $s\mathbf{u} \perp t\mathbf{v}$ for all $s, t \in \mathbb{R}$.

Recall (Section 3.2) that a *unit vector* is a vector of norm 1, and that for $\mathbf{u} \neq \mathbf{0}$ the unit vector $\mathbf{u}/\|\mathbf{u}\|$ is obtained by *normalizing* $\mathbf{u}$.

Definition. A set $S = \{\mathbf{v}_1, ..., \mathbf{v}_k\}$ of vectors in $\mathbb{R}^n$ is *orthogonal* if its vectors are pairwise orthogonal, $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ for all $i \neq j$. It is *orthonormal* if, for all $i, j = 1, ..., k$,

$$\mathbf{v}_i \cdot \mathbf{v}_j = \begin{cases} 0, & i \neq j, \\ 1, & i = j, \end{cases}$$

that is, if S is orthogonal and consists of unit vectors.

An orthogonal set in this sense may contain the zero vector; an orthonormal set cannot, since $\|\mathbf{0}\| = 0$. Normalizing each vector of an orthogonal set of nonzero vectors produces an orthonormal set (multiplication by scalars preserves orthogonality); an orthogonal set that contains $\mathbf{0}$ cannot be normalized. The book's own definition in Section 5.2 requires the vectors of an orthogonal set to be nonzero; where this matters we say "an orthogonal set of nonzero vectors".

Example. The set $\{(1,2,2)^T, (2,1,-2)^T, (2,-2,1)^T\}$ is orthogonal: the three pairwise dot products are $2+2-4$, $2-4+2$ and $4-2-2$, all equal to 0. It is not orthonormal, since each of its vectors has norm 3; dividing every vector by 3 gives an orthonormal set. The set $\{(1,3)^T, (-3,1)^T, \mathbf{0}\}$ in $\mathbb{R}^2$ is orthogonal, but it cannot be normalized.

Orthogonality of Sets

Let $L \subset \mathbb{R}^n$ be an arbitrary nonempty subset of vectors. A vector $\mathbf{x} \in \mathbb{R}^n$ is said to be *orthogonal* to the set L if $\mathbf{x} \cdot \mathbf{y} = 0$ for all $\mathbf{y} \in L$. Notation: $\mathbf{x} \perp L$. By definition, sets L and M are *orthogonal* if $\mathbf{x} \cdot \mathbf{y} = 0$ for any $\mathbf{x} \in L$ and $\mathbf{y} \in M$. Notation: $L \perp M$.

The set M of all vectors in $\mathbb{R}^n$, each of which is orthogonal to a given set L, is called its *orthogonal complement* in $\mathbb{R}^n$. Notation: $M = L^\perp$.

Proposition. For any set L, its orthogonal complement $L^\perp$ is a subspace. Moreover, $L \subset (L^\perp)^\perp$.

Proof. Let $\mathbf{x}, \mathbf{y} \in L^\perp$. Then $\mathbf{x} \cdot \mathbf{z} = \mathbf{y} \cdot \mathbf{z} = 0 \quad \forall \mathbf{z} \in L \Rightarrow (\alpha\mathbf{x} + \beta\mathbf{y}) \cdot \mathbf{z} = \alpha(\mathbf{x} \cdot \mathbf{z}) + \beta(\mathbf{y} \cdot \mathbf{z}) = 0 \quad \forall \mathbf{z} \in L \Rightarrow \alpha\mathbf{x} + \beta\mathbf{y} \in L^\perp$.

By definition, the set $(L^\perp)^\perp$ contains all vectors orthogonal to $L^\perp$, and hence all vectors from the set L. □

Orthogonality to a Subspace; $V^\perp = \text{Null}(\mathbf{A}^T)$ ADDED FOR MA1522
Let V be a subspace of $\mathbb{R}^n$. In the notation above, a vector $\mathbf{w}$ is *orthogonal to* V, $\mathbf{w} \perp V$, if $\mathbf{w} \cdot \mathbf{v} = 0$ for every $\mathbf{v} \in V$, and the *orthogonal complement* of V is

$$V^\perp = \{\mathbf{w} \in \mathbb{R}^n : \mathbf{w} \cdot \mathbf{v} = 0 \text{ for all } \mathbf{v} \in V\}.$$

The zero vector is orthogonal to every subspace. For example, if $\mathbf{a} \in \mathbb{R}^n$ is a nonzero vector, the solution set $V = \{\mathbf{v} \in \mathbb{R}^n : \mathbf{v} \cdot \mathbf{a} = 0\}$ of the single homogeneous equation $a_1v_1 + ... + a_nv_n = 0$ is a subspace, called a *hyperplane* (a plane through the origin if $n = 3$), and $\mathbf{a} \perp V$; the vector $\mathbf{a}$ is a *normal vector* of the hyperplane. Thus the plane $ax + by + cz = 0$ in $\mathbb{R}^3$ consists of the vectors orthogonal to its normal vector $(a, b, c)^T$.

Theorem (Orthogonality Test)**.** Let $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ be a spanning set for a subspace V of $\mathbb{R}^n$, $\text{span}(S) = V$. Then $\mathbf{w} \perp V$ if and only if $\mathbf{w} \cdot \mathbf{u}_i = 0$ for $i = 1, ..., k$.

Proof. If $\mathbf{w} \perp V$, then $\mathbf{w} \cdot \mathbf{u}_i = 0$ because $\mathbf{u}_i \in V$. Conversely, let $\mathbf{w} \cdot \mathbf{u}_i = 0$ for all i. Each $\mathbf{v} \in V$ can be written as $\mathbf{v} = c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k$, and then $\mathbf{w} \cdot \mathbf{v} = c_1(\mathbf{w} \cdot \mathbf{u}_1) + ... + c_k(\mathbf{w} \cdot \mathbf{u}_k) = 0$. □

So, to check that $\mathbf{w} \perp V$, it suffices to check that $\mathbf{w}$ is orthogonal to each vector of a spanning set (for example, of a basis) of V. These k conditions form one homogeneous linear system.

Theorem ($V^\perp = \text{Null}(\mathbf{A}^T)$)**.** Let $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ be a spanning set for a subspace V of $\mathbb{R}^n$, and let $\mathbf{A} = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \end{pmatrix}$ be the $n \times k$ matrix with these columns. Then

$$\mathbf{w} \perp V \quad \Leftrightarrow \quad \mathbf{A}^T\mathbf{w} = \mathbf{0} \quad \Leftrightarrow \quad \mathbf{w} \in \text{Null}(\mathbf{A}^T).$$

Consequently, the orthogonal complement of V is $V^\perp = \text{Null}(\mathbf{A}^T)$.

Proof. By block multiplication,

$$\mathbf{A}^T\mathbf{w} = \begin{pmatrix} \mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_k^T \end{pmatrix} \mathbf{w} = \begin{pmatrix} \mathbf{u}_1^T\mathbf{w} \\ \vdots \\ \mathbf{u}_k^T\mathbf{w} \end{pmatrix} = \begin{pmatrix} \mathbf{u}_1 \cdot \mathbf{w} \\ \vdots \\ \mathbf{u}_k \cdot \mathbf{w} \end{pmatrix},$$

so $\mathbf{A}^T\mathbf{w} = \mathbf{0}$ if and only if $\mathbf{w} \cdot \mathbf{u}_i = 0$ for all i, that is, by the orthogonality test, if and only if $\mathbf{w} \perp V$. The set of all such $\mathbf{w}$ is $V^\perp$. □

Algorithm (Orthogonal complement of a subspace).

1. Choose a spanning set $\{\mathbf{u}_1, ..., \mathbf{u}_k\}$ of V (for example, a basis) and form the $k \times n$ matrix $\mathbf{A}^T$ whose rows are $\mathbf{u}_1^T, ..., \mathbf{u}_k^T$.
2. Reduce $\mathbf{A}^T$ to RREF and write the general solution of the homogeneous system $\mathbf{A}^T\mathbf{w} = \mathbf{0}$.
3. The vectors multiplying the parameters in the general solution form a basis for $V^\perp = \mathrm{Null}(\mathbf{A}^T)$. A given vector $\mathbf{w}$ is orthogonal to V exactly when $\mathbf{A}^T\mathbf{w} = \mathbf{0}$.

Example. Let $V = \mathrm{span}\{(1,1,0,2)^T, (1,2,1,3)^T\} \subset \mathbb{R}^4$. A vector $\mathbf{w} = (w_1, w_2, w_3, w_4)^T$ is orthogonal to V if and only if $\mathbf{A}^T\mathbf{w} = \mathbf{0}$, where

$$\mathbf{A}^T = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 1 & 2 & 1 & 3 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}.$$

Hence $w_1 = s - t$, $w_2 = -s - t$, $w_3 = s$, $w_4 = t$ with $s, t \in \mathbb{R}$ arbitrary, and

$$V^\perp = \mathrm{span}\{(1,-1,1,0)^T, (-1,-1,0,1)^T\}.$$

Corollary. For every $m \times n$ matrix $\mathbf{A}$,

$$\mathrm{Row}(\mathbf{A})^\perp = \mathrm{Null}(\mathbf{A}), \quad \mathrm{Col}(\mathbf{A})^\perp = \mathrm{Null}(\mathbf{A}^T).$$

Proof. The columns of $\mathbf{A}$ span $\mathrm{Col}(\mathbf{A})$, so the theorem gives $\mathrm{Col}(\mathbf{A})^\perp = \mathrm{Null}(\mathbf{A}^T)$. The rows of $\mathbf{A}$ are the columns of $\mathbf{A}^T$, so $\mathrm{Row}(\mathbf{A}) = \mathrm{Col}(\mathbf{A}^T)$ and $\mathrm{Row}(\mathbf{A})^\perp = \mathrm{Null}\Big((\mathbf{A}^T)^T\Big) = \mathrm{Null}(\mathbf{A})$. □

Orthogonal Sum of Subspaces

Recall that the *sum of subspaces* $L_1, L_2, ..., L_m$ is the set L of all vectors of the form $\mathbf{x} = \mathbf{x}_1 + \mathbf{x}_2 + ... + \mathbf{x}_m$, where $\mathbf{x}_i \in L_i$ for all i (see Tyrtyshnikov, §11.6). It is elementary to verify that L is a subspace. Notation: $L = L_1 + ... + L_m$.

Recall also that L is called a *direct sum* (see Tyrtyshnikov, §12.4) if the subspaces L_i are nonzero and each vector $\mathbf{x} \in L$ has a unique decomposition of the form $\mathbf{x} = \mathbf{x}_1 + ... + \mathbf{x}_m$, where $\mathbf{x}_i \in L_i$ (if $\mathbf{x} = \mathbf{x}_1' + ... + \mathbf{x}_m'$ and $\mathbf{x}_i' \in L_i \quad \forall i$, then necessarily $\mathbf{x}_i' = \mathbf{x}_i \quad \forall i$).

A sum $L = L_1 + ... + L_m$ of nonzero subspaces is called an *orthogonal sum* if $L_i \perp L_j$ for $i \neq j$. Notation: $L = L_1 \oplus ... \oplus L_m$.

Proposition. An orthogonal sum of subspaces $L = L_1 \oplus ... \oplus L_m$ is a direct sum. Furthermore, if $\mathbf{x}_i \in L_i$, then

$$\|\mathbf{x}_1 + ... + \mathbf{x}_m\|^2 = \|\mathbf{x}_1\|^2 + ... + \|\mathbf{x}_m\|^2. \qquad (*)$$

Proof. Let us first prove $(*)$. Taking into account that $\mathbf{x}_i \cdot \mathbf{x}_j = 0$ for $i \neq j$, we find

$$\|\mathbf{x}_1 + ... + \mathbf{x}_m\|^2 = \sum_{i=1}^{m}\sum_{j=1}^{m} \mathbf{x}_i \cdot \mathbf{x}_j = \sum_{i=1}^{m} \mathbf{x}_i \cdot \mathbf{x}_i = \sum_{i=1}^{m} \|\mathbf{x}_i\|^2.$$

Next, suppose $\mathbf{x}_1 + ... + \mathbf{x}_m = \mathbf{x}'_1 + ... + \mathbf{x}'_m$, where $\mathbf{x}_i, \mathbf{x}'_i \in L_i \quad \forall i$. Then

$$0 = \|(\mathbf{x}_1 - \mathbf{x}'_1) + ... + (\mathbf{x}_m - \mathbf{x}'_m)\|^2 = \|\mathbf{x}_1 - \mathbf{x}'_1\|^2 + ... + \|\mathbf{x}_m - \mathbf{x}'_m\|^2$$
$$\Rightarrow \quad \mathbf{x}_i = \mathbf{x}'_i \quad \forall i. \qquad \square$$

Corollary 1. A finite set of nonzero pairwise orthogonal vectors is linearly independent.

Proof. Suppose the vectors $\mathbf{x}_1, ..., \mathbf{x}_m$ are pairwise orthogonal and nonzero. Then the sum of the spans $\text{span}\{\mathbf{x}_1\}, ..., \text{span}\{\mathbf{x}_m\}$ is an orthogonal sum and, if $\alpha_1\mathbf{x}_1 + ... + \alpha_m\mathbf{x}_m = \mathbf{0}$, then according to $(*)$

$$0 = \|\alpha_1\mathbf{x}_1 + ... + \alpha_m\mathbf{x}_m\|^2 = |\alpha_1|^2\|\mathbf{x}_1\|^2 + ... + |\alpha_m|^2\|\mathbf{x}_m\|^2$$
$$\Rightarrow \quad \alpha_1 = ... = \alpha_m = 0. \qquad \square$$

Corollary 2. If nonzero subspaces $L_1, ..., L_m$ are finite-dimensional and pairwise orthogonal, then

$$\dim(L_1 \oplus ... \oplus L_m) = \dim(L_1) + ... + \dim(L_m).$$

It suffices to recall that for a direct sum of finite-dimensional subspaces L_i, a basis is obtained as the union of bases for the subspaces L_i (see Tyrtyshnikov, §12.4).

5.2. Orthogonal and Orthonormal Bases

Orthogonal and Orthonormal Sets

A set $S = \{\mathbf{x}_1, ..., \mathbf{x}_n\}$ of nonzero vectors is called *orthogonal* if

$$\mathbf{x}_i \cdot \mathbf{x}_j = 0, \quad i \neq j, \qquad (*)$$

and *orthonormal* if, in addition, $\|\mathbf{x}_1\| = ... = \|\mathbf{x}_n\| = 1$.[6] Thus, the Gram matrix (see Section 5.3) for an orthogonal set is diagonal with positive diagonal entries, and for an orthonormal set it is the identity matrix.

Consider the space $\mathbb{C}^n$ with the natural inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^*\mathbf{x}$ (real case: $\mathbb{R}^n$ with the dot product, and $\mathbf{X}^T$ in place of $\mathbf{X}^*$ below) and an orthonormal set of column vectors:

$$\mathbf{x}_1 = \begin{pmatrix} x_{11} \\ \vdots \\ x_{n1} \end{pmatrix}, \quad ..., \quad \mathbf{x}_n = \begin{pmatrix} x_{1n} \\ \vdots \\ x_{nn} \end{pmatrix} \in \mathbb{C}^n.$$

Let us form from them the $n \times n$ matrix

[6] In MA1522 an orthogonal set may also contain the zero vector; here, as in the definition above, the vectors of an orthogonal set are nonzero.

$$\mathbf{X} = (\mathbf{x}_1 \ \cdots \ \mathbf{x}_n) = \begin{pmatrix} x_{11} & \dots & x_{1n} \\ \dots & \dots & \dots \\ x_{n1} & \dots & x_{nn} \end{pmatrix}$$

and note that the relations $(*)$ (together with normalization) are equivalent to the matrix equality

$$\mathbf{X}^*\mathbf{X} = \begin{pmatrix} \langle \mathbf{x}_1, \mathbf{x}_1 \rangle & \dots & \langle \mathbf{x}_n, \mathbf{x}_1 \rangle \\ \dots & \dots & \dots \\ \langle \mathbf{x}_1, \mathbf{x}_n \rangle & \dots & \langle \mathbf{x}_n, \mathbf{x}_n \rangle \end{pmatrix} = \mathbf{I}.$$

A matrix $\mathbf{X} \in \mathbb{C}^{n \times n}$ with the property $\mathbf{X}^*\mathbf{X} = \mathbf{I}$ is called *unitary*. Thus, any square matrix whose columns form an orthonormal set is unitary, and the columns of any unitary matrix form an orthonormal set. It is also clear that for a matrix to be unitary it is necessary and sufficient that its rows form an orthonormal set (prove this!).

A real unitary matrix is called *orthogonal*. In the next subsection we note that the set of all orthogonal matrices of order n is a group under the operation of matrix multiplication. The same holds with respect to the set of all unitary matrices of order n.

Orthogonal and Orthonormal Bases; Coordinates ADDED FOR MA1522

By Corollary 1 in Section 5.1 (Orthogonal Sum of Subspaces), *an orthogonal set of nonzero vectors is linearly independent*. Here is a direct argument. Let $\mathbf{u}_1, ..., \mathbf{u}_k$ be nonzero and pairwise orthogonal, and let $c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k = \mathbf{0}$. Taking the dot product with $\mathbf{u}_i$, we get

$$0 = \mathbf{u}_i \cdot (c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k) = c_i(\mathbf{u}_i \cdot \mathbf{u}_i) = c_i\|\mathbf{u}_i\|^2, \tag{1}$$

so $c_i = 0$ for every i. In particular, *every orthonormal set is linearly independent*. (An orthogonal set that contains $\mathbf{0}$ is, of course, linearly dependent.)

Definition. Let V be a subspace of $\mathbb{R}^n$. A set S is an *orthogonal basis* (an *orthonormal basis*) for V if S is a basis for V and S is an orthogonal (an orthonormal) set.

Since linear independence is automatic, any orthogonal set of k nonzero vectors in a subspace V of dimension k is an orthogonal basis for V (by the Basis Extension Lemma of Section 3.7, as every basis for V has k vectors). For example, the standard basis $E = \{\mathbf{e}_1, ..., \mathbf{e}_n\}$ is an orthonormal basis for $\mathbb{R}^n$. The coordinates of a vector relative to an orthogonal basis are found by dot products alone, without solving a linear system.

Theorem (Coordinates Relative to an Orthogonal Basis)**.** Let $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ be an orthogonal basis for a subspace V of $\mathbb{R}^n$. Then for every $\mathbf{v} \in V$

$$\mathbf{v} = \frac{\mathbf{v}\cdot\mathbf{u}_1}{\|\mathbf{u}_1\|^2}\mathbf{u}_1 + \frac{\mathbf{v}\cdot\mathbf{u}_2}{\|\mathbf{u}_2\|^2}\mathbf{u}_2 + \ldots + \frac{\mathbf{v}\cdot\mathbf{u}_k}{\|\mathbf{u}_k\|^2}\mathbf{u}_k, \tag{2}$$

that is, $[\mathbf{v}]_S = \left((\mathbf{v}\cdot\mathbf{u}_1)/\|\mathbf{u}_1\|^2, \ldots, (\mathbf{v}\cdot\mathbf{u}_k)/\|\mathbf{u}_k\|^2\right)^T$. If S is an orthonormal basis, then

$$\mathbf{v} = (\mathbf{v}\cdot\mathbf{u}_1)\mathbf{u}_1 + (\mathbf{v}\cdot\mathbf{u}_2)\mathbf{u}_2 + \ldots + (\mathbf{v}\cdot\mathbf{u}_k)\mathbf{u}_k, \quad [\mathbf{v}]_S = \begin{pmatrix} \mathbf{v}\cdot\mathbf{u}_1 \\ \vdots \\ \mathbf{v}\cdot\mathbf{u}_k \end{pmatrix}. \tag{3}$$

Proof. Write $\mathbf{v} = c_1\mathbf{u}_1 + \ldots + c_k\mathbf{u}_k$. Exactly as in (1), $\mathbf{v}\cdot\mathbf{u}_i = c_i\|\mathbf{u}_i\|^2$, whence $c_i = (\mathbf{v}\cdot\mathbf{u}_i)/\|\mathbf{u}_i\|^2$. For an orthonormal basis, $\|\mathbf{u}_i\| = 1$. □

For the standard basis, $\mathbf{v}\cdot\mathbf{e}_i = v_i$, and (3) gives $[\mathbf{v}]_E = \mathbf{v}$. The formulas use the orthogonality of S in an essential way: for the basis $S = \{(1,0)^T, (1,1)^T\}$ of $\mathbb{R}^2$ (which is not orthogonal) and $\mathbf{w} = (0,1)^T$, the right-hand side of (2) equals $0(1,0)^T + \frac{1}{2}(1,1)^T = (1/2, 1/2)^T \neq \mathbf{w}$.

Example. The vectors $\mathbf{u}_1 = (1,1,1)^T$ and $\mathbf{u}_2 = (1,-2,1)^T$ are orthogonal, so $S = \{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis for $V = \operatorname{span}(S)$, the plane $x = z$ in $\mathbb{R}^3$. For $\mathbf{v} = (4,-1,4)^T \in V$ we have $\mathbf{v}\cdot\mathbf{u}_1 = 7$, $\|\mathbf{u}_1\|^2 = 3$, $\mathbf{v}\cdot\mathbf{u}_2 = 10$, $\|\mathbf{u}_2\|^2 = 6$; hence

$$\mathbf{v} = \frac{7}{3}\mathbf{u}_1 + \frac{5}{3}\mathbf{u}_2, \quad [\mathbf{v}]_S = \begin{pmatrix} 7/3 \\ 5/3 \end{pmatrix}.$$

For the orthonormal basis $T = \left\{\mathbf{u}_1/\sqrt{3}, \mathbf{u}_2/\sqrt{6}\right\}$ of the same plane, (3) gives $[\mathbf{v}]_T = \left(7/\sqrt{3}, 10/\sqrt{6}\right)^T$.

Cartesian Systems and Orthogonal Matrices

Let $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ and $\tilde{\mathbf{e}}_1, \tilde{\mathbf{e}}_2, \tilde{\mathbf{e}}_3$ be the basis vectors of two Cartesian coordinate systems with a common origin. Let us express the vectors of the second system as linear combinations of the vectors of the first system:

$$\begin{aligned} \tilde{\mathbf{e}}_1 &= p_{11}\mathbf{e}_1 + p_{21}\mathbf{e}_2 + p_{31}\mathbf{e}_3, \\ \tilde{\mathbf{e}}_2 &= p_{12}\mathbf{e}_1 + p_{22}\mathbf{e}_2 + p_{32}\mathbf{e}_3, \\ \tilde{\mathbf{e}}_3 &= p_{13}\mathbf{e}_1 + p_{23}\mathbf{e}_2 + p_{33}\mathbf{e}_3, \end{aligned}$$

and observe that

$$\begin{pmatrix} p_{11} & p_{21} & p_{31} \\ p_{12} & p_{22} & p_{32} \\ p_{13} & p_{23} & p_{33} \end{pmatrix} \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{pmatrix} =$$

$$= \begin{pmatrix} \tilde{\mathbf{e}}_1\cdot\tilde{\mathbf{e}}_1 & \tilde{\mathbf{e}}_1\cdot\tilde{\mathbf{e}}_2 & \tilde{\mathbf{e}}_1\cdot\tilde{\mathbf{e}}_3 \\ \tilde{\mathbf{e}}_2\cdot\tilde{\mathbf{e}}_1 & \tilde{\mathbf{e}}_2\cdot\tilde{\mathbf{e}}_2 & \tilde{\mathbf{e}}_2\cdot\tilde{\mathbf{e}}_3 \\ \tilde{\mathbf{e}}_3\cdot\tilde{\mathbf{e}}_1 & \tilde{\mathbf{e}}_3\cdot\tilde{\mathbf{e}}_2 & \tilde{\mathbf{e}}_3\cdot\tilde{\mathbf{e}}_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus, the transition matrix $\mathbf{P}$ for the bases of two Cartesian coordinate systems satisfies the matrix equality

$$\mathbf{P}^T\mathbf{P} = \mathbf{I}. \qquad (**)$$

It is also clear that if a transition matrix possesses property $(**)$, then a Cartesian system passes into a Cartesian system.

Definition. A square real matrix $\mathbf{P}$ satisfying equality $(**)$ is called *orthogonal*.

Orthogonal matrices of order 2 effect the transition between bases of Cartesian systems in the plane. Such, in particular, are the transition matrices implementing a rotation (see Tyrtyshnikov, §19.3).

The definition we have given is also applicable to matrices of order greater than 3. According to $(**)$, for orthogonal matrices of arbitrary order, inversion reduces to transposition:

$$\mathbf{P}^{-1} = \mathbf{P}^T.$$

Moreover, the product of two orthogonal matrices remains an orthogonal matrix: if $\mathbf{P}^T\mathbf{P} = \mathbf{Q}^T\mathbf{Q} = \mathbf{I}$, then $(\mathbf{PQ})^T(\mathbf{PQ}) = \mathbf{Q}^T(\mathbf{P}^T\mathbf{P})\mathbf{Q} = \mathbf{Q}^T\mathbf{IQ} = \mathbf{Q}^T\mathbf{Q} = \mathbf{I}$. Obviously, the identity matrix $\mathbf{I}$ is an orthogonal matrix. Consequently, *the set of all orthogonal matrices of a fixed order forms a group with respect to the operation of matrix multiplication.*

Proposition. Let $\mathbf{y} = \mathbf{Px}$, where $\mathbf{P}$ is an arbitrary orthogonal matrix of order n and $\mathbf{x} \in \mathbb{R}^n$. Then the sum of the squares of the entries of the column vector $\mathbf{y}$ equals the sum of the squares of the entries of the column vector $\mathbf{x}$.

Proof. $y_1^2 + ... + y_n^2 = \mathbf{y}^T\mathbf{y} = (\mathbf{Px})^T(\mathbf{Px}) = \mathbf{x}^T(\mathbf{P}^T\mathbf{P})\mathbf{x} = \mathbf{x}^T\mathbf{x} = x_1^2 + ... + x_n^2$. □

Corollary. Let $\mathbf{B} = \mathbf{PAQ}$, where $\mathbf{P}, \mathbf{Q}$ are arbitrary orthogonal matrices of order n and $\mathbf{A}$ is an arbitrary real matrix of order n. Then the sum of the squares of the entries of the matrix $\mathbf{B}$ equals the sum of the squares of the entries of the matrix $\mathbf{A}$.

Orthogonal Matrices and Orthonormal Columns ADDED FOR MA1522

Let $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ be vectors in $\mathbb{R}^n$ and $\mathbf{Q} = \begin{pmatrix}\mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k\end{pmatrix}$ the $n \times k$ matrix with these columns. By block multiplication, the (i, j)-entry of the $k \times k$ matrix $\mathbf{Q}^T\mathbf{Q}$ is $\mathbf{u}_i^T\mathbf{u}_j = \mathbf{u}_i \cdot \mathbf{u}_j$:

$$\mathbf{Q}^T\mathbf{Q} = \begin{pmatrix}\mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_k^T\end{pmatrix}\begin{pmatrix}\mathbf{u}_1 & \cdots & \mathbf{u}_k\end{pmatrix} = \begin{pmatrix}\mathbf{u}_1 \cdot \mathbf{u}_1 & \cdots & \mathbf{u}_1 \cdot \mathbf{u}_k \\ \vdots & \ddots & \vdots \\ \mathbf{u}_k \cdot \mathbf{u}_1 & \cdots & \mathbf{u}_k \cdot \mathbf{u}_k\end{pmatrix}$$

(this is the Gram matrix of $\mathbf{u}_1, ..., \mathbf{u}_k$, see Section 5.3). Hence

$$S \text{ is orthogonal} \Leftrightarrow \mathbf{Q}^T\mathbf{Q} \text{ is diagonal}, \qquad S \text{ is orthonormal} \Leftrightarrow \mathbf{Q}^T\mathbf{Q} = \mathbf{I}_k.$$

In particular, if $k = n$, then S is orthonormal if and only if $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_n$, that is, $\mathbf{Q}^T = \mathbf{Q}^{-1}$. The course takes the latter equality as the definition.

Definition. A square matrix $\mathbf{A}$ of order n is *orthogonal* if $\mathbf{A}^T = \mathbf{A}^{-1}$, equivalently, $\mathbf{A}^T\mathbf{A} = \mathbf{I} = \mathbf{A}\mathbf{A}^T$.

This agrees with the definition in the preceding subsection, which asks only for $\mathbf{A}^T\mathbf{A} = \mathbf{I}$: for a square matrix this equality already implies $\mathbf{A}^T = \mathbf{A}^{-1}$. Indeed, $\det(\mathbf{A})^2 = \det(\mathbf{A}^T)\det(\mathbf{A}) = \det(\mathbf{I}) = 1$, so $\mathbf{A}$ is invertible (Sections 2.6 and 2.10), and multiplying $\mathbf{A}^T\mathbf{A} = \mathbf{I}$ on the right by $\mathbf{A}^{-1}$ gives $\mathbf{A}^T = \mathbf{A}^{-1}$.

Theorem. For a square matrix $\mathbf{A}$ of order n the following statements are equivalent:

(i) $\mathbf{A}$ is an orthogonal matrix;
(ii) the columns of $\mathbf{A}$ form an orthonormal basis for $\mathbb{R}^n$;
(iii) the rows of $\mathbf{A}$ form an orthonormal basis for $\mathbb{R}^n$.

Proof. By the computation above (with $\mathbf{Q} = \mathbf{A}$ and $k = n$), the columns of $\mathbf{A}$ form an orthonormal set if and only if $\mathbf{A}^T\mathbf{A} = \mathbf{I}$, that is, if and only if $\mathbf{A}$ is orthogonal; and n orthonormal vectors in $\mathbb{R}^n$ are linearly independent and therefore form a basis for $\mathbb{R}^n$. This proves (i) $\Leftrightarrow$ (ii). The rows of $\mathbf{A}$ are the columns of $\mathbf{A}^T$, and $\mathbf{A}$ is orthogonal if and only if $\mathbf{A}^T$ is, since $(\mathbf{A}^T)^T\mathbf{A}^T = \mathbf{A}\mathbf{A}^T$. Applying (i) $\Leftrightarrow$ (ii) to $\mathbf{A}^T$ gives (i) $\Leftrightarrow$ (iii). □

Example. Consider the matrices

$$\mathbf{A} = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}, \quad \mathbf{Q} = \frac{1}{3}\begin{pmatrix} 1 & 2 \\ 2 & 1 \\ 2 & -2 \end{pmatrix}.$$

The columns of $\mathbf{A}$ form the orthonormal set obtained in Section 5.1, so $\mathbf{A}$ is orthogonal: $\mathbf{A}^T\mathbf{A} = \mathbf{A}\mathbf{A}^T = \mathbf{I}_3$. The columns of $\mathbf{B}$ are orthogonal, but $\mathbf{B}^T\mathbf{B} = 5\mathbf{I}_2$, so $\mathbf{B}$ is not orthogonal; matrices whose columns are merely orthogonal are given no special name. The 3×2 matrix $\mathbf{Q}$ has orthonormal columns, $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_2$, but it is not square, hence not an orthogonal matrix; indeed $\mathbf{Q}\mathbf{Q}^T \neq \mathbf{I}_3$, since $\operatorname{rank}(\mathbf{Q}\mathbf{Q}^T) \leq 2$.

Remark. Some books call a square matrix with $\mathbf{A}^T = \mathbf{A}^{-1}$ an *orthonormal matrix*; the course, like most of the literature, calls it orthogonal.

5.3. Orthogonal Projection

Gram Matrix

Suppose we are given vectors $\mathbf{v}_1, ..., \mathbf{v}_n$ in a space with an inner product $\langle \cdot, \cdot \rangle$, and let

$$\mathbf{x} = \alpha_1 \mathbf{v}_1 + ... + \alpha_n \mathbf{v}_n, \quad \mathbf{y} = \beta_1 \mathbf{v}_1 + ... + \beta_n \mathbf{v}_n.$$

Then a direct calculation yields

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{n} \overline{\beta_i} \left(\sum_{j=1}^{n} \langle \mathbf{v}_j, \mathbf{v}_i \rangle \alpha_j \right) = \mathbf{b}^* \mathbf{G} \mathbf{a}, \qquad (*)$$

where

$$\mathbf{G} = \mathbf{G}(\mathbf{v}_1, ..., \mathbf{v}_n) = \begin{pmatrix} \langle \mathbf{v}_1, \mathbf{v}_1 \rangle & ... & \langle \mathbf{v}_n, \mathbf{v}_1 \rangle \\ ... & ... & ... \\ \langle \mathbf{v}_1, \mathbf{v}_n \rangle & ... & \langle \mathbf{v}_n, \mathbf{v}_n \rangle \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_n \end{pmatrix}.$$

The matrix $\mathbf{G}$ formed from the inner products of the vectors $\mathbf{v}_1, ..., \mathbf{v}_n$ is called their *Gram matrix*.[7] In the real case (in particular, in $\mathbb{R}^n$ with the dot product), $(*)$ reads $\mathbf{x} \cdot \mathbf{y} = \mathbf{b}^T \mathbf{G} \mathbf{a}$ with $\mathbf{G} = \left(\mathbf{v}_j \cdot \mathbf{v}_i \right)$, and $\mathbf{a}^*$ below becomes $\mathbf{a}^T$.

Theorem (Gram Matrix Theorem)**.** Vectors $\mathbf{v}_1, ..., \mathbf{v}_n$ are linearly dependent if and only if their Gram matrix is not invertible.

Proof. Let $\mathbf{x} = \alpha_1 \mathbf{v}_1 + ... + \alpha_n \mathbf{v}_n$. Using $(*)$ for $\mathbf{x} = \mathbf{y}$, we find

$$\langle \mathbf{x}, \mathbf{x} \rangle = \mathbf{a}^* \mathbf{G} \mathbf{a}, \quad \mathbf{a} = (\alpha_1, ..., \alpha_n)^T. \qquad (\#)$$

If $\mathbf{G}$ is not invertible, then there exists a column $\mathbf{a} \neq \mathbf{0}$ such that $\mathbf{G}\mathbf{a} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0} \Rightarrow$ the vectors $\mathbf{v}_1, ..., \mathbf{v}_n$ are linearly dependent.

Conversely, if these vectors are linearly dependent, then $\mathbf{x} = \mathbf{0}$ for some $\mathbf{a} \neq \mathbf{0}$. It is easy to see that $\mathbf{G}\mathbf{a} = \left(\langle \mathbf{x}, \mathbf{v}_1 \rangle, ..., \langle \mathbf{x}, \mathbf{v}_n \rangle \right)^T = \mathbf{0}$ is a nontrivial linear combination of the columns of the matrix $\mathbf{G}$ that is equal to zero $\Rightarrow$ the columns of $\mathbf{G}$ are linearly dependent $\Rightarrow$ $\mathbf{G}$ is not invertible. □

Problem 1. In a space with an inner product, two sets of vectors are given: $\mathbf{u}_1, ..., \mathbf{u}_m$ and $\mathbf{v}_1, ..., \mathbf{v}_m$. Suppose that $L^\perp \cap M = \{\mathbf{0}\}$, where L and M are the spans of the vectors of the first and second sets. Prove that at least one of these sets is linearly dependent if and only if the $m \times m$ matrix $\mathbf{A}$ with entries $a_{ij} = \langle \mathbf{v}_j, \mathbf{u}_i \rangle$ is not invertible.

Orthogonal Projection and Best Approximation

Let L be a subspace of $\mathbb{R}^n$ of dimension m. We already know (see Tyrtyshnikov, §23.6) that for any $\mathbf{x} \in \mathbb{R}^n$ there exists an element of best approximation $\mathbf{z}_0 \in L$—such that $\|\mathbf{x} - \mathbf{z}_0\| \leq \|\mathbf{x} - \mathbf{z}\|$ for all $\mathbf{z} \in L$. In this special case—for a norm

[7] Note that the entry in position i, j has the form $\langle \mathbf{v}_j, \mathbf{v}_i \rangle$. Often $\mathbf{G}^T$ is called the Gram matrix (in the real case, of course, $\mathbf{G}^T = \mathbf{G}$).

induced by an inner product—uniqueness of $\mathbf{z}_0$ holds, and there is a very simple way of obtaining it.

We proceed from the assumption that a basis $\{\mathbf{z}_1, ..., \mathbf{z}_m\}$ for L is given. Then

$$\mathbf{z}_0 = \alpha_1 \mathbf{z}_1 + ... + \alpha_m \mathbf{z}_m.$$

Let us find the coefficients $\alpha_1, ..., \alpha_m$ from the condition

$$\mathbf{x} - \mathbf{z}_0 \perp L \quad \Leftrightarrow \quad (\mathbf{x} - \mathbf{z}_0) \cdot \mathbf{z}_1 = 0, ..., (\mathbf{x} - \mathbf{z}_0) \cdot \mathbf{z}_m = 0 \quad \Leftrightarrow$$

$$\Leftrightarrow \quad \begin{cases} \alpha_1(\mathbf{z}_1 \cdot \mathbf{z}_1) + ... + \alpha_m(\mathbf{z}_m \cdot \mathbf{z}_1) = \mathbf{x} \cdot \mathbf{z}_1, \\ \alpha_1(\mathbf{z}_1 \cdot \mathbf{z}_2) + ... + \alpha_m(\mathbf{z}_m \cdot \mathbf{z}_2) = \mathbf{x} \cdot \mathbf{z}_2, \\ ... \\ \alpha_1(\mathbf{z}_1 \cdot \mathbf{z}_m) + ... + \alpha_m(\mathbf{z}_m \cdot \mathbf{z}_m) = \mathbf{x} \cdot \mathbf{z}_m. \end{cases}$$

Obviously, we have a linear system whose coefficient matrix coincides with the Gram matrix $\mathbf{G} = \mathbf{G}(\mathbf{z}_1, ..., \mathbf{z}_m)$ of the vectors $\mathbf{z}_1, ..., \mathbf{z}_m$. By the Gram matrix theorem for linearly independent vectors, it is invertible $\Rightarrow$ the linear system in $\alpha_1, ..., \alpha_m$ has a unique solution $\Rightarrow$ the vector $\mathbf{z}_0$ subject to the condition $\mathbf{x} - \mathbf{z}_0 \perp L$ exists and is unique.

The vector $\mathbf{x}_n \equiv \mathbf{x} - \mathbf{z}_0$ in the case $\mathbf{x}_n \perp L$, $\mathbf{z}_0 \in L$ is called the *perpendicular* dropped from $\mathbf{x}$ onto L, and $\mathbf{x}_p \equiv \mathbf{z}_0$ is called the *orthogonal projection* (or just *projection*) of the vector $\mathbf{x}$ onto L. Thus, $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$ with $\mathbf{x}_p \in L$ and $\mathbf{x}_n \perp L$.

Theorem (Orthogonal Projection Theorem)**.** For any vector $\mathbf{x}$ and any finite-dimensional subspace L, there exist a unique perpendicular $\mathbf{x}_n \perp L$ and a unique projection $\mathbf{x}_p \in L$ such that $\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n$. Moreover,

$$\|\mathbf{x}_n\| = \|\mathbf{x} - \mathbf{x}_p\| < \|\mathbf{x} - \mathbf{z}\| \quad \forall \mathbf{z} \in L, \quad \mathbf{z} \neq \mathbf{x}_p.$$

Proof. It remains to prove only that $\mathbf{x}_p$ is the uniquely determined element of best approximation in L for the vector $\mathbf{x}$. Let $\mathbf{z}$ be an arbitrary vector in L. Then $\mathbf{x} - \mathbf{z} = (\mathbf{x} - \mathbf{x}_p) + (\mathbf{x}_p - \mathbf{z})$, where $\mathbf{x} - \mathbf{x}_p \perp L$ and $\mathbf{x}_p - \mathbf{z} \in L$. From this it follows that $\mathbf{x} - \mathbf{x}_p$ and $\mathbf{x}_p - \mathbf{z}$ are the perpendicular and the orthogonal projection onto L for the vector $\mathbf{x} - \mathbf{z}$. By the Pythagorean theorem,

$$\|\mathbf{x} - \mathbf{z}\|^2 = \|\mathbf{x} - \mathbf{x}_p\|^2 + \|\mathbf{x}_p - \mathbf{z}\|^2 \quad \Rightarrow \quad \|\mathbf{x} - \mathbf{x}_p\| < \|\mathbf{x} - \mathbf{z}\| \quad \forall \mathbf{z} \neq \mathbf{x}_p. \ \square$$

Corollary. If L is a finite-dimensional subspace, then $L = (L^\perp)^\perp$.

Proof. We already know that $L \subset (L^\perp)^\perp$ (see Section 5.1). Take $\mathbf{x} \in (L^\perp)^\perp$ and drop the perpendicular $\mathbf{x}_n$ from it onto L. By the definition of the orthogonal complement, $\mathbf{x}_n \in L^\perp$. At the same time, $\mathbf{x}_n \perp L^\perp \Rightarrow \mathbf{x}_n \cdot \mathbf{x}_n = 0 \Rightarrow \mathbf{x}_n = \mathbf{0}$. Consequently, $\mathbf{x} \in L \Rightarrow (L^\perp)^\perp \subset L$. $\square$

Problem 2. Let L and M be subspaces in a finite-dimensional space V with an inner product. Are the equalities $L^{\perp} \cap M = \{\mathbf{0}\}$ and $L \cap M^{\perp} = \{\mathbf{0}\}$ equivalent?

Problem 3. In $\mathbb{R}^n$, vectors $\mathbf{a}_1, ..., \mathbf{a}_{n+1}$ are such that $\mathbf{a}_i \cdot \mathbf{a}_j < 0$ for $i \neq j$. Prove that any n of them are linearly independent.

Problem 4. Prove that in $\mathbb{R}^n$, any set of $n+2$ vectors contains a pair of vectors whose inner product is nonnegative.

Problem 5. Let P_n be the vector space of all real polynomials of degree at most n with the inner product $\langle f, g \rangle = \int_{-1}^{1} f(t)g(t)\,\mathrm{d}t$. Prove that the distance from the polynomial x^n to the subspace P_{n-1} does not exceed $\frac{\sqrt{2}}{2^n}$.

Projection Formula for an Orthogonal Basis ADDED FOR MA1522

In the notation of the course, the orthogonal projection theorem reads: *if V is a subspace of $\mathbb{R}^n$, then every $\mathbf{w} \in \mathbb{R}^n$ can be written uniquely as $\mathbf{w} = \mathbf{w}_p + \mathbf{w}_n$ with $\mathbf{w}_p \in V$ and $\mathbf{w}_n \perp V$*; the vector $\mathbf{w}_p$ is the *orthogonal projection* (or *projection*) of $\mathbf{w}$ onto V. The second part of the theorem is the *best approximation theorem*: $\mathbf{w}_p$ is the vector of V closest to $\mathbf{w}$,

$$\|\mathbf{w} - \mathbf{w}_p\| \leq \|\mathbf{w} - \mathbf{v}\| \quad \text{for all } \mathbf{v} \in V, \tag{1}$$

with equality only for $\mathbf{v} = \mathbf{w}_p$. An orthogonal basis for V makes the projection explicit: the Gram matrix of an orthogonal basis is diagonal, and the linear system for the coefficients of $\mathbf{w}_p$ splits into separate equations.

Theorem (Orthogonal Projection Formula)**.** Let $S = \{\mathbf{u}_1, ..., \mathbf{u}_k\}$ be an orthogonal basis for a subspace V of $\mathbb{R}^n$. Then for every $\mathbf{w} \in \mathbb{R}^n$

$$\mathbf{w}_p = \frac{\mathbf{w} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 + \frac{\mathbf{w} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2}\mathbf{u}_2 + ... + \frac{\mathbf{w} \cdot \mathbf{u}_k}{\mathbf{u}_k \cdot \mathbf{u}_k}\mathbf{u}_k. \tag{2}$$

If S is orthonormal, then $\mathbf{w}_p = (\mathbf{w} \cdot \mathbf{u}_1)\mathbf{u}_1 + (\mathbf{w} \cdot \mathbf{u}_2)\mathbf{u}_2 + ... + (\mathbf{w} \cdot \mathbf{u}_k)\mathbf{u}_k$.

Proof. Denote the right-hand side of (2) by $\mathbf{z}$; clearly $\mathbf{z} \in V$. Since $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ for $i \neq j$,

$$(\mathbf{w} - \mathbf{z}) \cdot \mathbf{u}_j = \mathbf{w} \cdot \mathbf{u}_j - \frac{\mathbf{w} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j}(\mathbf{u}_j \cdot \mathbf{u}_j) = 0, \quad j = 1, ..., k,$$

so $\mathbf{w} - \mathbf{z} \perp V$ by the orthogonality test of Section 5.1. By the uniqueness of the decomposition, $\mathbf{z} = \mathbf{w}_p$ (and $\mathbf{w} - \mathbf{z} = \mathbf{w}_n$). □

Several facts follow at once. The right-hand side of (2) does not depend on the choice of the orthogonal basis for V, because $\mathbf{w}_p$ is unique. For $k = 1$, (2) is the projection of $\mathbf{w}$ onto the line spanned by a nonzero vector $\mathbf{u}$, namely $\mathbf{w}_p = (\mathbf{w} \cdot \mathbf{u})/(\mathbf{u} \cdot \mathbf{u})\,\mathbf{u}$. If $\mathbf{w} \in V$, then $\mathbf{w}_p = \mathbf{w}$ and $\mathbf{w}_n = \mathbf{0}$, and (2) turns into the coordinate formula of Section 5.2. The distance from $\mathbf{w}$ to V is $\|\mathbf{w}_n\| = \|\mathbf{w} - \mathbf{w}_p\|$.

Example. Let V be the plane $x = z$ in $\mathbb{R}^3$ with the orthogonal basis $\mathbf{u}_1 = (1,1,1)^T$, $\mathbf{u}_2 = (1,-2,1)^T$ (Section 5.2), and let $\mathbf{w} = (1,2,5)^T$. Then $\mathbf{w} \cdot \mathbf{u}_1 = 8$ and $\mathbf{w} \cdot \mathbf{u}_2 = 2$, so

$$\mathbf{w}_p = \frac{8}{3}\mathbf{u}_1 + \frac{2}{6}\mathbf{u}_2 = \begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{w}_n = \mathbf{w} - \mathbf{w}_p = \begin{pmatrix} -2 \\ 0 \\ 2 \end{pmatrix}.$$

Indeed, $\mathbf{w}_n \cdot \mathbf{u}_1 = \mathbf{w}_n \cdot \mathbf{u}_2 = 0$. The vector of V closest to $\mathbf{w}$ is $(3,2,3)^T$, and the distance from $\mathbf{w}$ to V is $\|\mathbf{w}_n\| = 2\sqrt{2}$.

Gram–Schmidt Process

From the orthogonal projection theorem it immediately follows that in any finite-dimensional space V with an inner product (for example, in any subspace of $\mathbb{R}^m$) there exists an orthonormal basis.

Indeed, let us take an arbitrary basis $\{\mathbf{v}_1, ..., \mathbf{v}_n\}$ for V and assume that for the span $L_{n-1} = \operatorname{span}\{\mathbf{v}_1, ..., \mathbf{v}_{n-1}\}$ an orthonormal basis $\{\mathbf{q}_1, ..., \mathbf{q}_{n-1}\}$ has already been constructed (of course, $L_{n-1} = \operatorname{span}\{\mathbf{q}_1, ..., \mathbf{q}_{n-1}\}$). Let $\mathbf{h}_n$ be the perpendicular dropped from the vector $\mathbf{v}_n$ onto L_{n-1}. Clearly, $\mathbf{h}_n \neq \mathbf{0}$ (otherwise $\mathbf{v}_n \in L_{n-1} \Rightarrow$ the vectors $\mathbf{v}_1, ..., \mathbf{v}_n$ are linearly dependent). Set $\mathbf{q}_n = \frac{\mathbf{h}_n}{\|\mathbf{h}_n\|}$. Then $\{\mathbf{q}_1, ..., \mathbf{q}_n\}$ will be the desired orthonormal basis for V.

Note that in the constructed basis, for any $k = 1, ..., n$, the first k vectors $\mathbf{q}_1, ..., \mathbf{q}_k$ form an orthonormal basis for the span $L_k = \operatorname{span}\{\mathbf{v}_1, ..., \mathbf{v}_k\}$. Thus,

$$\operatorname{span}\{\mathbf{q}_1, ..., \mathbf{q}_k\} = \operatorname{span}\{\mathbf{v}_1, ..., \mathbf{v}_k\}, \quad k = 1, ..., n.$$

Actual computations begin with obtaining the vector $\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|}$. Then from the vector $\mathbf{v}_2$ the perpendicular $\mathbf{h}_2$ is dropped onto L_1 and normalized: $\mathbf{q}_2 = \frac{\mathbf{h}_2}{\|\mathbf{h}_2\|}$. And so on. When dropping the perpendicular onto L_k, it is reasonable to seek the expansion of the corresponding projection not in terms of the original vectors $\mathbf{v}_1, ..., \mathbf{v}_k$, but in terms of the already constructed orthonormal set $\mathbf{q}_1, ..., \mathbf{q}_k$. The advantage is obvious: the Gram matrix for $\mathbf{q}_1, ..., \mathbf{q}_k$ is the identity matrix!

This algorithm is called the *Gram–Schmidt process*. Here is its formal description:

$$\mathbf{h}_k = \mathbf{v}_k - \sum_{i=1}^{k-1} (\mathbf{v}_k \cdot \mathbf{q}_i)\mathbf{q}_i, \quad \mathbf{q}_k = \frac{\mathbf{h}_k}{\|\mathbf{h}_k\|}, \quad k = 1, ..., n.$$

Problem 6. In the space of real polynomials, an inner product $\langle f, g \rangle$ is defined in an arbitrary manner, but so that for any polynomials $f(x)$ and $g(x)$ the equality[8]

$$\langle xf(x), g(x) \rangle = \langle f(x), xg(x) \rangle$$

[8]Verify that it holds, for example, if the inner product is given by the formula $\langle f, g \rangle = \int_{-1}^{1} f(x)g(x)\,\mathrm{d}x$. In this case, the resulting orthogonal polynomials are called *Legendre polynomials*.

holds, and suppose that applying the Gram–Schmidt process to the polynomials $1, x, x^2, \dots, x^n$ yields polynomials $L_0(x), L_1(x), \dots, L_n(x)$. Prove that the three-term recurrence relations

$$L_k(x) = a_k x L_{k-1}(x) + b_k L_{k-1}(x) + c_k L_{k-2}(x),$$
$$2 \le k \le n, \quad a_k, b_k, c_k \in \mathbb{R},$$

hold.

Gram–Schmidt Process: Orthogonal Form ADDED FOR MA1522

In the course, the Gram–Schmidt process first produces an orthogonal set and postpones normalization to the end, which avoids square roots in the intermediate steps. Let $S = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be a linearly independent set in $\mathbb{R}^n$.

Algorithm (Gram–Schmidt process).

1. Set $\mathbf{v}_1 = \mathbf{u}_1$.
2. For $j = 2, \dots, k$ set
$$\mathbf{v}_j = \mathbf{u}_j - \frac{\mathbf{v}_1 \cdot \mathbf{u}_j}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_j}{\|\mathbf{v}_2\|^2}\mathbf{v}_2 - \dots - \frac{\mathbf{v}_{j-1} \cdot \mathbf{u}_j}{\|\mathbf{v}_{j-1}\|^2}\mathbf{v}_{j-1}. \tag{3}$$
3. Normalize: $\mathbf{q}_j = \mathbf{v}_j / \|\mathbf{v}_j\|$, $j = 1, \dots, k$.

Theorem. The vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are nonzero and pairwise orthogonal, and

$$\operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_j\} = \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_j\}, \quad j = 1, \dots, k.$$

Hence $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is an orthogonal basis and $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ is an orthonormal basis for $\operatorname{span}(S)$.

Proof. We use induction on j; for $j = 1$ there is nothing to prove. Suppose $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$ are nonzero, pairwise orthogonal, and span $V_{j-1} = \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_{j-1}\}$; then they form an orthogonal basis for V_{j-1}. By (2), the vector subtracted from $\mathbf{u}_j$ in (3) is the projection $\mathbf{p}$ of $\mathbf{u}_j$ onto V_{j-1}, so $\mathbf{v}_j = \mathbf{u}_j - \mathbf{p} \perp V_{j-1}$; in particular, $\mathbf{v}_j$ is orthogonal to $\mathbf{v}_1, \dots, \mathbf{v}_{j-1}$. If $\mathbf{v}_j = \mathbf{0}$, then $\mathbf{u}_j = \mathbf{p} \in V_{j-1}$, which contradicts the linear independence of S. Finally, $\mathbf{v}_j \in \operatorname{span}\{\mathbf{u}_j, \mathbf{v}_1, \dots, \mathbf{v}_{j-1}\} = \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_j\}$ and $\mathbf{u}_j = \mathbf{v}_j + \mathbf{p} \in \operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_j\}$, which gives the equality of the spans. □

This is the process of the preceding subsection written differently: if the vectors denoted there by $\mathbf{v}_1, \mathbf{v}_2, \dots$ (the input) are our $\mathbf{u}_1, \mathbf{u}_2, \dots$, then our $\mathbf{v}_j$ are exactly the perpendiculars $\mathbf{h}_j$, since $(\mathbf{v}_i \cdot \mathbf{u}_j)/\|\mathbf{v}_i\|^2\,\mathbf{v}_i = (\mathbf{u}_j \cdot \mathbf{q}_i)\mathbf{q}_i$ with $\mathbf{q}_i = \mathbf{v}_i/\|\mathbf{v}_i\|$, and the $\mathbf{q}_j$ are the same orthonormal vectors. In hand computations one may replace any $\mathbf{v}_j$ by a positive multiple (for instance, to clear fractions) before continuing: the terms of (3) in the later steps and the vectors $\mathbf{q}_j$ do not change.

If the process is applied to a set S that is not known to be linearly independent, the same argument shows that $\mathbf{v}_j = \mathbf{0}$ exactly when $\mathbf{u}_j \in \operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_{j-1}\}$. Such zero vectors are discarded (they must not enter (3) in the later steps), and the remaining nonzero $\mathbf{v}_j$ form an orthogonal basis for $\operatorname{span}(S)$. For instance, $\mathbf{v}_4 = \mathbf{0}$ means that $\mathbf{u}_4$ is a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$, so that S is linearly dependent.

Example. Apply the process to $\mathbf{u}_1 = (1,2,2)^T$, $\mathbf{u}_2 = (1,0,1)^T$, $\mathbf{u}_3 = (0,0,1)^T$:

$$\mathbf{v}_1 = (1,2,2)^T,$$
$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{3}{9}\mathbf{v}_1 = (2/3, -2/3, 1/3)^T,$$
$$\mathbf{v}_3 = \mathbf{u}_3 - \frac{2}{9}\mathbf{v}_1 - \frac{1/3}{1}\mathbf{v}_2 = (-4/9, -2/9, 4/9)^T.$$

Since $\|\mathbf{v}_1\| = 3$, $\|\mathbf{v}_2\| = 1$ and $\|\mathbf{v}_3\| = 2/3$, the resulting orthonormal basis for $\mathbb{R}^3$ is

$$\mathbf{q}_1 = \frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}, \quad \mathbf{q}_2 = \frac{1}{3}\begin{pmatrix}2\\-2\\1\end{pmatrix}, \quad \mathbf{q}_3 = \frac{1}{3}\begin{pmatrix}-2\\-1\\2\end{pmatrix}.$$

Completion to an Orthogonal Basis

Let V be a space of dimension n with an inner product (for example, $V = \mathbb{R}^n$ with the dot product).

Lemma (Completion to an Orthogonal Basis)**.** Any orthogonal (orthonormal) set of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in V$ can be completed by some vectors from V to an orthogonal (orthonormal) basis for V.

Proof. Complete $\mathbf{v}_1, \dots, \mathbf{v}_k$ by some vectors to a basis for V, and then apply the Gram–Schmidt process to the resulting basis. □

Corollary. If L_k is a subspace of dimension k, then $\dim\big(L_k^{\perp}\big) = n - k$. Moreover,

$$V = L_k \oplus L_k^{\perp}.$$

Proof. In V there exists an orthonormal basis $\{\mathbf{q}_1, \dots, \mathbf{q}_n\}$ such that $L_k = \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$. It is then obvious that any vector orthogonal to L_k is a linear combination of the vectors $\mathbf{q}_{k+1}, \dots, \mathbf{q}_n$. □

5.4. QR Factorization

QR Factorization of a Matrix

Suppose $\mathbf{A} \in \mathbb{C}^{n\times m}$ has linearly independent columns $\mathbf{a}_1, ..., \mathbf{a}_m \in \mathbb{C}^n$, and the Gram–Schmidt process is applied to them using the natural inner product (real case: $\mathbf{A} \in \mathbb{R}^{n\times m}$ and the dot product in $\mathbb{R}^n$). Suppose that as a result we obtain orthonormal vectors $\mathbf{q}_1, ..., \mathbf{q}_m \in \mathbb{C}^n$.

The relations $\mathbf{a}_k \in \operatorname{span}\{\mathbf{q}_1, ..., \mathbf{q}_k\}$ hold for $k = 1, ..., m$ and mean that for some numbers r_{ik} we have the equalities

$$\mathbf{a}_k = \sum_{i=1}^{k} r_{ik}\mathbf{q}_i, \quad k = 1, ..., m,$$

or, in matrix form,

$$\mathbf{A} = \mathbf{QR}, \quad \mathbf{Q} = (\mathbf{q}_1 \ \cdots \ \mathbf{q}_m), \quad \mathbf{R} = \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1m} \\ & r_{22} & \cdots & r_{2m} \\ & & \ddots & \vdots \\ 0 & & & r_{mm} \end{pmatrix}.$$

Definition. A factorization $\mathbf{A} = \mathbf{QR}$, where $\mathbf{Q}$ has orthonormal columns and $\mathbf{R}$ is an upper triangular matrix, is called a *QR factorization* of the matrix $\mathbf{A}$.[9]

Thus, we have just proved that *for any rectangular matrix with linearly independent columns there exists a QR factorization.* In particular, it exists for any invertible matrix. In fact, a more general result holds.

Theorem. Any rectangular matrix in which the number of rows is not less than the number of columns possesses a QR factorization with a matrix $\mathbf{R}$ in row-echelon form.

Proof. Let $\mathbf{a}_{i_1}$ be the first nonzero column of the matrix $\mathbf{A}$, $\mathbf{a}_{i_2}$ be the first column such that $\mathbf{a}_{i_2} \notin \operatorname{span}\{\mathbf{a}_{i_1}\}$, $\mathbf{a}_{i_3}$ be the first column such that $\mathbf{a}_{i_3} \notin \operatorname{span}\{\mathbf{a}_{i_1}, \mathbf{a}_{i_2}\}$, and so on. As a result, we obtain in $\mathbf{A}$ columns

$$\mathbf{a}_{i_1}, ..., \mathbf{a}_{i_r}, \quad i_1 < i_2 < ... < i_r,$$

forming a basis for $\operatorname{Col}(\mathbf{A})$ and possessing the following properties:

$$\begin{aligned} &\mathbf{a}_j = \mathbf{0} && \text{for } \ j < i_1; \\ &\mathbf{a}_j \in \operatorname{span}\{\mathbf{a}_{i_1}, ..., \mathbf{a}_{i_l}\} && \text{for } \ i_l < j < i_{l+1}, \quad l = 1, ..., r-1; \\ &\mathbf{a}_j \in \operatorname{span}\{\mathbf{a}_{i_1}, ..., \mathbf{a}_{i_r}\} && \text{for } \ i_r < j. \end{aligned}$$

[9] MA1522 defines the QR factorization only for a matrix with linearly independent columns, and requires in addition that $\mathbf{R}$ be invertible with positive diagonal entries; the definition here is broader.

Let us find the QR factorization:

$$\begin{pmatrix}\mathbf{a}_{i_1} & \cdots & \mathbf{a}_{i_r}\end{pmatrix} = \begin{pmatrix}\mathbf{q}_{i_1} & \cdots & \mathbf{q}_{i_r}\end{pmatrix}\mathbf{R}_r.$$

Let us complete the columns $\mathbf{q}_{i_1}, ..., \mathbf{q}_{i_r}$ to an orthonormal basis for the n-dimensional space of columns, and from the resulting columns form the matrix $\mathbf{Q}$, keeping the initial columns in positions $i_1, ..., i_r$.

Writing $\mathbf{A} = \mathbf{QR}$, we see that in the matrix $\mathbf{R}$ the first r entries of the i_l-th column are the same as in the l-th column of the matrix $\mathbf{R}_r$. At the same time, the j-th column for $i_l < j < i_{l+1}$ has zeros in positions below the i_l-th. □

Problem 1. Suppose $\mathbf{A} \in \mathbb{C}^{n\times n}$ has columns $\mathbf{a}_1, ..., \mathbf{a}_n \in \mathbb{C}^n$. Prove the inequality

$$|\det(\mathbf{A})| \le \prod_{j=1}^{n} \|\mathbf{a}_j\|.$$

Problem 2. Let $\mathbf{A}$ be a matrix of order n with entries $a_{ij} = \pm 1$. Prove that if $|\det(\mathbf{A})| = n^{\frac{n}{2}}$ (such matrices are called *Hadamard matrices*) and $n \ge 3$, then n is divisible by 4.

Computing a QR Factorization ADDED FOR MA1522

In the course, the QR factorization is considered for matrices with linearly independent columns and is normalized as follows.

Theorem (QR Factorization)**.** Let $\mathbf{A}$ be an $m \times n$ matrix with linearly independent columns, that is, $\operatorname{rank}(\mathbf{A}) = n$. Then

$$\mathbf{A} = \mathbf{QR},$$

where $\mathbf{Q}$ is an $m \times n$ matrix with $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_n$ (its columns are orthonormal) and $\mathbf{R}$ is an invertible upper triangular matrix of order n with positive diagonal entries. Such a decomposition is called a *QR factorization* of $\mathbf{A}$.

Proof. Let $\mathbf{A} = \begin{pmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n\end{pmatrix}$, let $\mathbf{v}_1, ..., \mathbf{v}_n$ and $\mathbf{q}_1, ..., \mathbf{q}_n$ be the vectors produced from $\mathbf{a}_1, ..., \mathbf{a}_n$ by the Gram–Schmidt process (Section 5.3), and let $\mathbf{Q} = \begin{pmatrix}\mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_n\end{pmatrix}$; then $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_n$ (Section 5.2). Since $(\mathbf{v}_i \cdot \mathbf{a}_j)/\|\mathbf{v}_i\|^2\, \mathbf{v}_i = (\mathbf{q}_i \cdot \mathbf{a}_j)\mathbf{q}_i$ and $\mathbf{v}_j = \|\mathbf{v}_j\|\mathbf{q}_j$, the Gram–Schmidt formula can be rewritten as

$$\mathbf{a}_j = (\mathbf{q}_1 \cdot \mathbf{a}_j)\mathbf{q}_1 + ... + (\mathbf{q}_{j-1} \cdot \mathbf{a}_j)\mathbf{q}_{j-1} + \|\mathbf{v}_j\|\mathbf{q}_j, \quad j = 1, ..., n.$$

Thus $\mathbf{A} = \mathbf{QR}$, where $\mathbf{R} = (r_{ij})$ is upper triangular with $r_{ij} = \mathbf{q}_i \cdot \mathbf{a}_j$ for $i < j$ and $r_{jj} = \|\mathbf{v}_j\| > 0$. Finally, $\mathbf{R}$ is invertible: if $\mathbf{Rx} = \mathbf{0}$, then $\mathbf{Ax} = \mathbf{QRx} = \mathbf{0}$, whence $\mathbf{x} = \mathbf{0}$ because the columns of $\mathbf{A}$ are linearly independent; so the columns of the square matrix $\mathbf{R}$ are linearly independent, and $\mathbf{R}$ is invertible (Section 2.6). □

Since $\mathbf{Q}^T\mathbf{Q} = \mathbf{I}_n$, multiplying $\mathbf{A} = \mathbf{QR}$ on the left by $\mathbf{Q}^T$ gives

$$\mathbf{R} = \mathbf{Q}^T\mathbf{A},$$

in agreement with $r_{ij} = \mathbf{q}_i \cdot \mathbf{a}_j$. (The entries $\mathbf{q}_i \cdot \mathbf{a}_j$ with $i > j$ vanish, since $\mathbf{a}_j \in \operatorname{span}\{\mathbf{q}_1, ..., \mathbf{q}_j\}$.) This gives a practical procedure for an $m \times n$ matrix $\mathbf{A}$ with linearly independent columns.

Algorithm (QR factorization).

1. Apply the Gram–Schmidt process to the columns $\mathbf{a}_1, ..., \mathbf{a}_n$ of $\mathbf{A}$ to obtain an orthonormal set $\{\mathbf{q}_1, ..., \mathbf{q}_n\}$.
2. Set $\mathbf{Q} = \begin{pmatrix} \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_n \end{pmatrix}$.
3. Compute $\mathbf{R} = \mathbf{Q}^T\mathbf{A}$. Then $\mathbf{A} = \mathbf{QR}$.

Example. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 2 \\ 1 & 1 & 0 \end{pmatrix}.$$

The Gram–Schmidt process applied to the columns of $\mathbf{A}$ gives

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \mathbf{a}_2 - \frac{8}{4}\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix}, \quad \mathbf{v}_3 = \mathbf{a}_3 - \frac{4}{4}\mathbf{v}_1 - \frac{0}{4}\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \end{pmatrix}.$$

All three vectors have norm 2, so

$$\mathbf{Q} = \frac{1}{2}\begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & -1 \end{pmatrix}, \quad \mathbf{R} = \mathbf{Q}^T\mathbf{A} = \begin{pmatrix} 2 & 4 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

and one checks directly that $\mathbf{QR} = \mathbf{A}$.

Corollary. Let $\mathbf{A}$ be an $m \times n$ matrix with linearly independent columns. Then $\mathbf{A}^T\mathbf{A}$ is invertible, and $\mathbf{A}$ has a left inverse: there is an $n \times m$ matrix $\mathbf{B}$ such that $\mathbf{BA} = \mathbf{I}_n$.

Proof. Let $\mathbf{A} = \mathbf{QR}$ be a QR factorization. Then $\mathbf{A}^T\mathbf{A} = \mathbf{R}^T\mathbf{Q}^T\mathbf{QR} = \mathbf{R}^T\mathbf{R}$ is a product of invertible matrices and hence invertible (Section 2.4). As $\mathbf{B}$ one can take $\mathbf{R}^{-1}\mathbf{Q}^T$, since $\mathbf{R}^{-1}\mathbf{Q}^T\mathbf{QR} = \mathbf{R}^{-1}\mathbf{R} = \mathbf{I}_n$; this matrix coincides with $(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T = (\mathbf{R}^T\mathbf{R})^{-1}\mathbf{R}^T\mathbf{Q}^T$. □

Remark. The invertibility of $\mathbf{A}^T\mathbf{A}$ can also be seen without the QR factorization. If $\mathbf{A}^T\mathbf{Ax} = \mathbf{0}$, then $\|\mathbf{Ax}\|^2 = \mathbf{x}^T\mathbf{A}^T\mathbf{Ax} = 0$, so $\mathbf{Ax} = \mathbf{0}$. Hence $\operatorname{Null}(\mathbf{A}^T\mathbf{A}) = \operatorname{Null}(\mathbf{A})$ for every matrix $\mathbf{A}$, and the square matrix $\mathbf{A}^T\mathbf{A}$ is invertible if and only if the columns of $\mathbf{A}$ are linearly independent.

5.5. Least Squares Approximation

Method of Least Squares

If a linear system $\mathbf{Ax} = \mathbf{b}$ is inconsistent, then the equality $\mathbf{Ax} = \mathbf{b}$ does not hold for any vector $\mathbf{x}$. In this case, nevertheless, one tries to consider those $\mathbf{x}$ for which the vector $\mathbf{b} - \mathbf{Ax}$ (it is called the *residual* for $\mathbf{x}$) has the minimum possible norm. A vector $\mathbf{x}$ is called a *least squares solution* of the system $\mathbf{Ax} = \mathbf{b}$ if

$$\|\mathbf{b} - \mathbf{Ax}\| = \min_{\mathbf{z}} \|\mathbf{b} - \mathbf{Az}\|.$$

In this method of defining a "generalized solution," in the real case one is indeed dealing with the minimum value of the sum of squares (whence the name of the method)

$$\|\mathbf{b} - \mathbf{Ax}\|^2 = \sum_{i=1}^{m} (b_i - a_{i1}x_1 - \ldots - a_{in}x_n)^2.$$

Below, the formulas are written for the real case; for a complex matrix $\mathbf{A}$, every transpose ($\mathbf{A}^T$, $\mathbf{V}^T$, $\mathbf{u}_k^T$) is to be replaced by the conjugate transpose ($\mathbf{A}^*$, $\mathbf{V}^*$, $\mathbf{u}_k^*$).

Proposition. Let $\mathbf{A}$ be an $m \times n$ matrix of rank r. The set of least squares solutions of the system $\mathbf{Ax} = \mathbf{b}$ is an affine set of dimension $n - r$, that is, a translate $\mathbf{x}_0 + W$ of a subspace W of dimension $n - r$.

Proof. Let $\mathbf{b}_n$ be the perpendicular dropped from the vector $\mathbf{b}$ onto the subspace $\mathrm{Col}(\mathbf{A})$, and let $\mathbf{b}_p \in \mathrm{Col}(\mathbf{A})$ be the corresponding orthogonal projection. Then the system $\mathbf{Az} = \mathbf{b}_p$ is consistent, and if $\mathbf{z}$ is an arbitrary solution of it, then $\|\mathbf{b}_n\| = \|\mathbf{b} - \mathbf{Az}\| < \|\mathbf{b} - \mathbf{Ax}\|$ for all $\mathbf{x}$ such that $\mathbf{Ax} \neq \mathbf{b}_p$. Hence, the set of least squares solutions coincides with the set of solutions of the consistent system $\mathbf{Az} = \mathbf{b}_p$. □

Corollary. The set of least squares solutions of the system $\mathbf{Ax} = \mathbf{b}$ coincides with the set of solutions of the system $\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}$ (the *normal equation*).

Proof. $\mathbf{b} - \mathbf{Az} \perp \mathrm{Col}(\mathbf{A}) \Leftrightarrow \mathbf{A}^T(\mathbf{b} - \mathbf{Az}) = \mathbf{0}$. □

Among all least squares solutions, the one of minimal norm, $\hat{\mathbf{x}}$, stands out—it is called the *least squares solution of minimum norm*. Geometrically, it is clear that $\hat{\mathbf{x}}$ is the perpendicular dropped onto $\mathrm{Null}(\mathbf{A})$ from any particular solution $\mathbf{z}$ of the consistent system $\mathbf{Az} = \mathbf{b}_p$ (the vector $\mathbf{b}_p$ is the orthogonal projection of the vector $\mathbf{b}$ onto $\mathrm{Col}(\mathbf{A})$). Thus, *the least squares solution of minimum norm exists and is unique.*

The singular value decomposition (see Section 6.5) makes it possible to give an explicit form of the least squares solution of minimum norm. Let $\mathbf{A} = \mathbf{U\Sigma V}^T$ be a singular value decomposition, where $\sigma_1 \geq \ldots \geq \sigma_r > 0$ are the

nonzero singular values of $\mathbf{A}$, and $\mathbf{u}_k$ and $\mathbf{v}_k$ are the columns of $\mathbf{U}$ and $\mathbf{V}$ (the left and right singular vectors). Then

$$\hat{\mathbf{x}} = \sum_{k=1}^{r} \frac{\mathbf{u}_k^T \mathbf{b}}{\sigma_k} \mathbf{v}_k. \qquad (*)$$

For the proof it suffices to verify that $\mathbf{b} - \mathbf{A}\hat{\mathbf{x}} \perp \mathrm{Col}(\mathbf{A})$ and $\hat{\mathbf{x}} \perp \mathrm{Null}(\mathbf{A})$.

The simplicity of the formula should not create the impression that there are no problems in computing $\hat{\mathbf{x}}$. The main problem, properly speaking, is that in the case $r < \min(m, n)$ the rank r can be increased by an arbitrarily small perturbation of the entries of the matrix, which means that the least squares solution of minimum norm, despite the fact of its existence and uniqueness, is not a continuous function of the entries of the matrix $\mathbf{A}$. For example, let $m = n = 1$ and consider the system $0 \cdot x = 1$. Its least squares solution of minimum norm is obviously $\hat{x} = 0$, whereas the least squares solution of minimum norm of the perturbed system $\varepsilon \cdot x = 1$ is $\hat{x}(\varepsilon) = \frac{1}{\varepsilon}$. As we see, $\hat{x}(\varepsilon)$ does not tend to $\hat{x}$ as $\varepsilon \to 0$. The very problem of computing such an unstable object does not seem terribly meaningful.

At the same time, problems of this kind constantly arise in applications, and some methods for solving them are required of us. When constructing such methods, one should bear in mind that they must be, above all, *methods for modifying the very formulation of the problem.* Such questions are connected with the so-called *regularization methods.*[10]

Problem 1. Find the least squares solution of minimum norm of the inconsistent system

$$x_1 + x_2 + \ldots + x_n = 1,$$
$$x_1 + x_2 + \ldots + x_n = 0.$$

Computing Least Squares Solutions ADDED FOR MA1522

Let $\mathbf{A}$ be an $m \times n$ matrix and $\mathbf{b} \in \mathbb{R}^m$. Recall that $\mathrm{Col}(\mathbf{A}) = \{\mathbf{Au} : \mathbf{u} \in \mathbb{R}^n\}$ is precisely the set of vectors $\mathbf{b}$ for which $\mathbf{Ax} = \mathbf{b}$ is consistent (Sections 2.3 and 4.1). If $\mathbf{b} \notin \mathrm{Col}(\mathbf{A})$, one looks for $\mathbf{u}$ such that $\mathbf{Au}$ is as close to $\mathbf{b}$ as possible: in the form used in the course, $\mathbf{u} \in \mathbb{R}^n$ is a *least squares solution* of $\mathbf{Ax} = \mathbf{b}$ if

$$\|\mathbf{Au} - \mathbf{b}\| \le \|\mathbf{Av} - \mathbf{b}\| \quad \text{for every } \mathbf{v} \in \mathbb{R}^n,$$

which is the definition given above.

Theorem. For a vector $\mathbf{u} \in \mathbb{R}^n$ the following statements are equivalent:
(i) $\mathbf{u}$ is a least squares solution of $\mathbf{Ax} = \mathbf{b}$;
(ii) $\mathbf{Au}$ is the orthogonal projection of $\mathbf{b}$ onto $\mathrm{Col}(\mathbf{A})$;

[10]The general theory of regularization methods was created by the founder of the Faculty of Computational Mathematics and Cybernetics at Moscow State University, Academician Andrei Nikolaevich Tikhonov.

(iii) $\mathbf{u}$ is a solution of the normal equation $\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}$.

Proof. (i) $\Leftrightarrow$ (ii). As $\mathbf{v}$ runs over $\mathbb{R}^n$, the vector $\mathbf{A}\mathbf{v}$ runs over $\mathrm{Col}(\mathbf{A})$. By the best approximation theorem (Section 5.3), the projection $\mathbf{b}_p$ of $\mathbf{b}$ onto $\mathrm{Col}(\mathbf{A})$ is the unique vector of $\mathrm{Col}(\mathbf{A})$ closest to $\mathbf{b}$; so $\mathbf{u}$ minimizes $\|\mathbf{A}\mathbf{u} - \mathbf{b}\|$ if and only if $\mathbf{A}\mathbf{u} = \mathbf{b}_p$. (This is the argument in the proof of the proposition above.)

(ii) $\Leftrightarrow$ (iii). Since $\mathbf{A}\mathbf{u} \in \mathrm{Col}(\mathbf{A})$, we have $\mathbf{A}\mathbf{u} = \mathbf{b}_p$ if and only if $\mathbf{b} - \mathbf{A}\mathbf{u} \perp \mathrm{Col}(\mathbf{A})$, that is, since $\mathrm{Col}(\mathbf{A})^\perp = \mathrm{Null}(\mathbf{A}^T)$ (Section 5.1), if and only if $\mathbf{A}^T(\mathbf{b} - \mathbf{A}\mathbf{u}) = \mathbf{0}$, which is $\mathbf{A}^T\mathbf{A}\mathbf{u} = \mathbf{A}^T\mathbf{b}$. □

Consequently, least squares solutions always exist, because the normal equation is always consistent, and the vector $\mathbf{A}\mathbf{u} = \mathbf{b}_p$ is the same for all least squares solutions $\mathbf{u}$. The least squares solution itself need not be unique: if $\mathbf{u}_0$ is one of them, all of them form the set $\mathbf{u}_0 + \mathrm{Null}(\mathbf{A})$ (Section 4.2), in agreement with the proposition above. If the system $\mathbf{A}\mathbf{x} = \mathbf{b}$ is consistent, then $\mathbf{b} \in \mathrm{Col}(\mathbf{A})$, $\mathbf{b}_p = \mathbf{b}$, and its least squares solutions are exactly its solutions.

Theorem. If the columns of $\mathbf{A}$ are linearly independent, that is, $\mathrm{rank}(\mathbf{A}) = n$, then $\mathbf{A}\mathbf{x} = \mathbf{b}$ has a unique least squares solution

$$\mathbf{u} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b}.$$

If, moreover, $\mathbf{A} = \mathbf{Q}\mathbf{R}$ is a QR factorization, then $\mathbf{u} = \mathbf{R}^{-1}\mathbf{Q}^T\mathbf{b}$; that is, $\mathbf{u}$ is the solution of the upper triangular system $\mathbf{R}\mathbf{x} = \mathbf{Q}^T\mathbf{b}$, which is found by back substitution.

Proof. By the corollary in Section 5.4, $\mathbf{A}^T\mathbf{A}$ is invertible, so the normal equation has the unique solution $(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b}$. If $\mathbf{A} = \mathbf{Q}\mathbf{R}$, then $\mathbf{A}^T\mathbf{A} = \mathbf{R}^T\mathbf{R}$ and $\mathbf{A}^T\mathbf{b} = \mathbf{R}^T\mathbf{Q}^T\mathbf{b}$, and multiplying $\mathbf{R}^T\mathbf{R}\mathbf{u} = \mathbf{R}^T\mathbf{Q}^T\mathbf{b}$ on the left by $(\mathbf{R}^T)^{-1}$ gives $\mathbf{R}\mathbf{u} = \mathbf{Q}^T\mathbf{b}$. □

Example (Non-Unique Least Squares Solutions)**.** Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 \\ 1 & -1 & 0 \\ 1 & 1 & 2 \\ 1 & -1 & 0 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ 4 \\ 2 \end{pmatrix}.$$

The system $\mathbf{A}\mathbf{x} = \mathbf{b}$ is inconsistent: its first and third equations have the same left-hand side but different right-hand sides. For the normal equation,

$$(\mathbf{A}^T\mathbf{A} \mid \mathbf{A}^T\mathbf{b}) = \left(\begin{array}{ccc|c} 4 & 0 & 4 & 8 \\ 0 & 4 & 4 & 4 \\ 4 & 4 & 8 & 12 \end{array}\right) \xrightarrow{\text{RREF}} \left(\begin{array}{ccc|c} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right),$$

so the least squares solutions are $\mathbf{u} = (2, 1, 0)^T + t(-1, -1, 1)^T$, $t \in \mathbb{R}$; there are infinitely many of them, as $\mathrm{rank}(\mathbf{A}) = 2 < 3$. For every one of

them $\mathbf{Au} = (3,1,3,1)^T = \mathbf{b}_p$, and the least possible value of $\|\mathbf{Au} - \mathbf{b}\|$ is $\|(1,1,-1,-1)^T\| = 2$.

Example (A Unique Least Squares Solution)**.** The matrix $\mathbf{A}$ of the example in Section 5.4 has linearly independent columns. For $\mathbf{b} = (1,2,3,4)^T$, the QR factorization found there gives $\mathbf{Q}^T\mathbf{b} = (5,-1,0)^T$, and back substitution in

$$\mathbf{Rx} = \begin{pmatrix} 2 & 4 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 0 \end{pmatrix}$$

yields $x_3 = 0$, $x_2 = -1/2$, $x_1 = 7/2$. Thus $\mathbf{u} = (7/2, -1/2, 0)^T$ is the unique least squares solution, and $\mathbf{b}_p = \mathbf{Au} = (2,3,2,3)^T$. The same $\mathbf{u}$ is obtained from the normal equation $\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}$, where

$$\mathbf{A}^T\mathbf{A} = \begin{pmatrix} 4 & 8 & 4 \\ 8 & 20 & 8 \\ 4 & 8 & 8 \end{pmatrix}, \quad \mathbf{A}^T\mathbf{b} = \begin{pmatrix} 10 \\ 18 \\ 10 \end{pmatrix}.$$

Least squares also give the orthogonal projection onto a subspace for which only an arbitrary (not necessarily orthogonal) basis is known.

Theorem. Let V be a subspace of $\mathbb{R}^n$ with a basis $\{\mathbf{u}_1, ..., \mathbf{u}_k\}$, and let $\mathbf{A} = (\mathbf{u}_1 \;\; \mathbf{u}_2 \;\; \cdots \;\; \mathbf{u}_k)$. Then the orthogonal projection of a vector $\mathbf{w} \in \mathbb{R}^n$ onto V is

$$\mathbf{w}_p = \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{w}.$$

If the basis is orthonormal, then $\mathbf{A}^T\mathbf{A} = \mathbf{I}_k$ and $\mathbf{w}_p = \mathbf{A}\mathbf{A}^T\mathbf{w}$.

Proof. Here $\mathrm{Col}(\mathbf{A}) = V$, and the columns of $\mathbf{A}$ are linearly independent. By the preceding theorem, $\mathbf{u} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{w}$ is the least squares solution of $\mathbf{Ax} = \mathbf{w}$, and by the first theorem of this subsection, $\mathbf{Au}$ is the projection of $\mathbf{w}$ onto $\mathrm{Col}(\mathbf{A}) = V$. For an orthonormal basis, $\mathbf{A}^T\mathbf{A} = \mathbf{I}_k$ (Section 5.2). □

Example. The vectors $(1,1,1)^T$ and $(2,0,2)^T$ form a (non-orthogonal) basis for the plane $x = z$ in $\mathbb{R}^3$ considered in Section 5.3. With

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 1 & 0 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{A}^T\mathbf{A} = \begin{pmatrix} 3 & 4 \\ 4 & 8 \end{pmatrix}, \quad \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T = \frac{1}{2}\begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix},$$

the projection of $\mathbf{w} = (w_1, w_2, w_3)^T$ onto the plane is

$$\mathbf{w}_p = \frac{1}{2}\begin{pmatrix} w_1 + w_3 \\ 2w_2 \\ w_1 + w_3 \end{pmatrix}.$$

> For $\mathbf{w} = (1, 2, 5)^T$ this is $(3, 2, 3)^T$, as found in Section 5.3 with an orthogonal basis.

Chapter 6. Eigenanalysis

6.1. Eigenvalues and Eigenvectors

Eigenvalues and Eigenvectors

Let $\mathbf{A}$ be a matrix of order n and let $\mathbf{P}$ be an invertible matrix with columns $\mathbf{p}_1, ..., \mathbf{p}_n$. It is easy to see that the equality $\mathbf{AP} = \mathbf{PD}$, where $\mathbf{D} = \text{diag}(\lambda_1, ..., \lambda_n)$, is equivalent to the system of equalities

$$\mathbf{A}\mathbf{p}_j = \lambda_j \mathbf{p}_j, \quad j = 1, ..., n.$$

These equalities lead us to the important concepts of an eigenvalue of a matrix and an eigenvector.

Definition. Let $\mathbf{A}$ be a matrix of order n. A scalar $\lambda \in \mathbb{C}$ and a nonzero vector $\mathbf{v} \in \mathbb{C}^n$ related by $\mathbf{Av} = \lambda\mathbf{v}$ are called an *eigenvalue* and an *eigenvector* of the matrix $\mathbf{A}$ (an eigenvector *associated to* λ). The pair $\lambda, \mathbf{v}$ is sometimes called an *eigenpair* of the matrix $\mathbf{A}$.

In MA1522 the matrix $\mathbf{A}$ is real, and only real eigenvalues $\lambda \in \mathbb{R}$, with eigenvectors $\mathbf{v} \in \mathbb{R}^n$, are considered.

Theorem. A matrix $\mathbf{A}$ of order n is diagonalizable (see Section 6.2) if and only if it has n linearly independent eigenvectors.

Proof. Let $\mathbf{p}_1, ..., \mathbf{p}_n$ be linearly independent eigenvectors of the matrix $\mathbf{A}$ associated to eigenvalues $\lambda_1, ..., \lambda_n$:

$$\mathbf{A}\mathbf{p}_j = \lambda_j \mathbf{p}_j, \quad j = 1, ..., n \quad \Leftrightarrow \quad \mathbf{AP} = \mathbf{P}\begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}, \quad \mathbf{P} = (\mathbf{p}_1 \ \cdots \ \mathbf{p}_n).$$

The matrix $\mathbf{P}$ is invertible—as a matrix with linearly independent columns. □

An example of a non-diagonalizable matrix: $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Suppose that

$$\begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}^{-1} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \quad \Rightarrow$$

$$\Rightarrow \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

From this,

$$\begin{pmatrix} p_{21} & p_{22} \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} p_{11}\lambda_1 & p_{12}\lambda_2 \\ p_{21}\lambda_1 & p_{22}\lambda_2 \end{pmatrix}.$$

At least one of the numbers λ_1, λ_2 must be nonzero. Suppose for definiteness that $\lambda_1 \neq 0 \Rightarrow p_{21} = 0 \Rightarrow p_{11} = 0$. We obtain a contradiction, since a matrix with a zero column cannot be invertible. □

The Characteristic Equation

Let λ be an arbitrary eigenvalue of a matrix $\mathbf{A}$. For a fixed λ, all eigenvectors $\mathbf{v}$ associated to it satisfy the homogeneous linear system

$$(\lambda\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}.$$

A scalar λ is an eigenvalue of the matrix $\mathbf{A} \Leftrightarrow$ this system has a nontrivial solution $\Leftrightarrow \det(\lambda\mathbf{I} - \mathbf{A}) = 0$.

Definition. The equation $\det(x\mathbf{I} - \mathbf{A}) = 0$ in x is called the *characteristic equation* of the matrix $\mathbf{A}$. The left-hand side of this equation is a polynomial of degree n in x, called the *characteristic polynomial* of the matrix $\mathbf{A}$ and denoted $\mathrm{char}(\mathbf{A}) = \det(x\mathbf{I} - \mathbf{A})$.

The full textbook uses $\det(\mathbf{A} - \lambda\mathbf{I})$ instead; the two polynomials differ by the factor $(-1)^n$ and have the same roots.

Proposition. The characteristic polynomial of a matrix $\mathbf{A}$ has the form

$$\mathrm{char}(\mathbf{A}) = x^n - s_{n-1}x^{n-1} + s_{n-2}x^{n-2} - \ldots + (-1)^n s_0,$$

where s_k is the sum of all $(n-k) \times (n-k)$ minors of the matrix $\mathbf{A}$ (determinants of $(n-k) \times (n-k)$ submatrices) located at the intersection of columns and rows with identical indices.

Proof. To obtain the coefficient of x^k, one must select among the $n!$ terms of the determinant

$$\det(x\mathbf{I} - \mathbf{A}) = \sum_{\sigma \in S_n} d_\sigma$$

those and only those terms d_σ that contain a product of exactly k diagonal entries of the form $x - a_{ii}$ (they and only they are polynomials of degree k in x), extract in each of them the leading-degree term of the form $x^k c_\sigma$, and sum the resulting coefficients c_σ. Obviously, the sum of all c_σ corresponding to k diagonal entries in fixed positions $i_1 < \ldots < i_k$ is equal to the determinant of the submatrix of $-\mathbf{A}$ located in the rows and columns complementary to the rows and columns with indices $i_1, \ldots, i_k$, that is, to $(-1)^{n-k}$ times the corresponding $(n-k) \times (n-k)$ minor of $\mathbf{A}$. Hence the coefficient of x^k is $(-1)^{n-k} s_k$. □

In particular, $s_{n-1} = a_{11} + ... + a_{nn}$ is the quantity called the *trace* of the matrix $\mathbf{A}$. Notation: $\operatorname{tr}(\mathbf{A})$. By Vieta's formulas, the trace equals the sum of all eigenvalues counting multiplicities. Note also that $s_0 = \det(\mathbf{A})$.

For $n \le 4$, the eigenvalues (as roots of a polynomial of degree $n \le 4$) can be expressed in radicals in terms of the coefficients of the characteristic polynomial and, consequently, in terms of the entries of the matrix. For $n \ge 5$, such formulas no longer exist (the celebrated result of Abel, Ruffini, and Galois).

Finding Eigenvalues; the Eigenvalue Zero ADDED FOR MA1522

A square matrix $\mathbf{A}$ of order n moves the vectors of $\mathbb{R}^n$ by the rule $\mathbf{u} \mapsto \mathbf{Au}$. Its eigenvectors are the nonzero vectors that are merely stretched, shrunk or reversed by this map, and the eigenvalue is the factor by which such a vector is scaled. The condition $\mathbf{v} \neq \mathbf{0}$ in the definition cannot be dropped: $\mathbf{A0} = \lambda\mathbf{0}$ holds for every λ, so admitting the zero vector would make every real number an eigenvalue.

By the criterion above, the eigenvalues of $\mathbf{A}$ are the real roots of $\operatorname{char}(\mathbf{A})$, and the eigenvectors associated to an eigenvalue λ are the nontrivial solutions of the homogeneous system $(\lambda\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0}$.

Example. For the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

we find

$$\operatorname{char}(\mathbf{A}) = \begin{vmatrix} x-1 & -2 \\ -2 & x-1 \end{vmatrix} = (x-1)^2 - 4 = (x-3)(x+1),$$

so the eigenvalues are 3 and -1. Indeed, $\mathbf{A}(1,1)^T = 3(1,1)^T$ and $\mathbf{A}(1,-1)^T = -(1,-1)^T$: the matrix stretches the line $y = x$ by the factor 3 and reverses the line $y = -x$.

Theorem. A square matrix $\mathbf{A}$ is invertible if and only if 0 is not an eigenvalue of $\mathbf{A}$.

Proof. The number 0 is an eigenvalue of $\mathbf{A}$ if and only if $\mathbf{Av} = 0\mathbf{v} = \mathbf{0}$ for some $\mathbf{v} \neq \mathbf{0}$, that is, if and only if $\det(0\mathbf{I} - \mathbf{A}) = (-1)^n \det(\mathbf{A}) = 0$. By Section 2.6, $\det(\mathbf{A}) = 0$ means exactly that $\mathbf{A}$ is not invertible. □

This adds one more item to the list of equivalent statements for invertibility assembled in Sections 2.6, 2.10, 3.6 and 4.2.

Theorem (equivalent statements for invertibility)**.** For a square matrix $\mathbf{A}$ of order n the following statements are equivalent.

(i) $\mathbf{A}$ is invertible.

(ii) $\mathbf{A}^T$ is invertible.
(iii) $\mathbf{A}$ has a left inverse: $\mathbf{BA} = \mathbf{I}$ for some matrix $\mathbf{B}$.
(iv) $\mathbf{A}$ has a right inverse: $\mathbf{AB} = \mathbf{I}$ for some matrix $\mathbf{B}$.
(v) The reduced row-echelon form of $\mathbf{A}$ is $\mathbf{I}_n$.
(vi) $\mathbf{A}$ is a product of elementary matrices.
(vii) The homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution.
(viii) For every $\mathbf{b} \in \mathbb{R}^n$ the system $\mathbf{Ax} = \mathbf{b}$ has exactly one solution.
(ix) $\det(\mathbf{A}) \neq 0$.
(x) The columns of $\mathbf{A}$ are linearly independent; equivalently, the rows of $\mathbf{A}$ are linearly independent.
(xi) The columns of $\mathbf{A}$ span $\mathbb{R}^n$; equivalently, the rows of $\mathbf{A}$ span $\mathbb{R}^n$.
(xii) $\operatorname{rank}(\mathbf{A}) = n$, that is, $\mathbf{A}$ has full rank.
(xiii) $\operatorname{nullity}(\mathbf{A}) = 0$.
(xiv) 0 is not an eigenvalue of $\mathbf{A}$.

Proof. The equivalence of (i)–(xiii) was established in Sections 2.6, 2.10, 3.6 and 4.2, and (i) ⇔ (xiv) is the preceding theorem. □

Algebraic Multiplicity of an Eigenvalue

The multiplicity of an eigenvalue λ as a root of the characteristic polynomial is called its *algebraic multiplicity* and is denoted r_λ: it is the largest integer r_λ such that $\operatorname{char}(\mathbf{A}) = (x-\lambda)^{r_\lambda} p(x)$ for some polynomial $p(x)$.

From the fundamental theorem of algebra we immediately obtain the following theorem.

Theorem. Every complex matrix $\mathbf{A}$ of order n has n complex eigenvalues counting algebraic multiplicities.

Real Eigenvalues and Triangular Matrices ADDED FOR MA1522

Over $\mathbb{R}$ the characteristic polynomial need not split into linear factors. If it does,

$$\operatorname{char}(\mathbf{A}) = (x-\lambda_1)^{r_1}(x-\lambda_2)^{r_2}\cdots(x-\lambda_k)^{r_k}, \quad r_1 + r_2 + \cdots + r_k = n,$$

with distinct real numbers $\lambda_1, ..., \lambda_k$, then these are the eigenvalues of $\mathbf{A}$, the algebraic multiplicity of λ_i is $r_{\lambda_i} = r_i$, and the algebraic multiplicities add up to n. In general $\operatorname{char}(\mathbf{A}) = (x-\lambda_1)^{r_1}\cdots(x-\lambda_k)^{r_k} q(x)$, where the polynomial $q(x)$ has no real roots; then $r_1 + \cdots + r_k \leq n$, with equality exactly when $\operatorname{char}(\mathbf{A})$ splits into linear factors.

Example. The zero matrix $\mathbf{0}$ of order n has $\operatorname{char}(\mathbf{0}) = \det(x\mathbf{I}) = x^n$; its only eigenvalue is 0, with $r_0 = n$. For the matrices

$$\mathbf{B} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

we get $\operatorname{char}(\mathbf{B}) = x^2 + 1$ and $\operatorname{char}(\mathbf{C}) = (x-2)(x^2+1)$. The matrix $\mathbf{B}$ is the matrix of the rotation through the angle $\pi/2$; $\operatorname{char}(\mathbf{B})$ has no real roots, and indeed no nonzero vector of $\mathbb{R}^2$ keeps its direction under a quarter-turn, so $\mathbf{B}$ has no eigenvalues at all. The only eigenvalue of $\mathbf{C}$ is 2, with $r_2 = 1 < 3$.

Theorem. The eigenvalues of a triangular matrix are its diagonal entries, and the algebraic multiplicity of each eigenvalue is the number of times it occurs on the diagonal.

Proof. If $\mathbf{A} = \left(a_{ij}\right)_{n\times n}$ is upper (lower) triangular, then so is $x\mathbf{I} - \mathbf{A}$, with diagonal entries $x - a_{11}, \ldots, x - a_{nn}$. The determinant of a triangular matrix is the product of its diagonal entries: expand it along the first column (for a lower triangular matrix, along the first row), whose only possibly nonzero entry is the diagonal one, and repeat with the remaining triangular submatrix. Hence

$$\operatorname{char}(\mathbf{A}) = (x - a_{11})(x - a_{22})\cdots(x - a_{nn}). \qquad \square$$

For example, the matrix

$$\begin{pmatrix} 2 & 5 & -1 \\ 0 & 3 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

has the eigenvalues 2 and 3, with $r_2 = 2$ and $r_3 = 1$.

Characteristic Polynomial and Similarity

Theorem. The characteristic polynomials of similar matrices coincide.

Proof. Let $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$, where $\mathbf{P}$ is an invertible matrix. Then

$$\begin{aligned} \det(x\mathbf{I} - \mathbf{B}) &= \det(x\mathbf{P}^{-1}\mathbf{P} - \mathbf{P}^{-1}\mathbf{A}\mathbf{P}) = \det(\mathbf{P}^{-1}(x\mathbf{I} - \mathbf{A})\mathbf{P}) = \\ &= \det(\mathbf{P}^{-1})\det(\mathbf{P})\det(x\mathbf{I} - \mathbf{A}) = \det(\mathbf{P}^{-1}\mathbf{P})\det(x\mathbf{I} - \mathbf{A}) = \\ &= \det(x\mathbf{I} - \mathbf{A}). \end{aligned}$$ $\square$

Corollary. The eigenvalues and their algebraic multiplicities for similar matrices coincide.

Problem 1. Find the characteristic polynomial of the matrix

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 1 & & & 0 \end{pmatrix}_{n\times n}.$$

Problem 2. Let $\mathbf{A}$ and $\mathbf{B}$ be square matrices of the same order. Prove that $\mathbf{AB}$ and $\mathbf{BA}$ have the same characteristic polynomials.

Geometric Multiplicity of an Eigenvalue

Fix an eigenvalue λ of a linear transformation $T : V \to V$ (that is, $T(\mathbf{v}) = \lambda\mathbf{v}$ for some $\mathbf{v} \neq \mathbf{0}$) and consider the set E_λ of all vectors $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \lambda\mathbf{v}$.

Proposition. The set E_λ is a subspace invariant under T.

Proof. Let $\mathbf{u}, \mathbf{v} \in E_\lambda \Rightarrow T(\mathbf{u}) = \lambda\mathbf{u}, T(\mathbf{v}) = \lambda\mathbf{v} \Rightarrow T(\alpha\mathbf{u} + \beta\mathbf{v}) = \lambda(\alpha\mathbf{u} + \beta\mathbf{v}) \Rightarrow \alpha\mathbf{u} + \beta\mathbf{v} \in E_\lambda$. The invariance of E_λ is obvious: if $\mathbf{v} \in E_\lambda$, then $T(\mathbf{v}) = \lambda\mathbf{v} \in E_\lambda$. □

Definition. The subspace E_λ is called the *eigenspace* associated to λ, and its dimension $\dim(E_\lambda)$ is called the *geometric multiplicity* of the eigenvalue λ.

For a square matrix $\mathbf{A}$ of order n, regarded as the linear transformation $T(\mathbf{u}) = \mathbf{A}\mathbf{u}$ of $\mathbb{R}^n$, the eigenspace is $E_\lambda = \{\mathbf{v} \in \mathbb{R}^n \mid \mathbf{A}\mathbf{v} = \lambda\mathbf{v}\} = \mathrm{Null}(\lambda\mathbf{I} - \mathbf{A})$, and the geometric multiplicity of λ is $\dim(E_\lambda) = \mathrm{nullity}(\lambda\mathbf{I} - \mathbf{A})$.

Computing Eigenspaces ADDED FOR MA1522

To find the eigenspace $E_\lambda = \mathrm{Null}(\lambda\mathbf{I} - \mathbf{A})$, reduce $\lambda\mathbf{I} - \mathbf{A}$ to reduced row-echelon form and read off a basis for the solution space of $(\lambda\mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0}$; the geometric multiplicity $\dim(E_\lambda)$ is the number of non-pivot columns. Since λ is an eigenvalue, $\dim(E_\lambda) \geq 1$.

Example. Let

$$\mathbf{A} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}.$$

Expanding along the first row,

$$\mathrm{char}(\mathbf{A}) = \begin{vmatrix} x-3 & 0 & 0 \\ 0 & x-1 & -2 \\ -2 & -2 & x-1 \end{vmatrix} = (x-3)\big((x-1)^2 - 4\big) = (x-3)^2(x+1),$$

so the eigenvalues are 3 and -1, with $r_3 = 2$ and $r_{-1} = 1$. Next,

$$3\mathbf{I} - \mathbf{A} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix},$$

$$-\mathbf{I} - \mathbf{A} = \begin{pmatrix} -4 & 0 & 0 \\ 0 & -2 & -2 \\ -2 & -2 & -2 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

hence $E_3 = \mathrm{span}\{(0,1,1)^T\}$ and $E_{-1} = \mathrm{span}\{(0,-1,1)^T\}$. Both eigenspaces are lines: $\dim(E_3) = 1 < r_3 = 2$ and $\dim(E_{-1}) = 1 = r_{-1}$.

Proposition. A square matrix $\mathbf{A}$ and its transpose $\mathbf{A}^T$ have the same characteristic polynomial, and hence the same eigenvalues with the same algebraic multiplicities. Moreover, every eigenvalue λ has the same geometric multiplicity for $\mathbf{A}$ as for $\mathbf{A}^T$.

Proof. Since $x\mathbf{I} - \mathbf{A}^T = (x\mathbf{I} - \mathbf{A})^T$ and a matrix and its transpose have equal determinants (Section 2.10), $\operatorname{char}(\mathbf{A}^T) = \operatorname{char}(\mathbf{A})$. Since a matrix and its transpose have equal ranks (Section 4.2), $\operatorname{rank}(\lambda\mathbf{I} - \mathbf{A}^T) = \operatorname{rank}((\lambda\mathbf{I} - \mathbf{A})^T) = \operatorname{rank}(\lambda\mathbf{I} - \mathbf{A})$, and therefore $\operatorname{nullity}(\lambda\mathbf{I} - \mathbf{A}^T) = \operatorname{nullity}(\lambda\mathbf{I} - \mathbf{A})$, since rank and nullity add up to n (Section 4.2). □

The eigenvectors of $\mathbf{A}$ and $\mathbf{A}^T$, however, are in general different.

6.2. Diagonalization

Similarity Transformation

Let $T : V \to V$ be a linear transformation of an n-dimensional vector space V and let $\mathbf{A}$ be its matrix when one and the same basis of V is chosen for the images and the preimages. In this case, the study of the transformation T reduces completely to the study of its matrix $\mathbf{A}$.

It is natural to try to choose a basis in such a way that the matrix $\mathbf{A}$ acquires the "simplest possible" form. If a linear transformation T is given by its matrix $\mathbf{A}$ in some basis, then choosing a new basis gives for the same transformation another matrix $\mathbf{B}$, which will be similar to the given matrix (see Section 7.1). The transition from $\mathbf{A}$ to a matrix $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ similar to it is called a *similarity transformation*.

The following question arises: to what "simplest possible" form can a given matrix be reduced by means of a similarity transformation? In effect, we are now beginning a not-so-simple journey toward a complete answer to this question.

Diagonalizable Matrices

Matrices similar to diagonal matrices are called *diagonalizable* (the full textbook also calls them *matrices of simple structure*). Thus a square matrix $\mathbf{A}$ is diagonalizable if there exists an invertible matrix $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$ is a diagonal matrix, or, equivalently, $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$.

Consider the problem of diagonalizing a (3×3)-matrix:

$$\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{D} \quad \Leftrightarrow \quad \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{pmatrix} =$$

$$= \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{pmatrix} \begin{pmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{pmatrix}.$$

This equality is equivalent to the three equalities:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} p_{1j} \\ p_{2j} \\ p_{3j} \end{pmatrix} = \lambda_j \begin{pmatrix} p_{1j} \\ p_{2j} \\ p_{3j} \end{pmatrix}, \quad j = 1, 2, 3.$$

Suppose that the value λ_j is known. Then the entries of the j-th column of the matrix $\mathbf{P}$ satisfy the homogeneous system

$$\begin{pmatrix} \lambda_j - a_{11} & -a_{12} & -a_{13} \\ -a_{21} & \lambda_j - a_{22} & -a_{23} \\ -a_{31} & -a_{32} & \lambda_j - a_{33} \end{pmatrix} \begin{pmatrix} p_{1j} \\ p_{2j} \\ p_{3j} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \qquad (*)$$

This system must have a nontrivial solution $\Leftrightarrow$ the determinant of the coefficient matrix is equal to zero. Thus, λ_j satisfies the following equation in x:

$$\det \begin{pmatrix} x - a_{11} & -a_{12} & -a_{13} \\ -a_{21} & x - a_{22} & -a_{23} \\ -a_{31} & -a_{32} & x - a_{33} \end{pmatrix} = 0. \qquad (\#)$$

This is a cubic equation of the form $x^3 - s_2 x^2 + s_1 x - s_0 = 0$, where, as is easy to see,

$$s_2 = a_{11} + a_{22} + a_{33},$$

$$s_1 = \det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} + \det \begin{pmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{pmatrix} + \det \begin{pmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{pmatrix},$$

$$s_0 = \det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

The question of diagonalizing matrices of order 3 also arises when studying quadric surfaces (see Tyrtyshnikov, §§20.1–20.5)—when searching for a coordinate system in which the matrix of the quadratic part (a real symmetric matrix of order 3) becomes diagonal.[11]

Eigenvectors for Distinct Eigenvalues

Theorem. Eigenvectors associated to pairwise distinct eigenvalues of a matrix are linearly independent.

Proof. Let $\mathbf{v}_1, ..., \mathbf{v}_m$ be eigenvectors for pairwise distinct eigenvalues $\lambda_1, ..., \lambda_m$ of a matrix $\mathbf{A}$. Suppose $\alpha_1 \mathbf{v}_1 + ... + \alpha_m \mathbf{v}_m = \mathbf{0}$. Multiply both sides on the left by the matrix $\mathbf{A}$:

[11] In Section 6.3 it is proved that *every real symmetric matrix* $\mathbf{A}$ *is orthogonally similar to a diagonal matrix* $\mathbf{D}$—this means that $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1} = \mathbf{P}\mathbf{D}\mathbf{P}^T$, where $\mathbf{P}$ is an orthogonal matrix, that is, $\mathbf{A}$ is orthogonally diagonalizable. This same statement will soon appear as a consequence of more general results.

$$\alpha_1\lambda_1\mathbf{v}_1 + ... + \alpha_m\lambda_m\mathbf{v}_m = \mathbf{0}.$$

From this equality subtract the preceding one multiplied by λ_m:

$$\alpha_1(\lambda_1 - \lambda_m)\mathbf{v}_1 + ... + \alpha_{m-1}(\lambda_{m-1} - \lambda_m)\mathbf{v}_{m-1} = \mathbf{0}.$$

From this it is clear that the linear independence of the vectors $\mathbf{v}_1, ..., \mathbf{v}_{m-1}$ would imply the linear independence of the vectors $\mathbf{v}_1, ..., \mathbf{v}_m$. The proof is completed by applying induction. □

Corollary. If a matrix of order n has n distinct eigenvalues, then it is diagonalizable.

Independence of Eigenspaces and the Two Multiplicities ADDED FOR MA1522

Theorem. Let $\lambda_1, ..., \lambda_p$ be distinct eigenvalues of a square matrix $\mathbf{A}$, and for each i let S_i be a linearly independent subset of the eigenspace E_{λ_i}. Then the union $S_1 \cup S_2 \cup \cdots \cup S_p$ is linearly independent.

Proof. Take a linear combination of the vectors of the union that equals $\mathbf{0}$, and group its terms by eigenspace: $\mathbf{w}_1 + \mathbf{w}_2 + \cdots + \mathbf{w}_p = \mathbf{0}$, where $\mathbf{w}_i \in E_{\lambda_i}$ is the part of the combination formed from the vectors of S_i. The nonzero vectors among $\mathbf{w}_1, ..., \mathbf{w}_p$ are eigenvectors associated to pairwise distinct eigenvalues, so by the theorem of the preceding subsection they are linearly independent, and their sum cannot be $\mathbf{0}$ unless there are none of them. Hence $\mathbf{w}_i = \mathbf{0}$ for every i, and since S_i is linearly independent, all the coefficients of the vectors of S_i vanish. □

In particular, the union of bases for $E_{\lambda_1}, ..., E_{\lambda_p}$ is a linearly independent set of $\dim\left(E_{\lambda_1}\right) + \cdots + \dim\left(E_{\lambda_p}\right)$ eigenvectors, so this sum never exceeds n.

Theorem. The geometric multiplicity of an eigenvalue λ of a square matrix $\mathbf{A}$ does not exceed its algebraic multiplicity: $1 \leq \dim(E_\lambda) \leq r_\lambda$.

Proof. Let $k = \dim(E_\lambda)$; then $k \geq 1$, since an eigenvalue has an eigenvector. Let $\{\mathbf{u}_1, ..., \mathbf{u}_k\}$ be a basis for E_λ, extend it to a basis $\{\mathbf{u}_1, ..., \mathbf{u}_n\}$ for $\mathbb{R}^n$ (Section 3.7), and let $\mathbf{Q} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_n)$. This matrix is invertible, and $\mathbf{Q}^{-1}\mathbf{u}_j = \mathbf{e}_j$ because $\mathbf{Q}\mathbf{e}_j = \mathbf{u}_j$. For $j \leq k$ we get $\mathbf{Q}^{-1}\mathbf{A}\mathbf{u}_j = \lambda\mathbf{Q}^{-1}\mathbf{u}_j = \lambda\mathbf{e}_j$, so the first k columns of $\mathbf{B} = \mathbf{Q}^{-1}\mathbf{A}\mathbf{Q}$ are $\lambda\mathbf{e}_1, ..., \lambda\mathbf{e}_k$:

$$\mathbf{B} = \begin{pmatrix} \lambda\mathbf{I}_k & \mathbf{C} \\ \mathbf{0} & \mathbf{F} \end{pmatrix}$$

for some blocks $\mathbf{C}$ and $\mathbf{F}$, with $\mathbf{F}$ square of order $n - k$. Expanding $\det(x\mathbf{I} - \mathbf{B})$ successively along its first k columns gives $\operatorname{char}(\mathbf{B}) = (x - \lambda)^k \operatorname{char}(\mathbf{F})$. Similar matrices have the same characteristic polynomial (Section 6.1), so $\operatorname{char}(\mathbf{A}) = (x - \lambda)^k \operatorname{char}(\mathbf{F})$, and therefore $r_\lambda \geq k$. □

Equivalent Statements for Diagonalizability ADDED FOR MA1522

Theorem. Let $\mathbf{A}$ be a square matrix of order n with distinct eigenvalues $\lambda_1, ..., \lambda_k$. The following statements are equivalent.

(i) $\mathbf{A}$ is diagonalizable.
(ii) There is a basis $\{\mathbf{u}_1, ..., \mathbf{u}_n\}$ for $\mathbb{R}^n$ consisting of eigenvectors of $\mathbf{A}$.
(iii) $\mathrm{char}(\mathbf{A})$ splits into linear factors, and the geometric multiplicity of each eigenvalue equals its algebraic multiplicity: $\dim\big(E_{\lambda_i}\big) = r_{\lambda_i}$, $i = 1, ..., k$.
(iv) $\dim\big(E_{\lambda_1}\big) + \dim\big(E_{\lambda_2}\big) + \cdots + \dim\big(E_{\lambda_k}\big) = n$.

Proof. (i) $\Leftrightarrow$ (ii) is the theorem of Section 6.1: $\mathbf{AP} = \mathbf{PD}$ with $\mathbf{P} = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{pmatrix}$ and $\mathbf{D} = \mathrm{diag}(\mu_1, ..., \mu_n)$ means exactly that $\mathbf{A}\mathbf{u}_j = \mu_j \mathbf{u}_j$ for all j, and $\mathbf{P}$ is invertible exactly when its columns form a basis for $\mathbb{R}^n$. Next, by the preceding theorem,

$$\dim\big(E_{\lambda_1}\big) + \cdots + \dim\big(E_{\lambda_k}\big) \leq r_{\lambda_1} + \cdots + r_{\lambda_k} \leq n,$$

where the first inequality holds term by term, and the second is an equality exactly when $\mathrm{char}(\mathbf{A})$ splits into linear factors (Section 6.1). The sum on the left equals n if and only if both inequalities are equalities, which proves (iii) $\Leftrightarrow$ (iv). If (iv) holds, the union of bases for the eigenspaces is a linearly independent set of n eigenvectors (by the theorem on independence of eigenspaces), that is, a basis for $\mathbb{R}^n$; this is (ii). Conversely, if (ii) holds, the vectors of the basis associated to λ_i form a linearly independent subset of E_{λ_i}, so there are at most $\dim\big(E_{\lambda_i}\big)$ of them; hence the sum in (iv) is at least n, and so equals n. □

Consequently, to show that a square matrix $\mathbf{A}$ is *not* diagonalizable, it suffices to show that $\mathrm{char}(\mathbf{A})$ does not split into linear factors over $\mathbb{R}$, or that some eigenvalue has $\dim(E_\lambda) < r_\lambda$. For example, the matrix $\mathbf{C}$ with $\mathrm{char}(\mathbf{C}) = (x-2)(x^2+1)$ considered in Section 6.1 is not diagonalizable, and neither is the matrix $\mathbf{A}$ of the example in the subsection "Computing Eigenspaces" of Section 6.1, for which $\dim(E_3) = 1 < 2 = r_3$. The corollary on n distinct eigenvalues is a special case of the theorem: then $1 \leq \dim\big(E_{\lambda_i}\big) \leq r_{\lambda_i} = 1$ for each of the n eigenvalues, and the sum in (iv) equals n.

Remark. If a matrix $\mathbf{A}$ of order n has only one eigenvalue λ, then $\mathbf{A}$ is diagonalizable if and only if $\mathbf{A} = \lambda \mathbf{I}_n$. Indeed, if $\mathbf{A} = \mathbf{PDP}^{-1}$, the diagonal entries of $\mathbf{D}$ are eigenvalues of $\mathbf{A}$, so $\mathbf{D} = \lambda \mathbf{I}$ and $\mathbf{A} = \mathbf{P}(\lambda \mathbf{I})\mathbf{P}^{-1} = \lambda \mathbf{I}$. Thus a triangular matrix whose diagonal entries are all equal is diagonalizable only if it is diagonal; for instance,

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

is not diagonalizable.

Algorithm for Diagonalization ADDED FOR MA1522

Algorithm (diagonalization).

Let $\mathbf{A}$ be a square matrix of order n.

1. Compute and factorize $\text{char}(\mathbf{A})$. If it does not split into linear factors, then $\mathbf{A}$ is not diagonalizable. Otherwise
$$\text{char}(\mathbf{A}) = (x-\lambda_1)^{r_{\lambda_1}} \cdots (x-\lambda_k)^{r_{\lambda_k}}$$
with distinct $\lambda_1, ..., \lambda_k$.
2. For each i, find a basis S_{λ_i} for E_{λ_i}, the solution space of $(\lambda_i \mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0}$. If $\dim\left(E_{\lambda_i}\right) < r_{\lambda_i}$ for some i, then $\mathbf{A}$ is not diagonalizable. (Eigenvalues with $r_\lambda = 1$ always pass this test, so it pays to treat first those with $r_\lambda > 1$.)
3. Otherwise $S = S_{\lambda_1} \cup \cdots \cup S_{\lambda_k} = \{\mathbf{u}_1, ..., \mathbf{u}_n\}$ is a basis for $\mathbb{R}^n$ consisting of eigenvectors of $\mathbf{A}$.
4. Set $\mathbf{P} = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{pmatrix}$ and $\mathbf{D} = \text{diag}(\mu_1, ..., \mu_n)$, where μ_j is the eigenvalue associated to $\mathbf{u}_j$. Then $\mathbf{A} = \mathbf{PDP}^{-1}$.

Example. Let
$$\mathbf{A} = \begin{pmatrix} 3 & 1 & 0 \\ -2 & 0 & 0 \\ 4 & 4 & 2 \end{pmatrix}.$$
Expanding along the third column,
$$\text{char}(\mathbf{A}) = (x-2)\begin{vmatrix} x-3 & -1 \\ 2 & x \end{vmatrix} = (x-2)(x^2-3x+2) = (x-1)(x-2)^2.$$
For the double eigenvalue 2,
$$2\mathbf{I} - \mathbf{A} = \begin{pmatrix} -1 & -1 & 0 \\ 2 & 2 & 0 \\ -4 & -4 & 0 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_2 = \text{span}\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\},$$
so $\dim(E_2) = 2 = r_2$. For the simple eigenvalue 1,
$$\mathbf{I} - \mathbf{A} = \begin{pmatrix} -2 & -1 & 0 \\ 2 & 1 & 0 \\ -4 & -4 & -1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & -\frac{1}{4} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}, \quad E_1 = \text{span}\left\{ \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} \right\}.$$
Hence $\mathbf{A}$ is diagonalizable, $\mathbf{A} = \mathbf{PDP}^{-1}$ with
$$\mathbf{P} = \begin{pmatrix} -1 & 0 & 1 \\ 1 & 0 & -2 \\ 0 & 1 & 4 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1} = \begin{pmatrix} -2 & -1 & 0 \\ 4 & 4 & 1 \\ -1 & -1 & 0 \end{pmatrix}.$$

The matrices $\mathbf{P}$ and $\mathbf{D}$ are not unique: the eigenvectors may be listed in another order (which permutes the diagonal entries of $\mathbf{D}$ in the same way),

and each S_{λ_i} may be replaced by any other basis for E_{λ_i}. For a given $\mathbf{P}$, however, $\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$ is determined.

6.3. Orthogonally Diagonalizable Matrices

Orthogonally Diagonalizable Matrices ADDED FOR MA1522

Recall that a square matrix $\mathbf{P}$ is orthogonal if $\mathbf{P}^T = \mathbf{P}^{-1}$, that is, if its columns form an orthonormal basis for $\mathbb{R}^n$ (Section 5.2).

Definition. A square matrix $\mathbf{A}$ is *orthogonally diagonalizable* if $\mathbf{A} = \mathbf{PDP}^T$ for some orthogonal matrix $\mathbf{P}$ and some diagonal matrix $\mathbf{D}$.

Since $\mathbf{P}^T = \mathbf{P}^{-1}$, an orthogonally diagonalizable matrix is diagonalizable, $\mathbf{A} = \mathbf{PDP}^{-1}$; what is required in addition is that the invertible matrix $\mathbf{P}$ be orthogonal.

Theorem (spectral theorem)**.** For a square matrix $\mathbf{A}$ of order n the following statements are equivalent.

(i) $\mathbf{A}$ is orthogonally diagonalizable.
(ii) There is an orthonormal basis $\{\mathbf{u}_1, ..., \mathbf{u}_n\}$ for $\mathbb{R}^n$ consisting of eigenvectors of $\mathbf{A}$.
(iii) $\mathbf{A}$ is symmetric, $\mathbf{A}^T = \mathbf{A}$.

Proof. (i) $\Leftrightarrow$ (ii): as in the diagonalization theorem (Section 6.2), $\mathbf{A} = \mathbf{PDP}^{-1}$ with $\mathbf{D}$ diagonal means that the columns of $\mathbf{P}$ are eigenvectors of $\mathbf{A}$ forming a basis for $\mathbb{R}^n$, and $\mathbf{P}$ is orthogonal exactly when this basis is orthonormal. (i) $\Rightarrow$ (iii): $\mathbf{A}^T = \left(\mathbf{PDP}^T\right)^T = \mathbf{PD}^T\mathbf{P}^T = \mathbf{PDP}^T = \mathbf{A}$, since a diagonal matrix is symmetric. (iii) $\Rightarrow$ (i) is the theorem on the diagonalization of real symmetric matrices, proved below in this section. □

Theorem. Eigenvectors of a symmetric matrix associated to distinct eigenvalues are orthogonal. Thus the eigenspaces of a symmetric matrix associated to distinct eigenvalues are orthogonal to each other.

Proof. Let $\mathbf{A}^T = \mathbf{A}$, $\mathbf{Av}_1 = \lambda_1\mathbf{v}_1$ and $\mathbf{Av}_2 = \lambda_2\mathbf{v}_2$ with $\lambda_1 \neq \lambda_2$. Then

$$\lambda_1(\mathbf{v}_1 \cdot \mathbf{v}_2) = (\mathbf{Av}_1)^T\mathbf{v}_2 = \mathbf{v}_1^T\mathbf{A}^T\mathbf{v}_2 = \mathbf{v}_1^T\mathbf{Av}_2 = \lambda_2(\mathbf{v}_1 \cdot \mathbf{v}_2),$$

so $(\lambda_1 - \lambda_2)(\mathbf{v}_1 \cdot \mathbf{v}_2) = 0$, and $\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$. □

Consequently, to build an orthonormal basis of eigenvectors one need not orthogonalize all n eigenvectors together: vectors from different eigenspaces are orthogonal already, and the Gram–Schmidt process has to be applied only within each eigenspace (see the algorithm at the end of this section).

Orthogonal Diagonalization of Real Symmetric Matrices

The following theorem is proved by the *method of rotations* (Jacobi's method): one annihilates off-diagonal entries one at a time by rotating the coordinate

system in coordinate planes. (For matrices of order 3 the method is worked out in detail in Tyrtyshnikov, §§20.3–20.5.)

Theorem (on the diagonalization of real symmetric matrices)**.** A real symmetric matrix $\mathbf{A}$ of arbitrary order n can be reduced to a diagonal matrix $\mathbf{D}$ by means of some orthogonal matrix $\mathbf{P}$:

$$\mathbf{D} = \mathbf{P}^T\mathbf{A}\mathbf{P}, \quad \text{that is,} \quad \mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^T.$$

In other words, every real symmetric matrix is orthogonally diagonalizable.

Proof. Starting with $\mathbf{A}_0 = \mathbf{A}$, let us construct a sequence of matrices $\mathbf{A}_k = \left(a_{ij}^{(k)}\right)$, $k = 0, 1, ...$, in which $\mathbf{A}_k$ is obtained from $\mathbf{A}_{k-1}$ by multiplying on the left and on the right by rotation matrices:

$$\mathbf{A}_k = \mathbf{R}_k^T\mathbf{A}_{k-1}\mathbf{R}_k, \qquad (*)$$

where $\mathbf{R}_k$ differs from the identity matrix $\mathbf{I}$ only in the four entries of the (2×2)-submatrix located at the intersection of the rows and columns with indices $i < j$ and equal to

$$\begin{pmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{pmatrix}.$$

The matrix $\mathbf{R}_k$ effects a rotation through the angle φ in the coordinate plane determined by the indices $i \neq j$. Every rotation matrix of this form is obviously orthogonal.

Clearly, the symmetry of the matrix $\mathbf{A}_0 = \mathbf{A}$ is inherited by all the matrices $\mathbf{A}_k$. Only the rows and columns with indices i and j change, and the (2×2)-submatrix of $\mathbf{A}_k$ in these rows and columns is obtained from the corresponding submatrix of $\mathbf{A}_{k-1}$ by the same rotation; its off-diagonal entry equals $a_{ij}^{(k-1)}\cos 2\varphi - \frac{1}{2}\left(a_{ii}^{(k-1)} - a_{jj}^{(k-1)}\right)\sin 2\varphi$. Hence φ can be chosen in such a way that $a_{ij}^{(k)} = a_{ji}^{(k)} = 0$ (if $a_{ij}^{(k-1)} \neq 0$, take φ with $\cot 2\varphi = \left(a_{ii}^{(k-1)} - a_{jj}^{(k-1)}\right)/\left(2a_{ij}^{(k-1)}\right)$). Denote by d_k and h_k the sums of the squares of the diagonal and off-diagonal entries of the matrix $\mathbf{A}_k$. An orthogonal transformation preserves the sum of the squares of the entries of a matrix (Section 5.2); applying this to the whole matrix and to the (2×2)-submatrix (which becomes diagonal), and noting that the other diagonal entries do not change, we get $d_k + h_k = d_{k-1} + h_{k-1}$ and

$$\left(a_{ii}^{(k)}\right)^2 + \left(a_{jj}^{(k)}\right)^2 = \left(a_{ii}^{(k-1)}\right)^2 + \left(a_{jj}^{(k-1)}\right)^2 + 2\left(a_{ij}^{(k-1)}\right)^2 \quad \Rightarrow$$

$$\Rightarrow \quad d_k - d_{k-1} = 2\left(a_{ij}^{(k-1)}\right)^2.$$

For each k, we shall choose the plane of rotation (the indices $i < j$) in such a way that the eliminated entry $a_{ij}^{(k-1)}$ is maximal in modulus among all off-

diagonal entries of the matrix $\mathbf{A}_{k-1}$. The total number of off-diagonal entries is $n^2 - n$. Therefore

$$\left(a_{ij}^{(k-1)}\right)^2 \geq \frac{h_{k-1}}{n^2 - n}.$$

Hence, taking into account the equality $d_k + h_k = d_{k-1} + h_{k-1}$, we obtain

$$h_k \leq h_{k-1} - \frac{2}{n^2 - n} h_{k-1} \leq \left(1 - \frac{2}{n^2 - n}\right)^k h_0 \to 0 \quad \text{as } k \to \infty.$$

From relations $(*)$ it follows that

$$\mathbf{A}_k = \mathbf{P}_k^T \mathbf{A} \mathbf{P}_k, \quad k = 1, 2, ...,$$

where for all k the matrices $\mathbf{P}_k = \left(p_{ij}^{(k)}\right)$ are orthogonal (as products of orthogonal matrices).

For any fixed i, j, the sequences $a_{ij}^{(k)}, p_{ij}^{(k)}$ are bounded. By the lemma on bounded sequences (from finitely many bounded sequences of numbers one can extract convergent subsequences with one and the same sequence of indices; this follows by applying the Bolzano–Weierstrass theorem to the sequences one after another, see Tyrtyshnikov, §20.4), there exists a sequence of indices $k_1 < k_2 < ...$ such that each of the subsequences $a_{ij}^{(k_l)}, p_{ij}^{(k_l)}$ is convergent. Note that $a_{ij}^{(k_l)} \to 0$ for $i \neq j$. Let

$$\lim_{l \to \infty} a_{ii}^{(k_l)} = \lambda_i, \quad i = 1, ..., n, \quad \lim_{l \to \infty} p_{ij}^{(k_l)} = p_{ij}, \quad i, j = 1, ..., n.$$

Introduce the matrices

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}, \quad \mathbf{P} = \left(p_{ij}\right).$$

For all $l = 1, 2, ...$ we have: $\mathbf{A}_{k_l} = \mathbf{P}_{k_l}^T \mathbf{A} \mathbf{P}_{k_l}$. Passing to the limit in the entrywise equalities, we obtain

$$\mathbf{D} = \mathbf{P}^T \mathbf{A} \mathbf{P}.$$

From the orthogonality conditions $\mathbf{P}_{k_l}^T \mathbf{P}_{k_l} = \mathbf{I}$ it follows that in the limit $\mathbf{P}^T \mathbf{P} = \mathbf{I}$. Hence the matrix $\mathbf{P}$ is orthogonal. □

We have just obtained one of the most important results—both for matrix theory itself and for its numerous applications. In our course we shall return to discussing it in connection with a number of fundamental concepts of linear algebra. Our proof is remarkable for its constructiveness: it simultaneously provides a method for the approximate computation of the matrices $\mathbf{D}$ and $\mathbf{P}$. This is one of the earliest practical methods of computational algebra, proposed by C. Jacobi in 1846.[12]

[12]The latest results in the study of the method of rotations belong to the very recent past: in the 1990s its special capabilities associated with the high-accuracy computation of entries of the matrix $\mathbf{D}$ that are small in modulus were discovered.

Problem 1. Given a symmetric matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ with a nonzero sum of the entries on the main diagonal, prove the existence of an orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{n\times n}$ such that in the matrix $\mathbf{Q}^T\mathbf{A}\mathbf{Q}$ all entries on the main diagonal are equal.

Hermitian Matrices

Recall that a matrix $\mathbf{A}$ is called Hermitian if $\mathbf{A}^* = \mathbf{A}$ (see Tyrtyshnikov, §25.2). A square complex matrix $\mathbf{A}$ is called *normal* if $\mathbf{A}^*\mathbf{A} = \mathbf{A}\mathbf{A}^*$; obviously, any Hermitian matrix is normal. A matrix is normal if and only if $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^*$ for some unitary matrix $\mathbf{Q}$ and some diagonal matrix $\mathbf{D} = \operatorname{diag}(\lambda_1, ..., \lambda_n)$, whose diagonal entries are then the eigenvalues of $\mathbf{A}$ (this is proved in Tyrtyshnikov, §33.1, by means of Schur's theorem, §30.8).

Proposition. A normal matrix is Hermitian if and only if all of its eigenvalues are real.

Proof. Let $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^*$ as above. Then $\mathbf{A}^* = \mathbf{Q}\mathbf{D}^*\mathbf{Q}^*$, so $\mathbf{A}^* = \mathbf{A} \Leftrightarrow \mathbf{D}^* = \mathbf{D} \Leftrightarrow \overline{\lambda}_i = \lambda_i$, $1 \le i \le n$. □

Remark. For a real matrix $\mathbf{A}$ we have $\mathbf{A}^* = \mathbf{A}^T$, so the real Hermitian matrices are exactly the real symmetric matrices, $\mathbf{A}^T = \mathbf{A}$; by the proposition, their eigenvalues are real. The real symmetric case is the *spectral theorem* of MA1522 (stated at the beginning of this section): a real square matrix $\mathbf{A}$ is orthogonally diagonalizable, $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^T$ with $\mathbf{P}$ orthogonal and $\mathbf{D}$ diagonal, if and only if $\mathbf{A}$ is symmetric. One direction is the theorem on the diagonalization of real symmetric matrices proved above; conversely, $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^T$ implies $\mathbf{A}^T = \mathbf{P}\mathbf{D}^T\mathbf{P}^T = \mathbf{P}\mathbf{D}\mathbf{P}^T = \mathbf{A}$.

Problem 2. It is known that $\mathbf{A}^2 = \mathbf{A}$ and $\operatorname{Null}(\mathbf{A}) \perp \operatorname{Col}(\mathbf{A})$ (orthogonality with respect to the standard inner product). Prove that $\mathbf{A} = \mathbf{A}^*$.

Problem 3. Given a subspace $V \subset \mathbb{C}^n$. Prove that among all matrices $\mathbf{A}$ such that $\operatorname{Col}(\mathbf{A}) = V$ and $\mathbf{A}^2 = \mathbf{A}$, the smallest value of the 2-norm is attained for some Hermitian matrix $\mathbf{A}$, and moreover for only one.

Problem 4. Prove that for any Hermitian matrix $\mathbf{H}$ the matrix $\mathbf{Q} = (\mathbf{I} - i\mathbf{H})^{-1}(\mathbf{I} + i\mathbf{H})$ is unitary. Can every unitary matrix be represented in this way?

Problem 5. Hermitian matrices $\mathbf{A}, \mathbf{B} \in \mathbb{C}^{n\times n}$ are such that $\mathbf{A}^2 = \mathbf{A}$ and $\mathbf{B}^2 = \mathbf{B}$. Prove that if $\|\mathbf{A} - \mathbf{B}\|_2 < 1$, then $\operatorname{rank}(\mathbf{A}) = \operatorname{rank}(\mathbf{B})$.

Algorithm for Orthogonal Diagonalization ADDED FOR MA1522

Algorithm (orthogonal diagonalization).

Let $\mathbf{A}$ be a symmetric matrix of order n.

1. Compute and factorize the characteristic polynomial,
$$\operatorname{char}(\mathbf{A}) = (x - \lambda_1)^{r_{\lambda_1}} \cdots (x - \lambda_k)^{r_{\lambda_k}},$$

with distinct $\lambda_1, ..., \lambda_k$.
2. For each i, find a basis S_{λ_i} for the eigenspace E_{λ_i}, the solution space of $(\lambda_i \mathbf{I} - \mathbf{A})\mathbf{x} = \mathbf{0}$.
3. Apply the Gram–Schmidt process (Section 5.3) to each S_{λ_i}, obtaining an orthonormal basis T_{λ_i} for E_{λ_i}. Then $T = T_{\lambda_1} \cup \cdots \cup T_{\lambda_k} = \{\mathbf{u}_1, ..., \mathbf{u}_n\}$ is an orthonormal basis for $\mathbb{R}^n$.
4. Set $\mathbf{P} = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{pmatrix}$ and $\mathbf{D} = \operatorname{diag}(\mu_1, ..., \mu_n)$, where μ_j is the eigenvalue associated to $\mathbf{u}_j$. Then $\mathbf{P}$ is orthogonal and $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^T$.

The algorithm always succeeds. By the spectral theorem, $\mathbf{A}$ is diagonalizable, so by Section 6.2 its characteristic polynomial splits into linear factors and $\dim\left(E_{\lambda_1}\right) + \cdots + \dim\left(E_{\lambda_k}\right) = n$; hence T consists of n unit vectors. Vectors of T taken from the same T_{λ_i} are orthogonal by construction, and vectors from different eigenspaces are orthogonal by the theorem at the beginning of this section. An orthonormal set of n vectors in $\mathbb{R}^n$ is a basis.

Example. Let

$$\mathbf{A} = \begin{pmatrix} 0 & -2 & -1 \\ -2 & 3 & 2 \\ -1 & 2 & 0 \end{pmatrix}.$$

A direct computation gives $\operatorname{char}(\mathbf{A}) = x^3 - 3x^2 - 9x - 5 = (x+1)^2(x-5)$, and

$$-\mathbf{I} - \mathbf{A} = \begin{pmatrix} -1 & 2 & 1 \\ 2 & -4 & -2 \\ 1 & -2 & -1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & -2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$E_{-1} = \operatorname{span}\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\},$$

$$5\mathbf{I} - \mathbf{A} = \begin{pmatrix} 5 & 2 & 1 \\ 2 & 2 & -2 \\ 1 & -2 & 5 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{pmatrix}, \quad E_5 = \operatorname{span}\left\{ \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} \right\}.$$

As the theorem predicts, $(-1, 2, 1)^T$ is orthogonal to both basis vectors $\mathbf{s}_1 = (2, 1, 0)^T$ and $\mathbf{s}_2 = (1, 0, 1)^T$ of E_{-1}, while these two are not orthogonal to each other. The Gram–Schmidt process in E_{-1} gives $\mathbf{w}_1 = \mathbf{s}_1$ and

$$\mathbf{w}_2 = \mathbf{s}_2 - \frac{\mathbf{s}_2 \cdot \mathbf{w}_1}{\|\mathbf{w}_1\|^2}\mathbf{w}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{2}{5}\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 1 \\ -2 \\ 5 \end{pmatrix}.$$

Normalizing $\mathbf{w}_1$, $\mathbf{w}_2$ and the eigenvector $(-1, 2, 1)^T$, we obtain $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^T$ with

$$\mathbf{P} = \begin{pmatrix} 2/\sqrt{5} & 1/\sqrt{30} & -1/\sqrt{6} \\ 1/\sqrt{5} & -2/\sqrt{30} & 2/\sqrt{6} \\ 0 & 5/\sqrt{30} & 1/\sqrt{6} \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

6.4. Application of Diagonalization: Markov Chains

Powers of Diagonalizable Matrices ADDED FOR MA1522

Theorem. If $\mathbf{A} = \mathbf{P}\mathbf{B}\mathbf{P}^{-1}$ with an invertible matrix $\mathbf{P}$, then $\mathbf{A}^k = \mathbf{P}\mathbf{B}^k\mathbf{P}^{-1}$ for every positive integer k. If, moreover, $\mathbf{A}$ is invertible, this holds for every integer k (with $\mathbf{A}^0 = \mathbf{I}$ and $\mathbf{A}^{-k} = (\mathbf{A}^{-1})^k$).

Proof. By induction on k: $\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A} = \mathbf{P}\mathbf{B}^k\mathbf{P}^{-1}\mathbf{P}\mathbf{B}\mathbf{P}^{-1} = \mathbf{P}\mathbf{B}^{k+1}\mathbf{P}^{-1}$. If $\mathbf{A}$ is invertible, then so is $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$, and $\mathbf{A}^{-1} = \mathbf{P}\mathbf{B}^{-1}\mathbf{P}^{-1}$; applying the first part to $\mathbf{A}^{-1}$ gives the negative powers, and $k = 0$ is trivial. □

The matrix $\mathbf{B}$ need not be diagonal here. When it is, its powers are easy to compute: the product of two diagonal matrices is the diagonal matrix of the products of their diagonal entries, so by induction

$$\operatorname{diag}(d_1, \ldots, d_n)^k = \operatorname{diag}\big(d_1^k, \ldots, d_n^k\big)$$

for every positive integer k, and for every integer k if all $d_i \neq 0$.

Corollary. If $\mathbf{A} = \mathbf{P}\operatorname{diag}(\mu_1, \ldots, \mu_n)\mathbf{P}^{-1}$, then

$$\mathbf{A}^k = \mathbf{P}\operatorname{diag}\big(\mu_1^k, \ldots, \mu_n^k\big)\mathbf{P}^{-1}$$

for every positive integer k, and for every integer k if $\mathbf{A}$ is invertible.

(If $\mathbf{A}$ is invertible, all $\mu_i \neq 0$, since 0 is not an eigenvalue of $\mathbf{A}$.) Thus once a matrix has been diagonalized, its powers cost no more than the powers of n numbers. In the same spirit:

Proposition. If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$, then $\mathbf{A}^k\mathbf{v} = \lambda^k\mathbf{v}$ for every positive integer k: an eigenvector of $\mathbf{A}$ associated to λ is an eigenvector of $\mathbf{A}^k$ associated to λ^k.

Proof. By induction: $\mathbf{A}^{k+1}\mathbf{v} = \mathbf{A}\big(\mathbf{A}^k\mathbf{v}\big) = \lambda^k\mathbf{A}\mathbf{v} = \lambda^{k+1}\mathbf{v}$. □

Stochastic Matrices and Markov Chains ADDED FOR MA1522

Definition. A vector $\mathbf{v} = (v_1, \ldots, v_n)^T \in \mathbb{R}^n$ is a *probability vector* if $v_i \geq 0$ for all i and $v_1 + v_2 + \cdots + v_n = 1$. A *stochastic matrix* is a square matrix whose columns are probability vectors. A *Markov chain* is a sequence of probability vectors $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \ldots$ together with a stochastic matrix $\mathbf{P}$ such that

$$\mathbf{x}_{k+1} = \mathbf{P}\mathbf{x}_k, \quad k = 0, 1, 2, \ldots$$

In applications, a Markov chain describes a system that at each step is in one of n states. The i-th entry of the *state vector* $\mathbf{x}_k$ is the probability that after k steps the system is in state i, and the entry p_{ij} of $\mathbf{P}$ is the probability of passing in one step from state j to state i. Clearly, $\mathbf{x}_k = \mathbf{P}^k\mathbf{x}_0$.

Let $\mathbf{1} = (1, 1, \ldots, 1)^T \in \mathbb{R}^n$. The sum of the entries of a vector $\mathbf{v}$ is $\mathbf{1}^T\mathbf{v}$, so a square matrix $\mathbf{P}$ with nonnegative entries is stochastic if and only if $\mathbf{1}^T\mathbf{P} = \mathbf{1}^T$, that is, $\mathbf{P}^T\mathbf{1} = \mathbf{1}$.

Proposition. If $\mathbf{P}$ is a stochastic matrix and $\mathbf{v}$ is a probability vector, then $\mathbf{P}\mathbf{v}$ is a probability vector. A product of stochastic matrices of the same order is stochastic; in particular, every power $\mathbf{P}^k$ of a stochastic matrix is stochastic.

Proof. The entries of $\mathbf{P}\mathbf{v}$ are sums of products of nonnegative numbers, and $\mathbf{1}^T(\mathbf{P}\mathbf{v}) = (\mathbf{1}^T\mathbf{P})\mathbf{v} = \mathbf{1}^T\mathbf{v} = 1$. The j-th column of a product $\mathbf{P}\mathbf{Q}$ is $\mathbf{P}$ times the j-th column of $\mathbf{Q}$. □

Thus a stochastic matrix $\mathbf{P}$ and any probability vector $\mathbf{x}_0$ generate a Markov chain. Its long-run behaviour can often be found by diagonalizing $\mathbf{P}$.

Example. Every working day a commuter travels either by bus (state 1) or by train (state 2). After a day on the bus she takes the bus again the next day with probability 0.9 and the train with probability 0.1; after a day on the train she switches to the bus with probability 0.2 and stays with the train with probability 0.8. The stochastic matrix of this Markov chain is

$$\mathbf{P} = \begin{pmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{pmatrix}.$$

Its characteristic polynomial is $\operatorname{char}(\mathbf{P}) = x^2 - 1.7x + 0.7 = (x - 1)(x - 0.7)$; the eigenspaces are $E_1 = \operatorname{span}\{(2, 1)^T\}$ and $E_{0.7} = \operatorname{span}\{(-1, 1)^T\}$, so that

$$\mathbf{P} = \mathbf{Q}\begin{pmatrix} 1 & 0 \\ 0 & 0.7 \end{pmatrix}\mathbf{Q}^{-1}, \quad \mathbf{Q} = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{Q}^{-1} = \frac{1}{3}\begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix},$$

and by the corollary of the preceding subsection

$$\mathbf{P}^k = \mathbf{Q}\begin{pmatrix} 1 & 0 \\ 0 & 0.7^k \end{pmatrix}\mathbf{Q}^{-1} = \frac{1}{3}\begin{pmatrix} 2 + 0.7^k & 2 - 2\cdot 0.7^k \\ 1 - 0.7^k & 1 + 2\cdot 0.7^k \end{pmatrix}.$$

If she takes the train today, $\mathbf{x}_0 = \mathbf{e}_2$, then three days later

$$\mathbf{x}_3 = \mathbf{P}^3\mathbf{e}_2 = \frac{1}{3}\begin{pmatrix} 2 - 2\cdot 0.343 \\ 1 + 2\cdot 0.343 \end{pmatrix} = \begin{pmatrix} 0.438 \\ 0.562 \end{pmatrix}.$$

Since $0.7^k \to 0$ as $k \to \infty$, for every probability vector $\mathbf{x}_0 = (a, b)^T$ we get

$$\mathbf{P}^k \mathbf{x}_0 \to \frac{1}{3}\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2/3 \\ 1/3 \end{pmatrix}$$

(because $a + b = 1$): in the long run she takes the bus on two days out of three, whatever she does today. The limit is the probability vector in E_1.

Equilibrium Vectors ADDED FOR MA1522

Definition. An *equilibrium vector* (or *steady-state vector*) of a stochastic matrix $\mathbf{P}$ is a probability vector $\mathbf{v}$ with $\mathbf{P}\mathbf{v} = \mathbf{v}$, that is, a probability vector that is an eigenvector of $\mathbf{P}$ associated to the eigenvalue 1.

Proposition. The number 1 is an eigenvalue of every stochastic matrix $\mathbf{P}$.

Proof. Since $\mathbf{P}^T\mathbf{1} = \mathbf{1}$, the number 1 is an eigenvalue of $\mathbf{P}^T$, and $\mathbf{P}$ and $\mathbf{P}^T$ have the same eigenvalues (Section 6.1). □

Theorem. Let $\mathbf{P}$ be a stochastic matrix and let $\mathbf{x}_{k+1} = \mathbf{P}\mathbf{x}_k$ be a Markov chain. If the state vectors $\mathbf{x}_k$ converge (entrywise) to a vector $\mathbf{v}$ as $k \to \infty$, then $\mathbf{v}$ is an equilibrium vector of $\mathbf{P}$.

Proof. As limits of nonnegative numbers, the entries of $\mathbf{v}$ are nonnegative, and their sum is the limit of the sums of the entries of $\mathbf{x}_k$, which all equal 1. Each entry of $\mathbf{P}\mathbf{x}_k$ is a fixed linear combination of the entries of $\mathbf{x}_k$, so $\mathbf{P}\mathbf{v} = \lim_{k\to\infty} \mathbf{P}\mathbf{x}_k = \lim_{k\to\infty} \mathbf{x}_{k+1} = \mathbf{v}$. □

Proposition. Every stochastic matrix has an equilibrium vector.

Proof. Take any probability vector $\mathbf{x}_0$ and let $\mathbf{y}_m = \frac{1}{m}(\mathbf{x}_0 + \mathbf{P}\mathbf{x}_0 + \cdots + \mathbf{P}^{m-1}\mathbf{x}_0)$, $m = 1, 2, \ldots$ Each $\mathbf{y}_m$ is a probability vector, and

$$\mathbf{P}\mathbf{y}_m - \mathbf{y}_m = \frac{1}{m}(\mathbf{P}^m\mathbf{x}_0 - \mathbf{x}_0)$$

has all its entries between $-1/m$ and $1/m$. The entries of the vectors $\mathbf{y}_m$ lie in $[0, 1]$; by the Bolzano–Weierstrass theorem, applied to the n coordinates one after another, some subsequence $\mathbf{y}_{m_1}, \mathbf{y}_{m_2}, \ldots$ converges to a vector $\mathbf{v}$. As in the preceding proof, $\mathbf{v}$ is a probability vector, and $\mathbf{P}\mathbf{v} - \mathbf{v} = \lim_{l\to\infty}\big(\mathbf{P}\mathbf{y}_{m_l} - \mathbf{y}_{m_l}\big) = \mathbf{0}$. □

An equilibrium vector is therefore found by solving a homogeneous linear system and rescaling.

Algorithm (equilibrium vector).

Let $\mathbf{P}$ be a stochastic matrix of order n.

1. Find a nonzero solution $\mathbf{u} = (u_1, ..., u_n)^T$ of $(\mathbf{I} - \mathbf{P})\mathbf{x} = \mathbf{0}$ with nonnegative entries, that is, such an eigenvector of $\mathbf{P}$ associated to 1. If $\dim(E_1) = 1$ (as for every regular stochastic matrix, see the next subsection), any nonzero solution will do, after a change of sign if necessary.
2. Set $\mathbf{v} = (u_1 + \cdots + u_n)^{-1}\mathbf{u}$. Then $\mathbf{v}$ is an equilibrium vector of $\mathbf{P}$.

Indeed, $\mathbf{v} \in E_1$ has nonnegative entries, and they add up to $(u_1 + \cdots + u_n)/(u_1 + \cdots + u_n) = 1$. When $\dim(E_1) = 1$, the line E_1 is spanned by an equilibrium vector (one exists by the proposition above), so every nonzero $\mathbf{u} \in E_1$ is a nonzero multiple of it and has all its entries ≥ 0 or all ≤ 0; in this case the equilibrium vector is unique, since two probability vectors on one line through $\mathbf{0}$ coincide. Equivalently, an equilibrium vector is a solution with nonnegative entries of the $(n+1) \times n$ linear system

$$\begin{pmatrix} \mathbf{P} - \mathbf{I}_n \\ \mathbf{1}^T \end{pmatrix} \mathbf{x} = \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix},$$

which has exactly one solution when $\dim(E_1) = 1$.

Regular Stochastic Matrices ADDED FOR MA1522

Definition. A stochastic matrix $\mathbf{P}$ is *regular* if some power $\mathbf{P}^k$, $k \geq 1$, has all its entries positive.

For example, the matrix $\mathbf{P}$ of the commuter example above is regular: already $\mathbf{P}^1$ has positive entries.

Lemma. If all the entries of a stochastic matrix $\mathbf{A} = \left(a_{ij}\right)_{n \times n}$ are positive, then the eigenvalue 1 of $\mathbf{A}$ has geometric multiplicity $\dim(E_1) = 1$.

Proof. By Section 6.1 it suffices to prove this for $\mathbf{A}^T$. Let $\mathbf{A}^T\mathbf{x} = \mathbf{x}$, $\mathbf{x} = (x_1, ..., x_n)^T$, and choose an index m with $x_m = \max_i x_i$. The m-th entry of the equality $\mathbf{A}^T\mathbf{x} = \mathbf{x}$ reads $a_{1m}x_1 + \cdots + a_{nm}x_n = x_m$; since $a_{1m} + \cdots + a_{nm} = 1$, this means

$$a_{1m}(x_m - x_1) + a_{2m}(x_m - x_2) + \cdots + a_{nm}(x_m - x_n) = 0.$$

All the terms are nonnegative and all $a_{im} > 0$, so $x_i = x_m$ for every i. Hence the eigenspace of $\mathbf{A}^T$ associated to 1 is $\operatorname{span}\{\mathbf{1}\}$, of dimension 1. □

Theorem. A regular stochastic matrix $\mathbf{P}$ has a unique equilibrium vector $\mathbf{v}$, and for every probability vector $\mathbf{x}_0$ the Markov chain $\mathbf{x}_{k+1} = \mathbf{P}\mathbf{x}_k$ converges to $\mathbf{v}$.

Proof. Let $\mathbf{B} = \mathbf{P}^k$ have positive entries. The matrix $\mathbf{B}$ is stochastic as a power of a stochastic matrix, so by the lemma its eigenspace associated to 1 is a line. Every eigenvector of $\mathbf{P}$ associated to 1 is an eigenvector of $\mathbf{B}$ associated to $1^k = 1$ (see the subsection on powers of matrices), so for $\mathbf{P}$ we get $1 \le \dim(E_1) \le 1$. By the preceding subsection, $\mathbf{P}$ has exactly one equilibrium vector $\mathbf{v}$.

To prove convergence, use the norm $\|\mathbf{y}\|_1 = |y_1| + \cdots + |y_n|$ on $\mathbb{R}^n$. For a stochastic matrix $\mathbf{Q} = \big(q_{ij}\big)$ and any $\mathbf{y} \in \mathbb{R}^n$,

$$\|\mathbf{Q}\mathbf{y}\|_1 = \sum_{i=1}^{n} \left| \sum_{j=1}^{n} q_{ij} y_j \right| \le \sum_{j=1}^{n} \left(\sum_{i=1}^{n} q_{ij} \right) |y_j| = \|\mathbf{y}\|_1,$$

and $\mathbf{1}^T\mathbf{Q}\mathbf{y} = \mathbf{1}^T\mathbf{y}$, so $\mathbf{Q}$ maps vectors with entry sum 0 to vectors with entry sum 0. Let $\delta > 0$ be the smallest entry of $\mathbf{B} = \big(b_{ij}\big)$; since the columns of $\mathbf{B}$ add up to 1, we have $n\delta \le 1$. If $\mathbf{1}^T\mathbf{y} = 0$, then $(\mathbf{B}\mathbf{y})_i = \sum_j \big(b_{ij} - \delta\big) y_j$, where $b_{ij} - \delta \ge 0$ and $\sum_i \big(b_{ij} - \delta\big) = 1 - n\delta$, and the same estimate gives

$$\|\mathbf{B}\mathbf{y}\|_1 \le (1 - n\delta)\|\mathbf{y}\|_1.$$

Now let $\mathbf{x}_0$ be a probability vector and $\mathbf{y} = \mathbf{x}_0 - \mathbf{v}$; then $\mathbf{1}^T\mathbf{y} = 0$ and $\mathbf{x}_m - \mathbf{v} = \mathbf{P}^m\mathbf{x}_0 - \mathbf{P}^m\mathbf{v} = \mathbf{P}^m\mathbf{y}$. Writing $m = qk + s$ with $0 \le s < k$, we obtain

$$\|\mathbf{x}_m - \mathbf{v}\|_1 = \|\mathbf{B}^q\mathbf{P}^s\mathbf{y}\|_1 \le (1 - n\delta)^q \|\mathbf{P}^s\mathbf{y}\|_1 \le (1 - n\delta)^q \|\mathbf{y}\|_1,$$

which tends to 0 as $m \to \infty$, because $0 \le 1 - n\delta < 1$. Hence $\mathbf{x}_m \to \mathbf{v}$. □

For the commuter example this explains the limit found there: $(2/3, 1/3)^T$ is the equilibrium vector of $\mathbf{P}$, and every Markov chain with this matrix converges to it.

Application: Ranking Web Pages ADDED FOR MA1522

The PageRank algorithm, on which the Google search engine of L. Page and S. Brin (1996) was originally based, ranks a set of web sites by the entries of an equilibrium vector. Its premise is that important sites receive more links from other sites. For sites $s_1, \ldots, s_n$, the *adjacency matrix* $\mathbf{A} = \big(a_{ij}\big)_{n \times n}$ has $a_{ij} = 1$ if site s_j has an outgoing link to site s_i, and $a_{ij} = 0$ otherwise. The sum of the i-th row of $\mathbf{A}$ is the number of links into s_i, and the sum of the j-th column is the number of links out of s_j. Assume that every site links to some other site, and divide each entry of $\mathbf{A}$ by the sum of its column; the result is the *probability transition matrix*

$$\mathbf{P} = \big(p_{ij}\big), \quad p_{ij} = \frac{a_{ij}}{a_{1j} + a_{2j} + \cdots + a_{nj}}.$$

It is a stochastic matrix: p_{ij} is the probability that a surfer who clicks a random link on s_j arrives at s_i, and $\mathbf{P}^k\mathbf{x}_0$ describes where a random surfer is after k clicks. If $\mathbf{P}$ is regular, these probabilities converge, from any starting distribution $\mathbf{x}_0$, to the unique equilibrium vector of $\mathbf{P}$, which may also be read as the share of all surfers visiting each site; the sites are ranked by its entries.

Example. Let s_1 link to s_2 and s_3; s_2 to s_3 and s_4; s_3 to s_1; and s_4 to s_1 and s_3. Then

$$\mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix} 0 & 0 & 1 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & 0 & 0 \end{pmatrix}.$$

A surfer starting at s_4 has the state vectors $\mathbf{x}_1 = \mathbf{P}\mathbf{e}_4 = \left(\frac{1}{2}, 0, \frac{1}{2}, 0\right)^T$ and $\mathbf{x}_2 = \mathbf{P}\mathbf{x}_1 = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}, 0\right)^T$. One checks that all the entries of $\mathbf{P}^5$ are positive, so $\mathbf{P}$ is regular. Next,

$$\mathbf{I} - \mathbf{P} = \begin{pmatrix} 1 & 0 & -1 & -\frac{1}{2} \\ -\frac{1}{2} & 1 & 0 & 0 \\ -\frac{1}{2} & -\frac{1}{2} & 1 & -\frac{1}{2} \\ 0 & -\frac{1}{2} & 0 & 1 \end{pmatrix} \xrightarrow{\text{RREF}} \begin{pmatrix} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -\frac{7}{2} \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

so E_1 is spanned by $\mathbf{u} = (8, 4, 7, 2)^T$, and the equilibrium vector is $\mathbf{v} = \frac{1}{21}(8, 4, 7, 2)^T$. The sites are ranked s_1, s_3, s_2, s_4. Note that s_3 receives the most links, yet s_1 comes first: the single link out of s_3 passes all of its weight to s_1.

6.5. Singular Value Decomposition

Singular Value Decomposition of a Matrix

In the 1870s, independently and almost simultaneously, Beltrami (1873) and Jordan (1874) discovered that any square matrix can be reduced to diagonal form by multiplying on the left and on the right by unitary matrices. Various questions connected with this discovery, including its generalizations, subsequently became the subject of a whole series of investigations. It would not be a great exaggeration to say that this fact turned out to be tremendously useful and one of the most sought-after results in matrix theory and applications of linear algebra.

In reality, the same holds for a rectangular matrix as well. By multiplication by unitary matrices, it is reduced to a rectangular matrix of the same dimensions having zeros everywhere except at the entries with indices $i = j$. We shall call such matrices *diagonal rectangular* matrices.

Thus, we are dealing with a decomposition of the form

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*, \tag{2}$$

[13]The full textbook writes this decomposition as $\mathbf{A} = \mathbf{V}\mathbf{\Sigma}\mathbf{U}^*$, with the letters $\mathbf{U}$ and $\mathbf{V}$, and the singular vectors $\mathbf{u}_i$ and $\mathbf{v}_i$, interchanged; this edition follows MA1522, where a real

where $\mathbf{A}$ is a given $(m \times n)$-matrix, $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices of orders m and n respectively,[13] and $\mathbf{\Sigma}$ is a diagonal rectangular $(m \times n)$-matrix having in the positions $(1,1), (2,2), \ldots$ the nonnegative numbers

$$\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_{\min(m,n)}.$$

The decomposition (2) is called the *singular value decomposition* of the matrix $\mathbf{A}$. The numbers σ_i are called the *singular values* of the matrix $\mathbf{A}$.

Theorem. A singular value decomposition $\mathbf{A} = \mathbf{U\Sigma V}^*$ exists for any complex rectangular matrix $\mathbf{A}$. If $\mathbf{A}$ is real, then the matrices $\mathbf{U}$ and $\mathbf{V}$ can be chosen to be real (that is, orthogonal), and then $\mathbf{A} = \mathbf{U\Sigma V}^T$.

Proof. The case $\mathbf{A} = \mathbf{0}$ is trivial, so we shall assume that $\mathbf{A} \neq \mathbf{0}$. Set $\sigma_1 = \|\mathbf{A}\|_2 = \sup_{\mathbf{x}\neq\mathbf{0}} \|\mathbf{Ax}\|/\|\mathbf{x}\|$ (the matrix norm induced by the length of vectors, called the *spectral norm*; see Tyrtyshnikov, §27.3). By the compactness of the unit sphere in $\mathbb{C}^n$, the continuity of the norm, and Weierstrass's theorem, there exists a vector $\mathbf{x}_1$ such that $\|\mathbf{Ax}_1\| = \|\mathbf{A}\|_2$ and $\|\mathbf{x}_1\| = 1$. Let $\mathbf{y}_1 = \mathbf{Ax}_1/\|\mathbf{Ax}_1\|$. Thus,

$$\mathbf{Ax}_1 = \sigma_1 \mathbf{y}_1, \quad \|\mathbf{x}_1\| = \|\mathbf{y}_1\|. \tag{3}$$

Let us complete $\mathbf{x}_1$ and $\mathbf{y}_1$ to orthonormal bases and form the unitary matrices

$$\mathbf{U}_1 = (\mathbf{y}_1 \;\; \mathbf{y}_2 \;\; \cdots \;\; \mathbf{y}_m), \quad \mathbf{V}_1 = (\mathbf{x}_1 \;\; \mathbf{x}_2 \;\; \cdots \;\; \mathbf{x}_n).$$

According to (3), the matrix $\mathbf{A}_1 \equiv \mathbf{U}_1^* \mathbf{A} \mathbf{V}_1$ has only one nonzero entry in the first column, equal to σ_1:

$$\mathbf{A}_1 = \mathbf{U}_1^* \mathbf{A} \mathbf{V}_1 = \begin{pmatrix} \sigma_1 & \mathbf{z}^* \\ \mathbf{0} & \mathbf{A}_2 \end{pmatrix}.$$

By the unitary invariance of the spectral norm (see Tyrtyshnikov, §27.7), $\|\mathbf{A}_1\|_2 = \sigma_1$. Therefore,

$$\sigma_1 \left\| \begin{pmatrix} \sigma_1 \\ \mathbf{z} \end{pmatrix} \right\| \geq \left\| \mathbf{A}_1 \begin{pmatrix} \sigma_1 \\ \mathbf{z} \end{pmatrix} \right\| \geq \sigma_1^2 + \|\mathbf{z}\|^2 \quad \Rightarrow \quad \sigma_1^2 \geq \sigma_1^2 + \|\mathbf{z}\|^2 \quad \Rightarrow$$

$$\Rightarrow \quad \mathbf{z} = \mathbf{0} \quad \Rightarrow \quad \mathbf{A}_1 = \begin{pmatrix} \sigma_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_2 \end{pmatrix}.$$

Next we argue by induction. If a singular value decomposition $\mathbf{A}_2 = \mathbf{U}_2 \mathbf{\Sigma}_2 \mathbf{V}_2^*$ is already available for $\mathbf{A}_2$, then a singular value decomposition for $\mathbf{A}$ is easily found. For this it suffices to take

$$\mathbf{U} = \mathbf{U}_1 \begin{pmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \mathbf{U}_2 \end{pmatrix}, \quad \mathbf{V} = \mathbf{V}_1 \begin{pmatrix} 1 & \mathbf{0} \\ \mathbf{0} & \mathbf{V}_2 \end{pmatrix},$$

matrix has $\mathbf{A} = \mathbf{U\Sigma V}^T$. (With the textbook's lettering, the Russian original inadvertently gives the orders of $\mathbf{U}$ and $\mathbf{V}$ as m and n.)

note that the matrices $\mathbf{U}$ and $\mathbf{V}$ are unitary (as products of unitary matrices), and verify that the equality

$$\mathbf{U}^*\mathbf{A}\mathbf{V} = \begin{pmatrix} \sigma_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{\Sigma}_2 \end{pmatrix}$$

holds.

It remains to note that the induction begins with constructing the singular value decomposition for matrices consisting of a single column or a single row.

Let $\mathbf{A} = (\mathbf{a}) \in \mathbb{C}^{m\times 1}$ be a column matrix. In this case we find in $\mathbb{C}^m$ an orthonormal basis $\mathbf{u}_1, ..., \mathbf{u}_m$ starting with $\mathbf{u}_1 = \mathbf{a}/\|\mathbf{a}\|$. Then

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*, \quad \mathbf{U} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_m), \quad \mathbf{\Sigma} = (\|\mathbf{a}\|, 0, ..., 0)^T, \quad \mathbf{V} = (1) \in \mathbb{C}^{1\times 1}.$$

For a row matrix, the singular value decomposition is obtained by transposition. □

Corollary 1. The spectral norm of a matrix is equal to its largest singular value.

Corollary 2. Let the matrix $\mathbf{A}$ be invertible and let σ_n be its smallest singular value. Then $\left\|\mathbf{A}^{-1}\right\|_2 = 1/\sigma_n$.

Now it is clear that the largest singular value of a matrix and the smallest singular value of an invertible matrix are uniquely determined. The same holds for the entire set of singular values, but we shall prove this later. The singular value decomposition and its important consequences deserve a more thorough discussion, to which we shall return later in this section.

Problem 1. Prove the inequality $\|\mathbf{A}\|_F \le \sqrt{\operatorname{rank}(\mathbf{A})}\|\mathbf{A}\|_2$, where $\|\mathbf{A}\|_F = \left(\sum_{i,j} |a_{ij}|^2\right)^{1/2}$ is the *Frobenius norm*.

Problem 2. Prove that

$$\max_{\mathbf{A}\in\mathbb{C}^{n\times n}, \mathbf{A}\neq\mathbf{0}} \frac{\|\mathbf{A}\|_2}{\|\mathbf{A}\|_1} = \sqrt{n},$$

where $\|\mathbf{A}\|_1 = \max_j \sum_i |a_{ij}|$ is the matrix norm induced by the vector norm $\|\mathbf{x}\|_1 = \sum_i |x_i|$ (see Tyrtyshnikov, §27.3).

Problem 3. Let $\mathbf{A} = (a_{ij})$ and $\mathbf{D} = (d_{ij})$ be complex matrices of order n, where $\mathbf{D}$ is a diagonal matrix with entries $d_{ii} = a_{ii}$ for $1 \le i \le n$. Prove that if $\|\mathbf{A}\|_2 = \|\mathbf{D}\|_2$, then the number of zero entries in the matrix $\mathbf{A}$ is at least $2n - 2$.

Problem 4. Let $\mathbf{L}$ be the lower triangular matrix with lower triangular part taken from a matrix $\mathbf{A} \in \mathbb{C}^{n\times n}$. Prove that

$$\|\mathbf{L}\|_2 \le \log_2(2n)\|\mathbf{A}\|_2.$$

Singular Values and Singular Vectors

Let $\mathbf{A} \in \mathbb{C}^{m\times n}$. Then $\mathbf{A}^*\mathbf{A} \in \mathbb{C}^{n\times n}$ is a Hermitian nonnegative definite matrix:

$$(\mathbf{A}^*\mathbf{A})^* = \mathbf{A}^*(\mathbf{A}^*)^* = \mathbf{A}^*\mathbf{A};$$

$$\mathbf{x}^*\mathbf{A}^*\mathbf{A}\mathbf{x} = \langle \mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{x}\rangle = \|\mathbf{A}\mathbf{x}\|^2 \geq 0 \quad \forall \mathbf{x} \in \mathbb{C}^n.$$

Therefore, all of its eigenvalues are nonnegative.

The nonnegative square roots of the eigenvalues of the matrix $\mathbf{A}^*\mathbf{A}$ are called the *singular values* of the matrix $\mathbf{A}$. It is customary to number the singular values $\sigma_i = \sigma_i(\mathbf{A})$ in nonincreasing order:

$$\sigma_1 \geq \sigma_2 \geq ... \geq \sigma_r > \sigma_{r+1} = ... = \sigma_n = 0.$$

We shall assume that $\mathbf{A}$ has r nonzero singular values.

Let $\mathbf{v}_1, ..., \mathbf{v}_n$ be an orthonormal basis of eigenvectors of the matrix $\mathbf{A}^*\mathbf{A}$ such that

$$\mathbf{A}^*\mathbf{A}\mathbf{v}_i = \begin{cases} \sigma_i^2\mathbf{v}_i, & 1 \leq i \leq r, \\ \mathbf{0}, & r+1 \leq i \leq n. \end{cases}$$

Set $\mathbf{u}_i = \frac{1}{\sigma_i}\mathbf{A}\mathbf{v}_i$, $1 \leq i \leq r$. Then $\langle \mathbf{u}_i, \mathbf{u}_j\rangle = 0$ for $i \neq j$ and $\langle \mathbf{u}_i, \mathbf{u}_i\rangle = 1$. Let us extend the set $\mathbf{u}_1, ..., \mathbf{u}_r$ by vectors $\mathbf{u}_{r+1}, ..., \mathbf{u}_m$ to an orthonormal basis in $\mathbb{C}^m$. Note also that for $j \geq r+1$,

$$\mathbf{A}^*\mathbf{A}\mathbf{v}_j = \mathbf{0} \Rightarrow \mathbf{v}_j^*\mathbf{A}^*\mathbf{A}\mathbf{v}_j = 0 \Rightarrow \big(\mathbf{A}\mathbf{v}_j\big)^*\big(\mathbf{A}\mathbf{v}_j\big) = 0 \Rightarrow \big\|\mathbf{A}\mathbf{v}_j\big\| = 0 \Rightarrow \mathbf{A}\mathbf{v}_j = \mathbf{0}.$$

As a result, we obtain

$$\mathbf{A}\begin{pmatrix}\mathbf{v}_1 & \cdots & \mathbf{v}_n\end{pmatrix} = \begin{pmatrix}\mathbf{u}_1 & \cdots & \mathbf{u}_m\end{pmatrix}\begin{pmatrix}\sigma_1 & & & 0 \\ & \ddots & & \\ & & \sigma_r & \\ 0 & & & 0\end{pmatrix} \Rightarrow \mathbf{A}\mathbf{V} = \mathbf{U}\mathbf{\Sigma},$$

where $\mathbf{V} = \begin{pmatrix}\mathbf{v}_1 & \cdots & \mathbf{v}_n\end{pmatrix}$ and $\mathbf{U} = \begin{pmatrix}\mathbf{u}_1 & \cdots & \mathbf{u}_m\end{pmatrix}$ are unitary matrices, and $\mathbf{\Sigma}$ is a diagonal rectangular matrix of the same dimensions as the matrix $\mathbf{A}$.

The columns of the matrices $\mathbf{U}$ and $\mathbf{V}$ form *singular bases* of the matrix $\mathbf{A}$. The columns of $\mathbf{V}$ are called *right singular vectors*, and the columns of $\mathbf{U}$ are called *left singular vectors* of the matrix $\mathbf{A}$. The relationship between the singular vectors and the nonzero singular values is established by the relations

$$\mathbf{A}\mathbf{v}_i = \sigma_i\mathbf{u}_i, \quad \mathbf{A}^*\mathbf{u}_i = \sigma_i\mathbf{v}_i, \quad 1 \leq i \leq r.$$

Moreover,

$$\mathbf{A}\mathbf{v}_i = \mathbf{0}, \quad r+1 \leq i \leq n, \quad \mathbf{A}^*\mathbf{u}_i = \mathbf{0}, \quad r+1 \leq i \leq m.$$

Thus, we have proved that *for any matrix* $\mathbf{A} \in \mathbb{C}^{m\times n}$ *there holds the equality*

$$\mathbf{A}\mathbf{V} = \mathbf{U}\mathbf{\Sigma} \tag{4}$$

for some unitary matrices $\mathbf{V} \in \mathbb{C}^{n\times n}$, $\mathbf{U} \in \mathbb{C}^{m\times m}$ *and a diagonal rectangular matrix of dimensions* $m \times n$ *with numbers* $\sigma_i \geq 0$ *for* $i = j$. Writing (4) in the form

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*, \tag{5}$$

we obtain a representation of the matrix called its *singular value decomposition.*[14]

If a decomposition (5) with unitary matrices $\mathbf{U}$ and $\mathbf{V}$ has been obtained in some way, then $\mathbf{A}^*\mathbf{A} = \mathbf{V}(\mathbf{\Sigma}^*\mathbf{\Sigma})\mathbf{V}^*$. Therefore, if $\mathbf{\Sigma}$ is a diagonal rectangular matrix with nonnegative entries, then its nonzero entries are uniquely determined.

For a real matrix $\mathbf{A}$ we have $\mathbf{A}^*\mathbf{A} = \mathbf{A}^T\mathbf{A}$, all the vectors above can be chosen real, and (5) becomes the decomposition $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ with orthogonal $\mathbf{U}$ and $\mathbf{V}$; this is the construction of the singular value decomposition used in MA1522.

Problem 5. Find the singular value decomposition of the $(2 \times n)$-matrix $\mathbf{A} = \begin{pmatrix} 1 & 1 & ... & 1 \\ 1 & 1 & ... & 1 \end{pmatrix}$.

Consequences of the Singular Value Decomposition

1) *The number of nonzero singular values r equals the rank of the matrix $\mathbf{A}$.*

2) *The singular value decomposition of the conjugate transpose matrix has the form*

$$\mathbf{A}^* = \mathbf{V}\mathbf{\Sigma}^T\mathbf{U}^*.$$

3) $\mathrm{Col}(\mathbf{A}) = \mathrm{span}\{\mathbf{u}_1, ..., \mathbf{u}_r\}$, $\mathrm{Null}(\mathbf{A}) = \mathrm{span}\{\mathbf{v}_{r+1}, ..., \mathbf{v}_n\}$.

4) $\mathrm{Col}(\mathbf{A}^*) = \mathrm{span}\{\mathbf{v}_1, ..., \mathbf{v}_r\}$, $\mathrm{Null}(\mathbf{A}^*) = \mathrm{span}\{\mathbf{u}_{r+1}, ..., \mathbf{u}_m\}$.

As a corollary, one can obtain representations of the spaces as orthogonal sums:

$$\mathbb{C}^n = \mathrm{Null}(\mathbf{A}) \oplus \mathrm{Col}(\mathbf{A}^*), \quad \mathbb{C}^m = \mathrm{Null}(\mathbf{A}^*) \oplus \mathrm{Col}(\mathbf{A}).$$

5) $\mathbf{A} = \sum_{k=1}^r \sigma_k \mathbf{u}_k \mathbf{v}_k^*$, $\mathbf{A}^* = \sum_{k=1}^r \sigma_k \mathbf{v}_k \mathbf{u}_k^*$.

6) *If $m = n = r$ (the matrix $\mathbf{A}$ is invertible), then*

$$\mathbf{A} = \sum_{k=1}^n \sigma_k \mathbf{u}_k \mathbf{v}_k^*, \quad \mathbf{A}^{-1} = \sum_{k=1}^n \frac{1}{\sigma_k} \mathbf{v}_k \mathbf{u}_k^*.$$

7) *Let $\sigma_1 \geq ... \geq \sigma_n$ be the singular values of an invertible matrix $\mathbf{A}$. Then $\sigma_n^{-1} \geq ... \geq \sigma_1^{-1}$ are the singular values of the matrix $\mathbf{A}^{-1}$.*

8) $\|\mathbf{A}\|_2 = \sigma_1$, $\|\mathbf{A}\|_F = \sqrt{\sigma_1^2 + ... + \sigma_r^2}$ (for the spectral and Frobenius norms, see the subsection "Singular Value Decomposition of a Matrix").

Proof. The spectral and Frobenius norms are unitarily invariant. Therefore $\|\mathbf{A}\|_2 = \|\mathbf{\Sigma}\|_2$ and $\|\mathbf{A}\|_F = \|\mathbf{\Sigma}\|_F$. Obviously, $\|\mathbf{\Sigma}\mathbf{x}\| \leq \sigma_1 \|\mathbf{x}\|$; equality is attained if $\mathbf{x}$ has 1 in the first position and 0 in the remaining ones.

It is also clear that $\|\mathbf{\Sigma}\|_F = \sqrt{\sigma_1^2 + ... + \sigma_r^2}$. □

[14] It was obtained by an altogether different method earlier in this section, in the subsection "Singular Value Decomposition of a Matrix".

Problem 6. Given a square matrix with norm $\|\mathbf{A}\|_2 \le 1$. Prove that there exist square matrices $\mathbf{B}, \mathbf{C}, \mathbf{D}$ such that the matrix $\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}$ is unitary.

Problem 7. Let $\mathbf{A} = \mathbf{A}^T \in \mathbb{C}^{n\times n}$. Prove that the matrix $\mathbf{A}$ possesses a singular value decomposition $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^*$ with the additional condition $\mathbf{V}^* = \mathbf{U}^T$.

Computing a Singular Value Decomposition ADDED FOR MA1522

For a real $m \times n$ matrix $\mathbf{A}$ the singular value decomposition is written in MA1522 as $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, with $\mathbf{U}$ an orthogonal matrix of order m, $\mathbf{V}$ an orthogonal matrix of order n, and

$$\mathbf{\Sigma} = \begin{pmatrix} \mathbf{D} & \mathbf{0}_{r\times(n-r)} \\ \mathbf{0}_{(m-r)\times r} & \mathbf{0}_{(m-r)\times(n-r)} \end{pmatrix},$$

$$\mathbf{D} = \operatorname{diag}(\sigma_1, ..., \sigma_r), \quad \sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_r > 0,$$

where $\sigma_1, ..., \sigma_r$ are the nonzero singular values of $\mathbf{A}$ and $r = \operatorname{rank}(\mathbf{A})$. The construction in the subsection "Singular Values and Singular Vectors" produces exactly such a decomposition. We restate its key step for real matrices, with a proof.

Lemma. Let $\{\mathbf{v}_1, ..., \mathbf{v}_n\}$ be an orthonormal basis for $\mathbb{R}^n$ consisting of eigenvectors of $\mathbf{A}^T\mathbf{A}$, with eigenvalues $\mu_1 \ge \cdots \ge \mu_r > 0 = \mu_{r+1} = \cdots = \mu_n$. Let $\sigma_i = \sqrt{\mu_i}$ and $\mathbf{u}_i = \frac{1}{\sigma_i}\mathbf{A}\mathbf{v}_i$, $i = 1, ..., r$. Then $\{\mathbf{u}_1, ..., \mathbf{u}_r\}$ is an orthonormal basis for $\operatorname{Col}(\mathbf{A})$; in particular, $\operatorname{rank}(\mathbf{A}) = r$.

Proof. First, $\|\mathbf{A}\mathbf{v}_i\|^2 = \mathbf{v}_i^T\mathbf{A}^T\mathbf{A}\mathbf{v}_i = \mu_i\mathbf{v}_i^T\mathbf{v}_i = \mu_i$ for every i; this shows once more that the eigenvalues of $\mathbf{A}^T\mathbf{A}$ are nonnegative, and that $\mathbf{A}\mathbf{v}_i = \mathbf{0}$ for $i > r$. For $i, j \le r$,

$$\mathbf{u}_i \cdot \mathbf{u}_j = \frac{1}{\sigma_i\sigma_j}\mathbf{v}_i^T\mathbf{A}^T\mathbf{A}\mathbf{v}_j = \frac{\mu_j}{\sigma_i\sigma_j}(\mathbf{v}_i \cdot \mathbf{v}_j),$$

which is 0 for $i \ne j$ and 1 for $i = j$. Each $\mathbf{u}_i$ lies in $\operatorname{Col}(\mathbf{A})$. Conversely, every vector of $\operatorname{Col}(\mathbf{A})$ has the form $\mathbf{A}\mathbf{x}$; writing $\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, we get $\mathbf{A}\mathbf{x} = c_1\sigma_1\mathbf{u}_1 + \cdots + c_r\sigma_r\mathbf{u}_r$. Thus the orthonormal, and hence linearly independent, set $\{\mathbf{u}_1, ..., \mathbf{u}_r\}$ spans $\operatorname{Col}(\mathbf{A})$. □

Extending $\{\mathbf{u}_1, ..., \mathbf{u}_r\}$ to an orthonormal basis $\{\mathbf{u}_1, ..., \mathbf{u}_m\}$ for $\mathbb{R}^m$ (Section 5.3) and setting $\mathbf{U} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_m)$, $\mathbf{V} = (\mathbf{v}_1 \ \cdots \ \mathbf{v}_n)$, we get

$$\mathbf{A}\mathbf{V} = (\sigma_1\mathbf{u}_1 \ \cdots \ \sigma_r\mathbf{u}_r \ \ \mathbf{0} \ \cdots \ \mathbf{0}) = \mathbf{U}\mathbf{\Sigma},$$

that is, $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$. This gives the following procedure.

Algorithm (singular value decomposition).

Let $\mathbf{A}$ be a real $m \times n$ matrix.

1. Find the eigenvalues of the symmetric matrix $\mathbf{A}^T\mathbf{A}$ and list them, with multiplicities, in decreasing order: $\mu_1 \geq \cdots \geq \mu_r > 0 = \mu_{r+1} = \cdots = \mu_n$; then $r = \text{rank}(\mathbf{A})$. Set $\sigma_i = \sqrt{\mu_i}$, $i = 1, ..., r$, and form $\mathbf{\Sigma}$ as above.
2. Orthogonally diagonalize $\mathbf{A}^T\mathbf{A}$ (Section 6.3): find an orthonormal basis for each eigenspace (including E_0, if $r < n$), and let $\mathbf{v}_i$ be the vector associated to μ_i, $i = 1, ..., n$. Set $\mathbf{V} = (\mathbf{v}_1 \ \cdots \ \mathbf{v}_n)$.
3. Set $\mathbf{u}_i = \frac{1}{\sigma_i}\mathbf{A}\mathbf{v}_i$, $i = 1, ..., r$. If $r < m$, extend $\{\mathbf{u}_1, ..., \mathbf{u}_r\}$ to an orthonormal basis for $\mathbb{R}^m$: find a basis for the solution space of $(\mathbf{u}_1 \ \cdots \ \mathbf{u}_r)^T\mathbf{x} = \mathbf{0}$ and apply the Gram–Schmidt process to it. Set $\mathbf{U} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_m)$.

Then $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$.

Example. Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \text{so that} \quad \mathbf{A}^T\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad \text{char}(\mathbf{A}^T\mathbf{A}) = (x-3)(x-1).$$

Thus $\mu_1 = 3$, $\mu_2 = 1$, $r = 2$, and

$$\mathbf{\Sigma} = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

From

$$3\mathbf{I} - \mathbf{A}^T\mathbf{A} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{I} - \mathbf{A}^T\mathbf{A} = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix}$$

we obtain the unit eigenvectors $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1,1)^T$ and $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(-1,1)^T$. Next, since $\sigma_1 = \sqrt{3}$ and $\sigma_2 = 1$,

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}}\mathbf{A}\mathbf{v}_1 = \frac{1}{\sqrt{6}}\begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{u}_2 = \mathbf{A}\mathbf{v}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

The system $(\mathbf{u}_1 \ \mathbf{u}_2)^T\mathbf{x} = \mathbf{0}$ is equivalent to $2x_1 + x_2 + x_3 = 0$, $x_2 - x_3 = 0$, whose solution space is $\text{span}\{(-1,1,1)^T\}$; normalizing, $\mathbf{u}_3 = \frac{1}{\sqrt{3}}(-1,1,1)^T$. Hence

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2/\sqrt{6} & 0 & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \end{pmatrix} \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}.$$

Chapter 7. Linear Transformations

7.1. Introduction to Linear Transformations

Mappings, Functions, Operators

All three words in the title of this section mean the same thing—we are speaking of a rule according to which to each element x of a set X there is assigned a uniquely determined element $y = f(x)$ of a set Y. Specifying such a rule is equivalent to choosing a subset

$$\Gamma = \{(x, f(x)) : x \in X\} \subset X \times Y,$$

called the *graph* of the mapping (function, operator) f.

The element $y = f(x)$ is called the *image* of the element x, and x is called a *preimage* of the element y under the mapping f. To emphasize that f acts from X into Y, one writes $f : X \to Y$; the set X is called the *domain* and Y the *codomain* of f.

The set $f(X) \equiv \{y : y = f(x) \text{ for some } x \in X\}$ is called the *range* (or *image*) of the mapping f.

If $M \subset Y$, then the set $f^{-1}(M) \equiv \{x : f(x) \in M\}$ is called the *complete preimage* of the set M. If $M = \{y\}$, one writes $f^{-1}(y) = f^{-1}(M)$.

A mapping $f : X \to Y$ is called *invertible* if there exists a mapping $g : Y \to X$ such that $f(g(y)) = y \ \forall y \in Y$ and $g(f(x)) = x \ \forall x \in X$. In this case, g is called the *inverse mapping* for f, and one writes $g = f^{-1}$.

A mapping f is called *bijective* if for every $y \in Y$ the complete preimage $f^{-1}(y)$ consists of exactly one element; that is, f is both *injective*, or *one-to-one* (each $f^{-1}(y)$ has at most one element: $f(x_1) = f(x_2)$ implies $x_1 = x_2$), and *surjective*, or *onto* (each $f^{-1}(y)$ is nonempty: $f(X) = Y$). (The full textbook calls a bijective mapping "one-to-one"; in this edition, as in the course, one-to-one means injective.) It is easy to show that invertibility is equivalent to being bijective.

Linear Transformations

Any matrix $\mathbf{A} \in \mathbb{C}^{m\times n}$ can be naturally regarded as a mapping that sends a vector $\mathbf{x} \in \mathbb{C}^n$ to the vector $\mathbf{Ax} \in \mathbb{C}^m$. Obviously, this mapping possesses the *linearity property*:[15]

$$\mathbf{A}(\alpha\mathbf{x} + \beta\mathbf{y}) = \alpha\mathbf{Ax} + \beta\mathbf{Ay} \quad \forall\alpha, \beta \in \mathbb{C}, \quad \forall\mathbf{x}, \mathbf{y} \in \mathbb{C}^n.$$

[15]This property can also be viewed as the property of *preserving operations* under a mapping of one vector space into another space over the same field. Such mappings are called *homomorphisms*.

(For a real $m \times n$ matrix $\mathbf{A}$ this is the linear transformation $T_{\mathbf{A}} : \mathbb{R}^n \to \mathbb{R}^m$, $T_{\mathbf{A}}(\mathbf{u}) = \mathbf{A}\mathbf{u}$, defined by $\mathbf{A}$.)

The linearity property holds for many very important mappings in vector spaces whose elements are functions united by some common feature (continuity, differentiability, etc.). Above all, one should mention mappings associated with differentiation and integration of functions. Thus, we have sufficient grounds to study the linearity property from a more general standpoint.

Definition. Let V and W be arbitrary vector spaces over the same field $\mathbb{P}$. A mapping $T : V \to W$ with the property

$$T(\alpha\mathbf{x} + \beta\mathbf{y}) = \alpha T(\mathbf{x}) + \beta T(\mathbf{y}) \quad \forall \alpha, \beta \in \mathbb{P}, \quad \forall \mathbf{x}, \mathbf{y} \in V$$

is called a *linear transformation* (or *linear operator*) from V to W. For linear transformations the argument is often written without parentheses, $T\mathbf{x} = T(\mathbf{x})$; in this edition we keep the parentheses, as the course does.

Linear Transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$ ADDED FOR MA1522

In the course $V = \mathbb{R}^n$ and $W = \mathbb{R}^m$ with real scalars, and vectors are written as columns. Thus a mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation if

$$T(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v}) \quad \text{for all } \mathbf{u}, \mathbf{v} \in \mathbb{R}^n \text{ and all scalars } \alpha, \beta;$$

$\mathbb{R}^n$ is its *domain* and $\mathbb{R}^m$ its *codomain*.

An $m \times n$ matrix $\mathbf{A}$ sends each $\mathbf{v} \in \mathbb{R}^n$ to $\mathbf{A}\mathbf{v} \in \mathbb{R}^m$, so it may be pictured as a mapping of $\mathbb{R}^n$ into $\mathbb{R}^m$, the linear transformation $T_{\mathbf{A}}$. Since $\mathbf{A}(\alpha\mathbf{u}) = \alpha\mathbf{A}\mathbf{u}$ and $\mathbf{A}(\mathbf{u} + \mathbf{v}) = \mathbf{A}\mathbf{u} + \mathbf{A}\mathbf{v}$, the image of a linear combination is the same linear combination of the images. Geometrically, the parallelogram with sides $\mathbf{u}$, $\mathbf{v}$ and diagonal $\mathbf{u} + \mathbf{v}$ is carried to the (possibly degenerate) parallelogram with sides $\mathbf{A}\mathbf{u}$, $\mathbf{A}\mathbf{v}$ and diagonal $\mathbf{A}\mathbf{u} + \mathbf{A}\mathbf{v}$.

Example (Matrices as mappings)**.**

1. The 3×2 matrix with columns $\mathbf{e}_1, \mathbf{e}_3 \in \mathbb{R}^3$ maps the plane onto the plane $y = 0$ in $\mathbb{R}^3$:
$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ 0 \\ y \end{pmatrix}.$$
2. The 2×3 matrix below projects $\mathbb{R}^3$ onto the plane $x = 0$, $(x, y, z)^T \mapsto (0, y, z)^T$, and identifies this plane with $\mathbb{R}^2$, $(0, y, z)^T \mapsto (y, z)^T$:
$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} y \\ z \end{pmatrix}.$$
3. In the plane, the matrix $\mathbf{R}_\theta$ below is the rotation about the origin through the angle θ, and $\mathbf{F}$ is the reflection in the x-axis:

$$\mathbf{R}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} : \quad \begin{pmatrix} r\cos\varphi \\ r\sin\varphi \end{pmatrix} \mapsto \begin{pmatrix} r\cos(\varphi+\theta) \\ r\sin(\varphi+\theta) \end{pmatrix};$$

$$\mathbf{F} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} : \quad \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x \\ -y \end{pmatrix}.$$

4. The zero matrix $\mathbf{0}_{m\times n}$ sends every vector of $\mathbb{R}^n$ to the zero vector of $\mathbb{R}^m$.

Proposition (Basic properties)**.** Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a mapping.

(a) T is linear if and only if it is *additive* and *homogeneous*:
$$T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}), \quad T(\alpha\mathbf{u}) = \alpha T(\mathbf{u}) \quad \text{for all } \mathbf{u},\mathbf{v} \in \mathbb{R}^n, \alpha \in \mathbb{R}.$$

(b) If T is linear, then for any $\mathbf{u}_1, ..., \mathbf{u}_k \in \mathbb{R}^n$ and scalars $c_1, ..., c_k$,
$$T(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + ... + c_k\mathbf{u}_k) = c_1T(\mathbf{u}_1) + c_2T(\mathbf{u}_2) + ... + c_kT(\mathbf{u}_k).$$

(c) If T is linear, then $T(\mathbf{0}) = \mathbf{0}$: the zero vector of $\mathbb{R}^n$ goes to the zero vector of $\mathbb{R}^m$.

Proof. (a) With $\alpha = \beta = 1$ the defining identity gives additivity, and with $\beta = 0$ it gives homogeneity. Conversely, additivity and then homogeneity give $T(\alpha\mathbf{u} + \beta\mathbf{v}) = T(\alpha\mathbf{u}) + T(\beta\mathbf{v}) = \alpha T(\mathbf{u}) + \beta T(\mathbf{v})$.

(b) Induction on k. For $k = 1$ this is homogeneity; for $k \geq 2$, by (a), $T(c_1\mathbf{u}_1 + ... + c_k\mathbf{u}_k) = T(c_1\mathbf{u}_1 + ... + c_{k-1}\mathbf{u}_{k-1}) + c_kT(\mathbf{u}_k)$, and the induction hypothesis applies to the first term.

(c) $T(\mathbf{0}) = T(0 \cdot \mathbf{0}) = 0 \cdot T(\mathbf{0}) = \mathbf{0}$. □

Consequently, a mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is *not* linear as soon as one of the following holds: $T(\mathbf{0}) \neq \mathbf{0}$; $T(\alpha\mathbf{u}) \neq \alpha T(\mathbf{u})$ for some scalar α and some $\mathbf{u}$; $T(\mathbf{u}+\mathbf{v}) \neq T(\mathbf{u}) + T(\mathbf{v})$ for some $\mathbf{u}$, $\mathbf{v}$. The condition $T(\mathbf{0}) = \mathbf{0}$ alone is necessary but not sufficient.

Example (Checking linearity)**.** The mapping $T : \mathbb{R}^3 \to \mathbb{R}^2$, $T(x,y,z)^T = (x - 2z, 3y + z)^T$, is linear: for $\mathbf{u}_i = (x_i, y_i, z_i)^T$, $i = 1, 2$,

$$T(\alpha\mathbf{u}_1 + \beta\mathbf{u}_2) = \begin{pmatrix} \alpha x_1 + \beta x_2 - 2(\alpha z_1 + \beta z_2) \\ 3(\alpha y_1 + \beta y_2) + \alpha z_1 + \beta z_2 \end{pmatrix}$$
$$= \alpha \begin{pmatrix} x_1 - 2z_1 \\ 3y_1 + z_1 \end{pmatrix} + \beta \begin{pmatrix} x_2 - 2z_2 \\ 3y_2 + z_2 \end{pmatrix} = \alpha T(\mathbf{u}_1) + \beta T(\mathbf{u}_2).$$

Example (Mappings that are not linear)**.**

1. The translation $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x,y)^T = (x+1, y)^T$, is not linear, because $T(\mathbf{0}) = (1,0)^T \neq \mathbf{0}$.
2. The norm $T : \mathbb{R}^2 \to \mathbb{R}$, $T(\mathbf{u}) = \|\mathbf{u}\|$, satisfies $T(\mathbf{0}) = 0$ but is not linear: $T(-\mathbf{e}_1) = 1 \neq -1 = -T(\mathbf{e}_1)$, and $T(\mathbf{e}_1 + \mathbf{e}_2) = \sqrt{2} \neq 2 = T(\mathbf{e}_1) + T(\mathbf{e}_2)$.

Matrix of a Linear Transformation

Consider a linear transformation $T : V_n \to V_m$, where V_n and V_m are vector spaces of dimensions n and m (over a common field $\mathbb{P}$).

Fix some basis $S = \{\mathbf{u}_1, ..., \mathbf{u}_n\}$ of V_n and some basis $C = \{\mathbf{v}_1, ..., \mathbf{v}_m\}$ of V_m. By the linearity of the transformation T,

$$T(x_1\mathbf{u}_1 + ... + x_n\mathbf{u}_n) = x_1T(\mathbf{u}_1) + ... + x_nT(\mathbf{u}_n). \tag{1}$$

Therefore T is completely determined by its action on the basis vectors $\mathbf{u}_1, ..., \mathbf{u}_n$. Let us expand the images of the basis vectors with respect to the basis of the codomain:

$$T(\mathbf{u}_j) = a_{1j}\mathbf{v}_1 + ... + a_{mj}\mathbf{v}_m, \quad j = 1, ..., n. \tag{2}$$

From (1) and (2) we obtain

$$T(x_1\mathbf{u}_1 + ... + x_n\mathbf{u}_n) = (a_{11}x_1 + ... + a_{1n}x_n)\mathbf{v}_1 + ... + \\ +(a_{m1}x_1 + ... + a_{mn}x_n)\mathbf{v}_m.$$

Consequently,

$$T(x_1\mathbf{u}_1 + ... + x_n\mathbf{u}_n) = y_1\mathbf{v}_1 + ... + y_m\mathbf{v}_m \quad \Leftrightarrow$$

$$\Leftrightarrow \quad \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} = \begin{pmatrix} a_{11} & ... & a_{1n} \\ \vdots & \vdots & \vdots \\ a_{m1} & ... & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

The matrix appearing on the right is called the *matrix of the linear transformation* T in the pair of bases S and C (its representation with respect to S and C); we denote it by $[T]_{S,C}$. Its j-th column is $\left[T(\mathbf{u}_j)\right]_C$, and the equivalence above reads $[T(\mathbf{x})]_C = [T]_{S,C}[\mathbf{x}]_S$ for every $\mathbf{x} \in V_n$.

For $T : \mathbb{R}^n \to \mathbb{R}^m$ and the standard basis $C = E$ of $\mathbb{R}^m$ (where $[\mathbf{w}]_E = \mathbf{w}$), this matrix is the *representation of* T *with respect to the basis* S, denoted in the course by $[T]_S$; if, in addition, $S = E = \{\mathbf{e}_1, ..., \mathbf{e}_n\}$ is the standard basis of $\mathbb{R}^n$, it is the *standard matrix* $\mathbf{A} = [T]_E$ of T:

$$[T]_S = [T]_{S,E} = (T(\mathbf{u}_1) \;\; T(\mathbf{u}_2) \;\; \cdots \;\; T(\mathbf{u}_n)),$$

$$\mathbf{A} = [T]_E = (T(\mathbf{e}_1) \;\; T(\mathbf{e}_2) \;\; \cdots \;\; T(\mathbf{e}_n)).$$

Thus $T(\mathbf{x}) = [T]_S[\mathbf{x}]_S$ and $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$. (Note that the course's $[T]_S$ is $[T]_{S,E}$, not $[T]_{S,S}$.)

Thus, any fixed pair of bases generates three *isomorphisms*, that is, bijective linear mappings (see Tyrtyshnikov, §12.2):

$$V_n \leftrightarrow \mathbb{P}^n, \quad V_m \leftrightarrow \mathbb{P}^m, \quad \mathcal{L}(V_n, V_m) \leftrightarrow \mathbb{P}^{m\times n},$$

where $\mathcal{L}(V_n, V_m)$ is the vector space of all linear transformations from the space V_n into the space V_m, and $\mathbb{P}^{m\times n}$ is the vector space of all $m \times n$ matrices with entries from the field $\mathbb{P}$. From this, in particular, it is seen that the dimension of the space of linear transformations $\mathcal{L}(V_n, V_m)$ is equal to mn.

Example. Let $D : V \to W$ be the differentiation transformation on the space of polynomials $V = W$ of degree 2 and lower. Consider the basis $S = \{1+t, 1-t, t^2\}$ of V and the basis $C = \{1, t, t^2\}$ of W. Obviously,

$$D(1+t) = 1, \quad D(1-t) = -1, \quad D(t^2) = 2t.$$

Therefore, in the pair of bases S and C, the matrix of the linear transformation D has the form

$$[D]_{S,C} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

What will be the matrix of the same transformation if the same basis S is chosen in $W = V$? For this we need to find the expansions of the images of the vectors $1+t$, $1-t$, t^2 of S with respect to the same vectors $1+t$, $1-t$, t^2:

$$\left.\begin{aligned} D(1+t) &= \tfrac{1}{2}(1+t) + \tfrac{1}{2}(1-t), \\ D(1-t) &= -\tfrac{1}{2}(1+t) - \tfrac{1}{2}(1-t), \\ D(t^2) &= (1+t) - (1-t) \end{aligned}\right\} \quad \Rightarrow \quad [D]_{S,S} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & 1 \\ \frac{1}{2} & -\frac{1}{2} & -1 \\ 0 & 0 & 0 \end{pmatrix}.$$

The Standard Matrix ADDED FOR MA1522

Theorem (Standard Matrix)**.** A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation if and only if there is an $m \times n$ matrix $\mathbf{A}$ such that

$$T(\mathbf{u}) = \mathbf{A}\mathbf{u} \quad \text{for all } \mathbf{u} \in \mathbb{R}^n.$$

Such a matrix is unique, namely

$$\mathbf{A} = \begin{pmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{pmatrix},$$

where $E = \{\mathbf{e}_1, ..., \mathbf{e}_n\}$ is the standard basis of $\mathbb{R}^n$: the j-th column of $\mathbf{A}$ is $T(\mathbf{e}_j)$.

Proof. If $T(\mathbf{u}) = \mathbf{A}\mathbf{u}$ for all $\mathbf{u}$, then T is linear by the linearity of matrix multiplication. Conversely, let T be linear. Every $\mathbf{u} = (u_1, ..., u_n)^T$ equals $u_1\mathbf{e}_1 + ... + u_n\mathbf{e}_n$, so by linearity

$$T(\mathbf{u}) = u_1 T(\mathbf{e}_1) + ... + u_n T(\mathbf{e}_n) = \begin{pmatrix} T(\mathbf{e}_1) & \cdots & T(\mathbf{e}_n) \end{pmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = \mathbf{A}\mathbf{u}.$$

If also $T(\mathbf{u}) = \mathbf{B}\mathbf{u}$ for all $\mathbf{u}$, then for each j the j-th columns $\mathbf{B}\mathbf{e}_j = T(\mathbf{e}_j) = \mathbf{A}\mathbf{e}_j$ coincide, so $\mathbf{B} = \mathbf{A}$. □

This matrix is the standard matrix $\mathbf{A} = [T]_E$ of T defined above. By the theorem, the rule $\mathbf{A} \mapsto T_{\mathbf{A}}$ is a bijective correspondence between the real $m \times n$ matrices and the linear transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$: every linear transformation is multiplication by exactly one matrix. If T is given by a

formula, its standard matrix is read off by collecting the coefficients of each variable; the coefficients of the j-th variable form the j-th column.

Example. For $T : \mathbb{R}^3 \to \mathbb{R}^2$ given by

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4x - y + 2z \\ x + 3z \end{pmatrix} = x\begin{pmatrix} 4 \\ 1 \end{pmatrix} + y\begin{pmatrix} -1 \\ 0 \end{pmatrix} + z\begin{pmatrix} 2 \\ 3 \end{pmatrix} =$$

$$= \begin{pmatrix} 4 & -1 & 2 \\ 1 & 0 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix},$$

the standard matrix is $\mathbf{A} = \begin{pmatrix} 4 & -1 & 2 \\ 1 & 0 & 3 \end{pmatrix}$; its columns are $T(\mathbf{e}_1)$, $T(\mathbf{e}_2)$, $T(\mathbf{e}_3)$.

Composition of Linear Transformations

The product of linear transformations $T_1 : V_n \to V_m$ and $T_2 : V_m \to V_k$ is defined as the composition of mappings: T_2T_1 is the transformation from V_n to V_k defined by the rule $(T_2T_1)(\mathbf{x}) = T_2(T_1(\mathbf{x}))$. It is elementary to verify that the product of linear transformations is a linear transformation.

Let $\mathbf{A} = [T_1]_{S,C}$ be the matrix of a linear transformation T_1 in a pair of bases S and C, and let $\mathbf{B} = [T_2]_{C,D}$ be the matrix of a linear transformation T_2 in a pair of bases C and D. Then *the product of the matrices* $\mathbf{BA}$ *is the matrix of the product of the transformations* T_2T_1 *in the pair of bases* S *and* D: $[T_2T_1]_{S,D} = [T_2]_{C,D}[T_1]_{S,C}$.

The proof reduces to a direct verification. Note that the full textbook actually began with the definition of matrix multiplication (reproduced here in Sections 2.1 and 2.2) and, in effect, with a discussion of the composition of linear mappings!

Change of Bases

Let $[T]_{S,C}$ be the matrix of a linear transformation T in a pair of bases $S = \{\mathbf{u}_1, ..., \mathbf{u}_n\}$ and $C = \{\mathbf{v}_1, ..., \mathbf{v}_m\}$. How can one find the matrix $[T]_{S',C'}$ of the same transformation in another pair of bases: $S' = \{\mathbf{u}'_1, ..., \mathbf{u}'_n\}$ and $C' = \{\mathbf{v}'_1, ..., \mathbf{v}'_m\}$?

Consider the equalities

$$T(x_1\mathbf{u}_1 + ... + x_n\mathbf{u}_n) = y_1\mathbf{v}_1 + ... + y_m\mathbf{v}_m,$$
$$T(z_1\mathbf{u}'_1 + ... + z_n\mathbf{u}'_n) = w_1\mathbf{v}'_1 + ... + w_m\mathbf{v}'_m.$$

According to the definition of the matrices $[T]_{S,C}$ and $[T]_{S',C'}$, we find

$$[T]_{S,C}\mathbf{x} = \mathbf{y}, \quad [T]_{S',C'}\mathbf{z} = \mathbf{w},$$

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}, \quad \mathbf{z} = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_m \end{pmatrix}.$$

Next we write

$$\mathbf{u}'_j = p_{1j}\mathbf{u}_1 + ... + p_{nj}\mathbf{u}_n, \quad 1 \le j \le n,$$
$$\mathbf{v}'_i = q_{1i}\mathbf{v}_1 + ... + q_{mi}\mathbf{v}_m, \quad 1 \le i \le m,$$

and introduce the transition matrices

$$\mathbf{P} = \begin{pmatrix} p_{11} & \cdots & p_{1n} \\ \vdots & \vdots & \vdots \\ p_{n1} & \cdots & p_{nn} \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} q_{11} & \cdots & q_{1m} \\ \vdots & \vdots & \vdots \\ q_{m1} & \cdots & q_{mm} \end{pmatrix}.$$

The columns of $\mathbf{P}$ are $\left[\mathbf{u}'_j\right]_S$, so $\mathbf{P}$ is the transition matrix from S' to S, and $\mathbf{Q}$ is the transition matrix from C' to C (see Section 3.8; some books, including the full textbook, name these matrices the other way round). Then $\mathbf{x} = \mathbf{Pz}$ and $\mathbf{y} = \mathbf{Qw}$. Consequently, $[T]_{S,C}(\mathbf{Pz}) = \mathbf{Qw} \Rightarrow \left(\mathbf{Q}^{-1}[T]_{S,C}\mathbf{P}\right)\mathbf{z} = \mathbf{w} \Rightarrow$

$$[T]_{S',C'} = \mathbf{Q}^{-1}[T]_{S,C}\mathbf{P}. \qquad (*)$$

For $T : \mathbb{R}^n \to \mathbb{R}^m$ with $C = C' = E$ (so that $\mathbf{Q} = \mathbf{I}_m$) and $S' = E$, formula $(*)$ becomes $\mathbf{A} = [T]_S\mathbf{P}$, where $\mathbf{A}$ is the standard matrix of T and $\mathbf{P}$ is the transition matrix from E to S.

Recall that matrices $\mathbf{A}$ and $\mathbf{B}$ are called *equivalent* if $\mathbf{B} = \mathbf{MAN}$ for some invertible $\mathbf{M}$ and $\mathbf{N}$; matrices of the same size are equivalent if and only if they have the same rank, and a matrix of rank r is equivalent to the matrix whose first r diagonal entries are 1 and all other entries are 0 (proved in Tyrtyshnikov, §8.6).

Proposition 1. Matrices are equivalent if and only if they are matrices of one and the same linear transformation in some pairs of bases.

Proof. Obviously, $(*)$ means the equivalence of the matrices $[T]_{S',C'}$ and $[T]_{S,C}$. If $\mathbf{B} = \mathbf{MAN}$, then $\mathbf{M}$ and $\mathbf{N}$ can be regarded as transition matrices for different pairs of bases. □

Corollary. For matrices of the same size to be matrices of one and the same linear transformation in some pairs of bases, it is necessary and sufficient that they have the same rank.

Let $\mathbf{A}$ be the matrix of a linear transformation T in some pair of bases. If $r = \operatorname{rank}(\mathbf{A})$, then $\mathbf{A}$ is equivalent to the matrix $\mathbf{B} = \left(b_{ij}\right)$ in which $b_{11} = ... = b_{rr} = 1$ and all other entries are equal to 0. Consequently, there exists a pair of "canonical" bases in which T is determined by the matrix $\mathbf{B}$.

Thus, by choosing a pair of bases, the matrix of a linear transformation can acquire a form so simple that it may turn out to be almost useless for studying

the individual properties of the given transformation. Therefore, studying a transformation cannot, in general, be reduced to studying its matrix.

If $V_n = V_m$, then it becomes possible to take $C = S$. Owing to the fact that images and preimages are considered in one and the same basis, the matrix of the transformation now contains everything needed for any detailed study of it. The same holds for any other pair of bases S' and C', provided $C' = S'$. In this case, of course, $\mathbf{Q} = \mathbf{P} \Rightarrow$

$$[T]_{S',S'} = \mathbf{P}^{-1}[T]_{S,S}\mathbf{P}. \qquad (**)$$

Matrices $\mathbf{A}$ and $\mathbf{B}$ are called *similar* if $\mathbf{B} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ for some invertible matrix $\mathbf{P}$. Obviously, the following proposition holds.

Proposition 2. Matrices are similar if and only if they are matrices of one and the same linear transformation in some bases, provided identical bases are chosen in the common domain and codomain.

Problem 1. Given arbitrary numbers $\beta_1, ..., \beta_{n-1}$, prove that any bidiagonal $n \times n$ matrix $\mathbf{A}$ of the form

$$\mathbf{A} = \begin{pmatrix} \alpha_1 & 1 & & & 0 \\ & \alpha_2 & 1 & & \\ & & \ddots & \ddots & \\ & & & \alpha_{n-1} & 1 \\ 0 & & & & \alpha_n \end{pmatrix}$$

is similar to some $n \times n$ matrix $\mathbf{B} = \left(b_{ij}\right)$ in which $b_{11} = \beta_1, ..., b_{n-1,n-1} = \beta_{n-1}$.

Recovering a Linear Transformation from the Images of a Basis

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Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be linear and $S = \{\mathbf{u}_1, ..., \mathbf{u}_n\}$ a basis of $\mathbb{R}^n$. We have seen that

$$T(\mathbf{v}) = [T]_S[\mathbf{v}]_S \quad \text{for all } \mathbf{v} \in \mathbb{R}^n, \quad \text{where} \quad [T]_S = \begin{pmatrix} T(\mathbf{u}_1) & \cdots & T(\mathbf{u}_n) \end{pmatrix},$$

and, by $(*)$, that the standard matrix is $\mathbf{A} = [T]_S\mathbf{P}$, where $\mathbf{P}$ is the transition matrix from E to S. Both facts are also immediate. If $\mathbf{v} = c_1\mathbf{u}_1 + ... + c_n\mathbf{u}_n$, then $[\mathbf{v}]_S = (c_1, ..., c_n)^T$ and, by linearity,

$$T(\mathbf{v}) = c_1T(\mathbf{u}_1) + ... + c_nT(\mathbf{u}_n) = [T]_S[\mathbf{v}]_S;$$

since $\mathbf{P}\mathbf{v} = \mathbf{P}[\mathbf{v}]_E = [\mathbf{v}]_S$, we get $\mathbf{A}\mathbf{v} = T(\mathbf{v}) = [T]_S\mathbf{P}\mathbf{v}$ for every $\mathbf{v}$, whence $\mathbf{A} = [T]_S\mathbf{P}$. The transition matrix from S to E is $\mathbf{U} = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{pmatrix}$, because $\left[\mathbf{u}_j\right]_E = \mathbf{u}_j$; hence $\mathbf{P} = \mathbf{U}^{-1}$ (Section 3.6). Thus *a linear transformation is completely determined by the images of the vectors of a basis*, and its formula is recovered as follows.

Algorithm (Standard matrix from the images of a basis).

1. Form the matrices $\mathbf{U} = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{pmatrix}$ and $[T]_S = \begin{pmatrix} T(\mathbf{u}_1) & \cdots & T(\mathbf{u}_n) \end{pmatrix}$.

2. Compute $\mathbf{U}^{-1}$, for instance by Gauss–Jordan elimination $(\mathbf{U} \mid \mathbf{I}) \to (\mathbf{I} \mid \mathbf{U}^{-1})$.
3. Then $\mathbf{A} = [T]_S \mathbf{U}^{-1}$, and $T(\mathbf{x}) = \mathbf{A}\mathbf{x}$ is the formula for T.

Example. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation with

$$T\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}1\\1\end{pmatrix}, \quad T\begin{pmatrix}0\\1\\1\end{pmatrix} = \begin{pmatrix}0\\4\end{pmatrix}, \quad T\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}3\\3\end{pmatrix}.$$

The three vectors form a basis S of $\mathbb{R}^3$, since the matrix $\mathbf{U}$ with these columns has $\det(\mathbf{U}) = 2 \neq 0$. Here

$$\mathbf{U} = \begin{pmatrix}1&0&1\\1&1&0\\0&1&1\end{pmatrix}, \quad [T]_S = \begin{pmatrix}1&0&3\\1&4&3\end{pmatrix}, \quad \mathbf{U}^{-1} = \frac{1}{2}\begin{pmatrix}1&1&-1\\-1&1&1\\1&-1&1\end{pmatrix},$$

and therefore

$$\mathbf{A} = [T]_S \mathbf{U}^{-1} = \begin{pmatrix}2&-1&1\\0&1&3\end{pmatrix}, \quad T\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}2x - y + z\\ y + 3z\end{pmatrix}.$$

Proposition (When the images determine T)**.** Let $\mathbf{u}_1, \dots, \mathbf{u}_k \in \mathbb{R}^n$. The images $T(\mathbf{u}_1), \dots, T(\mathbf{u}_k)$ determine a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ uniquely if and only if $\operatorname{span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\} = \mathbb{R}^n$.

Proof. If the vectors span $\mathbb{R}^n$, some of them form a basis of $\mathbb{R}^n$ (Section 3.6), and T is determined by its values on that basis. If they do not, the $n \times k$ matrix $\mathbf{U} = (\mathbf{u}_1 \ \cdots \ \mathbf{u}_k)$ has $\operatorname{rank}(\mathbf{U}^T) = \operatorname{rank}(\mathbf{U}) < n$, so by the Rank–Nullity Theorem (Section 4.2) there is a vector $\mathbf{y} \neq \mathbf{0}$ with $\mathbf{U}^T\mathbf{y} = \mathbf{0}$, that is, $\mathbf{y} \cdot \mathbf{u}_i = 0$ for all i. Fix any nonzero $\mathbf{c} \in \mathbb{R}^m$ and put $T'(\mathbf{x}) = T(\mathbf{x}) + (\mathbf{y} \cdot \mathbf{x})\mathbf{c}$. Then T' is linear (its standard matrix is $\mathbf{A} + \mathbf{c}\mathbf{y}^T$) and $T'(\mathbf{u}_i) = T(\mathbf{u}_i)$ for all i, but $T'(\mathbf{y}) = T(\mathbf{y}) + \|\mathbf{y}\|^2\mathbf{c} \neq T(\mathbf{y})$. $\square$

Example (Too little information)**.** Suppose a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ satisfies

$$T\begin{pmatrix}1\\0\\1\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix}, \quad T\begin{pmatrix}0\\1\\1\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix}, \quad T\begin{pmatrix}1\\1\\2\end{pmatrix} = \begin{pmatrix}1\\3\end{pmatrix}.$$

The third vector is the sum of the first two (and the data agree with this), so the three vectors are linearly dependent and span only a plane; T is not determined. Indeed, for any real s and t the linear transformation

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1-s & -s & s \\ 2-t & 1-t & t \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x+s(z-x-y) \\ 2x+y+t(z-x-y) \end{pmatrix}$$

takes the three prescribed values, since $z-x-y=0$ at each of the three given vectors.

7.2. Range and Kernel of a Linear Transformation

Kernel and Range of a Linear Transformation
The set $\ker(T) \equiv \{\mathbf{x} \in V_n : T(\mathbf{x}) = \mathbf{0}\}$ is called the *kernel* of the linear transformation T, and the set $\mathrm{R}(T) \equiv \{\mathbf{y} \in V_m : \mathbf{y} = T(\mathbf{x}) \text{ for some } \mathbf{x} \in V_n\}$ is called its *range* (or *image*). The dimension of the range is called the *rank* of the transformation T, and the dimension of the kernel is called its *nullity* (or *defect*). Notation: $\mathrm{rank}(T) = \dim(\mathrm{R}(T))$, $\mathrm{nullity}(T) = \dim(\ker(T))$.

Proposition 1. The kernel of a linear transformation $T : V \to W$ is a subspace of V, and its range is a subspace of W.

Proof. Let $\mathbf{x}, \mathbf{y} \in \ker(T)$. Then $T(\alpha\mathbf{x} + \beta\mathbf{y}) = \alpha T(\mathbf{x}) + \beta T(\mathbf{y}) = \alpha\mathbf{0} + \beta\mathbf{0} = \mathbf{0} \Rightarrow \alpha\mathbf{x} + \beta\mathbf{y} \in \ker(T)$. Let $\mathbf{x}, \mathbf{y} \in \mathrm{R}(T)$. Then $\mathbf{x} = T(\mathbf{u})$ and $\mathbf{y} = T(\mathbf{v})$ for some $\mathbf{u}, \mathbf{v} \in V \Rightarrow \alpha\mathbf{x} + \beta\mathbf{y} = T(\alpha\mathbf{u} + \beta\mathbf{v}) \Rightarrow \alpha\mathbf{x} + \beta\mathbf{y} \in \mathrm{R}(T)$. □

Theorem (Rank–Nullity Theorem)**.** Let V be finite-dimensional. Then

$$\dim(\ker(T)) + \dim(\mathrm{R}(T)) = \dim(V).$$

Proof. Let $\dim(\ker(T)) = k$ and let $\mathbf{v}_1, ..., \mathbf{v}_k$ be a basis of the subspace $\ker(T)$. Complete it with some vectors $\mathbf{v}_{k+1}, ..., \mathbf{v}_n$ to a basis of V. Obviously,

$$\mathrm{R}(T) = \mathrm{span}\{T(\mathbf{v}_{k+1}), ..., T(\mathbf{v}_n)\}.$$

It remains to prove that the vectors $T(\mathbf{v}_{k+1}), ..., T(\mathbf{v}_n)$ are linearly independent.

Suppose $\alpha_{k+1}T(\mathbf{v}_{k+1}) + ... + \alpha_n T(\mathbf{v}_n) = \mathbf{0} \Rightarrow T(\alpha_{k+1}\mathbf{v}_{k+1} + ... + \alpha_n\mathbf{v}_n) = \mathbf{0} \Rightarrow \alpha_{k+1}\mathbf{v}_{k+1} + ... + \alpha_n\mathbf{v}_n \in \ker(T) \Rightarrow \alpha_{k+1}\mathbf{v}_{k+1} + ... + \alpha_n\mathbf{v}_n = \beta_1\mathbf{v}_1 + ... + \beta_k\mathbf{v}_k$ for some scalars $\beta_1, ..., \beta_k \Rightarrow \alpha_{k+1} = ... = \alpha_n = 0$. □

Remark. This theorem could have been left unproved, since it is a consequence of a fact we already know: for any matrix $\mathbf{A}$, the sum of its rank and its nullity (the dimension of its nullspace $\mathrm{Null}(\mathbf{A})$, the solution space of the homogeneous system $\mathbf{Ax} = \mathbf{0}$) equals the number of its columns. We know that the rank coincides with the dimension of the column space $\mathrm{Col}(\mathbf{A})$, and the latter is obviously the range of $\mathbf{A}$ (regarded as the transformation of multiplication by the given matrix).

Proposition 2. Suppose a linear transformation T acts from V to V. Then its kernel and range are invariant under T.

Proof. The invariance of the kernel is obvious, since every vector in it is mapped to $\mathbf{0}$.

Let $\mathbf{x} \in \mathrm{R}(T)$. Then $\mathbf{x} = T(\mathbf{u}) \Rightarrow T(\mathbf{x}) = T(T(\mathbf{u})) \Rightarrow T(\mathbf{x}) \in \mathrm{R}(T)$. □

Problem 1. For two linear transformations acting from V to W, the sum of their kernels coincides with V. Prove that the range of the sum of these transformations equals the sum of their ranges. Does this hold in the case of three transformations?

Range and Kernel via the Standard Matrix ADDED FOR MA1522

For $T : \mathbb{R}^n \to \mathbb{R}^m$ the definitions above read

$$\mathrm{R}(T) = T(\mathbb{R}^n) = \{\mathbf{v} \in \mathbb{R}^m : \mathbf{v} = T(\mathbf{u}) \text{ for some } \mathbf{u} \in \mathbb{R}^n\},$$
$$\ker(T) = \{\mathbf{u} \in \mathbb{R}^n : T(\mathbf{u}) = \mathbf{0}\}.$$

The proof of Proposition 1 shows that both sets are closed under linear combinations; both also contain the zero vector, because $T(\mathbf{0}) = \mathbf{0}$ (Section 7.1). Hence $\mathrm{R}(T)$ is a subspace of the codomain $\mathbb{R}^m$ and $\ker(T)$ is a subspace of the domain $\mathbb{R}^n$.

Proposition 3. Let $\mathbf{A}$ be the standard matrix of a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$. Then

$$\mathrm{R}(T) = \mathrm{Col}(\mathbf{A}), \quad \ker(T) = \mathrm{Null}(\mathbf{A}).$$

Consequently $\mathrm{rank}(T) = \mathrm{rank}(\mathbf{A})$ and $\mathrm{nullity}(T) = \mathrm{nullity}(\mathbf{A})$; in particular, $\mathrm{rank}(T) + \mathrm{nullity}(T) = n$.

Proof. Write $\mathbf{A} = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_n)$. As $\mathbf{u} = (u_1, ..., u_n)^T$ runs over $\mathbb{R}^n$, the vector $T(\mathbf{u}) = \mathbf{A}\mathbf{u} = u_1\mathbf{a}_1 + ... + u_n\mathbf{a}_n$ runs over all linear combinations of the columns of $\mathbf{A}$; hence $\mathrm{R}(T) = \mathrm{Col}(\mathbf{A})$. Next, $T(\mathbf{u}) = \mathbf{0}$ if and only if $\mathbf{A}\mathbf{u} = \mathbf{0}$, so $\ker(T) = \mathrm{Null}(\mathbf{A})$. Taking dimensions gives the equalities for the rank and the nullity, and the last equality is the Rank–Nullity Theorem for matrices (Section 4.2). □

In particular, $\mathbf{v} \in \mathrm{R}(T)$ if and only if the system $\mathbf{A}\mathbf{x} = \mathbf{v}$ is consistent. So the range is described by row reducing $(\mathbf{A} \mid \mathbf{v})$ with an arbitrary $\mathbf{v}$, and the kernel is the solution set of $\mathbf{A}\mathbf{x} = \mathbf{0}$.

Example. Consider the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ given by

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 2y - z \\ 2x + 4y + z \\ x + 2y + 2z \end{pmatrix}.$$

The row operations $R_2 - 2R_1$, $R_3 - R_1$, $R_3 - R_2$ give

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & v_1 \\ 2 & 4 & 1 & v_2 \\ 1 & 2 & 2 & v_3 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 2 & -1 & v_1 \\ 0 & 0 & 3 & v_2 - 2v_1 \\ 0 & 0 & 0 & v_1 - v_2 + v_3 \end{array}\right).$$

Hence $\mathrm{R}(T)$ is the plane $v_1 - v_2 + v_3 = 0$, spanned by the first and third columns $(1,2,1)^T$, $(-1,1,2)^T$ of $\mathbf{A}$ (the pivot columns), and $\mathrm{rank}(T) = 2$. Solving $\mathbf{Ax} = \mathbf{0}$ gives $z = 0$, $x = -2y$, so $\ker(T) = \mathrm{span}\{(-2,1,0)^T\}$ and $\mathrm{nullity}(T) = 1$; indeed $2 + 1 = 3$.

For the transformation $T(x,y,z)^T = (4x - y + 2z, x + 3z)^T$ of Section 7.1,

$$\mathbf{A} = \begin{pmatrix} 4 & -1 & 2 \\ 1 & 0 & 3 \end{pmatrix} \quad \text{has the reduced row-echelon form} \quad \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 10 \end{pmatrix}.$$

So $\mathrm{rank}(T) = 2$ and therefore $\mathrm{R}(T) = \mathbb{R}^2$, while the kernel is the line spanned by $(-3,-10,1)^T$ and $\mathrm{nullity}(T) = 1$.

Injective and Surjective Linear Transformations ADDED FOR MA1522

Recall (Section 7.1) that $T : \mathbb{R}^n \to \mathbb{R}^m$ is *injective* (one-to-one) if $T(\mathbf{u}_1) = T(\mathbf{u}_2)$ implies $\mathbf{u}_1 = \mathbf{u}_2$, that is, every $\mathbf{v} \in \mathrm{R}(T)$ is the image of exactly one $\mathbf{u} \in \mathbb{R}^n$; and T is *surjective* (onto) if every $\mathbf{v} \in \mathbb{R}^m$ is the image of some $\mathbf{u} \in \mathbb{R}^n$, that is, $\mathrm{R}(T) = \mathbb{R}^m$.

Theorem (Injectivity)**.** A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is injective if and only if its kernel is trivial, $\ker(T) = \{\mathbf{0}\}$.

Proof. Let T be injective. If $\mathbf{u} \in \ker(T)$, then $T(\mathbf{u}) = \mathbf{0} = T(\mathbf{0})$, hence $\mathbf{u} = \mathbf{0}$. Conversely, let $\ker(T) = \{\mathbf{0}\}$ and $T(\mathbf{u}_1) = T(\mathbf{u}_2)$. Then $T(\mathbf{u}_1 - \mathbf{u}_2) = T(\mathbf{u}_1) - T(\mathbf{u}_2) = \mathbf{0}$, so $\mathbf{u}_1 - \mathbf{u}_2 \in \ker(T)$, that is, $\mathbf{u}_1 = \mathbf{u}_2$. □

In the language of linear systems: if $\mathbf{A}$ is the standard matrix of T, then $T(\mathbf{x}) = \mathbf{v}$ means $\mathbf{Ax} = \mathbf{v}$, and the solution set of a consistent system $\mathbf{Ax} = \mathbf{v}$ is $\mathbf{u} + \mathrm{Null}(\mathbf{A})$ for any particular solution $\mathbf{u}$ (Section 4.2). So the system has a unique solution for every $\mathbf{v} \in \mathrm{R}(T) = \mathrm{Col}(\mathbf{A})$ exactly when $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution.

Theorem (Surjectivity)**.** A linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ with standard matrix $\mathbf{A}$ is surjective if and only if $\mathrm{rank}(T) = m$, that is, $\mathrm{rank}(\mathbf{A}) = m$.

Proof. T is surjective if and only if $\mathrm{R}(T) = \mathbb{R}^m$, and then $\mathrm{rank}(T) = \dim(\mathbb{R}^m) = m$. Conversely, let $\dim(\mathrm{R}(T)) = m$. A basis of $\mathrm{R}(T)$ consists of m linearly independent vectors of $\mathbb{R}^m$; by the Basis Extension Lemma (Section 3.7) it can be completed to a basis of $\mathbb{R}^m$, which has exactly m vectors (Basis Theorem, Section 3.6), so nothing is added and $\mathrm{R}(T) = \mathbb{R}^m$. Finally, $\mathrm{rank}(T) = \mathrm{rank}(\mathbf{A})$ by Proposition 3. □

The two theorems extend the equivalent statements for matrices of full rank (Section 4.2) by one item each.

Theorem (Full Rank Equals Number of Columns)**.** Let $\mathbf{A}$ be an $m \times n$ matrix and $T : \mathbb{R}^n \to \mathbb{R}^m$, $T(\mathbf{u}) = \mathbf{Au}$, the linear transformation defined by $\mathbf{A}$. The following statements are equivalent.

(i) $\operatorname{rank}(\mathbf{A}) = n$: the rank equals the number of columns.
(ii) The rows of $\mathbf{A}$ span $\mathbb{R}^n$: $\operatorname{Row}(\mathbf{A}) = \mathbb{R}^n$.
(iii) The columns of $\mathbf{A}$ are linearly independent.
(iv) The homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution: $\operatorname{Null}(\mathbf{A}) = \{\mathbf{0}\}$.
(v) $\mathbf{A}^T\mathbf{A}$ is an invertible matrix of order n.
(vi) $\mathbf{A}$ has a left inverse: $\mathbf{BA} = \mathbf{I}_n$ for some $n \times m$ matrix $\mathbf{B}$.
(vii) T is injective.

Proof. Statements (i)–(vi) are equivalent by the theorem on full rank equal to the number of columns (Section 4.2). Since $\ker(T) = \operatorname{Null}(\mathbf{A})$ (Proposition 3), the injectivity theorem gives (iv) $\Leftrightarrow$ (vii). □

Theorem (Full Rank Equals Number of Rows)**.** Let $\mathbf{A}$ be an $m \times n$ matrix and $T : \mathbb{R}^n \to \mathbb{R}^m$, $T(\mathbf{u}) = \mathbf{Au}$, the linear transformation defined by $\mathbf{A}$. The following statements are equivalent.

(i) $\operatorname{rank}(\mathbf{A}) = m$: the rank equals the number of rows.
(ii) The columns of $\mathbf{A}$ span $\mathbb{R}^m$: $\operatorname{Col}(\mathbf{A}) = \mathbb{R}^m$.
(iii) The rows of $\mathbf{A}$ are linearly independent.
(iv) The system $\mathbf{Ax} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^m$.
(v) $\mathbf{AA}^T$ is an invertible matrix of order m.
(vi) $\mathbf{A}$ has a right inverse: $\mathbf{AB} = \mathbf{I}_m$ for some $n \times m$ matrix $\mathbf{B}$.
(vii) T is surjective.

Proof. Statements (i)–(vi) are equivalent by the theorem on full rank equal to the number of rows (Section 4.2). Since $\mathrm{R}(T) = \operatorname{Col}(\mathbf{A})$ (Proposition 3), T is surjective if and only if $\operatorname{Col}(\mathbf{A}) = \mathbb{R}^m$; thus (ii) $\Leftrightarrow$ (vii). □

Corollary. Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. If T is injective, then $n \leq m$; if T is surjective, then $n \geq m$. In particular, a bijective linear transformation $\mathbb{R}^n \to \mathbb{R}^m$ exists only for $n = m$.

Proof. Let $\mathbf{A}$ be the standard matrix of T; recall that $\operatorname{rank}(\mathbf{A}) \leq \min\{m, n\}$ (Section 4.2). If T is injective, then $n = \operatorname{rank}(\mathbf{A}) \leq m$; if T is surjective, then $m = \operatorname{rank}(\mathbf{A}) \leq n$. □

Example (Injective but not surjective)**.** Let $T : \mathbb{R}^2 \to \mathbb{R}^3$, $T(x, y)^T = (x + y, x - y, 2x + y)^T$. Row reduction of the augmented matrix of $T(\mathbf{x}) = \mathbf{v}$ gives

$$\left(\begin{array}{cc|c} 1 & 1 & v_1 \\ 1 & -1 & v_2 \\ 2 & 1 & v_3 \end{array}\right) \to \left(\begin{array}{cc|c} 1 & 1 & v_1 \\ 0 & -2 & v_2 - v_1 \\ 0 & 0 & 3v_1 + v_2 - 2v_3 \end{array}\right).$$

So $\mathbf{v} \in \mathrm{R}(T)$ if and only if $3v_1 + v_2 - 2v_3 = 0$; the range is a plane, and T is not surjective. For such $\mathbf{v}$ the solution is unique, $x = (v_1 + v_2)/2$, $y = (v_1 - v_2)/2$; thus T is injective, and $\mathbf{v} = \mathbf{0}$ gives $\ker(T) = \{\mathbf{0}\}$. Accordingly $\mathrm{rank}(\mathbf{A}) = 2 = n$, and a left inverse of $\mathbf{A}$ is

$$\mathbf{B} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T = \frac{1}{14}\begin{pmatrix} 1 & 5 & 4 \\ 4 & -8 & 2 \end{pmatrix}, \quad \mathbf{BA} = \mathbf{I}_2.$$

In the other direction, the transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ of the example on the standard matrix in Section 7.1 has $\mathrm{rank}(T) = 2 = m$ and $\mathrm{nullity}(T) = 1$ (see the previous subsection); it is surjective but not injective.

The Inverse of a Linear Transformation

A transformation $T : V \to W$ is called *invertible* if there exists a transformation $S : W \to V$ such that $T(S(\mathbf{y})) = \mathbf{y} \quad \forall \mathbf{y} \in W$ and $S(T(\mathbf{x})) = \mathbf{x} \quad \forall \mathbf{x} \in V$. In this case S is called the *inverse transformation* for T.

Proposition. If a linear transformation is invertible, then the inverse transformation is also linear.

Proof. Any vectors $\mathbf{y}_1, \mathbf{y}_2 \in W$ can be represented in the form $\mathbf{y}_1 = T(\mathbf{x}_1)$, $\mathbf{y}_2 = T(\mathbf{x}_2)$. Therefore,

$$S(\alpha\mathbf{y}_1 + \beta\mathbf{y}_2) = S(\alpha T(\mathbf{x}_1) + \beta T(\mathbf{x}_2)) = S(T(\alpha\mathbf{x}_1 + \beta\mathbf{x}_2)) = \alpha\mathbf{x}_1 + \beta\mathbf{x}_2.$$

It remains to take into account that

$$\mathbf{x}_1 = S(\mathbf{y}_1), \quad \mathbf{x}_2 = S(\mathbf{y}_2). \qquad \square$$

Theorem. Let $T : V \to W$ be a linear transformation, and let V and W be finite-dimensional spaces of the same dimension. The transformation T is invertible if and only if $\ker(T) = \{\mathbf{0}\}$.

Proof. Let $\dim(V) = \dim(W) = n$. By the rank–nullity theorem, if $\dim(\ker(T)) = 0$, then $\dim(\mathrm{R}(T)) = n$. This means that for each vector $\mathbf{y} \in W$ there exists $\mathbf{x} \in V$ such that $T(\mathbf{x}) = \mathbf{y}$ (T is surjective). Moreover, such a vector $\mathbf{x}$ is unique (otherwise the kernel would contain a nonzero vector), so T is also injective. Define the transformation $S : W \to V$ by the rule $S(\mathbf{y}) = \mathbf{x}$. Then $T(S(\mathbf{y})) = \mathbf{y}$ and $S(T(\mathbf{x})) = \mathbf{x} \Rightarrow S$ is the inverse transformation for T.

Conversely, if it is known that T is an invertible transformation, then its kernel can only be zero (if $\mathbf{x}_1 \neq \mathbf{x}_2$ and $T(\mathbf{x}_1) = T(\mathbf{x}_2)$, this contradicts the invertibility of the transformation T). $\square$

Remark. If a linear transformation $T : V \to W$ is invertible, then necessarily $W = \mathrm{R}(T)$, that is, T is surjective. At the same time, the condition $W = \mathrm{R}(T)$ is not sufficient for the invertibility of T.

Problem 2. Linear transformations S and T are such that the transformation $S + T$ is invertible. Prove that the transformations $P = (S + T)^{-1}S$ and $Q = (S + T)^{-1}T$ commute.

Problem 3. A transformation $T : \mathbb{C}^n \to \mathbb{C}^n$ preserves inner products. Prove that T is an invertible transformation. Must it be linear?

Problem 4. A linear transformation $T : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}$ preserves the determinant of a matrix. Prove that every such transformation is invertible.

Equivalent Statements for Invertibility ADDED FOR MA1522

Theorem (Equivalent Statements for Invertibility)**.** Let $\mathbf{A}$ be a square matrix of order n, and let $T : \mathbb{R}^n \to \mathbb{R}^n$, $T(\mathbf{u}) = \mathbf{Au}$, be the linear transformation defined by $\mathbf{A}$. The following statements are equivalent.

(i) $\mathbf{A}$ is invertible.
(ii) $\mathbf{A}^T$ is invertible.
(iii) (left inverse) $\mathbf{BA} = \mathbf{I}$ for some matrix $\mathbf{B}$.
(iv) (right inverse) $\mathbf{AB} = \mathbf{I}$ for some matrix $\mathbf{B}$.
(v) The reduced row-echelon form of $\mathbf{A}$ is $\mathbf{I}_n$.
(vi) $\mathbf{A}$ is a product of elementary matrices.
(vii) The homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution.
(viii) The system $\mathbf{Ax} = \mathbf{b}$ has a unique solution for every $\mathbf{b} \in \mathbb{R}^n$.
(ix) $\det(\mathbf{A}) \neq 0$.
(x) The columns of $\mathbf{A}$ are linearly independent; so are its rows.
(xi) The columns of $\mathbf{A}$ span $\mathbb{R}^n$; so do its rows.
(xii) $\mathrm{rank}(\mathbf{A}) = n$ ($\mathbf{A}$ has full rank).
(xiii) $\mathrm{nullity}(\mathbf{A}) = 0$.
(xiv) 0 is not an eigenvalue of $\mathbf{A}$.
(xv) T is injective.
(xvi) T is surjective.

Proof. The equivalence of (i)–(xiv) is the theorem on equivalent statements for invertibility in Section 6.1. For $m = n$, the two theorems on full rank above give (vii) $\Leftrightarrow$ (xv) (the homogeneous system has only the trivial solution if and only if T is injective) and (xi) $\Leftrightarrow$ (xvi) (the columns of $\mathbf{A}$ span $\mathbb{R}^n$ if and only if T is surjective). □

If $\mathbf{A}$ is invertible, then T is bijective and its inverse transformation is multiplication by $\mathbf{A}^{-1}$: $T(\mathbf{A}^{-1}\mathbf{v}) = \mathbf{v}$ and $\mathbf{A}^{-1}T(\mathbf{u}) = \mathbf{u}$. Conversely, let T have an inverse transformation S (by Section 7.1, this is the same as T being bijective). Then S is linear by the proposition above; if $\mathbf{B}$ is its standard matrix, then $\mathbf{ABu} = T(S(\mathbf{u})) = \mathbf{u}$ for all $\mathbf{u}$, so $\mathbf{AB} = \mathbf{I}$ and $\mathbf{A}$ is invertible with $\mathbf{B} = \mathbf{A}^{-1}$. Thus a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^n$ is bijective if and

only if its standard matrix $\mathbf{A}$ is invertible, and the standard matrix of T^{-1} is then $\mathbf{A}^{-1}$.

Example. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$, $T(x, y, z)^T = (x + 2y, y + z, x + 2y + z)^T$. Its standard matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix} \quad \text{has} \quad \det(\mathbf{A}) = 1, \quad \mathbf{A}^{-1} = \begin{pmatrix} -1 & -2 & 2 \\ 1 & 1 & -1 \\ -1 & 0 & 1 \end{pmatrix}.$$

Hence T is both injective and surjective: for every $\mathbf{v} = (v_1, v_2, v_3)^T$ the equation $T(\mathbf{x}) = \mathbf{v}$ has the unique solution

$$\mathbf{x} = T^{-1}(\mathbf{v}) = \mathbf{A}^{-1}\mathbf{v} = \begin{pmatrix} -v_1 - 2v_2 + 2v_3 \\ v_1 + v_2 - v_3 \\ -v_1 + v_3 \end{pmatrix}.$$